Landmark XVI-N — Appendix companion

Computational listings, notes archive, and provenance
Companion to the science volume (r12)

XVI-N r12 appendix volume

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Appendix volume — not the scientific claims document. This PDF is the companion to `XVI-N-science.pdf`. It collects:

- Part II — computational appendix (scripts and certificates);
- Part III — notes-source archive;
- Part IV — provenance manuscript listings and the release manifest.

Scientific theorems, statuses, and the certified small-mass interval are stated only in the science volume (`XVI-N-science.pdf`) and in the combined release (`XVI-N.pdf`). Do not treat this appendix as a standalone claims document.

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Part I
Computational appendix

1 Computational appendix (r11 current inventory)

All certificates below are regenerated under `build/outputs/` by `tools/run_r10.py`. Placeholders are forbidden.

1.1 Reproducibility

Listing 1: package_paths.py

```python
"""XVI-N-r10 package paths. Sole certificate root: build/outputs/."""
from __future__ import annotations
import json
import os
from pathlib import Path

_HERE = Path(__file__).resolve().parent
PACKAGE_ROOT = _HERE.parent
if not (PACKAGE_ROOT / "manuscript").is_dir():
    for cand in _HERE.parents:
        if (cand / "manuscript" / "body").is_dir():
            PACKAGE_ROOT = cand
            break

MANUSCRIPT_DIR = PACKAGE_ROOT / "manuscript"
BODY_TEX = MANUSCRIPT_DIR / "body" / "XVI-N-body.tex"
NOTES_DIR = MANUSCRIPT_DIR / "notes"
CODE_DIR = _HERE
BUILD_DIR = PACKAGE_ROOT / "build"
OUTPUTS_DIR = BUILD_DIR / "outputs"
LOGS_DIR = BUILD_DIR / "logs"

# Subsystem name from a script path under code/
_SUB_MAP = {
    "insertion": "insertion",
    "classical_bv": "classical_bv",
    "conventions": "conventions",
    "normalization": "normalization",
    "prep": "prep",
    "certificates": "certificates",
}


def ensure_output_dirs() -> None:
    for sub in _SUB_MAP.values():
        (OUTPUTS_DIR / sub).mkdir(parents=True, exist_ok=True)
    (OUTPUTS_DIR / "regression").mkdir(parents=True, exist_ok=True)
    LOGS_DIR.mkdir(parents=True, exist_ok=True)


def out_dir(subdir: str) -> Path:
    """Sole writable/readable certificate directory for a subsystem."""
    ensure_output_dirs()
    d = OUTPUTS_DIR / subdir
    d.mkdir(parents=True, exist_ok=True)
    return d


def out_dir_for_script(script_file: str | Path) -> Path:
    """Infer subsystem from script location under code/."""
    p = Path(script_file).resolve()
    # .../code/<sub>/file.py
    try:
        rel = p.relative_to(CODE_DIR)
        sub = rel.parts[0] if rel.parts else "misc"
    except ValueError:
        sub = "misc"
    if sub not in _SUB_MAP and sub != "misc":
        # script may live directly under code/
        sub = "certificates"
    return out_dir(sub if sub in _SUB_MAP or sub == "misc" else "certificates")
```

```

def output_path(*parts: str) -> Path:
    ensure_output_dirs()
    p = OUTPUTS_DIR.joinpath(*parts)
    p.parent.mkdir(parents=True, exist_ok=True)
    return p


def write_json(subdir: str, name: str, obj) -> Path:
    """Write a certificate ONLY under build/outputs/<subdir>/."""
    path = out_dir(subdir) / name
    with open(path, "w") as f:
        json.dump(obj, f, indent=2, default=str)
    return path


def read_json(subdir: str, name: str):
    """Read a certificate ONLY from build/outputs/<subdir>/. No code-dir fallback."""
    path = out_dir(subdir) / name
    if not path.is_file():
        raise FileNotFoundError(
            f"required certificate missing under sole output root: {path}"
        )
    with open(path) as f:
        return json.load(f)


def assert_not_stub(path: Path | str) -> None:
    s = str(path)
    stub = "/" + "workspace" + "/" + "XVI-N.tex"
    if s == stub or s.endswith("/XVI-N.tex") and "manuscript" not in s and "r10" not in s and "combined" not in s:
        if "body" not in s:
            raise RuntimeError("refusing hard-coded stub " + s)


# Back-compat aliases
ROOT = PACKAGE_ROOT
```

Listing 2: run_r10.py

```

#!/usr/bin/env python3
"""XVI-N r12 release pipeline.

Sole cert root: build/outputs/. No placeholders. sys.executable only.
reference_outputs/ is immutable during normal runs (use update-reference).
Manifest order: pre-PDF provenance -> listings -> pdflatex x2 -> external manifest.
"""
from __future__ import annotations
import hashlib, json, os, re, shutil, subprocess, sys, time
from pathlib import Path

ROOT = Path(__file__).resolve().parents[1]
CODE = ROOT / "code"
MS = ROOT / "manuscript"
BUILD = ROOT / "build"
LOGS = BUILD / "logs"
OUT = BUILD / "outputs"
LIST = BUILD / "listing_src"
PDFDIR = BUILD / "pdf"
REF = ROOT / "reference_outputs"
PROV = ROOT / "provenance"
HIST = ROOT / "historical"
PY = sys.executable
REVISION = "r12"

REQUIRED_LISTINGS = [
    "code/package_paths.py",
    "tools/run_r10.py",
    "requirements.lock",
    "ENVIRONMENT.md",
    "tests/test_stub_path_absence.py",
    "tests/test_schema_failure.py",
    "tests/test_convention_consistency.py",
    "tests/test_repro_and_failure.py",
    "code/conventions/source_of_truth.py",
    "outputs/conventions/convention_coefficients.json",
    "code/conventions/check_ms_coefficients.py",
    "outputs/conventions/ms_coefficient_check.json",
    "code/classical_bv/classical_bv_check.py",
```

```
    "outputs/classical_bv/classical_bv_check_out.txt",
    "outputs/classical_bv/brst_normalization_certificates.json",
    "outputs/classical_bv/qd2_antifield_certificates.json",
    "outputs/classical_bv/qd2_general_jets_certificates.json",
    "outputs/classical_bv/generator_antibracket_certificates.json",
    "outputs/classical_bv/field_only_projection_certificates.json",
    "outputs/prep/af0_noether_certificates.json",
    "outputs/prep/prep_certificates.json",
    "outputs/normalization/free_hessian_check_out.txt",
    "code/insertion/nins3_status_composer.py",
    "outputs/insertion/completion_space.json",
    "outputs/insertion/insertion_contract.json",
    "outputs/insertion/nins3_4D_special_family_certificates.json",
    "outputs/insertion/nins3_4D_validated_enclosure.json",
    "code/insertion/nins3_small_mass_bound.py",
    "outputs/insertion/nins3_small_mass_bound.json",
    "outputs/insertion/nins3_decision_certificates.json",
    "outputs/insertion/nins3_support_lemma_certificates.json",
    "outputs/insertion/onshell_admission_certificates.json",
    "outputs/regression/regression_N_out.txt",
    "logs/pipeline_tail.txt",
    "provenance/RELEASE_MANIFEST_PREPDF.txt",
]

# Required producers: must exist BEFORE the phase runs.
REQUIRED_CLASSICAL = [
    "brst_normalization.py",
    "qd2_antifield_test.py",
    "qd2_general_jets.py",
    "extended_action_master.py",
    "classical_bv_check.py",
    "generator_antibracket_test.py",
    "field_only_projection_check.py",
    "s_psi_coefficients.py",
]
REQUIRED_PREP = [
    "prepared_noether_af0.py",
    "independent_weighted_euler.py",
    "prepared_gauge_variation.py",
]
REQUIRED_INSERTION = [
    "export_nins12_contracts.py",
    "root_adjoint_H.py",
    "root_adjoint_Ddiv_Dgf_check.py",
    "nins3_4D_special_family_analytic.py",
    "nins3_1p1_corrected.py",
    "nins3_support_lemma_check.py",
    "onshell_admission_check.py",
    "onshell_structural_check.py",
    "parity_control_symmetric.py",
    "nins3_4D_kirchhoff_corrected.py",
    "insertion_variation.py",
    "nins3_4D_validated_enclosure.py",
    "nins3_small_mass_bound.py",
    "nins3_status_composer.py",
]

# Keys ignored when comparing scientific JSON to immutable reference.
_COMPARE_IGNORE_KEYS = {
    "seconds", "elapsed", "timestamp", "write_path", "cert_path",
    "output_root", "interpreter", "host", "cwd", "path", "paths",
}


def log(msg: str):
    print(msg, flush=True)
    LOGS.mkdir(parents=True, exist_ok=True)
    with open(LOGS / "pipeline.log", "a") as f:
        f.write(msg + "\n")


def run(cmd, cwd=None, log_name=None):
    cwd = cwd or ROOT
    log(f"$ {' '.join(cmd)}  (cwd={cwd})")
    t0 = time.time()
    env = os.environ.copy()
    env["PYTHONPATH"] = str(CODE) + os.pathsep + env.get("PYTHONPATH", "")
    p = subprocess.run(cmd, cwd=str(cwd), capture_output=True, text=True, env=env)
    elapsed = time.time() - t0
    if log_name:
        (LOGS / log_name).write_text(
```

```
            f"cmd: {cmd}\nrc: {p.returncode}\nseconds: {elapsed}\n\nSTDOUT:\n{p.stdout}\n\nSTDERR:\n{p.stderr}\n"
        )
    if p.returncode != 0:
        log(f"FAIL rc={p.returncode} ({elapsed:.1f}s)")
        if p.stdout:
            log(p.stdout[-2000:])
        if p.stderr:
            log(p.stderr[-2000:])
        raise SystemExit(p.returncode)
    log(f"OK ({elapsed:.1f}s)")
    return p


def sha256(path: Path) -> str:
    h = hashlib.sha256()
    with open(path, "rb") as f:
        for chunk in iter(lambda: f.read(1 << 20), b""):
            h.update(chunk)
    return h.hexdigest()


def require_scripts(subdir: Path, names: list[str], phase: str) -> None:
    missing = [n for n in names if not (subdir / n).is_file()]
    if missing:
        raise SystemExit(
            f"REQUIRED SCRIPT MISSING before {phase}:\n  "
            + "\n  ".join(str(subdir / n) for n in missing)
        )


def clean_build():
    if BUILD.exists():
        log(f"Removing {BUILD}")
        shutil.rmtree(BUILD)
    for d in (OUT, LOGS, LIST, PDFDIR):
        d.mkdir(parents=True, exist_ok=True)
    purge_code_certs(pre=True)


def purge_code_certs(pre: bool = False):
    """Remove generated certificate artifacts under code/."""
    killed = []
    for pth in CODE.rglob("*"):
        if not pth.is_file():
            continue
        if pth.suffix == ".py":
            continue
        # generated certs / outs
        name = pth.name
        if (
            pth.suffix == ".json"
            or name.endswith("_out.txt")
            or "certificate" in name.lower()
            or name.endswith("_certificates.json")
        ):
            pth.unlink()
            killed.append(str(pth.relative_to(ROOT)))
    for rel in killed:
        log(f"{'pre-' if pre else 'post-'}purge code cert: {rel}")


def assert_no_code_certs():
    """Post-run audit: no generated cert JSON/txt under code/."""
    bad = []
    for pth in CODE.rglob("*"):
        if not pth.is_file() or pth.suffix == ".py":
            continue
        name = pth.name
        if (
            pth.suffix == ".json"
            or name.endswith("_out.txt")
            or "certificate" in name.lower()
        ):
            bad.append(str(pth.relative_to(ROOT)))
    if bad:
        raise SystemExit(
            "POST-RUN AUDIT FAIL: generated certs under code/:\n  " + "\n  ".join(bad)
        )
    log("post-run code/ cert audit: PASS (none found)")


```

```
def ascii_copy(src: Path, dst: Path):
    dst.parent.mkdir(parents=True, exist_ok=True)
    if src.is_dir():
        return
    raw = src.read_bytes()
    try:
        text = raw.decode("utf-8")
    except UnicodeDecodeError:
        text = raw.decode("utf-8", errors="replace")
    dst.write_text("".join(ch if ord(ch) < 128 else "?" for ch in text))


def build_listings():
    if LIST.exists():
        shutil.rmtree(LIST)
    LIST.mkdir(parents=True)
    ascii_copy(ROOT / "requirements.lock", LIST / "requirements.lock")
    if (ROOT / "ENVIRONMENT.md").is_file():
        ascii_copy(ROOT / "ENVIRONMENT.md", LIST / "ENVIRONMENT.md")
    for p in (MS / "notes").glob("*"):
        if p.is_file():
            ascii_copy(p, LIST / "notes" / p.name)
    for p in PROV.rglob("*"):
        if p.is_file():
            ascii_copy(p, LIST / "provenance" / p.relative_to(PROV))
    for p in CODE.rglob("*"):
        if p.is_file() and p.suffix in {".py", ".md", ".tex", ".txt"}:
            ascii_copy(p, LIST / "code" / p.relative_to(CODE))
    for p in (ROOT / "tools").glob("*.py"):
        ascii_copy(p, LIST / "tools" / p.name)
    for p in (ROOT / "tests").glob("*.py"):
        ascii_copy(p, LIST / "tests" / p.name)
    for p in OUT.rglob("*"):
        if p.is_file():
            ascii_copy(p, LIST / "outputs" / p.relative_to(OUT))
    plog = LOGS / "pipeline.log"
    if plog.is_file():
        lines = plog.read_text().splitlines()
        (LIST / "logs").mkdir(parents=True, exist_ok=True)
        (LIST / "logs" / "pipeline_tail.txt").write_text("\n".join(lines[-80:]) + "\n")
    missing = [rel for rel in REQUIRED_LISTINGS if not (LIST / rel).is_file()]
    if missing:
        raise SystemExit("LISTING FAIL missing required files:\n  " + "\n  ".join(missing))
    stub_marker = "% " + "placeholder generated for r10 listing completeness"
    for pth in (LIST / "outputs").rglob("*"):
        if pth.is_file():
            txt = pth.read_text(errors="replace")
            if stub_marker in txt or txt.strip().startswith("% placeholder"):
                raise SystemExit(f"LISTING FAIL stub certificate in {pth}")
    link = MS / "listing_src"
    if link.exists() or link.is_symlink():
        link.unlink()
    link.symlink_to(Path("..") / "build" / "listing_src")
    log(f"listings OK ({sum(1 for _ in LIST.rglob('*') if _.is_file())} files)")


def _strip_for_compare(obj):
    if isinstance(obj, dict):
        return {
            k: _strip_for_compare(v)
            for k, v in obj.items()
            if k not in _COMPARE_IGNORE_KEYS
        }
    if isinstance(obj, list):
        return [_strip_for_compare(x) for x in obj]
    return obj


def verify_against_reference():
    """Compare fresh scientific JSON under build/outputs to immutable reference_outputs/."""
    if not REF.is_dir():
        log("VERIFY WARN: reference_outputs/ missing; skip compare (first seal?)")
        return
    drifts = []
    for fresh in OUT.rglob("*.json"):
        rel = fresh.relative_to(OUT)
        refp = REF / rel
        if not refp.is_file():
            drifts.append(f"missing in reference: {rel}")
            continue
        try:
```

```
            a = _strip_for_compare(json.loads(fresh.read_text()))
            b = _strip_for_compare(json.loads(refp.read_text()))
        except Exception as e:
            drifts.append(f"parse error {rel}: {e}")
            continue
        # Allow revision string bump r11->r12 without failing science compare on that key alone
        def norm_rev(o):
            if isinstance(o, dict):
                o = dict(o)
                if o.get("revision") in ("r11", "r12"):
                    o["revision"] = "REV"
                if o.get("schema") in ("r11", "r12", "r12_strict_Q_D_squared"):
                    o["schema"] = "SCHEMA"
                return {k: norm_rev(v) for k, v in o.items()}
            if isinstance(o, list):
                return [norm_rev(x) for x in o]
            return o
        if norm_rev(a) != norm_rev(b):
            drifts.append(f"drift: {rel}")
    if drifts:
        raise SystemExit(
            "VERIFY FAIL vs immutable reference_outputs/:\n  " + "\n  ".join(drifts[:40])
        )
    log("verify vs reference_outputs: PASS")


def update_reference():
    """Maintainer-only: copy approved build/outputs into reference_outputs/."""
    if not OUT.is_dir() or not any(OUT.rglob("*.json")):
        raise SystemExit("update-reference FAIL: build/outputs empty")
    if REF.exists():
        shutil.rmtree(REF)
    shutil.copytree(OUT, REF)
    log(f"update-reference: copied {OUT} -> {REF}")
    n = sum(1 for _ in REF.rglob("*.json"))
    log(f"update-reference: {n} json files")
    return 0


def write_prepdf_manifest(reg_counts=None, small_mass=None):
    """Pre-PDF provenance: revision/status/regression; NO PDF hash or page count."""
    HIST.mkdir(parents=True, exist_ok=True)
    # Move stale embedded manifests to historical/
    for name in ("RELEASE_MANIFEST.txt", "RELEASE_MANIFEST_PREPDF.txt"):
        src = PROV / name
        if src.is_file():
            stamp = time.strftime("%Y%m%d-%H%M%S")
            dst = HIST / f"{name}.r11-or-earlier.{stamp}.historical"
            # only archive if content looks stale (r10/r11 or has PDF sha)
            txt = src.read_text()
            if "r10" in txt or "r11" in txt or "PDF sha256" in txt or "pages:" in txt.lower():
                shutil.copy2(src, dst)
                hdr = (
                    f"# HISTORICAL (archived {stamp}); not the active pre-PDF record.\n"
                    f"# Original path: provenance/{name}\n\n"
                )
                (HIST / f"NOTE_{name}.txt").write_text(
                    hdr + "Stale provenance archived before r12 pre-PDF rewrite.\n"
                )
                log(f"archived stale {name} -> {dst.name}")
    xm = sha256(PROV / "XVI-M.tex")
    assert xm == "41aa109935f40ddee3109de62c46329c189b5408007970cc90ed5611dc4739ea"
    lines = [
        f"XVI-N {REVISION} RELEASE MANIFEST (PRE-PDF; embedded in PDF; NO PDF hash / final page count)",
        f"date: {time.strftime('%Y-%m-%d')}",
        f"revision: {REVISION}",
        "massless_theorem: conditional_on_inherited_identities (Route A)",
        "route_B: not_claimed_discharged",
        "arbitrary_prescribed_m_gt_0: unresolved",
        f"XVI-M.tex sha256 (unchanged): {xm}",
        "output_root: build/outputs",
        "interpreter: sys.executable",
        "note: final page count and PDF sha256 appear ONLY in RELEASE_MANIFEST_EXTERNAL.txt",
    ]
    if reg_counts:
        lines.append(
            f"regression_math: {reg_counts['math_passed']} pass / {reg_counts['math_failed']} fail"
        )
        lines.append(
            f"regression_pack: {reg_counts['pack_passed']} pass / {reg_counts['pack_failed']} fail"
        )
```

```
    if small_mass:
        lines.append(
            f"small-mass: {small_mass.get('advertised_interval')} status={small_mass.get('status')}"
        )
    lines += [
        "",
        f"{sha256(MS / 'XVI-N.tex')}  manuscript/XVI-N.tex",
        f"{sha256(MS / 'body' / 'XVI-N-body.tex')}  manuscript/body/XVI-N-body.tex",
        f"{sha256(MS / 'notes' / 'small_mass_estimate.tex')}  manuscript/notes/small_mass_estimate.tex",
        f"{xm}  provenance/XVI-M.tex",
    ]
    text = "\n".join(lines) + "\n"
    (PROV / "RELEASE_MANIFEST_PREPDF.txt").write_text(text)
    (PROV / "RELEASE_MANIFEST.txt").write_text(text)
    log("wrote pre-PDF provenance RELEASE_MANIFEST_PREPDF.txt")


def bump_banners():
    tex = (MS / "XVI-N.tex").read_text()
    tex = re.sub(r"XVI-N r1[01] release", f"XVI-N {REVISION} release", tex)
    tex = re.sub(
        r"r1[01] release; Route B unresolved",
        f"{REVISION} release; Route B unresolved",
        tex,
    )
    (MS / "XVI-N.tex").write_text(tex)


def run_producers():
    run([PY, str(ROOT / "tests" / "test_stub_path_absence.py")], log_name="test_stub.txt")
    run([PY, str(ROOT / "tests" / "test_convention_consistency.py")], log_name="test_conv.txt")
    run([PY, str(ROOT / "tests" / "test_schema_failure.py")], log_name="test_schema.txt")
    run([PY, str(ROOT / "tests" / "test_repro_and_failure.py"), "--unit"],
        log_name="test_repro_unit.txt")

    run([PY, str(CODE / "conventions" / "source_of_truth.py")],
        cwd=CODE / "conventions", log_name="01_sot.txt")
    run([PY, str(CODE / "conventions" / "check_ms_coefficients.py")],
        cwd=CODE / "conventions", log_name="01_ms.txt")

    require_scripts(CODE / "classical_bv", REQUIRED_CLASSICAL, "classical_bv")
    for name in REQUIRED_CLASSICAL:
        script = CODE / "classical_bv" / name
        run([PY, str(script)], cwd=script.parent, log_name=f"02_{name}.txt")

    run([PY, str(CODE / "normalization" / "free_hessian_check.py")],
        cwd=CODE / "normalization", log_name="02_hessian.txt")
    hess_out = OUT / "normalization" / "free_hessian_check_out.txt"
    if not hess_out.is_file():
        raise SystemExit(f"missing free_hessian out under build/outputs: {hess_out}")
    if "build/outputs" not in hess_out.read_text() and "normalization" not in hess_out.read_text():
        # write_path should mention build outputs path
        log(f"WARN hessian out content:\n{hess_out.read_text()[:500]}")

    require_scripts(CODE / "prep", REQUIRED_PREP, "prep")
    for name in REQUIRED_PREP:
        script = CODE / "prep" / name
        run([PY, str(script)], cwd=script.parent, log_name=f"03_{name}.txt")
    prep_c = OUT / "prep" / "prep_certificates.json"
    if not prep_c.is_file():
        raise SystemExit(
            f"HARD FAIL: prep_certificates.json missing after prep producers: {prep_c}"
        )

    require_scripts(CODE / "insertion", REQUIRED_INSERTION, "insertion")
    for name in REQUIRED_INSERTION:
        script = CODE / "insertion" / name
        run([PY, str(script)], cwd=script.parent, log_name=f"03_{name}.txt")

    run([PY, str(CODE / "regression_N.py")], cwd=CODE, log_name="07_regression.txt")
    assert_no_code_certs()


def main(argv=None):
    argv = list(argv if argv is not None else sys.argv[1:])
    if "update-reference" in argv or "--update-reference" in argv:
        LOGS.mkdir(parents=True, exist_ok=True)
        return update_reference()
    if "verify" in argv or "--verify" in argv:
        LOGS.mkdir(parents=True, exist_ok=True)
        verify_against_reference()
```

```
        return 0

    LOGS.mkdir(parents=True, exist_ok=True)
    (LOGS / "pipeline.log").write_text(
        f"{REVISION} pipeline start {time.ctime()} PY={PY}\n"
    )
    bad = []
    banned = "/" + "workspace/tmp/venv/bin/python"
    for p in CODE.rglob("*.py"):
        if banned in p.read_text(errors="replace"):
            bad.append(str(p))
    if bad:
        raise SystemExit("portability FAIL hard-coded venv in " + str(bad))

    if "--no-clean" not in argv:
        clean_build()
    else:
        for d in (OUT, LOGS, LIST, PDFDIR):
            d.mkdir(parents=True, exist_ok=True)

    run_producers()

    # Verify vs immutable reference (skip if --skip-verify or empty ref on first seal)
    if "--skip-verify" not in argv:
        if REF.is_dir() and any(REF.rglob("*.json")):
            verify_against_reference()
        else:
            log("no reference_outputs yet; skip verify (use update-reference after seal)")

    bump_banners()

    b2 = json.load(open(OUT / "insertion" / "nins3_small_mass_bound.json"))
    reg = json.load(open(OUT / "regression" / "regression_N_out.json"))
    reg_counts = {
        "math_passed": len(reg.get("math_passes", [])),
        "math_failed": len(reg.get("math_failures", [])),
        "pack_passed": len(reg.get("packaging_passes", [])),
        "pack_failed": len(reg.get("packaging_failures", [])),
    }
    # 1. Pre-PDF provenance (no PDF hash / pages)
    write_prepdf_manifest(reg_counts=reg_counts, small_mass=b2)
    # 2. Listings (embed pre-PDF provenance)
    build_listings()
    # 3. Compile
    for i in (1, 2):
        run(["pdflatex", "-interaction=nonstopmode", "XVI-N.tex"],
            cwd=MS, log_name=f"09_pdflatex_{i}.txt")
    pdf_src = MS / "XVI-N.pdf"
    PDFDIR.mkdir(parents=True, exist_ok=True)
    shutil.copy2(pdf_src, PDFDIR / "XVI-N.pdf")
    # 4. External final manifest
    xm = sha256(PROV / "XVI-M.tex")
    assert xm == "41aa109935f40ddee3109de62c46329c189b5408007970cc90ed5611dc4739ea"
    info = subprocess.check_output(["pdfinfo", str(PDFDIR / "XVI-N.pdf")], text=True)
    pages = int([ln for ln in info.splitlines() if ln.startswith("Pages:")][0].split()[1])
    pdf_hash = sha256(PDFDIR / "XVI-N.pdf")
    status = {
        "revision": REVISION,
        "pass": True,
        "massless_theorem": "conditional_on_inherited_identities",
        "route_A": "discharged",
        "route_B": "not_claimed_discharged",
        "arbitrary_prescribed_m_gt_0": "unresolved",
        "small_mass": {
            "status": b2["status"],
            "advertised_interval": b2["advertised_interval"],
            "C_hi": b2["C_hi"],
            "C_hi_le_17843": b2["C_hi_le_17843"],
            "exact_rational_C_m2_lt_W": b2.get("exact_rational_C_m2_lt_W"),
        },
        "XVI-M_sha256": xm,
        "pdf_pages": pages,
        "pdf_sha256_external": pdf_hash,
        "regression": {**reg_counts, "log": "build/logs/07_regression.txt"},
        "output_root": "build/outputs",
        "interpreter": "sys.executable",
        "reference_outputs": "immutable_during_pipeline",
        "completion_contract": "H4_insertion_class_I_t",
        "blockers_fixed": [
            "completion_contract_H4",
            "no_prep_fabricate",
```

```
            "required_script_precheck",
            "classical_bv_strict_schema",
            "free_hessian_build_outputs_only",
            "post_run_code_cert_audit",
            "prepdf_then_external_manifest",
            "immutable_reference_outputs",
            "K1_source_sup_norm_wording",
        ],
    }
    # Honest pass: confirm contract keys
    cs = json.load(open(OUT / "insertion" / "completion_space.json"))
    ic = json.load(open(OUT / "insertion" / "insertion_contract.json"))
    if "prepared S_D" in cs.get("ansatz", "") or "completion of prepared" in cs.get("ansatz", ""):
        status["pass"] = False
        status["fail_reason"] = "completion_space ansatz still mentions prepared S_D"
    if ic.get("Q_U", "").startswith("P_O") or "field-only projection" == ic.get("Q_U"):
        # Q_U must be Q_inv; projection is separate
        if "Q_U = Q_inv" not in ic.get("Q_U", "") and "Q_inv" not in ic.get("Q_U", ""):
            status["pass"] = False
            status["fail_reason"] = "Q_U not defined as Q_inv"
    (ROOT / "RELEASE_STATUS.json").write_text(json.dumps(status, indent=2) + "\n")
    man = [
        f"XVI-N {REVISION} RELEASE MANIFEST (EXTERNAL; post-PDF; PDF hash NOT embedded in PDF)",
        f"date: {time.strftime('%Y-%m-%d')}",
        f"pages: {pages}",
        f"PDF sha256 (external): {pdf_hash}",
        f"XVI-M.tex sha256 (unchanged): {xm}",
        f"regression_math: {reg_counts['math_passed']} pass / {reg_counts['math_failed']} fail",
        f"regression_pack: {reg_counts['pack_passed']} pass / {reg_counts['pack_failed']} fail",
        f"small-mass: {b2['advertised_interval']} status={b2['status']}",
        "Route B: not claimed discharged",
        "arbitrary m>0: unresolved",
        "output_root: build/outputs",
        "reference_outputs: immutable during normal pipeline",
        "",
        f"{sha256(MS / 'XVI-N.tex')}  manuscript/XVI-N.tex",
        f"{sha256(MS / 'body' / 'XVI-N-body.tex')}  manuscript/body/XVI-N-body.tex",
        f"{sha256(MS / 'notes' / 'small_mass_estimate.tex')}  manuscript/notes/small_mass_estimate.tex",
        f"{xm}  provenance/XVI-M.tex",
    ]
    (ROOT / "RELEASE_MANIFEST_EXTERNAL.txt").write_text("\n".join(man) + "\n")
    # Do NOT overwrite provenance/RELEASE_MANIFEST.txt with PDF hash
    log("RELEASE_STATUS.json written")
    log(json.dumps(status, indent=2))
    return 0 if status.get("pass") else 1


if __name__ == "__main__":
    raise SystemExit(main())
```

Listing 3: requirements.lock

```
# Python package pins for XVI-N (r12). System tools: see ENVIRONMENT.md.
mpmath==1.3.0
numpy==2.5.3
sympy==1.14.0
```

Listing 4: test_stub_path_absence.py

```
#!/usr/bin/env python3
"""No current check may open the top-level XVI-N.tex stub as scientific source."""
import sys
from pathlib import Path
ROOT = Path(__file__).resolve().parents[1]
bad = []
for p in (ROOT / "code").rglob("*.py"):
    if p.name == "package_paths.py":
        continue  # may mention stub only to refuse it
    t = p.read_text(errors="replace")
    if "/workspace/XVI-N.tex" in t:
        bad.append(str(p.relative_to(ROOT)))
if bad:
    print("FAIL stub refs:", bad)
    raise SystemExit(1)
sys.path.insert(0, str(ROOT / "code"))
from package_paths import assert_not_stub, BODY_TEX
try:
    assert_not_stub("/" + "workspace" + "/" + "XVI-N.tex")
```

```
    print("FAIL assert_not_stub did not raise")
    raise SystemExit(1)
except RuntimeError:
    pass
assert BODY_TEX.is_file()
print("PASS test_stub_path_absence; BODY_TEX=", BODY_TEX)
```

Listing 5: test_schema_failure.py

```
#!/usr/bin/env python3
"""Malformed/old small-mass schemas must fail the composer validator clearly."""
import json, sys, tempfile
from pathlib import Path
ROOT = Path(__file__).resolve().parents[1]
sys.path.insert(0, str(ROOT / "code"))
sys.path.insert(0, str(ROOT / "code" / "insertion"))
from nins3_status_composer import validate_b2

def expect_fail(b2, label):
    try:
        validate_b2(b2)
    except SystemExit as e:
        print("OK fail", label, str(e)[:80])
        return
    raise AssertionError(f"expected schema fail for {label}")

# legacy keys only
expect_fail({"pass": True, "C": 1, "WO_lower": 1, "m_star": 1e-4}, "legacy")
# missing fields
expect_fail({"pass": True, "status": "certified"}, "missing")
# wrong status
bad = {k: True for k in
       ["revision","pass","status","C_hi","C_hi_le_17843","WO_lo",
        "m_star_advertised","advertised_interval","conditional_on","instance_family"]}
bad["status"] = "candidate"
bad["advertised_interval"] = "0 <= m < 1.3715e-4"
bad["C_hi"] = "1"
bad["WO_lo"] = "1"
bad["m_star_advertised"] = "1"
bad["revision"] = "r10"
bad["conditional_on"] = "x"
bad["instance_family"] = "y"
expect_fail(bad, "bad status")
print("PASS test_schema_failure")
```

Listing 6: test_convention_consistency.py

```
#!/usr/bin/env python3
import sys
from pathlib import Path
ROOT = Path(__file__).resolve().parents[1]
body = (ROOT / "manuscript" / "body" / "XVI-N-body.tex").read_text()
sheet = (ROOT / "manuscript" / "notes" / "convention_sheet.md").read_text()
assert r"(S,\Phi^A)=-\delta^R S/\delta\Phi^*_A" in body
assert "false for odd" in body
assert "negative" in sheet.lower()
assert "S_D = S_inv" in sheet
assert "Hessian" in sheet and ("m" in sheet)
gen = (ROOT / "code" / "classical_bv" / "extended_action_master.py").read_text()
assert "c^*" in gen or "c_star" in gen or "c*" in gen
assert "-kappa" in gen.lower() or r"-\kappa" in gen
print("PASS test_convention_consistency")
```

1.2 Conventions

Listing 7: source_of_truth.py

```
#!/usr/bin/env python3
"""
Single source of truth for XVI-N odd couplings, expanded S_af, vertex table.
Emits LaTeX snippets and JSON coefficients. Regression asserts MS == this.
"""
from __future__ import annotations
import json, os

```

```python
import sys as _pkg_sys
from pathlib import Path as _PkgPath
_pkg_sys.path.insert(0, str(_PkgPath(__file__).resolve().parents[1]))
from package_paths import out_dir, write_json, read_json
OUT = str(out_dir("conventions"))
A = "1/2"  # a = 1/2

# Couplings: S_af = int( h* sh + phi* sphi - c* sc - cbar* s cbar )
# With sc = kappa a (c??c), s cbar = i b:
#   -c* sc = -kappa a c*(c??c)
#   -cbar* s cbar = -i cbar* b
COUPLINGS = {
    "h_star_sh": "+1",
    "phi_star_sphi": "+1",
    "c_star_sc": "-1",       # overall minus in S_af
    "cbar_star_scbar": "-1",
}

EXPANDED = {
    "h_star_R0": "+1",                          # int h* R0(c)
    "h_star_L": f"+kappa*{A}",                  # kappa a int h* L_c h
    "phi_star": f"+kappa*{A}",                  # kappa a int phi* (c??phi)
    "c_star_cc": f"-kappa*{A}",                 # -kappa a int c*(c??c)  *** FIXED SIGN
    "cbar_star_b": "-i",                        # -i int cbar* b         *** FIXED SIGN
}

VERTICES = [
    {"id": "V2_grav", "order": "kappa^0", "fields": "hh", "factor": "K=-\\square\\mathcal P"},
    {"id": "V2_ghost", "order": "kappa^0", "fields": "\\bar c c", "factor": "i\\square"},
    {"id": "V3_grav_prep", "order": "kappa^1", "fields": "hhh", "factor": "\\kappa\\sigma_\\varepsilon\\ell_3"},
    {"id": "V3_cbar_c_h", "order": "kappa^1", "fields": "\\bar c c h", "factor": "\\kappa\\cdot 2ia"},
    {"id": "V_af_h", "order": "kappa^{0,1}", "fields": "h^*c(+h)", "factor": "h^*(R_0(c)+\\kappa a\\mathcal L_c h)"},
    {"id": "V_af_c", "order": "kappa^1", "fields": "c^*cc", "factor": "-\\kappa a\\,c^*(c\\cdot\\partial c)"},
    {"id": "V_af_cbar", "order": "kappa^0", "fields": "\\bar c^*b", "factor": "-i\\,\\bar c^*b"},
]

MASTER_SIGN = {
    "delta_xi_S": "-kappa int xi B",
    "xi": "a c = c/2",
    "SS_Phi_star_0": "2 s S_field",
    "SS_equals": "-kappa int c B",  # FIXED connecting prose
    "SD_SD": "kappa int c (B_g - B) + O(kappa^2)",  # displayed combined
    "Bg_location": "prepared/switched breaking",  # not invariant baseline
}

ANTIFIELD_TRANSFORM = "s Phi*_A = (S, Phi*_A)"  # NOT (Phi*, S)


def emit_latex_saf():
    return r"""\begin{equation}
\label{eq:Saf}
\boxed{S_{\mathrm{af}}
=\int\bigl(h^{*\mu\nu}(sh)_{\mu\nu}+\phi^*(s\phi)
-c^*_\mu(sc^\mu)-\bar c^{*\mu}(s\bar c_\mu)\bigr).}
\end{equation}"""


def emit_latex_expanded():
    return r"""\begin{align}
\label{eq:Saf-expanded}
S_{\mathrm{af}}
&=\int h^{*\mu\nu}\bigl(R_0(c)_{\mu\nu}+\kappa a(\mathcal L_ch)_{\mu\nu}\bigr)
+\kappa a\int\phi^*(c^\rho\partial_\rho\phi)
\nonumber\\
&\quad
-\kappa a\int c^*_\mu\,(c^\rho\partial_\rho c^\mu)
-i\int\bar c^{*\mu}b_\mu.
\end{align}"""


def emit_latex_vertices():
    rows = "\n".join(
        f"${v['id']}$ & ${v['order']}$ & ${v['fields']}$ & ${v['factor']}$ \\\\"
        for v in VERTICES
    )
    return (
        r"\begin{center}\begin{tabular}{llll}" "\n"
        r"ID & Order & Fields & Factor / density \\" "\n"
        r"\midrule" "\n"
        + rows + "\n"
        r"\end{tabular}\end{center}"
```

```
    )


def main():
    data = {
        "couplings": COUPLINGS,
        "expanded": EXPANDED,
        "vertices": VERTICES,
        "master_sign": MASTER_SIGN,
        "antifield_transform": ANTIFIELD_TRANSFORM,
        "a": A,
        "latex_saf": emit_latex_saf(),
        "latex_expanded": emit_latex_expanded(),
        "latex_vertices": emit_latex_vertices(),
    }
    json.dump(data, open(os.path.join(OUT, "convention_coefficients.json"), "w"), indent=2)
    # write includable snippets
    open(os.path.join(OUT, "snippet_Saf.tex"), "w").write(emit_latex_saf())
    open(os.path.join(OUT, "snippet_Saf_expanded.tex"), "w").write(emit_latex_expanded())
    open(os.path.join(OUT, "snippet_vertices.tex"), "w").write(emit_latex_vertices())
    print("WROTE convention source of truth")
    print("c_star_cc", EXPANDED["c_star_cc"], "cbar_star_b", EXPANDED["cbar_star_b"])
    return 0


if __name__ == "__main__":
    raise SystemExit(main())
```

Listing 8: convention_coefficients.json

```
{
  "couplings": {
    "h_star_sh": "+1",
    "phi_star_sphi": "+1",
    "c_star_sc": "-1",
    "cbar_star_scbar": "-1"
  },
  "expanded": {
    "h_star_R0": "+1",
    "h_star_L": "+kappa*1/2",
    "phi_star": "+kappa*1/2",
    "c_star_cc": "-kappa*1/2",
    "cbar_star_b": "-i"
  },
  "vertices": [
    {
      "id": "V2_grav",
      "order": "kappa^0",
      "fields": "hh",
      "factor": "K=-\\square\\mathcal P"
    },
    {
      "id": "V2_ghost",
      "order": "kappa^0",
      "fields": "\\bar c c",
      "factor": "i\\square"
    },
    {
      "id": "V3_grav_prep",
      "order": "kappa^1",
      "fields": "hhh",
      "factor": "\\kappa\\sigma_\\varepsilon\\ell_3"
    },
    {
      "id": "V3_cbar_c_h",
      "order": "kappa^1",
      "fields": "\\bar c c h",
      "factor": "\\kappa\\cdot 2ia"
    },
    {
      "id": "V_af_h",
      "order": "kappa^{0,1}",
      "fields": "h^*c(+h)",
      "factor": "h^*(R_0(c)+\\kappa a\\mathcal L_c h)"
    },
    {
      "id": "V_af_c",
      "order": "kappa^1",
      "fields": "c^*cc",
      "factor": "-\\kappa a\\,c^*(c\\cdot\\partial c)"
```

```
    },
    {
      "id": "V_af_cbar",
      "order": "kappa^0",
      "fields": "\\bar c^*b",
      "factor": "-i\\,\\bar c^*b"
    }
  ],
  "master_sign": {
    "delta_xi_S": "-kappa int xi B",
    "xi": "a c = c/2",
    "SS_Phi_star_O": "2 s S_field",
    "SS_equals": "-kappa int c B",
    "SD_SD": "kappa int c (B_g - B) + O(kappa^2)",
    "Bg_location": "prepared/switched breaking"
  },
  "antifield_transform": "s Phi*_A = (S, Phi*_A)",
  "a": "1/2",
  "latex_saf": "\\begin{equation}\n\\label{eq:Saf}\n\\boxed{S_{\\mathrm{af}}\n=\\int\\bigl(h^{*\\mu\\nu}(sh)_{\\mu\\nu}+\\phi
    ^*(s\\phi)\n-c^*_\\mu(sc^\\mu)-\\bar c^{*\\mu}(s\\bar c_\\mu)\\bigr).}\n\\end{equation}",
  "latex_expanded": "\\begin{align}\n\\label{eq:Saf-expanded}\nS_{\\mathrm{af}}\n&=\\int h^{*\\mu\\nu}\\bigl(R_0(c)_{\\mu\\nu
    }+\\kappa a(\\mathcal L_ch)_{\\mu\\nu}\\bigr)\n+\\kappa a\\int\\phi^*(c^\\rho\\partial_\\rho\\phi)\n\\nonumber\\\\n&\\
    quad\n-\\kappa a\\int c^*_\\mu\\,(c^\\rho\\partial_\\rho c^\\mu)\n-i\\int\\bar c^{*\\mu}b_\\mu.\n\\end{align}",
  "latex_vertices": "\\begin{center}\\begin{tabular}{llll}\nID & Order & Fields & Factor / density \\\\\n\\midrule\n$V2_grav$
    & $kappa^0$ & $hh$ & $K=-\\square\\mathcal P$ \\\\\n$V2_ghost$ & $kappa^0$ & $\\bar c c$ & $i\\square$ \\\\\
    n$V3_grav_prep$ & $kappa^1$ & $hhh$ & $\\kappa\\sigma_\\varepsilon\\ell_3$ \\\\\n$V3_cbar_c_h$ & $kappa^1$ & $\\bar c c
    h$ & $\\kappa\\cdot 2ia$ \\\\\n$V_af_h$ & $kappa^{0,1}$ & $h^*c(+h)$ & $h^*(R_0(c)+\\kappa a\\mathcal L_c h)$ \\\\\
    n$V_af_c$ & $kappa^1$ & $c^*cc$ & $-\\kappa a\\,c^*(c\\cdot\\partial c)$ \\\\\n$V_af_cbar$ & $kappa^0$ & $\\bar c^*b$ &
    $-i\\,\\bar c^*b$ \\\\\n\\end{tabular}\\end{center}"
}
```

Listing 9: check_ms_coefficients.py

```
#!/usr/bin/env python3
from __future__ import annotations
import json, os, sys
from pathlib import Path
HERE = Path(__file__).resolve().parent
sys.path.insert(0, str(HERE.parent))
from package_paths import BODY_TEX, output_path, assert_not_stub

assert_not_stub(BODY_TEX)
TEX = open(BODY_TEX).read()
fails = []

def need(cond, msg):
    if not cond:
        fails.append(msg); print("FAIL", msg)
    else:
        print("OK", msg)

need("-\\kappa a\\int c^*_\\mu" in TEX, "expanded -kappa a int c*")
need("-i\\int\\bar c^{*\\mu}b_\\mu" in TEX, "expanded -i int cbar* b")
need("-\\kappa a\\,c^*" in TEX, "vertex -kappa a c*")
need("-i\\,\\bar c^*b" in TEX, "vertex -i cbar*b")
need("s\\Phi^*_A=(S,\\Phi^*_A)" in TEX, "antifield (S,Phi*)")
need("(S,S)=-\\kappa\\int c" in TEX, "(S,S)=-kappa int c B")
need("prepared/switched" in TEX, "Bg prepared/switched")
need("a_{\\mathrm{root}}" in TEX, "a_root")
need("\\Rp(" in TEX, "R0^pred / Rp")
# Convention: boxed field extraction present
need("(S,\\Phi^A)=-\\delta^R S/\\delta\\Phi^*_A" in TEX,
     "field extraction (S,Phi)= - delta^R S / delta Phi*")
bad = [ln.strip() for ln in TEX.splitlines()
       if "K\\xi" in ln and "never write" not in ln and "not write" not in ln]
need(len(bad) == 0, f"no ill-typed K xi in identities: {bad[:3]}")
need("+\\kappa\\int c^*_\\mu\\,(a" not in TEX, "no wrong +kappa int c*(a")
out = {"pass": len(fails) == 0, "failures": fails, "body": str(BODY_TEX)}
json.dump(out, open(output_path("conventions", "ms_coefficient_check.json"), "w"), indent=2)
print("pass", out["pass"], fails)
raise SystemExit(0 if not fails else 1)
```

Listing 10: ms_coefficient_check.json

```
{
  "pass": true,
  "failures": [],
  "body": "/workspace/XVI-N-r10/manuscript/body/XVI-N-body.tex"
}
```

1.3 Classical BV

Listing 11: classical_bv_check.py

```python
#!/usr/bin/env python3
"""Orchestrate classical BV revision checks. Certs ONLY under build/outputs/classical_bv/."""
from __future__ import annotations
import json, os, subprocess, sys
from pathlib import Path

SCRIPT_DIR = Path(__file__).resolve().parent
sys.path.insert(0, str(SCRIPT_DIR.parent))
from package_paths import out_dir, write_json

OUT = str(out_dir("classical_bv"))
PYBIN = sys.executable

REQUIRED_MASTER_KEYS = (
    "Q_D_squared",
    "full_retained_order_broken_master_identity",
)
REQUIRED_QD_KEYS = ("definition", "explicit_nonzero", "status")
REQUIRED_ENZ_KEYS = ("conclusion",)


def run(script):
    r = subprocess.run(
        [PYBIN, str(SCRIPT_DIR / script)],
        cwd=str(SCRIPT_DIR),
        capture_output=True,
        text=True,
    )
    print(r.stdout)
    if r.returncode != 0:
        print(r.stderr, file=sys.stderr)
    return r.returncode


def qd2_explicit_nonzero_ok(master: dict) -> bool:
    """Fail-closed: exact schema, no whole-blob 'nonzero' search."""
    missing = [k for k in REQUIRED_MASTER_KEYS if k not in master]
    if missing:
        raise SystemExit(f"SCHEMA FAIL classical_master missing keys: {missing}")
    qd = master["Q_D_squared"]
    if not isinstance(qd, dict):
        raise SystemExit("SCHEMA FAIL: Q_D_squared must be an object")
    missing_qd = [k for k in REQUIRED_QD_KEYS if k not in qd]
    if missing_qd:
        raise SystemExit(f"SCHEMA FAIL Q_D_squared missing keys: {missing_qd}")
    enz = qd["explicit_nonzero"]
    if not isinstance(enz, dict):
        raise SystemExit("SCHEMA FAIL: Q_D_squared.explicit_nonzero must be an object")
    missing_enz = [k for k in REQUIRED_ENZ_KEYS if k not in enz]
    if missing_enz:
        raise SystemExit(
            f"SCHEMA FAIL Q_D_squared.explicit_nonzero missing keys: {missing_enz}"
        )
    conclusion = enz["conclusion"]
    if not isinstance(conclusion, str) or not conclusion.strip():
        raise SystemExit("SCHEMA FAIL: explicit_nonzero.conclusion empty")
    # Exact expected scientific conclusion markers (not a blob scan)
    markers = ("NOT the zero", "not a BV complex")
    if not any(m in conclusion for m in markers):
        raise SystemExit(
            "SCHEMA FAIL: explicit_nonzero.conclusion lacks required markers "
            f"{markers}; got {conclusion!r}"
        )
    return True


def main():
    rc1 = run("brst_normalization.py")
    rc2 = run("extended_action_master.py")
    brst_path = os.path.join(OUT, "brst_normalization_certificates.json")
    master_path = os.path.join(OUT, "classical_master_certificates.json")
    if not os.path.isfile(brst_path):
        raise SystemExit(f"missing {brst_path}")
    if not os.path.isfile(master_path):
        raise SystemExit(f"missing {master_path}")
    brst = json.load(open(brst_path))
```

```
    master = json.load(open(master_path))
    s2_ok = brst["s2_checks"]["interacting_unrestricted_2jets_correct"]["pass"]
    neg_ok = brst["s2_checks"]["negative_controls_ok"]
    qd2_ok = qd2_explicit_nonzero_ok(master)
    full = master["full_retained_order_broken_master_identity"]
    interface_ok = s2_ok and neg_ok and qd2_ok and rc1 == 0 and rc2 == 0
    cert = {
        "gate": "N-CME (revised)",
        "revision": "r12",
        "verdict": "PASS_INTERFACE" if interface_ok else "FAIL",
        "meaning": (
            "constructs the unprepared classical BV complex and its explicitly broken "
            "prepared-action extension through the declared order"
        ),
        "s2_unrestricted_2jets": s2_ok,
        "negative_controls": neg_ok,
        "Q_D_squared_nonzero": qd2_ok,
        "full_retained_order_broken_master_identity": full,
        "N_A_release": "Provisional" if full == "Unresolved" else "Constructed",
        "pass": bool(interface_ok),
        "schema": "r12_strict_Q_D_squared",
    }
    with open(os.path.join(OUT, "classical_bv_check_out.txt"), "w") as f:
        for k, v in cert.items():
            f.write(f"{k}: {v}\n")
    write_json("classical_bv", "classical_bv_check_certificates.json", cert)
    print(json.dumps(cert, indent=2))
    return 0 if interface_ok else 1


if __name__ == "__main__":
    raise SystemExit(main())
```

Listing 12: classical_bv_check_out.txt

```
gate: N-CME (revised)
revision: r12
verdict: PASS_INTERFACE
meaning: constructs the unprepared classical BV complex and its explicitly broken prepared-action extension through the
     declared order
s2_unrestricted_2jets: True
negative_controls: True
Q_D_squared_nonzero: True
full_retained_order_broken_master_identity: Constructed
N_A_release: Constructed
pass: True
schema: r12_strict_Q_D_squared
```

Listing 13: brst_normalization_certificates.json

```
{
  "normalization": {
    "a": "1/2",
    "xi_mu": "a * c^mu",
    "symmetrization_includes_half": true,
    "R_0(c)": "partial_(mu c_nu)",
    "s0_h": "R_0(c)",
    "s1_h": "a L_c h",
    "s0_c": "0",
    "s1_c": "a c^rho partial_rho c^mu",
    "s0_cbar": "i b",
    "s1_cbar": "0",
    "s0_b": "0",
    "s1_b": "0",
    "equivalent_form": "s h = (1/2) L_c (eta + kappa h)",
    "predecessor_K_xi_note": "XI-A/XII-C delta_xi gamma = K_pred xi + kappa L_xi gamma; R0_pred(xi):=K_pred xi renamed; R_0(c)
     =R0_pred(a c); K:=-Box P only.",
    "matches_XIV_A_def_kin": true
  },
  "s2_checks": {
    "free_s0": {
      "s0_h": "R_0(c)",
      "s0_c": "0",
      "s0_cbar": "i b",
      "s0_b": "0",
      "s0_squared": true,
      "evidence": "written derivation (XI-B lem:nilpotence); trivial on generators"
    },
```

```
    "interacting_unrestricted_2jets_correct": {
      "n_trials": 30,
      "failures": 0,
      "pass": true,
      "jet_order": "unrestricted 2-jets (independent rational components)",
      "a": "1/2",
      "sign": 1
    },
    "negative_wrong_a": {
      "n_trials": 10,
      "failures": 10,
      "pass": false,
      "jet_order": "unrestricted 2-jets (independent rational components)",
      "a": "1",
      "sign": 1
    },
    "negative_wrong_sign": {
      "n_trials": 10,
      "failures": 10,
      "pass": false,
      "jet_order": "unrestricted 2-jets (independent rational components)",
      "a": "1/2",
      "sign": -1
    },
    "negative_controls_ok": true
  },
  "evidence_kind": "reported computation (exact Q unrestricted 2-jet samples) + written free nilpotence",
  "scope_note": "Samples probe generic unrestricted 2-jets; not a closed-form coefficientwise identity printout for every
     monomial. Free s0^2 is exact on generators."
}
```


Listing 14: qd2_antifield_certificates.json


```
{
  "gate": "Q_D2_antifield_smeared",
  "pass": true,
  "formula": "Q_D^2 F = (B_tot, F); F=int f_mu c^{*mu}; (B_tot,F)=(-kappa/2) int f_mu B^mu +O(kappa^2)",
  "master_square_sign": "(S,S)= -kappa int c^nu B_nu + O(kappa^2)",
  "trials_ok": 6,
  "trials": 6,
  "samples": [
    {
      "pairing_density_coeff": "-8909/54",
      "B": [
        "8909/27",
        "-2717/36",
        "86731/216",
        "5009/54"
      ]
    },
    {
      "pairing_density_coeff": "147551/432",
      "B": [
        "-147551/216",
        "-9974/27",
        "-26761/216",
        "-8531/108"
      ]
    }
  ]
}
```


Listing 15: qd2_general_jets_certificates.json


```
{
  "gate": "Q_D^2_general_jets",
  "pass": true,
  "formula": {
    "general": "Q_D^2 F = (B_tot, F) with B_tot = (S,S)/2",
    "at_O_kappa": "B_tot = (kappa/2) int c^nu B_nu + O(kappa^2)  [or without 1/2 per XIV-A reporting of (S,S)]",
    "B_nu": "B_nu = NO_nu(Z) = -w Q_nu(A h) + R_sigma,nu,  Z = w J3 + Dsource",
    "nonzero": "prepared theory is not a BV complex whenever B != 0"
  },
  "jet_order": "unrestricted rational 3-jets for B_nu; structural antibracket identity",
  "stats": {
    "trials": 12,
    "B_eq_minus_wQ_plus_R": 12,
    "nonzero_B": 12,
```

```
    "nat_B0": 12,
    "af_action_cstar": 12
  },
  "samples": [
    {
      "B": [
        "-255/2",
        "151/12",
        "2189/18",
        "-615/2"
      ],
      "w": "1"
    },
    {
      "B": [
        "7087/36",
        "8507/36",
        "-631/12",
        "-331/6"
      ],
      "w": "1/2"
    }
  ],
  "spot_vs_general": "supersedes single pure-trace spot config; holds on random unrestricted jets"
}
```


Listing 16: generator_antibracket_certificates.json


```
{
  "gate": "generator_antibracket_r6",
  "pass": true,
  "fields": {
    "h": {
      "sigma": 1,
      "s_H_over_s_table": 1,
      "match": true,
      "coupling": "+h* sh"
    },
    "phi": {
      "sigma": 1,
      "s_H_over_s_table": 1,
      "match": true,
      "coupling": "+phi* sphi"
    },
    "c": {
      "sigma": -1,
      "s_H_over_s_table": 1,
      "match": true,
      "coupling": "-c* sc"
    },
    "cbar": {
      "sigma": -1,
      "s_H_over_s_table": 1,
      "match": true,
      "coupling": "-cbar* scbar"
    },
    "b": {
      "match": true,
      "note": "sb=0; no b* in S_af"
    }
  },
  "antifields": {
    "rule": "s Phi*_A = (S, Phi*_A)",
    "forbidden": "s Phi* = (Phi*, S)",
    "pass": true,
    "fields_tested": [
      "h*",
      "phi*",
      "c*",
      "cbar*",
      "b*"
    ],
    "note": "For antifield-linear S=S_field+S_af, (S, Phi*_A) extracts delta S_field/delta Phi^A (KT) with graded signs from
     S_af. Regression records orientation; full jet KT check deferred to classical_bv Q^2 suite on S_inv."
  },
  "expanded_signs": {
    "c_star_cc": "-kappa*1/2",
    "cbar_star_b": "-i",
    "c_star_negative": true,
    "cbar_star_negative": true,
```

```
    "pass": true
  },
  "master_sign": {
    "SS_equals": "-kappa int c B",
    "SD_SD": "kappa int c (B_g - B) + O(kappa^2)",
    "Bg_location": "prepared/switched breaking",
    "pass": true
  },
  "Q2_Sinv_retained_order": {
    "pass": true,
    "source": "brst_normalization_certificates.json"
  }
}
```


Listing 17: field_only_projection_certificates.json


```
{
  "gate": "A4_field_only_projection",
  "revision": "r11",
  "N_sites": 3,
  "F1_(SO,Phi)=sOPhi_on_generators": true,
  "F2_PO(SO,AO)=<Eh,U>+<Ephi,V>_with_Ephi=-M_phi": true,
  "F2_equiv_<Ah,U>-<(Box+m2)phi,V>": true,
  "F2_(SO,AO)_has_antifield_terms": true,
  "F3_PO(SO,c*W)=O_via_hstar_in_deltaSO_deltac": true,
  "F4_(SO,F)=sOF": true,
  "F5_field_only_part_zero_on_free_shell": true,
  "F5_control_nonzero_off_shell": true,
  "F6_reversed_bracket_gives_minus": true,
  "pass": true
}
```


1.4 Preparation / normalization

Listing 18: af0_noether_certificates.json


```
{
  "gate": "af0_prepared_Noether",
  "pass": true,
  "elapsed_sec": 59.32280445098877,
  "jet_order": "unrestricted rational 3-jets (h,\u2202h,\u2202\u2202h,\u2202\u2202\u2202h independent) \u2014 covers O(\u03ba)
      EH cubic densities needing \u22643 derivatives; plus degree-4 sparse polynomial families at random rational basepoints
     (retained-jet order 3 unrestricted + poly locus order 4)",
  "method2": "NO(A)=0; NO(J3)+Q(A)=0; B=NO(Z) with Z=w J3+Dsource; const-w B=-w Q(A)",
  "method1": "product-rule IBP densities for A, J3, Ew=sigma J3+Ds (Euler-of-divergence=0)",
  "XII_C_match": "B_nu = -w Q_nu(A h) + R_sigma,nu with R_sigma = B + w Q(A); delta_xi S = \u222b \u03be^\u03bd (-\u03ba B_\
     u03bd) + O(\u03ba\u00b2); no sign/factor discrepancy observed vs XII-C",
  "discrepancy_with_XII_C": null,
  "numeric_stats": {
    "trials": 20,
    "NOA": 20,
    "lem": 20,
    "prA": 20,
    "prJ": 20,
    "prW": 20,
    "nat": 20,
    "bulk": 19,
    "neg": 19,
    "B_eq_bulk_plus_R": 0,
    "sample_B": [
      "-255/2",
      "151/12",
      "2189/18",
      "-615/2"
    ]
  },
  "numeric_failures_head": [],
  "poly_stats": {
    "seeds": 3,
    "points": 6,
    "ok_points": 6,
    "n_failures": 0
  },
  "method2_spot": {
    "lem": true,
    "const_w_B_eq_minus_wQ": true,
```

```
    "w": "1/2"
  },
  "controls": {
    "naturality_sigma1": true,
    "bulk_minus_wQ_nonzero": true,
    "negative_drop_wQ": true,
    "const_w_neq1": true
  }
}
```


Listing 19: prep_certificates.json


```
{
  "gate": "N-Prep",
  "verdict": "PASS",
  "chosen_route": "C",
  "routes": {
    "A": {
      "name": "Route A \u2014 preparation belongs to the insertion",
      "preferred_if_provable": true,
      "predecessor_definitions": {
        "XII-C_eq:selected-action": "S_grav,D = S_EH + kappa int (sigma-1) ell_3  (ACTION-level cubic weighting)",
        "XVI-M_graph_vertex": "V(h) = int chi_I ell_3(h) with chi_I independent of sigma_eps (graph insertion switch)",
        "XVI-M_conventions": "relation of chi_I to action weighting kappa int sigma ell_3 is NOT fixed (Unresolved in XVI-M)"
      },
      "verdict": "REFUTED as primary identification",
      "reason": "XII-C places sigma in the selected gravitational ACTION (cubic-only weight), not merely in a composite insertion. The graph switch chi_I is a separate object (XVI-K/XVI-L/XVI-M). Predecessor definitions do not identify preparation with the insertion.",
      "proved": false
    },
    "B": {
      "name": "Route B \u2014 external transforming background",
      "verdict": "NOT ADOPTED for Phase I",
      "reason": "Would require introducing a transformation law for sigma and an invariant completion, then specializing to the fixed ramp. XII-C already treats sigma as a fixed external weight and computes the resulting classical breaking; constructing a new invariant completion is a different theory than the actual prepared action.",
      "note": "Declaring the ramp a scalar is not a proof that the existing cubic modification is invariant (plan N.1).",
      "proved": false
    },
    "C": {
      "name": "Route C \u2014 preparation explicitly breaks gauge symmetry",
      "verdict": "ADOPTED",
      "justification": "XII-C Theorem thm:noether (Changed Noether identity) already proves the actual prepared action has a known classical breaking, distinct from any quantum anomaly. Breaking includes: (i) ramp-derivative terms Rampdef_sigma from Dsource; (ii) bulk coefficient mismatch -w Q_nu(A gamma) from weighting only the cubic while leaving the quadratic unweighted; (iii) separate gauge-fixing breaking N(e_gf) when S_gf is included. Relates to XI-A thm:noether B_nu ( switching) but is not a relabeling of it.",
      "classical_breaking_terms": [
        "kappa (-w Q_nu(A gamma))",
        "kappa Rampdef_{sigma,nu}",
        "kappa^2 Q_nu(Z_sigma)",
        "N_nu(e_gf) if gauge-fixed (XI-A prop:gfbad; linear breaking -Box D^nu)"
      ],
      "distinct_from_quantum_anomaly": true,
      "proved": true,
      "predecessor_labels": [
        "XII-C thm:noether / eq:changed-noether",
        "XII-C thm:actualsource / eq:actualsource",
        "XIII-C eq:Rampdef",
        "XI-A thm:noether / eq:B (related switching pattern, different object)"
      ]
    }
  },
  "symbolic_certificates": {
    "Dsource_vanishes_on_plateau": true,
    "trace_Dsource_matches_XII_C": true,
    "trace_Dsource_expr": "-6*psi(t)*Derivative(psi(t), t)*Derivative(sigma(t), t)",
    "bulk_mismatch_for_constant_w": true,
    "bulk_mismatch_expr_const_sigma": "6*c*Derivative(psi(t), t)*Derivative(psi(t), (t, 2)) - 6*Derivative(psi(t), t)* Derivative(psi(t), (t, 2))",
    "Q0_A_pure_trace": "-6*Derivative(psi(t), t)*Derivative(psi(t), (t, 2))",
    "plateau_preparation_breaking_vanishes": true,
    "chi_I": {
      "in_S_grav_D": false,
      "in_graph_vertex_V": true,
      "independence": "chi_I independent of sigma_eps, f, g (XVI-K R hypotheses / XVI-M)",
      "localized_identity": "switch-derivative terms supported in supp(d chi_I); bulk where chi_I=1 drops them"
    },
    "variation_formula": {
```

```
      "order": "O(kappa) cubic, pure metric (no scalar)",
      "delta_xi_S_EH": "O (naturality / bulk diffeomorphism invariance)",
      "delta_xi_kappa_int_w_ell3": "kappa int Z_sigma^{mu nu} (K xi + kappa L_xi gamma)_{mu nu} = -kappa int xi^nu [N_{0,nu}(
     Z_sigma) + kappa Q_nu(Z_sigma)]",
      "NO_Z": "N_{0,nu}(Z_sigma) = -w Q_nu(A gamma) + Rampdef_{sigma,nu}",
      "Z_sigma": "w J_3 + Dsource, Dsource = 2 C_2 dd sigma - 2 M_2 d sigma",
      "retained_identity": "N_nu(e_D) = kappa [ -w Q_nu(A gamma) + Rampdef_{sigma,nu} ] + O(kappa^2)",
      "preparation_derivatives_retained": true
    },
    "Dsource_00": "-3*psi(t)*Derivative(psi(t), t)*Derivative(sigma(t), t)",
    "Dsource_11": "psi(t)*Derivative(psi(t), t)*Derivative(sigma(t), t)"
  },
  "relation_to_XI_A_B_nu": {
    "XI_A_thm_noether": "N_nu(E) = kappa (E_phi d_nu phi - B_nu) with B_nu switching breaking",
    "XII_C_distinction": "Changed identity is NOT XI-A with relabeled source; adds -w Q_nu(A gamma) bulk mismatch and
     Rampdef_sigma",
    "gauge_fixing": "Separate N_nu(e_gf); not made invariant by the ramp"
  },
  "interaction_switch": {
    "chi_I": "independent of preparation; not part of S_grav,D",
    "localized_identity_support": "supp(d chi_I) for switch-derivative terms; bulk chi_I=1 stated separately"
  },
  "stop_rule": "If N-Prep fails, stop before N.2. Here: PASS under Route C."
}
```

Listing 20: free_hessian_check_out.txt

```
Gate N-Norm / free_hessian_check
write_path: /workspace/XVI-N-r10/build/outputs/normalization/free_hessian_check_out.txt
cert_path: /workspace/XVI-N-r10/build/outputs/normalization/propagator_normalization_certificates.json
identity_holds: True
P^2=id: True
K = -Box P already includes S_gf: True
XI G_gr^R = -P E_0^R  NOT equated with  X-sheet Delta_R^gr = -4 P E_0^R
D_F = -P (h_0 + i E_0^A); K D_F = i id
S = (1/(4 pi^2))(Q - i0)^{-1}
All 16 Hessian components match -Box P.
```

1.5 Insertion / N-Ins

Listing 21: nins3_status_composer.py

```python
#!/usr/bin/env python3
"""r12 status composer. Consumes r10/r11 small-mass schema from build/outputs only."""
from __future__ import annotations
import json, sys
from pathlib import Path

sys.path.insert(0, str(Path(__file__).resolve().parents[1]))
from package_paths import read_json, write_json, OUTPUTS_DIR

REQUIRED = [
    "revision", "pass", "status", "C_hi", "C_hi_le_17843",
    "W0_lo", "m_star_advertised", "advertised_interval",
    "conditional_on", "instance_family",
]


def validate_b2(b2: dict) -> None:
    missing = [k for k in REQUIRED if k not in b2]
    if missing:
        raise SystemExit(f"SCHEMA FAIL: small-mass cert missing fields {missing}")
    if b2.get("pass") is not True:
        raise SystemExit("SCHEMA FAIL: pass is not True")
    if b2.get("status") != "certified":
        raise SystemExit(f"SCHEMA FAIL: status={b2.get('status')!r} != 'certified'")
    if b2.get("C_hi_le_17843") is not True:
        raise SystemExit("SCHEMA FAIL: C_hi_le_17843 is not True")
    if b2.get("advertised_interval") != "0 <= m < 1.3715e-4":
        raise SystemExit(f"SCHEMA FAIL: advertised_interval={b2.get('advertised_interval')!r}")


def main():
    b2 = read_json("insertion", "nins3_small_mass_bound.json")
    validate_b2(b2)

```

```
    def L(name):
        return read_json("insertion", name)

    rc = L("root_adjoint_Ddiv_Dgf_certificates.json")
    sf = L("nins3_4D_special_family_certificates.json")
    en = L("nins3_4D_validated_enclosure.json")
    p1 = L("nins3_1p1_corrected_certificates.json")
    a2 = L("nins3_support_lemma_certificates.json")
    a3 = L("onshell_admission_certificates.json")
    a4 = read_json("classical_bv", "field_only_projection_certificates.json")

    ok = all(x.get("pass") for x in (rc, sf, en, p1, a2, a3, a4, b2))
    status = (
        "Analytic 4D nonvanishing for the massless special family, conditional on the "
        "inherited reduction identities and configuration-space admissibility "
        "(Route A sufficient; Route B not claimed discharged)"
    )
    dec = {
        "gate": "N-Ins.3_decision",
        "revision": "r12",
        "pass": bool(ok),
        "route_A": "discharged",
        "route_B": "not claimed discharged",
        "arbitrary_prescribed_m": "unresolved",
        "support_lemma": a2.get("pass"),
        "admission": a3.get("pass"),
        "projection": a4.get("pass"),
        "small_mass": {k: b2[k] for k in REQUIRED},
        "decision": {
            "N-Ins.3": status + f"; certified small-mass {b2['advertised_interval']}",
            "N-Ins.4": "Unresolved",
        },
        "schema": "r12",
        "output_root": str(OUTPUTS_DIR),
    }
    ic = {
        "revision": "r12",
        "milestones": {
            "N-Ins.3": {"status": dec["decision"]["N-Ins.3"]},
            "N-Ins.4": {"status": "Unresolved"},
        },
        "statuses": {
            "proved_obstructed_claim_withdrawn": True,
            "toy_witness_removed": True,
            "inadmissible_asymmetric_witness_withdrawn": True,
            "route_B_not_claimed_discharged": True,
            "arbitrary_prescribed_m_unresolved": True,
        },
    }
    red = {
        "revision": "r12",
        "inadmissible_nonzero_withdrawn": {"status": "withdrawn_as_NIns_witness"},
        "N_Ins_3": dec["decision"]["N-Ins.3"],
        "N_Ins_4": "Unresolved",
        "route_B": "not claimed discharged",
    }
    write_json("insertion", "nins3_decision_certificates.json", dec)
    write_json("insertion", "insertion_certificates.json", ic)
    write_json("insertion", "onshell_reduction_certificates.json", red)
    print(json.dumps(dec["decision"], indent=1), "pass", ok)
    return 0 if ok else 1


if __name__ == "__main__":
    raise SystemExit(main())
```

Listing 22: completion_space.json

```
{
  "revision": "r12",
  "gate": "N-Ins.1",
  "status": "Constructed",
  "minimal_sector": "metric + real scalar + diffeomorphism ghosts (frozen classical BV)",
  "ansatz": "regular, seed-preserving, antifield-linear insertion completion I_t = F_t + A_0 + kappa A_1 in the Gate N-Ins
     kernel class, with A_0 = int(h^{*mu nu} U_{mu nu} + phi* V + c*_mu W^mu), gh(U,V,W)=(1,1,2), and seed I_t|_{c=cbar=b=Phi
     *=0} = F_t",
  "ghost_numbers": {
    "h": 0,
    "phi": 0,
    "c": 1,
```

```
    "cbar": -1,
    "b": 0
  },
  "U_V_W_ghost_numbers": {
    "U": 1,
    "V": 1,
    "W": 2
  },
  "seed_preservation": "I_t|_{c=cbar=b=Phi*=0} = F_t (independent of b); t_D / E_F / J_t as in XVI-I / Gate N-Ins.2",
  "degrees": {
    "antifield_polynomial": "linear",
    "kappa": "through O(kappa)"
  },
  "derivative_bounds": "finite-order classical densities as declared for the Gate N-Ins kernel class (H4 / body Sec. N-Ins.4); no new analytic bound is introduced by this export",
  "kernels": "inherited XII-D / XVI-H compact root; not re-derived",
  "support": "compactly supported cutoffs as in Lem. support / XII-D",
  "regularity": "C^infty",
  "inverse_operators": "advanced/retarded Green operators E_m^A inherited only; scalar vs tensor inverses distinguished as in the inherited root; no new inverse operators",
  "equivalence": "no new equivalence relation; any field-redefinition / free-equation language refers only to the inherited factorization scaffold (Lemma P0Q0A0 / Gate N-Ins), not a newly declared relation",
  "equations": "insertion closure: Q_inv I_t = O(kappa^2); degree-zero: P_0 Q_0 A_0 = -s_0 F_t (field-only projection of Q_0 A_0; Lemma P0Q0A0). N-Ins.4 remains Unresolved as a universal completion statement.",
  "outcome_with_onshell_witness": "N-Ins.3 conditional massless obstruction (Route A) + certified small-mass interval; Route B and arbitrary prescribed m>0 unresolved; N-Ins.4 Unresolved universally",
  "pass": true,
  "pass_means": "contract text agrees with H4 / Lemma P0Q0A0 / Thm massless hypotheses; does not claim universal N-Ins.4 completion"
}
```


Listing 23: insertion_contract.json


```
{
  "revision": "r12",
  "gate": "N-Ins.1/2",
  "option": "insertion completion interface (Gate N-Ins)",
  "functional_seed": "F_t / t_D / E_F / J_t as constructed",
  "external_data": "g, sigma, chi_I fixed-external policy",
  "source": "XII-C / XII-D / XVI-I inheritance",
  "completion_space": "see completion_space.json (insertion class I_t=F_t+A_0+kappa A_1)",
  "analytic_domain": "Route A ambient-smooth configurations; Route B not claimed",
  "Q_U": "Q_U = Q_inv (invariant differential). Separate operation: field-only projection P_0 Q_0 A_0 (Lemma P0Q0A0).",
  "ansatz": "regular, seed-preserving, antifield-linear insertion completion I_t = F_t + A_0 + kappa A_1 in the Gate N-Ins kernel class, with A_0 = int(h^{*mu nu} U_{mu nu} + phi* V + c*_mu W^mu), gh(U,V,W)=(1,1,2), and seed I_t|_{c=cbar=b=Phi*=0} = F_t",
  "equations": "insertion closure: Q_inv I_t = O(kappa^2); degree-zero: P_0 Q_0 A_0 = -s_0 F_t (field-only projection of Q_0 A_0; Lemma P0Q0A0). N-Ins.4 remains Unresolved as a universal completion statement.",
  "pass": true,
  "pass_means": "contract text agrees with H4 / Lemma P0Q0A0 / Thm massless hypotheses; does not claim universal N-Ins.4 completion"
}
```


Listing 24: nins3_4D_special_family_certificates.json


```
{
  "gate": "N-Ins.3_4D_special_family_analytic",
  "A_special_family_factorization_alpha_eq_(g-sigma)a^{mu1}d_mu v_f": true,
  "B_Ddiv_alpha_source_identity": true,
  "B_formula": "(D_div alpha)^b = d_a(g-sigma) a^{mu1,ab} d_mu v_f - 1/2 (g-sigma) eta^{b1} Box v_f ;  Box v_f = f (m=0)",
  "B_support": "supp D_div alpha c (supp d(g-sigma) n supp dv_f) u (supp(g-sigma) n supp f); ramp-only iff the second set is empty, e.g. supp f c Int{g=sigma}",
  "B_trace_alpha_eq_(g-sigma)d^1 v_f": true,
  "B_note_old_vs_new": "D_div P alpha = D_div alpha - 1/2 grad tr alpha, tr alpha=(g-sigma) d^1 v_f != 0: the superseded source differs by a gradient",
  "C": {
    "J_on_cones": [
      "x*Derivative(S(t), t)",
      "3*S(t)",
      "0",
      "0"
    ],
    "J_old_on_cones": [
      "0",
      "4*S(t)",
      "0",
      "0"
    ],
```

```
    "w_corrected_on_U": [
      "-m0*x",
      "-3*(m0*t - m1)",
      "0",
      "0"
    ],
    "w_old_on_U_superseded": [
      "0",
      "-4*(m0*t - m1)",
      "0",
      "0"
    ],
    "Box_w_zero_on_U": true,
    "Box_a_root_zero_on_U": true,
    "alpha_on_U": "0 (S=0 for t<2, so g=sigma=1 and alpha=(g-sigma)(...)=0 on a neighbourhood of closure(U))",
    "H_direct_eq_H_red_on_U": true,
    "H_direct_eq_H_old_on_U": false,
    "H_on_U": {
      "01": "2*m0"
    },
    "H_old_on_U_superseded": {
      "01": "4*m0"
    },
    "D_div_H_zero_on_U": true
  },
  "h": "h_12=h_21=p(t-z), p(u)=p1 u+p2 u^2 (TT plane wave; G_lin h=0; curvature ~ p2)",
  "h_on_shell": true,
  "C_on_U": [
    "0",
    "0",
    "-4*m0*(-p1 - 2*p2*t + 2*p2*z)",
    "0"
  ],
  "C_old_on_U_superseded": [
    "0",
    "0",
    "-8*m0*(-p1 - 2*p2*t + 2*p2*z)",
    "0"
  ],
  "C_on_U_nonzero": true,
  "D_IBP_identity_-<H,L_xi h>=<C,xi>+div": true,
  "D_C_equals_XIIC_Q_nu(H)": true,
  "D_L_xi_h_convention": "L_xi h_{mu nu}=xi^rho d_rho h_{mu nu}+h_{rho nu}d_mu xi^rho+h_{mu rho}d_nu xi^rho (linear in h; R_0
     xi is the separate kappa^0 term s_0 h)",
  "E": {
    "E0_J3_copy_matches_grav_density": true,
    "E1_A_h_zero_identically": true,
    "E2_cubicvar_pointwise_2J3pol(Rxi,h)=-A(L_xi h)": true,
    "E2_points": 6,
    "E3_control_drop_transport_term_fails": true,
    "E4_R0xi_term_immaterial_since_A_R0=0": true
  },
  "F": {
    "box": [
      [
        "19/20",
        "21/20"
      ],
      [
        "-1/20",
        "1/20"
      ],
      [
        "-1/20",
        "1/20"
      ],
      [
        "-1/20",
        "1/20"
      ]
    ],
    "xi": "xi = e_2 * prod_i ((X_i-lo_i)(hi_i-X_i))^3 on the box (C^2, vanishing to 2nd order at the boundary)",
    "I_A_-<a_root,A L_xi h>": "m0*(p1 + 2*p2)/960400000000000000000000000000000000",
    "I_H_-<H,L_xi h>": "m0*(p1 + 2*p2)/960400000000000000000000000000000000",
    "I_C_<C,xi>": "m0*(p1 + 2*p2)/960400000000000000000000000000000000",
    "I_closed_4m0_int_pprime_chi": "m0*(p1 + 2*p2)/9604000000000000000000000000000000000",
    "I_A_eq_I_H_eq_I_C_eq_closed": true,
    "I_orig_profile": "p1=1, p2=1/2",
    "I_exact_at_profile_m0_coeff": "1/480200000000000000000000000000000000",
    "I_orig_2int_a:J3pol(Rxi,h)_m0_coeff": "2.0824656393169512703040399833402748854 59e-36",
    "I_orig_m1_coeff": "5.0753637592829937782754337208885728 6377e-75",
```

```
    "I_orig_vs_exact_rel_dev": "2.457e-39",
    "I_orig_matches": true,
    "gauss_seconds": 267.9
  },
  "conclusion": "On U (special family, m=0 instance, admissible C^infty data): H|_U = 2 m0 (e_0 e_1 + e_1 e_0), m0 = int S > 0; C|_U = 4 m0 p'(t-z) dx^2 (only C_2 != 0). For every xi in C_c^infty(U;TM): sOF_t = 4 m0 int p'(t-z) xi^2 d^4x. Choosing p'>0 on supp xi and xi^2>=0, xi^2 != 0 gives sOF_t>0. Superseded vector w_old gives H_old = 4 m0 (e0e1+e1e0) ( factor 2 too large, same sign).",
  "pass": true,
  "elapsed_s": 527.7
}
```

Listing 25: nins3_4D_validated_enclosure.json

```
{
  "gate": "N-Ins.3_4D_validated_enclosure",
  "revision": "r11",
  "I_b_enclosure": "[1.20665032243787617533623799633048339480439142, 1.20715032243787617533623799633048339486281753]",
  "m0_enclosure": "[0.60332516121893808766811899816524169740219 5708, 0.60357516121893808766811899816524169743140 8763]",
  "int_xi2_enclosure": "[0.00013908995636540791544870385501199584490959 4766,
     0.00013932063865964463372602728717865578162185700 8]",
  "sOFt_enclosure": "[0.00033566588139237917791699265940453415007814 9078, 0.00033636190776048171281926622192826364024458 4952]",
  "sOFt_enclosure_lo_rational": "335665881392379177916992659404524150078 1/10000000000000000000000000000000000000000000000",
  "W0_certified_lower_rational": "335665881392379177916992659404524150078 1/10000000000000000000000000000000000000000000000",
  "partition": {
    "I_b_cells": 4000,
    "method": "monotone lower/upper Riemann sums; outward iv evaluation",
    "export": "lo_rational = Fraction(nstr(L))-1e-35 (downward pad)"
  },
  "zero_excluded": true,
  "error_budget": {
    "quadrature": "monotone lower/upper Riemann sums with outward-rounded interval evaluation (rigorous)",
    "support": "exact by construction",
    "regularity": "all data C^infty",
    "root": "closed forms in m0; no numerical root error",
    "endpoint_export": "Rational downward-padded from iv lower endpoint"
  },
  "conditional_on": "inherited reduction chain; m=0 special family",
  "pass": true
}
```

Listing 26: nins3_small_mass_bound.py

```
#!/usr/bin/env python3
"""
nins3_small_mass_bound.py  (r11)  --  inclusion-preserving small-mass bound.

Goal: produce outward-rounded C_hi and downward-rounded W0_lo, m_star_hi such that
    |W(m)-W(0)| <= C_hi m^2   for 0 <= m <= 1,
hence W(m) > 0 for 0 <= m < m_star with m_star = sqrt(W0_lo/C_hi) (downward rounded).

Method (no float leakage in the certified path):
  - exact sympy Rational coefficients for densities and cone constants;
  - mpmath.iv interval arithmetic for all bounds, norms, and the final sum;
  - Rational -> iv via iv.mpf(p)/iv.mpf(q) (integer endpoints);
  - partition cells with Rational endpoints covering the exact domain;
  - tails kept in iv; final C_hi = outward upper bound of the iv sum;
  - W0_lo taken as the archived enclosure lower endpoint (never larger).

Advertised safe interval (once C_hi <= 17843):  0 <= m < 1.3715e-4.
"""
from __future__ import annotations
import itertools, json, os, sys, time
from pathlib import Path
import sympy as sp
from mpmath import iv, mp

HERE = os.path.dirname(os.path.abspath(__file__))
import sys as _sys
_sys.path.insert(0, os.path.join(HERE, ".."))
from package_paths import out_dir, read_json
OUT = str(out_dir("insertion"))
sys.path.insert(0, HERE)
sys.path.insert(0, os.path.join(HERE, "..", "classical_bv"))
# Ddiv helpers live next to this script historically; also try package outputs path
# imports: co-located modules only (certs live under build/outputs/)

```

```
# Import from co-located or code tree
try:
    from root_adjoint_Ddiv_Dgf_check import P, G_lin, eta, ETA, box
except ImportError:
    sys.path.insert(0, os.path.join(os.path.dirname(HERE), "insertion"))
    from root_adjoint_Ddiv_Dgf_check import P, G_lin, eta, ETA, box

t, x, y, z = sp.symbols("t x y z", real=True)
X = (t, x, y, z); R4 = range(4)
m = sp.Symbol("m", nonnegative=True)
ts = sp.Rational(5, 2)
Q = sp.Rational
iv.dps = 40
mp.dps = 40


def fa(mu, nu, a, b):
    return Q(1, 2) * eta(mu, nu) * eta(a, b) - Q(1, 2) * (eta(mu, a) * eta(b, nu) + eta(mu, b) * eta(a, nu))


def rat_to_iv(r):
    """Exact Rational (or int) -> closed iv singleton (no float).

    Do NOT call nsimplify first: it can rewrite Rationals as radical Muls
    (e.g. 463/12500), after which int(r) collapses to 0.
    """
    if isinstance(r, sp.Rational):
        p, q = int(r.p), int(r.q)
        if q < 0:
            p, q = -p, -q
        if q == 0:
            raise ZeroDivisionError(r)
        return iv.mpf(p) / iv.mpf(q)
    if isinstance(r, sp.Integer):
        return iv.mpf(int(r))
    if isinstance(r, int):
        return iv.mpf(r)
    # last resort: force Rational without radical rewrite
    r = sp.Rational(sp.sympify(r))
    p, q = int(r.p), int(r.q)
    if q < 0:
        p, q = -p, -q
    return iv.mpf(p) / iv.mpf(q)


def _mpi_lo(v):
    """Lower endpoint of an iv as mp.mpf (full limb precision via make_mpf)."""
    if hasattr(v, "_mpi_"):
        return mp.make_mpf(v._mpi_[0])
    return mp.mpf(v)


def _mpi_hi(v):
    """Upper endpoint of an iv as mp.mpf."""
    if hasattr(v, "_mpi_"):
        return mp.make_mpf(v._mpi_[1])
    return mp.mpf(v)


def iv_cover(a, b):
    """Closed iv covering the union of two iv/mpf values (ordered endpoints)."""
    lo = min(_mpi_lo(a), _mpi_hi(a), _mpi_lo(b), _mpi_hi(b))
    hi = max(_mpi_lo(a), _mpi_hi(a), _mpi_lo(b), _mpi_hi(b))
    return iv.mpf([lo, hi])


def iv_abs_upper(v):
    """Outward [0, ub] containing |v|."""
    ub = max(abs(_mpi_lo(v)), abs(_mpi_hi(v)))
    return iv.mpf([mp.mpf(0), ub])


def iv_nonneg_ub(v):
    """Collapse an iv to [0, upper(v)] (for accumulating upper bounds)."""
    return iv.mpf([mp.mpf(0), _mpi_hi(v)])


# ---------------------------------------------------------------- densities (exact symbolic)
def densities():
    psi = sp.Function("psi")(*X)
    S = sp.Function("S")(t)
```

```
    phi = x * sp.cos(m * (t - ts))
    dphi = [sp.diff(phi, v) for v in X]
    dd = [sp.diff(phi - x, v) for v in X]
    dA = [[0] * 4 for _ in R4]
    for a, b in itertools.product(R4, R4):
        e = -sum(fa(l, mu, a, b) * sp.diff(psi * dd[mu], X[l]) for l in R4 for mu in R4)
        e += (1 - S) * sum(fa(mu, nu, a, b) * sp.diff(psi, X[mu]) * dd[nu] for mu in R4 for nu in R4)
        e += -m**2 / 2 * (1 - S) * eta(a, b) * psi * phi
        dA[a][b] = sp.expand(e)
    PdA = P(dA)
    Sig = [sp.expand((box(psi) + m**2 * psi) * dphi[mu] - box(psi * dphi[mu])) for mu in R4]
    Tl = []
    for l in R4:
        Tab = [[sp.expand(sum(fa(l, mu, a, b) * Sig[mu] for mu in R4)) for b in R4] for a in R4]
        Tl.append(P(Tab))
    Cq, Sq, P0, Pt, Px, Py, Pz = sp.symbols("Cq Sq P0 Pt Px Py Pz")
    Ssym = sp.Symbol("Ssym")
    rep_d = {sp.Derivative(psi, t): Pt, sp.Derivative(psi, x): Px,
             sp.Derivative(psi, y): Py, sp.Derivative(psi, z): Pz}

    def clean(e):
        e = sp.expand(e)
        e = e.xreplace(rep_d).xreplace({psi: P0, S: Ssym})
        e = sp.expand(e)
        phase = sp.expand(m * (t - ts))
        e = e.subs({sp.cos(phase): 1 - m**2 * Cq, sp.sin(phase): m * Sq,
                    sp.cos(-phase): 1 - m**2 * Cq, sp.sin(-phase): -m * Sq})
        e = sp.expand(e)
        assert not e.has(sp.Derivative), e
        assert not e.has(sp.cos) and not e.has(sp.sin), e
        return e
    return ([[clean(PdA[a][b]) for b in R4] for a in R4],
            [[[clean(Tl[l][a][b]) for b in R4] for a in R4] for l in R4],
            (Cq, Sq, P0, Pt, Px, Py, Pz, Ssym))


# ---------------------------------------------------------------- chi' and one_d_norm in iv
def chi_iv(s):
    # s is an iv in (0,1)
    a = iv.exp(-1 / s); b = iv.exp(-1 / (1 - s))
    return a / (a + b)


def chiprime_sup_iv(n=4000, delta=Q(1, 50)):
    """iv upper bound of sup chi' on (0,1). Rational cells cover [delta,1-delta]; analytic tail."""
    best = iv.mpf(0)
    width_r = 1 - 2 * delta  # Rational
    for i in range(n):
        lo_r = delta + width_r * Q(i, n)
        hi_r = delta + width_r * Q(i + 1, n)
        s = iv_cover(rat_to_iv(lo_r), rat_to_iv(hi_r))
        c = chi_iv(s)
        v = c * (1 - c) * (1 / s**2 + 1 / (1 - s)**2)
        best = iv.mpf([mp.mpf(0), max(_mpi_hi(best), _mpi_hi(v))])
    d = rat_to_iv(delta)
    tail = d**(-2) * iv.exp(-1 / d) * (1 + 4 * d**2) * iv.exp(1 / (1 - d))
    return iv.mpf([mp.mpf(0), max(_mpi_hi(best), _mpi_hi(tail))])


def b_Q(kmax=5):
    v, s = sp.symbols("v s")
    Qs = [sp.Integer(1)]
    for _ in range(kmax):
        q = Qs[-1]
        Qs.append(sp.expand(sp.diff(q, v) + 2 * v * s**2 * sp.diff(q, s) - 2 * v * s**2 * q))
    return v, s, Qs


def one_d_norm_iv(weight_poly, k, v, s, Qk, ncell=1500, delta=Q(1, 100)):
    """iv upper bound of int_{-1}^1 |w(v) b^{(k)}(v)| dv; Rational partition (exact endpoints)."""
    wp = sp.Poly(sp.expand(weight_poly), v)
    qp = sp.Poly(sp.expand(Qk), v, s)
    tot = iv.mpf(0)
    L_r = 2 * (1 - delta)  # Rational
    for i in range(ncell):
        lo_r = -1 + delta + L_r * Q(i, ncell)
        hi_r = -1 + delta + L_r * Q(i + 1, ncell)
        assert lo_r < hi_r
        vv = iv_cover(rat_to_iv(lo_r), rat_to_iv(hi_r))
        # v^2 range over the cell from Rational endpoint squares
        squares = [lo_r**2, hi_r**2]
```

```
        if lo_r <= 0 <= hi_r:
            v2_lo_r, v2_hi_r = Q(0), max(squares)
        else:
            v2_lo_r, v2_hi_r = min(squares), max(squares)
        v2 = iv_cover(rat_to_iv(v2_lo_r), rat_to_iv(v2_hi_r))
        ss = 1 / (1 - v2)
        wv = iv.mpf(0)
        for (deg,), coef in wp.terms():
            wv += rat_to_iv(coef) * (vv ** int(deg))
        qv = iv.mpf(0)
        for (dv, ds), coef in qp.terms():
            qv += rat_to_iv(coef) * (vv ** int(dv)) * (ss ** int(ds))
        val = iv_abs_upper(wv) * iv_abs_upper(qv) * iv.exp(1 - ss)
        cell_w = rat_to_iv(hi_r - lo_r)  # exact positive Rational width
        tot += cell_w * iv_abs_upper(val)
    # tails |v| in [1-delta, 1)
    dlt = rat_to_iv(delta)
    one_md = rat_to_iv(1 - delta)
    s0 = 1 / (1 - one_md**2)
    if wp.degree() >= 0:
        wmax = sum((abs(rat_to_iv(c)) for c in wp.coeffs()), iv.mpf(0))
    else:
        wmax = abs(rat_to_iv(weight_poly))
    wmax_u = iv_abs_upper(wmax)
    qb = iv.mpf(0)
    max_ds = 0
    for (dv, ds), c in qp.terms():
        max_ds = max(max_ds, int(ds))
        qb += abs(rat_to_iv(c)) * (s0 ** int(ds))
    assert _mpi_lo(s0) > max_ds
    tail = 2 * dlt * wmax_u * iv_abs_upper(qb) * iv.exp(1 - s0)
    return tot + iv_abs_upper(tail)


# ---------------------------------------------------------------------- Y jets
def Y_terms():
    Bt, Bx, By, Bz = [sp.Function(n) for n in ("Bt", "Bx", "By", "Bz")]
    chi = Bt(t) * Bx(x) * By(y) * Bz(z)
    xi = [0, 0, chi, 0]
    h = [[0] * 4 for _ in R4]; h[1][2] = h[2][1] = t - z
    L = [[sp.expand(sum(xi[r] * sp.diff(h[a][b], X[r]) + h[r][b] * sp.diff(xi[r], X[a])
                        + h[a][r] * sp.diff(xi[r], X[b]) for r in R4)) for b in R4] for a in R4]
    Lup = [[ETA[a] * ETA[b] * L[a][b] for b in R4] for a in R4]
    Y = [[2 * c for c in row] for row in G_lin(Lup)]
    dY = [[[sp.expand(sp.diff(Y[a][b], X[l])) for b in R4] for a in R4] for l in R4]
    return Y, dY, (Bt, Bx, By, Bz)


def L1_bound_iv(expr, fns, norms):
    Bt, Bx, By, Bz = fns
    expr = sp.expand(expr)
    if expr == 0:
        return iv.mpf(0)
    tot = iv.mpf(0)
    for term in sp.Add.make_args(expr):
        coef = sp.Integer(1); orders = {"t": 0, "x": 0, "y": 0, "z": 0}; pa = {"t": 0, "z": 0}
        for f in sp.Mul.make_args(term):
            base, ex = f.as_base_exp()
            if f.is_number:
                coef *= f
            elif base in (t, z):
                pa[str(base)] += int(ex)
            elif isinstance(f, sp.Derivative):
                v_ = f.variables[0]; orders[str(v_)] = len(f.variables)
            elif isinstance(f, sp.Function) or isinstance(base, sp.Function):
                pass
            else:
                raise ValueError(f)
        cf = sp.nsimplify(coef)
        assert isinstance(cf, (sp.Rational, sp.Integer)), cf
        tot += abs(rat_to_iv(cf)) * norms[("t", orders["t"], pa["t"])] * norms[("x", orders["x"], 0)] \
            * norms[("y", orders["y"], 0)] * norms[("z", orders["z"], pa["z"])]
    return tot


def supbound_iv(e, bnd):
    """iv upper bound of |e|/m^2 for 0<m<=1 (triangle inequality on exact expansion)."""
    e = sp.expand(e)
    if e == 0:
        return iv.mpf(0)
    tot = iv.mpf(0)
```

```
    for term in sp.Add.make_args(e):
        c, rest = term.as_coeff_Mul()
        val = abs(rat_to_iv(sp.nsimplify(c)))
        pm = 0
        for f in sp.Mul.make_args(rest):
            base, ex = f.as_base_exp()
            if base == m:
                pm += int(ex)
            elif base in bnd:
                val *= bnd[base] ** int(ex)
            elif f == 1:
                pass
            else:
                raise ValueError(f"unbounded atom {f}")
        assert pm >= 2, term
        tot += val  # m^{pm} <= m^2 <= 1
    return tot


def main():
    t0 = time.time()
    out = {"gate": "B2_small_mass_bound", "revision": "r10", "t_star": "5/2", "m_max": 1,
           "method": "inclusion-preserving iv; Rational coeffs; no float in certified path"}
    PdA, Tl, syms = densities()
    Cq, Sq, P0, Pt, Px, Py, Pz, Ssym = syms
    zero0 = all(sp.expand(e.subs(m, 0)) == 0 for row in PdA for e in row) and \
        all(sp.expand(e.subs(m, 0)) == 0 for T in Tl for row in T for e in row)
    out["delta_vanishes_at_m0"] = bool(zero0)

    cps = chiprime_sup_iv()
    # geometric bounds (exact Rational)
    psi_max = rat_to_iv(Q(25, 4))          # 6.25 = (2.5)^2
    # |d_t psi| <= chi'_sup / 0.1 * 6.25
    psi_t_max = cps / rat_to_iv(Q(1, 10)) * psi_max
    # |d_i psi| crude envelope
    psi_x_max = iv.mpf([mp.mpf(0), max(
        _mpi_hi(cps / rat_to_iv(Q(1, 10)) * rat_to_iv(Q(49, 100)) + rat_to_iv(Q(7, 5))),
        _mpi_hi(rat_to_iv(Q(23, 5))),
        _mpi_hi(cps / rat_to_iv(Q(1, 5)) * psi_max + rat_to_iv(5)),
    )])
    bnd = {
        P0: psi_max, Pt: psi_t_max, Px: psi_x_max, Py: psi_x_max, Pz: psi_x_max,
        Ssym: rat_to_iv(1),
        Cq: rat_to_iv(Q(49, 100)) / 2,   # 0.7^2 / 2
        Sq: rat_to_iv(Q(7, 10)),
        x: rat_to_iv(Q(5, 2)),
    }
    out["validated_sup_bounds"] = {
        "chi_prime_sup_iv": str(cps),
        "sup|psi|": str(psi_max),
        "sup|d_t psi|": str(psi_t_max),
        "sup|d_i psi|": str(psi_x_max),
        "|tau|<=": "7/10", "|x|<=": "5/2",
        "|1-cos|/m^2<=": str(bnd[Cq]), "|sin|/m<=": "7/10",
    }

    # remove float() guard leftover in chiprime ? recompute cleanly without float
    # (chiprime_sup_iv already uses only Rational/iv; the float line was removed in final version)

    supA = [[supbound_iv(PdA[a][b], bnd) for b in R4] for a in R4]
    supT = [[[supbound_iv(Tl[l][a][b], bnd) for b in R4] for a in R4] for l in R4]

    # cone constants exact
    K1 = Q(1113, 500)   # Phi(91/100)
    # recompute K2 exactly as before
    R_ = sp.symbols("R", nonnegative=True)
    tpv = Q(91, 100)
    K2 = sp.integrate(R_ * (Q(1, 2) * ((Q(32, 10) - tpv - R_)**2 - (Q(18, 10) - tpv - R_)**2 if False else 0)), (R_, 0, 1))
    # use the known closed form from r9
    K2 = Q(11197151, 10000000)
    # verify by direct integrate
    Phi_full = lambda tt: Q(1, 2) * ((Q(32, 10) - tt)**2 - (Q(18, 10) - tt)**2)
    K2 = sp.integrate(R_ * Phi_full(tpv + R_), (R_, 0, Q(18, 10) - tpv)) + \
        sp.integrate(R_ * Q(1, 2) * (Q(32, 10) - tpv - R_)**2, (R_, Q(18, 10) - tpv, Q(32, 10) - tpv))
    K2 = sp.nsimplify(K2)
    out["cone_constants"] = {"K1": str(K1), "K2": str(K2)}
    K1iv, K2iv = rat_to_iv(K1), rat_to_iv(K2)

    Y, dY, fns = Y_terms()
    v, s, Qs = b_Q(5)
    d = Q(9, 100)
```

```
    norms = {}
    for var in ("t", "x", "y", "z"):
        for k in range(5):
            for pw in range(3):
                if var in ("x", "y") and pw:
                    continue
                if var == "t":
                    w = (1 + d * v)**pw
                elif var == "z":
                    w = (d * v)**pw
                else:
                    w = sp.Integer(1)
                # int |w B^{(k)}| ds = d^{1-k} int |w(v) b^{(k)}(v)| dv
                nrm = (rat_to_iv(d) ** (1 - k)) * one_d_norm_iv(sp.expand(w), k, v, s, Qs[k])
                # slight outward padding by (1+1e-12) in iv
                pad = 1 + rat_to_iv(Q(1, 10**12))
                norms[(var, k, pw)] = nrm * pad

    nY = [[L1_bound_iv(Y[a][b], fns, norms) for b in R4] for a in R4]
    ndY = [[[L1_bound_iv(dY[l][a][b], fns, norms) for b in R4] for a in R4] for l in R4]

    term_loc = sum((K1iv * supA[a][b] * nY[a][b] for a in R4 for b in R4), iv.mpf(0))
    term_nl = sum((K2iv * supT[l][a][b] * ndY[l][a][b] for l in R4 for a in R4 for b in R4), iv.mpf(0))
    C_iv = (term_loc + term_nl) * (1 + rat_to_iv(Q(1, 10**9)))
    C_hi = _mpi_hi(C_iv)  # outward upper (mp.mpf)

    # ---- Exact release constants (Rational; no float endpoint arithmetic) ----
    from fractions import Fraction
    from package_paths import read_json, out_dir
    C_release = Fraction(17843)
    W_release = Fraction(33566, 10**8)
    m_release = Fraction(13715, 10**8)
    exact_rational_ok = (C_release * m_release * m_release) < W_release
    assert exact_rational_ok, "C_release m_release^2 < W_release failed"

    # Enclosure ONLY from build/outputs (no code-dir fallback)
    enc = read_json("insertion", "nins3_4D_validated_enclosure.json")
    # Prefer explicit certified lower if present
    if "W0_certified_lower_rational" in enc:
        W0_certified = Fraction(enc["W0_certified_lower_rational"])
        enc_audit = "rational_field"
    elif "sOFt_enclosure_lo_rational" in enc:
        W0_certified = Fraction(enc["sOFt_enclosure_lo_rational"])
        enc_audit = "rational_field"
    else:
        lo_s, hi_s = enc["sOFt_enclosure"].strip("[]").split(",")
        lo_s = lo_s.strip()
        # Import decimal string as Fraction via exact decimal parse (not float)
        W0_certified = Fraction(lo_s)
        enc_audit = "decimal_string_as_Fraction_audit_required"
        # Record that string import is not automatic certification of the enclosure method
        out["enclosure_endpoint_audit"] = {
            "method": enc_audit,
            "note": "Importing a decimal string is not automatic certification of the upstream partition; "
                    "see enclosure script export fields and partition metadata.",
            "enclosure_keys": sorted(enc.keys()),
        }
    W_release_le_W0 = (W_release <= W0_certified)
    # Compare C_iv upper to C_release using mp.mpf then Fraction from exact decimal nstr
    C_hi_mpf = _mpi_hi(C_iv)
    # Outward: take string of upper with enough digits, parse as Fraction, compare
    C_hi_str = mp.nstr(C_hi_mpf, 40, strip_zeros=False)
    # Ensure we do not understate C_hi: use ceil via next decimal digit pad
    C_hi_frac = Fraction(C_hi_str)
    # If nstr truncated, pad outward by 10^{-30}
    C_hi_frac = C_hi_frac + Fraction(1, 10**30)
    C_iv_upper_le_release = (C_hi_frac <= C_release)

    certified = bool(exact_rational_ok and W_release_le_W0 and C_iv_upper_le_release and zero0)

    # Diagnostic raw sqrt only (NOT a downward-rounding claim)
    ratio_diag = mp.mpf(W0_certified.numerator) / mp.mpf(W0_certified.denominator)
    ratio_diag = ratio_diag / (C_hi_mpf if C_hi_mpf > 0 else mp.mpf(1))
    m_star_diag = mp.sqrt(ratio_diag)

    out["revision"] = "r10"
    out["C_release"] = str(C_release)
    out["W_release"] = str(W_release)
    out["m_release"] = str(m_release)
    out["exact_rational_C_m2_lt_W"] = bool(exact_rational_ok)
    out["C_iv_upper"] = C_hi_str
```

```
    out["C_hi"] = mp.nstr(C_hi_mpf, 20, strip_zeros=False)
    out["C_hi_le_17843"] = bool(C_iv_upper_le_release)
    out["W0_lo"] = str(W0_certified)
    out["W0_certified_lower"] = str(W0_certified)
    out["W_release_le_W0"] = bool(W_release_le_W0)
    out["m_star_advertised"] = "0.00013715"
    out["advertised_interval"] = "0 <= m < 1.3715e-4" if certified else "candidate"
    out["m_star_raw_diagnostic"] = mp.nstr(m_star_diag, 20, strip_zeros=False)
    out["m_star_diagnostic_note"] = "raw mp.sqrt diagnostic only; not a downward-rounding claim"
    out["status"] = "certified" if certified else "candidate/partial"
    out["conditional_on"] = (
        "inherited reduction identities; configuration-space admissibility (Route A); "
        "massless enclosure lower bound; written small_mass_estimate.tex"
    )
    out["instance_family"] = (
        "f_m=(Box+m^2)psi, phi_m=x^1 cos(m(t-5/2)), v_f=psi; h,xi,psi,g,sigma,U fixed"
    )
    out["pass"] = bool(certified)
    out["statement"] = (
        f"Exact Rational: C_release*m_release^2 < W_release; "
        f"C_iv_upper<=C_release; W_release<=W0_lo; "
        f"hence W(m)>0 on {out['advertised_interval']}. "
        + out["conditional_on"]
    )
    out["float_leakage"] = False
    out["seconds"] = round(time.time() - t0, 1)

    dests = [OUT]
    seen = set()
    for dest in dests:
        if dest in seen:
            continue
        seen.add(dest)
        os.makedirs(dest, exist_ok=True)
        json.dump(out, open(os.path.join(dest, "nins3_small_mass_bound.json"), "w"), indent=2, default=str)
    keys = ["pass", "status", "C_hi", "C_hi_le_17843", "W0_lo", "m_star_advertised",
            "advertised_interval", "exact_rational_C_m2_lt_W", "W_release_le_W0",
            "float_leakage", "seconds"]
    print(json.dumps({k: out[k] for k in keys if k in out}, indent=1))
    return 0 if out["pass"] else 1


if __name__ == "__main__":
    raise SystemExit(main())
```

Listing 27: nins3_small_mass_bound.json

```
{
  "gate": "B2_small_mass_bound",
  "revision": "r10",
  "t_star": "5/2",
  "m_max": 1,
  "method": "inclusion-preserving iv; Rational coeffs; no float in certified path",
  "delta_vanishes_at_m0": true,
  "validated_sup_bounds": {
    "chi_prime_sup_iv": "[0.0, 2.004805778523837403896525968914072291680994 17]",
    "sup|psi|": "[6.25, 6.25]",
    "sup|d_t psi|": "[0.0, 125.30036115773983774353287305712951823006613]",
    "sup|d_i psi|": "[0.0, 67.650180578869918871766436528564759115033799 9]",
    "|tau|<=": "7/10",
    "|x|<=": "5/2",
    "|1-cos|/m^2<=": "[0.24499999999999999999999999999999999999999999999196, 0.24500000000000000000000000000000000000000000002066]",
    "|sin|/m<=": "7/10"
  },
  "cone_constants": {
    "K1": "1113/500",
    "K2": "11197151/10000000"
  },
  "C_release": "17843",
  "W_release": "16783/50000000",
  "m_release": "2743/20000000",
  "exact_rational_C_m2_lt_W": true,
  "C_iv_upper": "17842.202596755819240553950443574758547 33",
  "C_hi": "17842.202596755819241",
  "C_hi_le_17843": true,
  "W0_lo": "335665881392379177916992659404524150078 1/10000000000000000000000000000000000000000000000000",
  "W0_certified_lower": "335665881392379177916992659404524150078 1/10000000000000000000000000000000000000000000000000",
  "W_release_le_W0": true,
  "m_star_advertised": "0.00013715",
  "advertised_interval": "0 <= m < 1.3715e-4",
```

```
  "m_star_raw_diagnostic": "0.00013716059698465440588",
  "m_star_diagnostic_note": "raw mp.sqrt diagnostic only; not a downward-rounding claim",
  "status": "certified",
  "conditional_on": "inherited reduction identities; configuration-space admissibility (Route A); massless enclosure lower bound; written small_mass_estimate.tex",
  "instance_family": "f_m=(Box+m^2)psi, phi_m=x^1 cos(m(t-5/2)), v_f=psi; h,xi,psi,g,sigma,U fixed",
  "pass": true,
  "statement": "Exact Rational: C_release*m_release^2 < W_release; C_iv_upper<=C_release; W_release<=W0_lo; hence W(m)>0 on 0 <= m < 1.3715e-4. inherited reduction identities; configuration-space admissibility (Route A); massless enclosure lower bound; written small_mass_estimate.tex",
  "float_leakage": false,
  "seconds": 28.9
}
```


Listing 28: nins3_decision_certificates.json


```
{
  "gate": "N-Ins.3_decision",
  "revision": "r12",
  "pass": true,
  "route_A": "discharged",
  "route_B": "not claimed discharged",
  "arbitrary_prescribed_m": "unresolved",
  "support_lemma": true,
  "admission": true,
  "projection": true,
  "small_mass": {
    "revision": "r10",
    "pass": true,
    "status": "certified",
    "C_hi": "17842.202596755819241",
    "C_hi_le_17843": true,
    "W0_lo": "335665881392379177916992659404524150078 1/10000000000000000000000000000000000000000000000000",
    "m_star_advertised": "0.00013715",
    "advertised_interval": "0 <= m < 1.3715e-4",
    "conditional_on": "inherited reduction identities; configuration-space admissibility (Route A); massless enclosure lower bound; written small_mass_estimate.tex",
    "instance_family": "f_m=(Box+m^2)psi, phi_m=x^1 cos(m(t-5/2)), v_f=psi; h,xi,psi,g,sigma,U fixed"
  },
  "decision": {
    "N-Ins.3": "Analytic 4D nonvanishing for the massless special family, conditional on the inherited reduction identities and configuration-space admissibility (Route A sufficient; Route B not claimed discharged); certified small-mass 0 <= m < 1.3715e-4",
    "N-Ins.4": "Unresolved"
  },
  "schema": "r12",
  "output_root": "/workspace/XVI-N-r10/build/outputs"
}
```


Listing 29: nins3_support_lemma_certificates.json


```
{
  "gate": "A2_support_lemma",
  "revision": "r9",
  "P1_exact_containments": {
    "kap0_one_on_suppbeta": true,
    "mu_one_on_supplam": true,
    "sigma_one_on_suppbeta": true,
    "beta_one_on_suppS": true,
    "supp_g_in_open_slab": true,
    "Ubar_in_plateau": true,
    "cone_R_range_from_Ubar": [
      "9/10",
      "21/10"
    ],
    "cone_r_range": [
      "0.72679492",
      "2.2732051"
    ],
    "cone_r_range_strictly_in_lam_plateau": true,
    "cone_t_range_strictly_in_beta_plateau": true,
    "supp_alpha_time_in_suppS": true,
    "pass": true
  },
  "P2_symbolic_actual_eq_model": {
    "alpha_actual_eq_model_on_plateau_set": true,
    "Ddiv_alpha_actual_eq_model_on_plateau_set": true,
    "J_mod_eq_Ddiv_alpha_mod": true,
```

```
    "alpha_on_N_proportional_to_S": true,
    "J_mod": [
      "x*Derivative(S(t), t)",
      "3*S(t)",
      "0",
      "0"
    ],
    "pass": true
  },
  "P3_numeric_corroboration": {
    "max_dev_actual_vs_model_J_at_cone_points": "3.39e-21",
    "points": [
      {
        "q": [
          "2.57331",
          "0.340029",
          "0.0014548",
          "1.61218"
        ],
        "dev": "3.39e-21"
      },
      {
        "q": [
          "2.53776",
          "-0.762752",
          "0.053831",
          "-1.18132"
        ],
        "dev": "2.12e-22"
      },
      {
        "q": [
          "2.55099",
          "-0.594195",
          "-0.947123",
          "-0.973439"
        ],
        "dev": "0.0"
      },
      {
        "q": [
          "1.94382",
          "-0.364892",
          "0.0902458",
          "-0.980115"
        ],
        "dev": "0.0"
      },
      {
        "q": [
          "2.43873",
          "-0.692007",
          "0.321307",
          "-1.04667"
        ],
        "dev": "0.0"
      },
      {
        "q": [
          "2.3041",
          "-0.0454924",
          "-0.0290805",
          "-1.28264"
        ],
        "dev": "3.39e-21"
      }
    ],
    "alpha_actual_zero_on_U_point": true,
    "w_quadrature": [
      -0.0241380064487576,
      2.661215210975533
    ],
    "w_closed_form": [
      -0.024138006448757524,
      2.661215210975517
    ],
    "w_rel_dev": [
      3.16214319956437e-15,
      6.0074854106751665e-15
    ],
    "pass": true,
```

```
    "note": "numerical corroboration only; the lemma is P1+P2"
  },
  "vartheta": "vartheta(y^0)=1 on a neighbourhood of [0,T] (XII-D eq:compact-root); U and all cone points used lie in (0,T)",
  "chi_I": "chi_I does not occur in t_D (XVI-I eq:push / XVI-H thm:trep / XII-D B_{f,g}); it enters only the XVI-J/K
     interaction V(h)=int chi_I l_3 used for extension/ambiguity terms, which are outside F_t",
  "pass": true
}
```


Listing 30: onshell_admission_certificates.json


```
{
  "gate": "A3_onshell_admission",
  "revision": "r9",
  "K_riemann_symmetries": true,
  "G1_divergence_free_identically": true,
  "G2_Glin_zero_mod_BoxPhi_all_10_components": true,
  "G2_Glin_not_identically_zero_offshell": true,
  "G3_plane_wave_components": {
    "12": "Subs(Derivative(F(_xi_1), (_xi_1, 2)), _xi_1, t - z)/2",
    "21": "Subs(Derivative(F(_xi_1), (_xi_1, 2)), _xi_1, t - z)/2"
  },
  "G3_trace_free": true,
  "G3_only_12_21": true,
  "G3_coefficient_c": "1/2",
  "G4_control_without_Riemann_symmetry_breaks_G1": true,
  "scalar_phi_m_on_shell": true,
  "scalar_phi0_eq_x": true,
  "pass": true
}
```


1.6 Regression log (current)

Listing 31: regression_N_out.txt


```
{
  "math_passes": [
    "N-Norm_free",
    "generator",
    "source_of_truth",
    "ms_coefficients_match_SOT",
    "root_adjoint_H",
    "H_formula_recorded",
    "Ddiv_Dgf_certificate",
    "H_direct_eq_reduced_traceful",
    "EA_commutation",
    "H_formula_corrected_in_cert",
    "special_family_4D_pass",
    "corrected_w_on_U",
    "H_on_U_2m0",
    "complete_original_pairing",
    "cubicvar_pointwise",
    "validated_enclosure_excludes_0",
    "1p1_corrected",
    "nins3_decision_pass",
    "nins3_status_r9_conditional",
    "nins4_Unresolved",
    "A2_support_lemma",
    "A2_cone_in_plateau",
    "A3_admission",
    "A3_Glin_mod_BoxPhi",
    "A4_projection",
    "A4_shell_vanishing",
    "B2_schema_pass",
    "B2_status_certified",
    "B2_C_hi_le_17843",
    "B2_advertised_interval",
    "B2_has_conditional_on",
    "B2_no_legacy_only",
    "sPsi",
    "af0",
    "onshell_structural",
    "J3_pol_nonzero_on_shell",
    "A_zero_TT",
    "not_toy",
    "parity_control_recompute",
    "parity_I0_zero",
```

```
    "classical_bv",
    "BRST_s2",
    "audit_inadmissible_withdrawn",
    "NIns3_status_in_reduction_cert"
  ],
  "math_failures": [],
  "packaging_passes": [
    "no_reversed_antifield_rule",
    "S_inv_bullet",
    "corrected_w_in_tex",
    "no_exact_obstruction_1p1",
    "1p1_label",
    "status_downgraded_r9",
    "route_A_sufficient",
    "bump_membership_supp_closed",
    "support_lemma_present",
    "thm_massless_present",
    "admission_section",
    "projection_lemma",
    "small_mass_section",
    "scaling_unavailable",
    "Ddiv_cert_present",
    "nins3_decision_cert",
    "factorization_md",
    "a_root_in_tex",
    "Saf_expanded_minus_c",
    "toy_witness_absent",
    "failing_witness_cert_absent",
    "tD_definition_md",
    "admissibility_md",
    "seed_deriv",
    "Jt_tex",
    "N-Ins.3_status_r9",
    "N-Ins.4_Unresolved",
    "proved_obstructed_withdrawn",
    "toy_removed_flag",
    "inadmissible_flag",
    "tex_Unresolved",
    "tex_admissible",
    "tex_actual_tD",
    "tex_Saf",
    "tex_Bg",
    "tex_RELEASE",
    "no_inadmissible_claim_as_proved"
  ],
  "packaging_failures": [],
  "passed": 81,
  "failed": 0
}
```

Listing 32: pipeline.log excerpt

```
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/tests/test_stub_path_absence.py  (cwd=/workspace/XVI-N-r10)
OK (0.0s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/tests/test_convention_consistency.py  (cwd=/workspace/XVI-N-r10)
OK (0.0s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/tests/test_schema_failure.py  (cwd=/workspace/XVI-N-r10)
OK (0.0s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/tests/test_repro_and_failure.py --unit  (cwd=/workspace/XVI-N-r10)
OK (0.0s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/conventions/source_of_truth.py  (cwd=/workspace/XVI-N-r10/code/
    conventions)
OK (0.0s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/conventions/check_ms_coefficients.py  (cwd=/workspace/XVI-N-r10/
    code/conventions)
OK (0.0s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/classical_bv/brst_normalization.py  (cwd=/workspace/XVI-N-r10/code/
    classical_bv)
OK (4.5s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/classical_bv/qd2_antifield_test.py  (cwd=/workspace/XVI-N-r10/code/
    classical_bv)
OK (13.5s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/classical_bv/qd2_general_jets.py  (cwd=/workspace/XVI-N-r10/code/
    classical_bv)
OK (27.8s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/classical_bv/extended_action_master.py  (cwd=/workspace/XVI-N-r10/
    code/classical_bv)
OK (0.3s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/classical_bv/classical_bv_check.py  (cwd=/workspace/XVI-N-r10/code/
    classical_bv)
```

```
OK (4.7s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/classical_bv/generator_antibracket_test.py  (cwd=/workspace/XVI-N-r10/code/classical_bv)
OK (0.0s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/classical_bv/field_only_projection_check.py  (cwd=/workspace/XVI-N-r10/code/classical_bv)
OK (8.8s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/classical_bv/s_psi_coefficients.py  (cwd=/workspace/XVI-N-r10/code/classical_bv)
OK (0.0s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/normalization/free_hessian_check.py  (cwd=/workspace/XVI-N-r10/code/normalization)
OK (0.4s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/prep/prepared_noether_af0.py  (cwd=/workspace/XVI-N-r10/code/prep)
OK (60.9s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/prep/independent_weighted_euler.py  (cwd=/workspace/XVI-N-r10/code/prep)
OK (23.4s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/prep/prepared_gauge_variation.py  (cwd=/workspace/XVI-N-r10/code/prep)
OK (0.4s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/insertion/export_nins12_contracts.py  (cwd=/workspace/XVI-N-r10/code/insertion)
OK (0.0s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/insertion/root_adjoint_H.py  (cwd=/workspace/XVI-N-r10/code/insertion)
OK (3.0s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/insertion/root_adjoint_Ddiv_Dgf_check.py  (cwd=/workspace/XVI-N-r10/code/insertion)
OK (3.2s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/insertion/nins3_4D_special_family_analytic.py  (cwd=/workspace/XVI-N-r10/code/insertion)
OK (532.0s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/insertion/nins3_1p1_corrected.py  (cwd=/workspace/XVI-N-r10/code/insertion)
OK (156.1s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/insertion/nins3_support_lemma_check.py  (cwd=/workspace/XVI-N-r10/code/insertion)
OK (1.8s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/insertion/onshell_admission_check.py  (cwd=/workspace/XVI-N-r10/code/insertion)
OK (0.6s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/insertion/onshell_structural_check.py  (cwd=/workspace/XVI-N-r10/code/insertion)
OK (0.4s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/insertion/parity_control_symmetric.py  (cwd=/workspace/XVI-N-r10/code/insertion)
OK (6.8s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/insertion/nins3_4D_kirchhoff_corrected.py  (cwd=/workspace/XVI-N-r10/code/insertion)
OK (2.4s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/insertion/insertion_variation.py  (cwd=/workspace/XVI-N-r10/code/insertion)
OK (0.3s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/insertion/nins3_4D_validated_enclosure.py  (cwd=/workspace/XVI-N-r10/code/insertion)
OK (0.4s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/insertion/nins3_small_mass_bound.py  (cwd=/workspace/XVI-N-r10/code/insertion)
OK (29.3s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/insertion/nins3_status_composer.py  (cwd=/workspace/XVI-N-r10/code/insertion)
OK (0.0s)
$ /workspace/tmp/venv/bin/python /workspace/XVI-N-r10/code/regression_N.py  (cwd=/workspace/XVI-N-r10/code)
OK (90.9s)
post-run code/ cert audit: PASS (none found)
archived stale RELEASE_MANIFEST.txt -> RELEASE_MANIFEST.txt.r11-or-earlier.20261007-042408.historical
wrote pre-PDF provenance RELEASE_MANIFEST_PREPDF.txt
```

Part II
Notes appendix

2 Notes and fragment sources

Supporting notes and TeX fragments. Items already integrated into the main body via `\input` (support lemma, hypotheses, admission, projection lemma, mass audit, scaling audit) are repeated here as source-file listings for archival completeness.

2.1 N_insertion_sector_decision.md

Listing 33: N_insertion_sector_decision.md

```
# N.3 sector decision ? Option B

## Decision
**Option B: preserve predecessor F_t = ?t_D, h ? h ? ?? from XVI-K/XVI-I; reinstate scalar sector.**

## Reasons
1. **Continuity with XVI-K / XVI-I.** The classical insertion already constructed and used in the reduced-graph programme is
     F_t with a scalar third slot. Dropping ? would make F_t|_{?=0}=0 identically, erasing the seed.
2. **Option A rejected:** pure-metric-only insertion ?t_D, h?h? is not the predecessor object; would break XVI-K kernel
     continuity and WF/support statements tied to the three-slot kernel.
3. **Option C rejected:** inventing a new metric-only composite without predecessor identification would be a model change
     without a theorem (XV-F-patch documentary vs frozen-object distinction).

## Consequences
- Reinstate minimally coupled scalar action S_g~{sc} (XI-A/XII-C) with switch g, mass m, and B_g terms in the Noether identity
    .
- s ? = ? a c~? ?_? ? (match ?_? ? = ? ? ? ?? with ?=ac).
- Include ?* (s ?) in the extended action.
- Master identity recovers E_? ?_? ? and B_{g,?} from XII-C eq:changed-noether.
- Preparation ?_?, interaction switch ?_I, and kernel t_D remain distinct external data.
```

2.2 admissibility_verdict.md

Listing 34: admissibility_verdict.md

```
# Admissibility verdict (r8)

## Frozen 4D requirements (XII-D/XI-B; XI-A for the mass)
g?C_c~?(M_T;[0,1]), ??Int{g=1}?{?<t<T}, g_a=C_c~?(U;TM).
Mass: XI-A/XI-B/XII-C state P_free=?+m?, **m?0, fixed for each instance of the model**.
m=0 is therefore a permitted instance; a massless computation is a statement about the
m=0 instance only (a model restriction relative to any fixed m>0 instance).

## Corrected root source (r8)
w=E~A(D_div ?), H=?+2P D_div~* w (certificate root_adjoint_Ddiv_Dgf_certificates.json).
The r7 vector E~A(D_div P?) is superseded (differs by a trace gradient).

## 4D special family, m=0 instance ? admissible C~? data
nins3_4D_special_family_certificates.json / nins3_4D_validated_enclosure.json:
U=(0.9,1.1)?(?0.1,0.1)?, g=1?S(t) on supp ?, S?C_c~?((2,3);[0,1]), ?=r? on the cone window,
?=x, ?=1 on supp ? ? U. Result on U: H=2m?(e?e?+e?e?), C=4m?p'(t?z)dx?, m?=?S>0;
s?F_t=4m??p'(t?z)?? for every ??C_c~?(U;TM). Explicit C~? ghost: s?F_t?[3.3566e?4, 3.3637e?4].
Status: **rigorous (analytic) nonvanishing for the special family, m=0 instance**, conditional on the
inherited reduction chain (XVI-I eq:s0Ft; XI-E thm:cubicvar, re-checked pointwise).
Fixed m>0 instances: **rigorous admissible 4D nonvanishing certificate pending**.

## 1+1 planar auxiliary model
r7 configuration (h??=t?x, s=??): corrected value **0** (structural); r7 value 307/4054050 was computed
for the superseded functional and is **withdrawn**. Time-only ramp s=?(t?2)(3?t): corrected values
2/4725 (h??=t?x, ??) and ?1891/4989600 (h??=t, ??).
Label: *Exact nonzero variation in a 1+1 planar auxiliary model; no frozen-4D completion obstruction established.*
NOT a frozen-4D certificate (non-compact planar g; 1+1 Green function; C? bumps; no lower-dimensional BV theory/class
     specified).

Inadmissible (g?1) on ghost support remains withdrawn.
```

2.3 convention_sheet.md

Listing 35: convention_sheet.md

```
# XVI-N frozen convention sheet (r10 release)

Source of truth: `code/conventions/source_of_truth.py` and body Sec. conv.
Any sheet with positive odd couplings or the blanket rule `s?=(S,?)=?^L S/??*` is **HISTORICAL/SUPERSEDED**.

## Metric / ghost
- ? = diag(1,-1,-1,-1); q = ? + ? h; P = trace reversal.
- a = 1/2; ?^? = a c^? (rescaled gauge parameter).
- Actual vector generating the diffeomorphism of q is ? ? = ? a c.
- R_0(c)_{??} := (1/2)(?_? c_? + ?_? c_?) with c_? = ?_{??} c^?.
- R_0^{pred}(?)_{??} := ?_? ?_? + ?_? ?_?; then R_0(c) = R_0^{pred}(a c).
- sh = a L_c q; sq = ? a L_c q. (Never write a L_c q / ?.)
- s ? = ? a c^? ?_? ?; s c^? = ? a c^? ?_? c^?; s c?_? = i b_?; s b = 0.
- Free scalar Hessian (XI-B): ?(?+m?); E_?^{(0)} = ?(?+m?)?.
- ?_?, ?_I, g, t_D: s-inert under fixed-external-data policy.

## Antibracket / Q / master square
- Antibracket (HT):
  (F,G) = ?_A [ (?^R F/??^A)(?^L G/??*_A) ? (?^R F/??*_A)(?^L G/??^A) ].
- Left derivatives on odd variables; right on the complementary slot as in HT.
- Q F := (S, F).
- **Field extraction (boxed; parity-dependent):**
  (S, ?^A) = ? ?^R S / ??*_A .
  The overall minus and right derivative arise from the ordered antifield couplings
  and the HT bracket; do **not** use the blanket identity s? = ?^L S/??*.
- **Antifield transformations (boxed):** s ?*_A = (S, ?*_A).
- Ordered antifield couplings in S (boxed S_af; **negative** odd couplings):
  ? h^{*??} (s h)_{??} + ?* (s ?) ? c*_? (s c^?) ? c?^{*?} (s c?_?).
  Expanded: ?? a ? c*_? (c^? ?_? c^?) ? i ? c?^{*?} b_?.
  Retained scalar sector: ?* (s ?) is present (not dropped).
- Action decomposition: **S_D = S_inv + V_? + V_g** (with S_inv = S_D ? V_? ? V_g boxed).
- Symmetric tensor pairing: (? T)_{??} = (1/2)(T_{??}+T_{??}).
- For antifield-independent field action S_field and linear antifield truncation,
  (S,S)|_{?*=0} = 2 s S_field.
- Target: **(S_D, S_D) = ?? ? c^? B_? + O(??)** with B_? = N_{0,?}(Z).

## Decomposition of (S,S)
By polynomial degree in antifields (not antifield number):
- deg 0: action variation (prepared Noether);
- deg 1: closure / s? coefficients;
- deg ?2: absent for antifield-linear off-shell-closed construction.

## Gauge fermion
- ? = ?i ? c?_? (2 D^?[h] ? b^?).
- Graded Leibniz: s(c? X) = (s c?) X ? c? (s X).
- Gauge fixing = anticanonical T_?; (S_{D,?}, S_{D,?}) = T_?(S_D,S_D) = (S_D,S_D)+O(??).
```

2.4 factorization_P0Q0A0.md

Listing 36: factorization_P0Q0A0.md

```
# Factorization / four projected equations (parallel start)

## Degree-zero equation
\[
P_0 Q_0 A_0 = -s_0 F_t
\]
in the free BV complex, with completion ansatz (Gate N-Ins kernel class)
\[
A_0 = \int\bigl(h^{*\mu\nu}U_{\mu\nu}+\phi^*V+c^*_\mu W^\mu\bigr),
\]
ghost numbers \(\mathrm{gh}(U,V,W)=(1,1,2)\), regular differential polynomials
in the retained jets, **no new inverse operators** beyond those already in the
inherited root/kernel construction (\(E^{\mathrm A}\), etc.).

## EOM-ideal structure
On the free shell, \(Q_0\) acting on antifield-linear \(A_0\) produces at
antifield degree 0 only terms in the free EOM ideal
\((G_{\mathrm{lin}}h,\,(\square+m^2)\phi,\,\ldots)\), which vanish on shell.
Hence a regular seed-preserving antifield-linear completion can cancel \(s_OF_t\)
**only if** \(s_OF_t\) itself lies in that ideal (on-shell vanishing of the
obstruction class). Conversely, an **admissible** on-shell nonzero \(s_OF_t\)
```

```
obstructs exactly this class ? not every kernel in the inherited family.

## Corrected root current (r8)
With \(H=\alpha+2\mathcal P D_{\rm div}^*w\), \(w=E^{\mathrm A}(D_{\rm div}\alpha)\), on \(U\) (special family, \(\alpha|_U
    =0\)):
\(s_OF_t=-\langle H,\mathcal L_\xi h\rangle=-2\langle w,D_{\rm gf}(\mathcal L_\xi h)\rangle=\langle\mathcal C_\rho,\xi^\rho\
    rangle\),
\(\mathcal C_\rho=2\partial_\mu(H^{\mu\nu}h_{\rho\nu})-H^{\mu\nu}\partial_\rho h_{\mu\nu}\) (= XII-C \(Q_\rho(H)\)).
The superseded r7 form used \(w_{\rm old}=E^{\mathrm A}(D_{\rm div}\mathcal P\alpha)\).
Admissible m=0 special-family data give \(\mathcal C|_U=4m_0p'(t-z)\,dx^2\neq0\)
(nins3_4D_special_family_certificates.json). The field-only part of \(Q_0A_0\) is the
Koszul--Tate image \(\int(\delta S_0/\delta h)U+\int(\delta S_0/\delta\phi)V\), which vanishes on shell;
hence such a nonzero on-shell value obstructs the declared class via the EOM cancellation requirement
(not via local cohomology). Written as a scaffold argument here, not as a theorem.

## Status
Scaffold only. No solution of the four projected equations claimed.
```

2.5 field_only_projection_lemma.tex

Listing 37: field_only_projection_lemma.tex

```
% field_only_projection_lemma.tex  (r10)
\begin{lemma}[Field-only projection]
\label{lem:P0Q0A0}
Let
\[
A_0=\int\bigl(h^{*\mu\nu}U_{\mu\nu}+\phi^*V+c^*_\mu W^\mu\bigr)
\]
be antifield-linear of ghost numbers $\mathrm{gh}(U,V,W)=(1,1,2)$, with
coefficients $U,V,W$ regular multilocal functionals of the fields (no antifields)
in the Gate~N-Ins kernel class. Write $P_0$ for the projection onto antifield number
zero (field-only part) and $Q_0=(S_0,\cdot)$ for the free differential of the
minimal free extended action
\[
S_0=\tfrac12\langle h,Ah\rangle-\tfrac12\langle\phi,(\square+m^2)\phi\rangle
+\int h^{*\mu\nu}\Ro(c)_{\mu\nu}
\]
(XI-B: the free scalar Hessian is $-(\square+m^2)$). Then, under the frozen antibracket
of Section~\ref{sec:conv},
\begin{equation}
\label{eq:P0Q0A0}
P_0Q_0A_0
=\bigl\langle Ah,U\bigr\rangle
-\bigl\langle(\square+m^2)\phi,V\bigr\rangle
=\bigl\langle E_h^{(0)},U\bigr\rangle
+\bigl\langle E_\phi^{(0)},V\bigr\rangle,
\end{equation}
where $E_h^{(0)}=\delta S_0/\delta h=Ah$ and
$E_\phi^{(0)}=\delta S_0/\delta\phi=-(\square+m^2)\phi$. In particular
\begin{equation}
\label{eq:P0Q0A0shell}
P_0Q_0A_0\big|_{\mathrm{free\ shell}}=0.
\end{equation}
\end{lemma}
\begin{proof}
Write the frozen antibracket
$(F,G)=\sum_A\bigl(\tfrac{\delta^R F}{\delta\Phi^A}\tfrac{\delta^L G}{\delta\Phi^*_A}
-\tfrac{\delta^R F}{\delta\Phi^*_A}\tfrac{\delta^L G}{\delta\Phi^A}\bigr)$.
Expanding $Q_0A_0=(S_0,A_0)$:
\begin{itemize}
\item The terms $\int h^*U$ and $\int\phi^*V$ contribute, through the antifield derivatives
  of $A_0$ paired against the free Euler operators of $S_0$, the pairings
  $\langle Ah,U\rangle+\langle E_\phi^{(0)},V\rangle
  =\langle Ah,U\rangle-\langle(\square+m^2)\phi,V\rangle$,
  plus antifield-linear remainders coming from $Q_0$ acting on the field dependence of
  $U,V$. Those remainders have antifield number $\ge1$ and are discarded by $P_0$.
\item The term $\int c^*W$ contributes through
  $(\delta^R S_0/\delta c)\,(\delta^L A_0/\delta c^*)$.
  The free minimal sector has $\delta S_0/\delta c$ containing $h^*$ (from
  $\int h^{*}\Ro(c)$), so this product has antifield number $\ge1$ and is killed by the
  antifield-zero projection $P_0$. (It is \emph{not} claimed that $S_0$ has no $c^*$
  dependence in every extended model; the killing mechanism is the antifield content of
  $\delta S_0/\delta c$.)
\item Permitted kernel operations (compactly supported multilocal products, pullbacks,
  and constant-coefficient differential operators already present in the inherited root)
  preserve the form of $A_0$ and the domain of the pairings, so they preserve
```

```
  \eqref{eq:POQOA0}.
\end{itemize}
The pairings are defined on admitted configurations~\ref{H:config} (Route~A) because $U,V$ have
compact support in the observation variables and the free Euler operators are continuous
in the smooth topology. Coefficients remain regular on the free shell by the regularity
hypothesis of the class. Exactly one antifield appears in each summand of $A_0$, so
antifield number is well-defined and $P_0$ is the stated projection.
Restricting to $Ah=0$ and $(\square+m^2)\phi=0$ gives \eqref{eq:POQOA0shell}.
\end{proof}
\paragraph{Certificate.}
A graded (Grassmann) exact check on a three-site lattice truncation of the minimal sector,
with generic polynomial coefficients $U,V,W$ and a nontrivial free-shell kernel, confirms
\eqref{eq:POQOA0}--\eqref{eq:POQOA0shell} with the corrected scalar sign
$E_\phi=-M\phi$, the vanishing of the field-only part of the $c^*W$ contribution via the
$h^*$-content of $\delta S_0/\delta c$, the orientation of the frozen bracket, and a
negative control with the reversed bracket
(\texttt{classical\_bv/field\_only\_projection\_certificates.json}). The continuum statement
is Lemma~\ref{lem:POQOA0}; the lattice check is corroboration of the graded algebra, not a
substitute for the continuum proof.
```

2.6 free_obstruction_current.tex

Listing 38: free_obstruction_current.tex

```
% s0 F_t reduction chain (hypotheses per step):
% s0 F_t = 2 <a_root, J_D(R0(c),h)>     [XVI-I; fixed prep]
%        = 2 <a_root, J3_pol(R0^pred(xi),h)>  [sigma=1 on U drops D_sigma in THIS pairing]
%        = -<a_root, A(L_xi h)>          [XI-E cubicvar; Ah=0]
%        = -<H, L_xi h>,  H:=A a_root    [A self-adjoint; xi compact in U]
% a_root = -E^A(P alpha);  H = alpha + 2 P D_div^* E^A(D_div alpha)   [r8: D_div vs D_gf=D_div P distinguished]
% w = E^A(D_div alpha) (NOT E^A(D_div P alpha)); general family: H|_U not proved zero.
% Special family m=0 admissible data: H|_U = 2 m0 (e0 e1 + e1 e0), C = Q(H) = 4 m0 p'(t-z) dz^2.
% s0 F_t = -<H, L_xi h> = <C_rho, xi^rho>,  C_rho = 2 d_mu(H^{mu nu} h_{rho nu}) - H^{mu nu} d_rho h_{mu nu}.
% Causal-plateau assumption kept separate from g=1 on U.
% K denotes only -Box P; gauge map is R0^pred(xi).
```

2.7 general_derivation_af0.md

Listing 39: general_derivation_af0.md

```
# General derivation of the O(?) prepared Noether / master-square identity

## Imported theorem (XII-C thm:noether), pure-metric specialization

XII-C states off shell (eq:changed-noether), for the selected action
S_D = S_EH + ? ? (??1) ?_3 + (scalar sector if present):

  N_?(e_D) = ? [ E_? ?_? ? ? B_{g,?} ? w Q_?(A ?) + R_{?,?} ] + ?? Q_?(Z_?) + ?

with w=??1, Z = w J_3 + D_?, B_?^{metric} := N_{0,?}(Z) = ?w Q_?(A?) + R_{?,?}
(XII-C lem:divergence / eq:Zdivergence).

**Pure-metric specialization (this sector, Option B later reinstates scalar):**
set ?=0 and omit switch g ? drop E_? ?? and B_g. Retained:

  N_?(e_D) = ? [ ?w Q_?(A h) + R_{?,?} ] + ?? Q_?(Z) + O(??)
           = ? B_? + ?? Q_?(Z) + O(??),   B_? := N_{0,?}(Z).

## Variation identity

For compact ?, XII-C / XI-A adjoint identity gives

  ?_? S_field = ? ?^? ( ? N_?(e_D) ) d?x
              = ?? ? ?^? B_? + O(??).

## BRST identification

With ? = a c, a=1/2, and s acting as the diffeomorphism BRST differential
on fields (preparation inert):

  s S_field = ?_{?=ac} S_field = ?(?/2) ? c^? B_? + O(??).

## Antibracket square (convention sheet)

```

```
With Q F = (S,F), ordered couplings ?* s?, and (S,S)|_{?*=0} = 2 s S_field:

  (S_D, S_D) = 2 s S_field + (deg?1 antifield terms)
             = ?? ? c^? B_? + O(??)   at polynomial antifield degree 0,

with deg-1 vanishing by s?=0 and deg?2 absent for antifield-linear truncation.

## Match to XIV-A

XIV-A thm:breaking writes (S_ext,S_ext)=? ? c^? K_? with
K_? = B_{g,?} + w Q_?(A?) ? R_{?,?} ? ? Q_?(Z).
Pure-metric: K_? = ? B_? ? ? Q_?(Z) + O(??) wait: wQ ? R = ?(?wQ+R) = ?B.
So ? ? c K = ?? ? c B + O(??). **Sign agreement.**

## Independent route 2 (density-first)

Displayed density ? ?_3 with ?_3=?2(r_3+b_1 r_2+b_2 r_1) (eq:ell3).
XII-C lem:weighted-curvature / thm:actualsource: Euler_h(? ?_3)=? J_3+D_?.
Regression: exact-arithmetic sampling corroborates N_0(J_3)+Q(A)=0 and
B=N_0(Z); Method-1 IBP is machinery only. independent_euler_certificates.json
records density cubic homogeneity + lem corroboration.

## Status wording

Universal retained-order identity: **Constructed** by imported theorem +
complete convention-conversion derivation above.
Sampling: regression corroboration only (?exact-arithmetic sampling of
unrestricted retained jets passed?).
```

2.8 insertion_completion.tex

Listing 40: insertion_completion.tex

```
% insertion_completion.tex
\paragraph{Ansatz.}
Seek $I_t=I_{t,0}+\kappa I_{t,1}$ in the declared completion space of
\texttt{completion\_ansatz.json}, with seed $I_t\big|_{c=\bar c=b=\Phi^*=0}=F_t$.
\paragraph{Obstruction (antifield-independent).}
$Q_0 F_t=\pm s_0 F_t\neq0$ (XVI-I). No antifield-independent completion of the bare seed.
\paragraph{Status.}
Proved obstructed on that subspace; Unresolved for antifield-linear extension.
```

2.9 insertion_seed.tex

Listing 41: insertion_seed.tex

```
% insertion_seed.tex ? Option B seed
\begin{equation}
F_t(h,\phi)=\langle t_{\mathrm D},\,h\otimes h\otimes\phi\rangle
\qquad\text{(XVI-I prop:Ft / XVI-K).}
\end{equation}
Fixed-external-data: $s\,t_{\mathrm D}=0$. With $a=\tfrac12$,
$s_0 h=R_0(c)$, $s_0\phi=0$, $s_1\phi=a\,c^\rho\partial_\rho\phi$,
\begin{equation}
s_0 F_t=2\langle t_{\mathrm D},\,(R_0 c)\otimes h\otimes\phi\rangle,
\qquad
s_1 F_t=2\langle t_{\mathrm D},\,(a\mathcal L_ch)\otimes h\otimes\phi\rangle
+\langle t_{\mathrm D},\,h\otimes h\otimes(a c^\rho\partial_\rho\phi)\rangle.
\end{equation}
In particular $s_0 F_t\not\equiv0$ (XVI-I prop:s0Ft; cf.\ XI-D thm:freewitness).
```

2.10 invariant_baseline_certificate.tex

Listing 42: invariant_baseline_certificate.tex

```
% invariant_baseline_certificate.tex
\subsection*{Invariant baseline}
\[
S_{\mathrm{inv}}^{[1]}
=
```

```
S_{\mathrm{EH}}^{[2]}+S_{\mathrm{EH}}^{[3]}
+S^{\mathrm{sc}}_{\mathrm{inv}}
+\int\Bigl[
 h^{*\mu\nu}(sh)_{\mu\nu}
 +\phi^*(s\phi)
 -c^*_\mu(sc^\mu)
 -\bar c^{*\mu}(s\bar c_\mu)
\Bigr],
\]
with unswitched invariant scalar (no $g$-breaking).
\[
(S_{\mathrm{inv}}^{[1]},S_{\mathrm{inv}}^{[1]})=0(\kappa^2),
\qquad
Q_{\mathrm{inv}}=(S_{\mathrm{inv}}^{[1]},\cdot).
\]
Generator tests: $(S_{\mathrm{inv}},\Phi^A)=s\Phi^A$ for all generators
(\texttt{generator\_antibracket\_certificates.json}).

Deformations:
$V_\sigma=\kappa\int(\sigeps-1)\ell_3$,\quad
$V_g=$ switched-scalar deformation (encodes $\mathcal B_g$).
\[
S_D=S_{\mathrm{inv}}+V_\sigma+V_g.
\]
Full square (convention sheet r3):
\[
\boxed{(S_D,S_D)=\kappa\int c^\nu\bigl(\mathcal B_{g,\nu}-B_\nu\bigr)+0(\kappa^2),}
\]
$B_\nu=N_{0,\nu}(Z)$.
The $E_\phi\partial_\nu\phi$ terms from metric variation of the scalar stress and from
$\delta_\xi S^{\mathrm{sc}}$ cancel in the Noether identity (XII-C/XI-A), leaving
$\mathcal B_g$ and the prepared ramp $-B$.
Setting $\sigeps=1$ removes $V_\sigma$ but \textbf{not} $\mathcal B_g$.
```

2.11 kernel_admissibility.md

Listing 43: kernel_admissibility.md

```
# Analytic admissibility of insertion terms (classical, N.3)

| Term | Pairing exists | Functional derivatives | t_D derivatives | Support | WF | New diagonal products |
|------|----------------|------------------------|-----------------|---------|-----|------------------------|
| F_t = ?t_D,h?h??? | Yes (compact t_D; XVI-I prop:continuity) | Yes on smooth fields | None (t_D fixed) | supp(t_D) | WF?Z_3 | None at seed |
| s? F_t | Yes | Yes | None | supp(t_D) | same | None |
| s? F_t | Yes | Yes | None | supp(t_D) | same | None |
| Antifield-linear local completion (schematic) | Conditional on declared local density | Yes | via coefficients | local ? supp(t_D) | needs check | **Defer singular extensions to N.5** |

Exclusions: no loops; no renormalized identities; no graph closure; no claim that all completion candidates are admitted.
```

2.12 mass_dependence_audit.tex

Listing 44: mass_dependence_audit.tex

```
% mass_dependence_audit.tex  (r9, Track B1)
\paragraph{Mass-dependence audit.}
\label{sec:mass-audit}
\begin{center}
\begin{tabular}{@{}llp{7.2cm}@{}}
\toprule
Object & Mass dependence & Source \\
\midrule
Scalar operator & $P_m=\square+m^2$, $m\ge0$ fixed per instance & XII-D D0, XI-A \\
$v_f$ defining inverse & $v_f=E_m^{\mathrm A}f$ (massive) & XII-D after eq:alpha \\
Coordinate inverse in $\alpha_{\mathrm{co}}$ & $E_0^{\mathrm A}$ (massless) & XII-D eq:alpha, $E_0^{\mathrm A}$ \\
Tensor root inverse & $E_0^{\mathrm A}$ (massless); $a=-\vartheta E_0^{\mathrm A}\Pc\alpha$ & XII-D eq:beta, eq:compact-root \\
$\alpha_{\mathrm{sc}}$ mass term & $-\tfrac{m^2}{2}g\,\eta^{ab}v_f\phi$ & XII-D eq:alpha \\
$\mathcal J_{\mathrm D}$, $A$, $H$ & mass-independent (gravitational) & XII-C / XI-A \\
Kernel $t_{\mathrm D}$, $c_{\alpha\beta}$ & independent of $m$ at fixed $B_{f,g}$ & XVI-I eq:push \\
$B_{f,g}$ through $\alpha$ & inherits $m$ via $v_f$ and the $m^2$ term & XII-D \\
\bottomrule
\end{tabular}
```

```
\end{center}
\textbf{Do not replace every $E^{\mathrm A}$ by $E_m^{\mathrm A}$.}
The Green operator must invert the differential operator it accompanies.
In particular the tensor root and the coordinate completion remain massless;
only the scalar leg $v_f$ and the explicit $m^2$ term in $\alpha_{\mathrm{sc}}$
carry the mass. For the Track~B2 deformation one may choose the observation
$f_m=(\square+m^2)\psi$ so that $v_{f_m}=E_m^{\mathrm A}f_m=\psi$ identically
(unique advanced solution), eliminating every massive Green operator from the
pairing while retaining the explicit $m^2$ term and the scalar deformation
$\phi_m=x^1\cos(m(t-t_*))$.
```

2.13 massless_obstruction_hypotheses.tex

Listing 45: massless_obstruction_hypotheses.tex

```
% massless_obstruction_hypotheses.tex  (r9, Track A1)
% Hypothesis block for the conditional massless obstruction theorem.
\paragraph{Hypotheses of the conditional massless obstruction.}
\label{hyp:massless}
\begin{enumerate}
\item[\textbf{(H1)}]\label{H:tD}\textbf{Inherited seed.}
  $t_{\mathrm D}$ is the distribution of XVI-I Proposition~\texttt{prop:test}, eq:push
  (sha256 \texttt{5c99e57543cbe65134e28810b7de605e8ded2b267e633f069aec834cc534bd80}):
  \[
  t_{\mathrm D}
  =\sum_{|\alpha|+|\beta|\le 2}
  \partial_1^\alpha\partial_2^\beta\,\iota_*(c_{\alpha\beta}B_{f,g})
  \in\mathcal E'(M^3),
  \]
  with $B_{f,g}$ the compact root of XII-D Theorem~\texttt{thm:root} /
  Lemma~\texttt{lem:root-kernel}, and $c_{\alpha\beta}$ the source coefficient
  tensors of XVI-I eq:j-delta. Equivalently XVI-H Theorem~\texttt{thm:trep}.
  The seed functional is
  $F_t=\langle t_{\mathrm D},h\otimes h\otimes\phi\rangle=\int a_{\mathrm{root}}:
  \mathcal J_{\mathrm D}[h]$ with $a_{\mathrm{root}}=a_{f,g}[\phi]$.
\item[\textbf{(H2)}]\label{H:reduction}\textbf{Reduction identities.}
  On the invariant free shell $Ah=0$, for $\xi\in\mathfrak g_a$ with
  $\operatorname{supp}\xi\subset U$ and with the plateau $\sigeps=1$ on $U$
  (so that $\mathcal D_\sigma$ drops \emph{in this pairing only}):
  \[
  s_OF_t
  =2\int a_{\mathrm{root}}:\mathcal J_{\mathrm D}(\Ro(c),h)
  =-\langle a_{\mathrm{root}},A\,\mathcal L_\xi h\rangle
  =\langle\mathcal C,\xi\rangle,
  \]
  citing XVI-I eq:sOFt and XI-E Theorem~\texttt{thm:cubicvar}
  (the latter re-checked pointwise in this manuscript).
\item[\textbf{(H3)}]\label{H:config}\textbf{Admitted free configuration space.}
  The free classical configurations are ambient smooth solutions of
  $Ah=0$ and $(\square+m^2)\phi=0$ on a neighbourhood of the closed slab,
  with \emph{no spatial-decay requirement} (XII-D DO:
  ``Fields are ambient smooth configurations. No spatial decay is required.'';
  XI-B: ``No spatial decay was imposed on classical configurations.'';
  XII-C: ``Metric Cauchy data are unrestricted.'').
  Compact ghosts $\xi\in\mathfrak g_a=C_c^\infty(U;TM)$ are required by the frozen
  support (XII-D/XI-B). The displayed free solutions $\phi=x^1$ and
  $h_{12}=p(t-z)$ are therefore admitted by Route~A of
  Section~\ref{sec:admission}. Route~B (spatially compact Cauchy data) is
  \textbf{not} claimed discharged in this revision (support defect recorded there).
\item[\textbf{(H4)}]\label{H:class}\textbf{Completion class.}
  Regular, seed-preserving, antifield-linear completions
  $\mathcal I_t=F_t+A_0+\kappa A_1$ in the Gate~N-Ins kernel class, with
  \[
  A_0=\int\bigl(h^{*\mu\nu}U_{\mu\nu}+\phi^*V+c^*_\mu W^\mu\bigr),
  \quad\mathrm{gh}(U,V,W)=(1,1,2),
  \]
  and with no new inverse operators beyond those already in the inherited root
  construction (Gate~N-Ins.1; \texttt{insertion/completion\_space.json}).
\item[\textbf{(H5)}]\label{H:plateau}\textbf{Plateau and root-cutoff requirements.}
  $g\in C_c^\infty(M_T;[0,1])$,
  $\overline U\subset\operatorname{Int}\{g=1\}\cap\{\varepsilon<t<T\}$,
  $\sigeps(t)=\chi(t/\varepsilon)$ with $\chi=1$ for $s\ge1$ (so $\sigeps=1$ and
  $\sigeps'=\sigeps''=0$ on $U$). The collar $\vartheta\in C_c^\infty((-\delta,T+\delta))$
  of XII-D eq:compact-root equals $1$ on a neighbourhood of $[0,T]$.
  The independent switch $\chi_{\mathrm I}$ of XVI-J/K multiplies interaction densities
  outside $F_t$ and does not enter $t_{\mathrm D}$. Explicit support bounds for the
  special-family cutoffs $\beta,\lambda,\kappa_0,\mu$ are those of Lemma~\ref{lem:support}.
```

```
\end{enumerate}

\paragraph{Scope.}
The theorem of Section~\ref{thm:massless} concerns \textbf{one specified kernel instance}:
the massless special-family preparation of Proposition~\ref{prop:4Dsf}, with the explicit
cutoffs of Lemma~\ref{lem:support}. An obstruction for that instance does not automatically
establish obstruction for every inherited kernel, nor for a fixed $m>0$ instance without
further work (Track~B).
```

2.14 obstruction_or_completion.md

Listing 46: obstruction_or_completion.md

```
# N-Ins.3/4 decision (r8)

## Verdict
- **N-Ins.3:** special family, m=0 instance ? *Constructed*: analytic 4D nonvanishing for admissible
  C^? data + validated enclosure s?F_t?[3.3566e?4, 3.3637e?4] (conditional on the inherited reduction chain).
  Fixed m>0 instances and the general family: **Rigorous admissible 4D nonvanishing certificate pending**.
- **N-Ins.4:** **Unresolved** (see below).

## Corrected root source
D_div T=?_?T^{??}, D_gf r=D_div P r, D_div^*v=??^{(?}v^{?)}, D_gf^*=P D_div^*,
A=??P?2D_gf^*D_gf=??P?2P D_div^*D_div P, H=?+2P D_div^*E^A(D_div ?), w=E^A(D_div ?).

## Superseded r7 values (computed for the wrong vector E^A(D_div P?); withdrawn)
- 1+1 planar 307/4054050 ? corrected 0 for that configuration.
- 4D Kirchhoff ?2.913?10?? (16?10), ?2.669?10?? (12?8): generating script not in the artifact tree;
  reconstruction (nins3_4D_kirchhoff_corrected.json) gives corrected ?1.312?10?? (16?10), ?1.277?10?? (12?8)
  vs superseded-functional reconstruction ?2.498?10??, ?2.709?10??. Corroboration only.
- Structural ?w,0??_U=?1/1296 for FREE w=t?x: unaffected (w free), structural only.

## Completion class (N-Ins.4)
For the declared regular, seed-preserving, antifield-linear class, the field-only part of Q?A? is
?(?S?/?h)U+?(?S?/??)V (Koszul?Tate image), which vanishes on the invariant free shell.
A nonzero admissible on-shell s?F_t therefore obstructs this class through the equation-of-motion
cancellation requirement ? not through an automatic appeal to local cohomology (F_t is multilocal).
The m=0 special-family input is now rigorous, but N-Ins.4 is kept **Unresolved** pending: (i) a written
proof (not scaffold) of the EOM-ideal lemma in the kernel class, (ii) independent re-derivation of the
inherited reduction chain, (iii) the on-shell class question (?=x and the TT plane wave are not spatially
compact; only their restrictions to compact sets enter).
```

2.15 onshell_configuration_admission.tex

Listing 47: onshell_configuration_admission.tex

```
% onshell_configuration_admission.tex  (r10)
\paragraph{Configuration-space admission.}
\label{sec:admission}
\paragraph{Route A (quoted; discharged).}
XII-D DO: ``Fields are ambient smooth configurations. No spatial decay is required.''
XI-B: ``No spatial decay was imposed on classical configurations.''
XII-C: ``Metric Cauchy data are unrestricted.''
Positive-order \emph{scalar} coefficients in the Einstein jet have the X-A zero Cauchy data
(XI-A def:jet), but the free background $(\gamma_0,\phi_0)$ is unrestricted.
Thus $\phi=x^1$ and $h_{12}=p(t-z)$ (smooth free solutions of $\square\phi=0$ and
$G_{\mathrm{lin}}h=0$) are admitted by the frozen contract. Route~A is therefore available
and sufficient for the massless theorem: the displayed fields of Proposition~\ref{prop:4Dsf}
qualify. Hypothesis~\ref{H:config} is discharged by Route~A alone.

\paragraph{Route B (spatially compact Cauchy data; \textbf{not claimed discharged}).}
\label{route:B}
A previous draft proposed domain-of-dependence cutoffs relative to
$\overline{\operatorname{supp}f\cup\operatorname{supp}\xi\cup\operatorname{supp}(g-\sigma)\cup U}$.
That set is \emph{not} spatially compact: for $t>\varepsilon$ one has $\sigma=1$, while
$g=0$ outside the compact spatial support of $\kappa_0\mu$, so $g-\sigma=-1$ on an unbounded
spatial region. Cauchy shadows of $\operatorname{supp}(g-\sigma)$ are therefore not
spatially compact, and the draft construction is invalid as written.

A repair would replace $\operatorname{supp}(g-\sigma)$ by an explicitly identified
\emph{compact} region $K_{\mathrm{rel}}$ that actually enters the seed and root pairing
(e.g.\ a closed neighbourhood of $\overline U$ together with the compact cone-meeting set
in $N$ from Lemma~\ref{lem:support}, on which $\alpha=-S\,\mathfrak a^{\mu1}\partial_\mu\psi$
is already supported), and prove that free solutions agreeing with $(\phi,h)=(x^1,p(t-z))$
```

```
on $K_{\mathrm{rel}}$ preserve $H|_U$ and $s_OF_t$. That repair is \textbf{not} carried out
in this revision. Route~B is therefore \textbf{not claimed discharged}. The massless
obstruction uses Route~A only.

The algebraic gravity gadget below remains on record as a constraint-compatible TT
generator (certificate \texttt{onshell\_admission\_certificates.json}), but it does not
by itself discharge Route~B without the compact-region repair just described.
\textbf{Gravity gadget (not a Route~B discharge).}
Let $K_{\mu\alpha\nu\beta}$ be the constant $4$-tensor
\[
K=\tfrac12\bigl((e_1\wedge e_0)\otimes(e_2\wedge e_0)+(e_2\wedge e_0)\otimes(e_1\wedge e_0)\bigr)
\]
(Riemann symmetries). For any scalar $\Phi$, set
$Z_{\mu\alpha\nu\beta}=\Phi\,K_{\mu\alpha\nu\beta}$ and
$\bar h_{\mu\nu}=\partial^\alpha\partial^\beta Z_{\mu\alpha\nu\beta}$. Then
$\partial^\mu\bar h_{\mu\nu}=0$ identically, and
$G_{\mathrm{lin}}(P\bar h)=0$ whenever $\square\Phi=0$ (all ten components). For
$\Phi=F(t-z)$, one recovers the TT plane wave $h_{12}=p(t-z)$ upon choosing $F''=2p$.
```

2.16 scaling_audit.tex

Listing 48: scaling_audit.tex

```
% scaling_audit.tex  (r9, Track B3)
\paragraph{Scaling audit (arbitrary prescribed mass).}
\label{sec:scaling}
The frozen contract fixes the preparation data $f,g,\sigeps$ and the observation
region $U$ for each instance (XII-D/XI-B plateau conditions; Gate~N-Ins.1). It does
\emph{not} declare a free length-scale parameter $L$ that may be varied after the
instance is chosen. Rescaling to $X=(x-x_*)/L$ would change $\operatorname{supp}f$,
$\operatorname{supp}g$, and the relative geometry of the $g$-dip and the $\sigma=1$
plateau; whether such a change stays inside a single frozen instance is not
granted by the contract.

\textbf{Verdict.} The localized scaling route of Track~B3 is \textbf{unavailable}
as an automatic extension of the present instance. A future proof that shrinking
the geometry is permitted, and that a normalized constant
$|W(m,L)/W(0,L)-1|\le C(mL)^2$ stays controlled, would reopen it. Until then,
arbitrary prescribed $m>0$ remains outside the small-mass interval of
Section~\ref{sec:smallmass}.
```

2.17 seed_functional_derivatives.tex

Listing 49: seed_functional_derivatives.tex

```
% seed_functional_derivatives.tex --- ACTUAL t_D
\subsection*{Definition (quoted)}
From XVI-I eq:push / XVI-H thm:trep,
\[
t_{\mathrm D}
=
\sum_{|\alpha|+|\beta|\le 2}
\partial_1^\alpha\partial_2^\beta\,
\iota_*(c_{\alpha\beta}B_{f,g}),
\qquad
F_t=\langle t_{\mathrm D},h\otimes h\otimes\phi\rangle
=\int a_{f,g}[\phi]_{\mu\nu}\,
 \mathcal J_{\mathrm D}^{\mu\nu}[h]\,\mathrm d^4y.
\]
(cf.\ XII-D eq:root-identity). Both metric slots are retained until symmetry is used.

\subsection*{Functional derivative}
Varying $h\to h+\delta h$,
\[
E_F^{\mu\nu}(x)
:=\frac{\delta F_t}{\delta h_{\mu\nu}(x)}
=
2\int a_{f,g}[\phi]_{\alpha\beta}(y)\,
 \mathcal J_{\mathrm D}^{\alpha\beta}\bigl(\delta_x^{\mu\nu},h\bigr)(y)\,
 \mathrm d^4y,
\]
where $\mathcal J_{\mathrm D}(\,\cdot\,,\,\cdot\,,)$ is the bilinear polarization of the
quadratic source $\mathcal J_{\mathrm D}[h]$ (XVI-L / XII-C: $\sigeps J_3+\mathcal D_{\sigeps}$),
and the factor $2$ is metric-slot exchange symmetry of $t_{\mathrm D}$ / $\mathcal O_{\mathrm{pol}}$.
```

```
Equivalently, from the kernel form and XVI-I eq:test, derivatives $\partial_1^\alpha$ hit
$\delta h$ in the first slot (and symmetrically the second).

Fixed-external-data: $s\,t_{\mathrm D}=0$ (not ``$\mathrm d t_{\mathrm D}=0$'').
Kernel/$B_{f,g}$/$c_{\alpha\beta}$/$a_{f,g}$ derivatives appear when $\partial$ acts on $E_F$.
```

2.18 snippet_Saf.tex

Listing 50: snippet_Saf.tex

```
\begin{equation}
\label{eq:Saf}
\boxed{S_{\mathrm{af}}
=\int\bigl(h^{*\mu\nu}(sh)_{\mu\nu}+\phi^*(s\phi)
-c^*_\mu(sc^\mu)-\bar c^{*\mu}(s\bar c_\mu)\bigr).}
\end{equation}
```

2.19 snippet_Saf_expanded.tex

Listing 51: snippet_Saf_expanded.tex

```
\begin{align}
\label{eq:Saf-expanded}
S_{\mathrm{af}}
&=\int h^{*\mu\nu}\bigl(R_0(c)_{\mu\nu}+\kappa a(\mathcal L_ch)_{\mu\nu}\bigr)
+\kappa a\int\phi^*(c^\rho\partial_\rho\phi)
\nonumber\\
&\quad
-\kappa a\int c^*_\mu\,(c^\rho\partial_\rho c^\mu)
-i\int\bar c^{*\mu}b_\mu.
\end{align}
```

2.20 snippet_vertices.tex

Listing 52: snippet_vertices.tex

```
\begin{center}\begin{tabular}{llll}
ID & Order & Fields & Factor / density \\
\midrule
$V2_grav$ & $kappa^0$ & $hh$ & $K=-\square\mathcal P$ \\
$V2_ghost$ & $kappa^0$ & $\bar c c$ & $i\square$ \\
$V3_grav_prep$ & $kappa^1$ & $hhh$ & $\kappa\sigma_\varepsilon\ell_3$ \\
$V3_cbar_c_h$ & $kappa^1$ & $\bar c c h$ & $\kappa\cdot 2ia$ \\
$V_af_h$ & $kappa^{0,1}$ & $h^*c(+h)$ & $h^*(R_0(c)+\kappa a\mathcal L_c h)$ \\
$V_af_c$ & $kappa^1$ & $c^*cc$ & $-\kappa a\,c^*(c\cdot\partial c)$ \\
$V_af_cbar$ & $kappa^0$ & $\bar c^*b$ & $-i\,\bar c^*b$ \\
\end{tabular}\end{center}
```

2.21 source_master_identity.tex

Listing 53: source_master_identity.tex

```
% source_master_identity.tex
\begin{equation}
S_{D,\lambda}=S_D+\lambda I_t,
\qquad
\tfrac12(S_{D,\lambda},S_{D,\lambda})
=B_D+\lambda(S_D,I_t)+O(\lambda^2),
\end{equation}
where $B_D=\tfrac12(S_D,S_D)=-\frac\kappa2\int c^\nu B_\nu+\cdots$ (convention sheet).
If $Q_U I_t=0$ with $Q_U$ from $S_U$, then
$(S_D,I_t)=(V_\sigma,I_t)+O(\kappa^2)$.
For antifield-independent $I_t$ and $V_\sigma$, the displayed antibracket gives
$(V_\sigma,I_t)=0$ identically (both antifield derivatives vanish). Any nonzero
contribution in the $\lambda$ identity must come from antifield-dependent pieces of
the completion, not from the antifield-independent seed alone.
Present revision: completion $Q_U I_t=0$ Unresolved, so the $\lambda$ identity is
recorded structurally only.
```

2.22 support_lemma.tex

Listing 54: support_lemma.tex

```
% support_lemma.tex  (r10) -- inserted before Proposition prop:4Dsf
\begin{lemma}[Actual-source support]
\label{lem:support}
Let $T=4$, $\varepsilon=\tfrac12$, $U=(0.9,1.1)\times(-0.1,0.1)^3$, and let
$\chi$ be the XII-D ramp profile. Define
\begin{align*}
S(u)&=b(2u-5),\qquad
b(v)=e^{1-1/(1-v^2)}\ (|v|<1),\ 0\ \text{else},\\
\beta(t)&=\chi\bigl((t-1.8)/0.1\bigr)\chi\bigl((3.2-t)/0.1\bigr),\\
\lambda(r)&=\chi\bigl((r-0.6)/0.1\bigr)\chi\bigl((2.5-r)/0.2\bigr),\\
\kappa_0(t)&=\chi\bigl((t-0.55)/0.05\bigr)\chi\bigl((3.6-t)/0.1\bigr),\\
\mu(r)&=\chi\bigl((3.2-r)/0.2\bigr),
\end{align*}
so $S\in C_c^\infty(\mathbb{R};[0,1])$ with $\operatorname{supp}S=[2,3]$ (not claimed
inside $(2,3)$),
$\operatorname{supp}\beta=[1.8,3.2]$ with $\beta=1$ on $[1.9,3.1]$,
$\operatorname{supp}\lambda=[0.6,2.5]$ with $\lambda=1$ on $[0.7,2.3]$,
$\kappa_0=1$ on $[0.6,3.5]\supset\operatorname{supp}\beta$, and
$\mu=1$ on $\{r\le3\}\supset\operatorname{supp}\lambda$.
Set $\psi=\beta(t)\lambda(r)r^2$, $f=\square\psi$, $v_f=E_0^{\mathrm A}f=\psi$,
$g=\kappa_0(t)\mu(r)(1-S(t))$, and $\sigeps(t)=\chi(t/\varepsilon)$.
Then:
\begin{enumerate}
\item $g\in C_c^\infty(M_T;[0,1])$, $g=\sigma=1$ on a neighbourhood of $\overline U$,
  and $\overline U\subset\operatorname{Int}\{g=1\}\cap\{\varepsilon<t<T\}$.
\item \textbf{(Global support.)}
  On a neighbourhood of $\operatorname{supp}d\psi$ one has $\kappa_0=\mu=\sigma=1$, hence
  $g-\sigma=-S$. Outside $\operatorname{supp}d\psi$, the massless special-family identity
  $\alpha=(g-\sigma)\,\mathfrak a^{\mu1}\partial_\mu v_f$ (XII-D eq:alpha at $m=0$,
  $\phi=x^1$, $v_f=\psi$) forces $\alpha=0$. Therefore
  \[
  \alpha=-S(t)\,\mathfrak a^{\mu1}\partial_\mu\psi
  \qquad\text{globally on }M_T,
  \]
  and $\operatorname{supp}_t(D_{\mathrm{div}}\alpha)\subset[2,3]$.
\item On the open set $N=\{t\in(1.9,3.1)\}\times\{0.7<r<2.3\}$ one has
  $\beta=\lambda=\kappa_0=\mu=\sigma=1$ with all derivatives vanishing, hence
  $\psi=r^2$ and
  \[
  D_{\mathrm{div}}\alpha=(S'x,\,3S,\,0,\,0)
  \]
  on $N$. Combined with item~2, $D_{\mathrm{div}}\alpha$ vanishes off the time slab
  $[2,3]$: temporal cutoff edges of $\beta$ and the $\kappa_0/\mu$ transitions
  contribute no source; radial cutoff contributions of $\lambda$ during $[2,3]$
  do not meet the relevant advanced cones from $\overline U$ (item~4).
  (No claim is made that radial edges contribute no source globally.)
\item Every advanced cone from a point of $\overline U$ meets
  $\{t\in\operatorname{supp}S\}$ only at radii in $[2-1.1,\,3-0.9]=[0.9,2.1]$,
  hence at spatial radii in $[0.726,2.274]\subset(0.7,2.3)$. Those cone points
  therefore lie in $N$. By items~2--3 the future cones from $U$ see exactly the
  model source $D_{\mathrm{div}}\alpha=(S'x,3S,0,0)$ (no exterior-$N$ contribution).
\item The collar $\vartheta$ of XII-D eq:compact-root equals $1$ on a neighbourhood
  of $[0,T]$, hence on $U$ and on every cone point used above; insertion of
  $\vartheta$ changes nothing (XII-D thm:root). The independent switch
  $\chi_{\mathrm I}$ of XVI-J/K does not occur in $t_{\mathrm D}$ (XVI-I eq:push /
  XVI-H thm:trep / XII-D $B_{f,g}$) and does not affect $F_t$.
\item Consequently the local root formulas on $U$,
  \[
  w=(-m_0x,\;3(m_1-m_0t),\;0,\;0),\qquad
  H^{01}=2m_0,
  \]
  follow from the actual compact construction by $4$D Huygens (cone support of
  $E_0^{\mathrm A}$), not from polynomial surrogate sources.
\end{enumerate}
\end{lemma}
\begin{proof}
Item~1 is an exact containment of the support intervals.
For item~2: $\operatorname{supp}d\psi\subset\operatorname{supp}\beta\times\operatorname{supp}\lambda$
up to the radial factor $r^2$, and on that set $\kappa_0=\mu=1$ by construction while
$\sigma=1$ because $t\ge1.8>\varepsilon$; thus $g-\sigma=-S$ on a neighbourhood of
$\operatorname{supp}d\psi$. Off $\operatorname{supp}d\psi$ one has $\partial\psi=0$, so the
special-family factorisation $\alpha=(g-\sigma)\mathfrak a^{\mu1}\partial_\mu\psi$ yields
$\alpha=0$. The global formula and the time support of $D_{\mathrm{div}}\alpha$ follow.
Item~3 is the symbolic reduction of XII-D eq:alpha on the plateau set $N$
(\texttt{nins3\_support\_lemma\_certificates.json}, Parts~P1--P2), together with item~2
excluding all source outside the time slab $[2,3]$.
```

```
Item~4 is the cone-radius computation
$R\in[2-1.1,\,3-0.9]$ and
$r\in[R-\sqrt3\cdot0.1,\,R+\sqrt3\cdot0.1]\subset(0.7,2.3)$, so every contributing
cone point lies in $N$ where the model source holds.
Item~5 quotes XII-D thm:root and the XVI-K inheritance recorded in
Section~\ref{sec:tD}. Item~6 is then the Kirchhoff evaluation of
Proposition~\ref{prop:4Dsf}. Numeric corroboration (Part~P3 of the certificate)
reproduces $J_{\mathrm{actual}}=J_{\mathrm{model}}$ at random cone points to
$3\times10^{-21}$ and the closed forms for $w$ on $U$ to relative
$10^{-14}$; it is corroboration only.
\end{proof}
```

2.23 tD_definition.md

Listing 55: tD_definition.md

```
# Actual t_D definition (quoted)

## Primary constructive definition

**XVI-I.tex** Proposition prop:test, eq:push
(sha256 ‘5c99e57543cbe65134e28810b7de605e8ded2b267e633f069aec834cc534bd80‘):

‘‘‘
t_D = sum_{|?|+|?|?2} ?_1^? ?_2^? ?_*(c_{??} B_{f,g})
‘‘‘

with ?(y,x)=(y,y,x), B_{f,g} the compact-root kernel (XII-D lem:root-kernel / XVI-H),
and c_{??} the source coefficient tensors from j_D (eq:j-delta), |?|+|?|?2.
The tensors c^{J_3}_{??} are implicitly specified by ell_3/J_3 and the source-kernel
identity; they are **not explicitly expanded** (XVI-I prop:test).

Equivalently **XVI-H.tex** thm:trep / eq:O-t
(enter-companion/raw/tex copy sha256
‘1552ea91d0df41bf88b1b8e7c341bfefedfc27441233e2e7fd28bfc7112e8489‘):

‘‘‘
O_pol(a,c;?) = ?t_D, a ? c ? ??
‘‘‘

and by eq:O-root / XII-D thm:root eq:root-identity
(enter-companion/raw/tex XII-D sha256
‘ff1f5da45640c895ccd6603bf96c6a5dcb966e781fa391368e92789476228fd9‘):

‘‘‘
C_{3,D}(f,g)[?,?] = ? a_{f,g}[?]_{??}(y) J_D^{??}[?](y) d?y
O_pol(a,c;?) = ? a_{f,g}[?]_{??} J_D^{??}(a,c)(y) d?y
‘‘‘

**XVI-K.tex** (sha256 ‘6264bc8ec68534b8486c418596139fd20610d7f49c2f09b5113171e0be0789a8‘)
does **not** redefine t_D; it inherits the XVI-I/XVI-H/XII-D off-shell kernel and
the XVI-J construction data: preparation f,g,?_?; compact switch ?_I; polarized
source Q; metric-dual basis B^i=(G^{-1})^{ii}B_i with
G=diag(1,-2,-2,-2,1,2,2,1,2,1); imported compact root B_{f,g} under
Hypothesis H_root (R1 composition exists; R2 compact support;
R3 WF(B_{f,g})?Z_2).

**XVI-J.tex** (sha256 ‘dcba2602ab4af7487ce03d3de8a7bac801ca106a036f65b9afbae77d1f699841‘)
supplies the tensor time-ordered / root-attached insertion coefficients that
XVI-K retains.

## Properties
- Compactly supported: yes (thm:trep; supp ? {y1=y2?C_K}?O_{f,g})
- Tensor structure: first two slots = symmetric contravariant 2-tensors; third = scalar ?
- Slot-symmetric in the two metric slots
- WF(t_D) ? Z_3 (XII-D thm:admission / XVI-I eq:t-bounds)
- **Fully specified constructively** via B_{f,g}, j_D, ?_*, but the component tensors
  c^{J_3}_{??} are **implicitly** specified and ?not explicitly expanded?
- Contains ?_I only through XVI-K attachment coefficients / ambiguity Leibniz terms;
  the XVI-I/XVI-H seed t_D itself is built from B_{f,g} and source coefficients
  (preparation ?_? enters J_D and c_{??})

## Seed
F_t = ?t_D, h?h??? = C_{3,D}(f,g)[h,?] = O_pol(h,h;?)  (XVI-I eq:Ft)

## Continuum obstruction identity
s_O F_t = 2 ? a_{f,g}[?]_{??} J_D^{??}(R_O(c), h) d?y = -? c_? J_t^?
```

```
with E_F^{??} = 2 ? a_{f,g}[?] : J_D(?^{??}, h) and J_t^? = ?_? E_F^{??}.
```

Part III
Provenance and references

3 Provenance manuscripts (full source listings)

The XVI-M baseline sha256 is 41aa109935f40ddee3109de62c46329c189b5408007970cc90ed5611dc4739ea (verified unchanged against the package baseline).

3.1 Baseline hashes and manifests

```
41aa109935f40ddee3109de62c46329c189b5408007970cc90ed5611dc4739ea  XVI-M.tex
653fe2431040bec75a845fb52db523f3171bfd29049b124eaf612773e1002ff0  XVI-M-package.zip
4accad8e258eaa5e90f33650acdddc1a236c330c5a5bcd35471a71706e1deb84  XVI-M-artifacts/regression_full_out.txt
db5f027ef08e08ebfadb8a64edae72447cfb55297f67aca816b0e825e4764866  XVI-L.tex
56105068f12b68204013f5f686536aaae0d4ebd53a11e801bd1fdf4b3f1e770b  XVI-L-package.zip
6264bc8ec68534b8486c418596139fd20610d7f49c2f09b5113171e0be0789a8  XVI-K.tex
cb1d8285fa30439fbd04484e3f207ddf9afb1ee7533dc23b718ae4882d791791  XVI-K-package.zip
d7d838e6cc621ffa8dbd5e20f44b3bace65544123361a7555f968d57642e9abb  XVI-M-artifacts/conventions.json
cf4af14a5ad75000bea73c29ef1b45c99adb90ebb504f6799b38483c201da7b6  XVI-M-artifacts/completion_ledger_M89.json
ce840f99f6df6ba31f6d415d78ec050a5ba09bb0c3153192d94c829f585ce490  XVI-K-artifacts/hessian_full_out.txt
f0359494d7e7b799c8055ed3a5d9e14b32452eb68893c93cbf0ef4d7bea220b7  enter-companion/raw/tex/XI-A.tex
b19723e9bec65086b685a19c53b5f8b1431fa8d68961513a3739dd3afabfbad2  enter-companion/raw/tex/XI-B.tex
83c6d0383b0aa6817ad1d2d8db570423b0a33ec7953c4adc0aa0d572f2ab7cb5  ms/XII-C.tex
d686b6ec595e1afa01857560d140423ff8bcd1352acd4a339f59c55d648216b4  ms/XI-E.tex
2e2911f0fa38aedde66b5da909388b595ad6e265a7f3addecc75d1fd5dd46e8e  ms/XI-G.tex
dcba2602ab4af7487ce03d3de8a7bac801ca106a036f65b9afbae77d1f699841  XVI-J.tex
bf3618ce0825a5e96f67ea48a02aca4c081cbdfb15ed6276a6f7d5b484c41876  /home/box/agent-data/agents/ae9192b4-f3cc-4cd8-9725-63
     ba20a35856/attachments/bf3618ce0825a5e96f67ea48a02aca4c081cbdfb15ed6276a6f7d5b484c41876.txt
455390b7cee1d6df3203464aec98b6a6d8c57967b965bbd6f04e4615f638858e  /home/box/agent-data/agents/ae9192b4-f3cc-4cd8-9725-63
     ba20a35856/attachments/455390b7cee1d6df3203464aec98b6a6d8c57967b965bbd6f04e4615f638858e.txt
```

```
667c4c8fa10cf8457bc4d31bd284ad945d89d746cb505ee8b093d19463478e73  XVI-N.tex
2f4d949b234b6030d07466e2ca53ed0ba326f978a06e403bd0e88e9e8ff5fac2  XVI-N.pdf
```


```
{
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  "author": "Cheung",
  "programme": "enter",
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  "ledgers": {
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        "sha256_package": "653fe2431040bec75a845fb52db523f3171bfd29049b124eaf612773e1002ff0",
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        "status": "mathematical dependency"
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      {
        "id": "XVI-K",
        "role": "Hessian Lpq; insertion F_t; root B_f,g R1-R3; chi_I independence",
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        "status": "mathematical dependency"
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      {
        "id": "XVI-J",
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        "status": "mathematical dependency"
```

```
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      {
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        "role": "S_grav,D; ell_3; thm:actualsource; thm:noether (changed); sigma_eps freeze",
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        "status": "mathematical dependency"
      },
      {
        "id": "XI-A",
        "role": "S_EH; Ricci; S_gf; thm:noether; B_nu; gamma=H/kappa conventions via q=eta+kappa gamma",
        "sha256_tex": "f0359494d7e7b799c8055ed3a5d9e14b32452eb68893c93cbf0ef4d7bea220b7",
        "status": "mathematical dependency"
      },
      {
        "id": "XI-B",
        "role": "gauge fermion Psi; s_0; lem:nilpotence; def:algebra; Green G_gr=-P E; no interacting BRST class",
        "sha256_tex": "b19723e9bec65086b685a19c53b5f8b1431fa8d68961513a3739dd3afabfbad2",
        "status": "mathematical dependency"
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    "supplementary_corroboration": [
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        "note": "corroboration of propagator interface; not a new UV claim"
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      {
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      {
        "id": "corpus_excerpt_455390b7",
        "role": "additional corpus excerpts (XVI-L L.8 EFT bridge context)",
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      {
        "id": "enter-companion",
        "role": "read-only fetch of XI-A/XI-B tex from local clone; GitHub not modified",
        "note": "cheungngo/enter-companion read-only"
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        "value": "finite vertex through O(kappa) cubic + free nonminimal BV sector",
        "status": "Chosen in XVI-N"
      },
      {
        "item": "preparation symmetry route",
        "value": "Route C (explicit classical breaking; XII-C thm:noether)",
        "status": "Chosen; Route A refuted as primary; Route B not adopted"
      },
      {
        "item": "field variable name",
        "value": "h ? gamma (XI-A); q = eta + kappa h",
        "status": "Reconciled; no rescaling"
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      {
        "item": "kappa / hbar / i / loop grading",
        "value": "see N_normalization_ledger.json (resolves XVI-M Unresolved items for this sector)",
        "status": "Chosen for Phase I sector"
      },
      {
        "item": "master identity coefficients defining closure",
        "value": "free CME; O(kappa) classical breaking coeffs; free gauge-fermion sector",
        "status": "Chosen; higher / quantum coefficients excluded until N.3+"
      }
    ]
  },
  "hard_constraints_honored": [
    "eta = diag(1,-1,-1,-1)",
    "P r = r - (1/2) eta tr_eta r",
    "do NOT equate XI G_gr^R=-P E_0^R with X-sheet Delta_R^gr=-4 P E_0^R",
    "no epsilon->0",
    "no UV claims",
    "no interacting BRST/Ward claimed unless proved here",
    "Enter/qubit readout has no role",
    "never modify XVI-A..XVI-M or GitHub"
  ]
}
```

```
{
  "manuscript": "Landmark XVI-M (draft: integration of work packages A-C, final verification; Gate S-bv checked (written check of XVI-J eq:rational, corroboration not frozen XVI-L premise); M-C_curv proved obstructed; broad M-C not achieved (varpi unspecified by XVI-L thm:L6); M-C_data satisfied)",
  "frozen_dependency": "XVI-L.tex and XVI-L-artifacts/ (read only); XVI-K-artifacts/hessian_full_out.txt (read only)",
  "signature": "eta = diag(1,-1,-1,-1)",
  "trace_reversal": "P r = r - (1/2) eta tr_eta r, P^2 = id",
  "symmetric_basis_order": [
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    "(01)",
    "(02)",
    "(03)",
    "(11)",
    "(12)",
    "(13)",
    "(22)",
    "(23)",
    "(33)"
  ],
  "gram_matrix": "G = diag(1,-2,-2,-2,1,2,2,1,2,1) for <A,C>_eta = A_{mu nu} C^{mu nu}",
  "dual_basis": "B^i = (G^{-1})^{ii} B_i ; Hessian on field directions B_i ; propagator/source contraction on dual slots -P B^i",
  "placement_sign": "d_w^p [u(y-w)] = (-1)^{|p|} (d^p u)(y-w) (XVI-K eq:sign; XVI-L lem:Lsign)",
  "graph_factor": "(1/2!)*2 = 1 (XVI-L prop:Lnorm, re-derived there from XVI-I prop:signs(6) and eq:Dh); carries no power of kappa or hbar",
  "normalization_table": [
    {
      "item": "metric fluctuation",
      "value": "q_{mu nu} = eta_{mu nu} + kappa gamma_{mu nu}",
      "source": "XII-C (Section 'selected formal action', display before eq:curvature)",
      "status": "Imported"
    },
    {
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      "value": "h is a dummy real symmetric tensor of the same species as gamma; no rescaling inserted",
      "source": "XVI-I prop:Ft",
      "status": "Imported"
    },
    {
      "item": "ell_3 vs gravitational action",
      "value": "ell_3 = coefficient of kappa in the unintegrated signed Einstein-Hilbert density; S_EH = -(2/kappa^2) int sqrt(-q) R[q]; S_grav,D = S_EH + kappa int (sigma-1) ell_3",
      "source": "XII-C eq:ell3, eq:selected-action",
      "status": "Imported"
    },
    {
      "item": "graph vertex",
      "value": "V(h) = int chi_I ell_3(h) with chi_I independent of sigma_eps, f, g",
      "source": "XVI-I prop:second; XVI-K/XVI-L",
      "status": "Imported; relation of chi_I to the action weighting kappa int sigma ell_3 is NOT fixed (Unresolved)"
    },
    {
      "item": "quadratic kinetic operator",
      "value": "K = -Box P = A + 2 D^*D, A = 2 G_lin, S_gf = int D_nu gamma D^nu gamma",
      "source": "XII-C (Unchanged free handoff); XVI-I (K=-Box P, XI-B L_gamma)",
      "status": "Imported"
    },
    {
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      "source": "XVI-I prop:DF; XVI-K eq:S",
      "status": "Imported"
    },
    {
      "item": "position-space form of S on R^4\\{0}",
      "value": "S = h_0 + i E_0^A = -W_0 + i E_0^A = c (Q - i0)^{-1} on R^4\\{0}, Q = eta(z,z), c = 1/(4*pi^2), boundary side Q - i0 (including across the nonzero null cone); W_0 = -(1/(4 pi^2)) (Q - i0 sgn z^0)^{-1}; E_0^A = (1/(2 pi)) theta(-z^0) delta(Q)",
      "source": "XVI-M Theorem thm:wpA-Sbv (work package A; wpA/kernel_identification.tex, wpA/kernel_conventions.json); Euclidean Remark rem:Sbv-motivation is motivation only",
      "status": "Certified (Gate S-bv, Theorem thm:wpA-Sbv); no extension across z=0; M.2-M.4/M.7 graded results unconditional on Omega (only c != 0 and the single side are used)",
      "c": "1/(4*pi^2)",
      "boundary_side": "Q-i0"
    },
    {
```

```
      "item": "S^2 residue and scale units",
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      "source": "XVI-M thm:M6-S2, sec:wpB",
      "status": "Constructed; not a beta function"
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      "value": "action cubic vertex carries kappa; graph coefficients a_lambda and the graph factor carry none",
      "source": "XII-C eq:selected-action; XVI-L prop:Lnorm",
      "status": "Overall kappa order of the graph amplitude: Unresolved (Target II)"
    },
    {
      "item": "powers of hbar",
      "value": "No power of hbar is assigned (XVI-I prop:sector graph table)",
      "source": "XVI-I",
      "status": "Unresolved"
    },
    {
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      "value": "i in D_F fixed by K D_F = i id; no i-epsilon prescription and no limit epsilon->0 (XVI-I prop:signs(5)); vertex/time-ordering factors of i (e.g. i/hbar per vertex) not assigned",
      "source": "XVI-I prop:DF, prop:signs",
      "status": "Partially imported; remaining factors Unresolved"
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    {
      "item": "insertion normalization",
      "value": "F_t(h,phi) = <t_D, h (x) h (x) phi> = C_{3,D}(f,g)[h,phi]; D_h^2 F_t[a,c] = 2 <t_D, a (x) c (x) phi>; source J_D = sigma_eps J_3 + 2 C_2 dd sigma_eps - 2 M_2 d sigma_eps",
      "source": "XVI-I prop:Ft, eq:Dh; XII-C eq:actualsource; XVI-L eq:L-JD",
      "status": "Imported"
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    {
      "item": "loop order",
      "value": "A count of one cycle is not a loop order of an interacting observable (XVI-I)",
      "source": "XVI-I prop:sector",
      "status": "Unresolved"
    }
  ],
  "preparation": "fixed 0<eps<T, fixed f, g, sigma_eps (time-only ramp); never varied; no eps->0",
  "evidence_kinds": [
    "symbolic identity in independent jets",
    "complete exact matrix calculation over Q",
    "analytic distributional argument (manuscript)",
    "finite exact test",
    "bookkeeping",
    "imported assumption",
    "independent sympy Euler-operator route (jet_euler.py)"
  ],
  "boundary_value_vs_epsilon": "The distributional boundary-value notation (Q - i0)^{-m} is unrelated to the preparation parameter epsilon (fixed). 'No epsilon->0' does not forbid the i0 notation; no i-epsilon prescription is introduced and no definition is changed.",
  "gate_V_variational_contract": {
    "status": "Chosen Phase II convention (XVI-M Gate V); NOT certified by the normalization ledger",
    "field_variable": "symmetric lowered residual field h_{mu nu}; no rescaling; no trace reversal",
    "variation": "h -> h + t, t compactly supported symmetric",
    "pairing": "delta I[h](t) = int d^4y t_{mu nu} E_I^{mu nu}[h], full symmetric-index contraction over all 16 (mu,nu)",
    "ten_component_form": "sum_{mu<=nu} w_{mu nu} t_{mu nu} E^{mu nu}, w=1 for mu=nu, w=2 for mu<nu; Euler: E^{cd} = (1/w_cd) sum_beta (-D)^beta dL/d(d^beta h_cd)",
    "external_data": "sigma_eps and chi_I fixed under field variation; their derivatives retained in integration by parts",
    "order": "h-linear collapsed tensor; quadratic functional ansatz modulo field-independent terms",
    "helmholtz_test": "int r (DE_C[h].t) = int t (DE_C[h].r); for linear E_C: int <r, E_C[t]> = int <t, E_C[r]>"
  },
  "contact_collapse": "E_C^{mu nu}(y) = sum C d_w^gamma (bar_a_b^{mu nu}(y,w;h))|_{w=y} with bar_a = chi_I * ahatbar (single switch; preferred) = int dw G(C)(y,w); equiv. d_w^gamma(chi_I * ahatbar)",
  "readings": {
    "formal": "sigma, chi, h jets independent (universal; equivalent to: identity for all smooth external data)",
    "chi_plateau": "chi_I = 1 locally (all chi derivatives 0), sigma formal",
    "prep_plateau": "sigma_eps = 1 locally (nonzero sigma-jet coeffs dropped; only J_3), chi formal/variable switch",
    "double_plateau": "chi_I = 1 and sigma_eps = 1 locally: only J_3 terms, constant coefficients",
    "fixed": "actual sigma_eps, chi_I: characterized by injectivity criterion + region ledger (Stage 2); concrete ramp closed form Unresolved"
  },
  "deterministic_seeds": {
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          "-1/2",
```

```
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        ],
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          "-2/5",
          "-9/5",
          "-1/2",
          "2"
        ]
      ]
    },
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      "purpose": "mod-p rank cross-check of formal contact-map blocks"
    },
    "helmholtz_check.py_unit_Lagrangians": {
      "random.seed": 7,
      "purpose": "random quadratic Lagrangian samples for adjoint vs jet_euler unit tests"
    },
    "m4_common.block_kernel": {
      "default_seed_argument": 0,
      "purpose": "deterministic column ordering / pivot search inside exact sparse kernel routine; per-call seed may be overridden"
    }
  },
  "gate_Sbv_status": {
    "status": "Certified (Theorem thm:wpA-Sbv)",
    "c": "1/(4*pi^2)",
    "boundary_side": "Q-i0",
    "domain": "D'(R^4\\{0}), including across the nonzero null cone",
    "extension_across_z0": "The predecessor S is globally defined (XVI-J); what is not established here is its position-space identification at the origin",
    "items_established": [
      "distributional representation of actual S=h_0+i E_0^A on Omega",
      "boundary-value side matching T_1 (Q-i0)",
      "normalization c = 1/(4 pi^2) != 0",
      "equality across nonzero null cone (not only Q!=0)"
    ],
    "alternative": "direct certification of the 185 relations: no longer needed",
    "euclidean_motivation": "motivates c=1/(4 pi^2) only; is NOT the certification (the certification is the real-time argument of thm:wpA-Sbv)",
    "discharged_by_matching": false,
    "history": "Explicit Input, not certified, in the M.2-M.7 drafts; certified in the wpA-C integration cycle",
    "extension_across_z0_audit2_historical": "not claimed",
    "predecessor_print": "XVI-J eq:rational prints d=-(1/4pi^2)(Q_geom-i0)^{-1} on z!=0 incl. nonzero null points, S=-d (eq:DF); thm:wpA-Sbv is a written check, not a new extension",
    "damping_parameter": "tau (renamed from epsilon in audit 3; epsilon is the preparation width)"
  },
  "third_order_double_plateau": {
    "dim_SA_D3": 12,
    "dim_ker_Coll_among_SA_D3": 4,
    "dim_variational_image_D3": 8,
    "character": "pure third-order operators on residual field h (field multi-index order exactly 3)",
    "origin": "N=2 coherent contact assignments with |gamma|=1 (and linear combinations)",
    "not_gravitational_scalar_counterterms": true,
    "note": "Not Euler tensors of int R, R^2, Ric^2, or other Lorentz-scalar curvature polynomials at this order; recorded in M.5 ledger (1) without scalar-CT claim"
  },
  "stage2_fixed_data": {
    "status": "fixed-data criteria, universal region comparisons and universal time-only classification constructed; actual-ramp transfer on (0,eps)xR^3 is Proposition prop:to-actual with explicit hypotheses (H1)-(H3); actual chi_I non-constant regions and transitions Unresolved",
    "gate_V": "unchanged",
    "S_bv": "Certified (Theorem thm:wpA-Sbv)",
    "imported": "sigma_eps = XII-C/XVI-L time-only ramp; chi_I = compact switch",
    "regions": {
      "1_full": "sufficient criteria: injectivity of external-data evaluation on coefficient spaces of (i) G, (ii) Coll, (iii) adjoint-defect map ((iii) needed for SA equality); time-only Helmholtz analysis now exact (thm:to-timeonly); transfer under (H1)-(H3) (prop:to-actual)",
      "2_prep_plateau": "UNIVERSAL formal-chi reading: exact ker Coll=21; SA=(26,0,0,4,17); variational image=26 ultralocal only; actual fixed chi_I Unresolved",
      "3_double_plateau": "exact (M.4): ker Coll=22; SA=1347; var image=1325",
      "4_transitions": "Unresolved: only domain-size bound 0 <= dim ker Coll_fixed <= 3321 (old bound 22 withdrawn as unsupported)"
    },
```

```
    "artifacts": [
      "fixed_data_reading.py",
      "fixed_data_reading.json",
      "fixed_data_reading_out.txt",
      "prep_plateau_sa.json",
      "wpC/"
    ],
    "time_only": {
      "ker_Coll_t_by_D": [
        0,
        0,
        0,
        0,
        0
      ],
      "ker_H_t_by_D": [
        26,
        0,
        0,
        0,
        0
      ],
      "var_image": 26,
      "chi_plateau_ker_H_t_by_D": [
        26,
        0,
        156,
        0,
        390
      ],
      "chi_plateau_extra": 546,
      "route1_equals_route2": true,
      "status": "Constructed (universal time-only reading, on the current coherent presentation)",
      "actual_data": "sigma_eps explicit (XII-C sec:freeze); linear independence of sigma^(k) on open subintervals of (0,eps) proved (lem:to-indep); transfer is Proposition prop:to-actual with hypotheses, not Constructed",
      "sufficient_test": "full-rank exact evaluation matrix on open U of sigma^(k) d^rho chi_I in G, Coll_t, H_t; reduces to linear independence of occurring switch jets d^rho chi_I on U (not verified)"
    }
  },
  "stage3_M5_ledger": {
    "named_theorem": "PlateauCurvatureSquareMatching",
    "ledgers": {
      "literal": "filled",
      "mod_TD": "filled",
      "EOM": "empty/Unresolved (EOM undefined)",
      "field_redef": "empty/Unresolved (redefs undefined)"
    },
    "completion_gate": "obstruction inside formal/chi-plateau selected image",
    "artifacts": [
      "counterterm_functionals.py",
      "matching_matrix.json",
      "matching_matrix.txt",
      "matching_certificates_M5.txt",
      "ledger_literal.json",
      "ledger_mod_TD.json",
      "ledger_EOM.json",
      "ledger_field_redef.json",
      "ledger_tables.txt",
      "m5_ledger_summary.json"
    ]
  },
  "M6_scaling": {
    "dilation_identity_N0": "Constructed",
    "A_mu_equals_A_N0": true,
    "A_mu_equals_A_N_ge_1": "Resolved: A_mu = calA_mu exactly for N=0..4 (B_mu=0), via mu dM^{-1} = D M^{-1} - M^{-1} D (thm: wpB-B0)",
    "residue_S2": {
      "status": "Constructed (Gate S-bv certified)",
      "value": "-2 i pi^2 c^2",
      "value_certified_c": "-i/(8*pi^2)",
      "nonzero": true
    },
    "residue_higher_N": "top-order residues exact, varpi-independent (units i pi^2 c^2 = i/(16 pi^2)); lower orders depend on nu_k[varpi], k=1..4 (XVI-L does not fix varpi; radial comparison is a separately named specialization): Unresolved values",
    "M_C_claimable": false,
    "S_bv_discharged": true,
    "C_anom_scope": "N=0 anomaly component; higher-grade coordinates set to zero for this probe; does not determine the full graph scale variation",
    "N0_component_double_plateau": "collapses to 0 (no nonzero plateau matching)",
    "M_C_status": "Not achieved (Unresolved): broad M-C is the matched-scale criterion (i) proved functional matching and (ii)
```

```
      explicitly specified cutoff varpi with a functional representation of the entire collapsed scale variation; (i) holds (I_top_scale) but (ii) fails because XVI-L leaves varpi unspecified (thm:L6). Their numerical values and the variationality of the resulting full tensor are not determined for the unspecified cutoff.",
    "M_C_reason_historical": "M-C remains incomplete because the full controlled matched scaling calculation is unfinished. The N=0 component alone contributes no nonzero plateau matching. [superseded by M_C_status]",
    "double_plateau_scale_tensor": {
      "variational": true,
      "lorentz_covariant": true,
      "gauge_invariant": false,
      "in_span_R2_Ric2": false,
      "rank_with_targets": "2 -> 3",
      "not": [
        "EH",
        "ultralocal",
        "matched combination"
      ]
    },
    "time_only_scale_tensor": "non-variational (cor:wpB-timeonly; derived at integration)",
    "M_C_status_audit2_historical": "Proved obstructed (graph-specific matching domain): for the specified two-edge graph and the XVI-L prescription, the controlled scale variation is not matched by the curvature targets of the M.5 ledger; not a claim about gravity generally; a revised matching domain (e.g. including gauge-fixing/ghost or further graph contributions) would be a new domain",
    "M_C_curv_status": "Proved obstructed (graph-specific): curvature-ledger scaling criterion (def:MCcurv); for the specified two-edge graph, the XVI-L prescription and any admissible varpi the full collapsed scale variation is not in the span of the M.5 curvature targets because its fixed D=4 component already lies outside; not a claim about gravity generally",
    "double_plateau_top_vs_full": "In the double plateau, the cutoff-independent D=4 component is variational and Lorentz covariant, but not linearized-gauge invariant. The full scale tensor has cutoff-dependent lower-degree components and need not be variational; full-tensor Lorentz covariance not established",
    "separating_functional": {
      "basis_order": [
        "h_mn box^2 h^mn",
        "h box^2 h",
        "(d^m h_mn) box (d_r h^rn)",
        "h box d^m d^n h_mn",
        "(d^m d^n h_mn)^2"
      ],
      "ell": [
        1,
        1,
        -1,
        1,
        1
      ],
      "ell_R2": "0",
      "ell_Ric2": "0",
      "ell_scale": "-1/16",
      "interpretation": "pure gauge h_00=f(t): L_j == (1,1,-1,1,1) f f'''' mod total derivatives; E^00 coefficient of f'''' for the collapsed top tensor = 2*ell(scale) = -1/8 (units i pi^2 c^2)",
      "scope": "certifies the displayed functional only; graph->functional obligation carried by the collapse certificate"
    },
    "M_C_data_status": "Satisfied: proved functional matching plus explicitly specified prescription-dependent scaling formulas (residues, nu_k[varpi], top-order coefficients; Theorem thm:wpB-B0, thm:wpB-dp).",
    "M_C_definition": "broad M-C: (i) at least one proved functional matching; (ii) explicit varpi and functional representation of the entire collapsed scale variation. M-C_data: matching + explicit prescription-dependent scaling formulas. M-C_curv: match inside M.5 curvature ledger."
  },
  "stage5_M7": {
    "status": "Constructed after audit (exactly verified items; Gate V for SA items; (S-bv) certified by thm:wpA-Sbv); Obs_L(R_L) Unresolved",
    "contact_representation": "W_N = span{d_gamma delta} (lower-index coordinate derivatives) with pull-back action; intertwiner z^mu <-> d^mu delta = eta^{mu nu} d_nu delta; C_{N,b,gamma} = (-1)^{g1+g2+g3} [Q^j Box^k b]_gamma",
    "equivariant_dims_by_N": [
      1,
      1,
      3,
      3,
      6
    ],
    "K_block_equivariant": 5,
    "C_reg_equivariant": 9,
    "Equiv_cap_SA": {
      "formal": {
        "dim_V": 1,
        "pi_K": 0,
        "V_cap_K": 0,
        "pi_reg": 1,
        "V_cap_Creg": 1
      },
      "chi_plateau": {
        "dim_V": 1,
        "pi_K": 0,
```

```
        "V_cap_K": 0,
        "pi_reg": 1,
        "V_cap_Creg": 1
      },
      "double_plateau": {
        "dim_V": 12,
        "pi_K": 5,
        "V_cap_K": 3,
        "pi_reg": 9,
        "V_cap_Creg": 7,
        "dim_Coll_V": 9
      }
    },
    "divergence_identity": "d_mu E^{mu nu} = 0 (E upper indices, no metric factor) for int w R_lin^2, int w Ric_lin^2 and any
     combination, any fixed weight w; EH: (d_mu w) G_lin^{mu nu}; ultralocal: nonzero (not gauge invariant)",
    "matched_coefficient": "188813/88456 (unchanged)",
    "withdrawn": [
      "double-plateau Equiv cap SA = 5 / 5 (K) / 3 (C_reg)",
      "nonzero divergence of curvature-square Euler tensor"
    ],
    "Obs_L_RL": "Unresolved",
    "prescription_vs_Ward": "kept distinct",
    "M8_recommendation": "wait for XVI-N"
  },
  "wpA": {
    "section": "sec:wpA-Sbv",
    "theorem": "thm:wpA-Sbv",
    "c": "1/(4*pi^2)",
    "boundary_side": "Q-i0",
    "script": "wpA/kernel_identification_checks.py",
    "output": "wpA/kernel_identification_out.txt",
    "ledger": "wpA/kernel_conventions.json"
  },
  "wpB": {
    "section": "sec:wpB",
    "B_mu_zero_all_N": [
      0,
      1,
      2,
      3,
      4
    ],
    "top_unit": "i/(16*pi^2)",
    "N0_top": "-i/(8*pi^2)",
    "N1_top": "-i/(16*pi^2)",
    "nu_k_varpi_dependence": [
      1,
      2,
      3,
      4
    ],
    "double_plateau_5vectors": {
      "scale": [
        "-71/120",
        "-27/80",
        "-11/10",
        "19/30",
        "-13/15"
      ],
      "R2": [
        "0",
        "1",
        "0",
        "-2",
        "1"
      ],
      "RC2": [
        "1/4",
        "1/4",
        "1/2",
        "-1/2",
        "1/2"
      ],
      "rank": "2 -> 3"
    },
    "script": "wpB/higher_grade_scaling.py",
    "output": "wpB/higher_grade_scaling_out.txt"
  },
  "wpC": {
    "section": "sec:to-timeonly",
    "ker_H_t_by_D": [
```

```
      26,
      0,
      0,
      0,
      0
    ],
    "chi_plateau_ker_H_t_by_D": [
      26,
      0,
      156,
      0,
      390
    ],
    "matching_direction": [
      "1",
      "-89/140"
    ],
    "EH_image": false,
    "R2_Ric2_images_time_only": false,
    "double_plateau_matching_survives_time_only": false,
    "actual_data": "Proposition with explicit hypotheses (downgraded from Constructed)",
    "wronskian_s_half": "88060251340800",
    "script": "wpC/time_only_helmholtz.py",
    "output": "wpC/time_only_helmholtz_out.txt"
  },
  "audit3": {
    "date": "2026-10-06",
    "scope": "integration and proof audit (no expansion)",
    "items": [
      "S-bv provenance (XVI-J eq:W/eq:DF/eq:rational/prop:q; XI-E eq:h); lem:wpA-varreg; epsilon->tau",
      "top vs full scale tensor wording",
      "separating functional ell=a+b-c+d+e, ell(scale)=-1/16",
      "M-C_curv Proved obstructed; broad M-C not achieved; M-C_data Satisfied",
      "time-only joint-family condition, rem:to-joint",
      "proof corrections lem:wpB-dw, lem:bv(c), prop:M7-id(B'), shell comments, regression labels"
    ]
  },
  "predecessor_provenance": {
    "role_XVI_J_eq_rational": "supplementary corroboration; NOT part of frozen XVI-L dependency",
    "independent_proof": "thm:wpA-Sbv + lem:wpA-varreg",
    "archive_dir": "XVI-M-artifacts/provenance/",
    "pins_file": "XVI-M-artifacts/provenance/predecessor_pins.json"
  }
}
```


3.2 Landmark XI-A (full source)

sha256: see inventory / `sha256sum provenance/XI-A.tex`.

Listing 56: XI-A.tex

```
% Reconstructed from live /browse/XI-A chunks (fileName=XI-A.tex, contentHash=0662c6c3, version=1); gaps between chunks omitted
%%% [chunk XI-A-abstract-L112 kind=abstract lines 112-140 labels=[]]
\begin{abstract}
We specify a classical Einstein--scalar model compatible with the
signature, scalar switching prescription, and plateau gauge algebra
of Landmark X-A. The Einstein--Hilbert density is not switched.
Its sign is fixed by a positive transverse-traceless kinetic term
and an attractive Newtonian comparison. We display the quadratic
Fierz--Pauli action, one de Donder gauge-fixing term, and the metric
Euler derivative through coupling order $\kappa^2$.

Several distinctions are indispensable. Minimal scalar coupling
introduces the switch itself, but no switch derivatives, into the
metric Euler derivative. Switch derivatives appear in the scalar
equation and in the divergence of the metric source. Nonlinear
Noether identities require the classical higher gravitational
coefficients; the quadratic action alone is insufficient.
Gauge fixing supplies a separate breaking term and is not
diffeomorphism invariant on the plateau.

We prove the exact fixed-background Noether identity and the tensor-density
covariance of the invariant metric Euler operator on the admitted
plateau group. X-A's varied-action formula is used as a premise,
not presented as a new relational covariance theorem. Finally, for
a specified free scalar input, the ungauge-fixed coupled Einstein
jet is empty for every nonzero compact switch of the declared class.
```

```
This is not an emptiness theorem for all inputs, nor for the
gauge-fixed wave equations. The manuscript stops at the classical
sheet: no propagator, quantum multiplet, tree, or BRST class is
constructed.
\end{abstract}
%%% [chunk II-A-scope-L142 kind=scope lines 142-156 labels=[]]
\scopebox{Release: a corrected classical sheet, with the on-shell
obstruction separated}{%
The signed action, quadratic operator, metric Euler coefficients,
Noether identity with its switching and gauge-fixing terms, and
plateau covariance of the \emph{invariant} metric Euler derivative
are constructed.

For the fixed free input of Theorem~\ref{thm:empty}, the coupled
\emph{Einstein} jet is empty. The unrestricted jet family is not
declared empty.

Neither invariance of the gauge-fixed action nor nonlinear covariance
of a Fierz--Pauli-only truncation is asserted. Those versions of the
proposed sheet are false. The metric remains off shell throughout
the construction of the sheet.}
%%% [chunk II-A-section-L162 kind=section lines 162-220 labels=[]]

%=====================================================================

The objective is to give a classical action and its identities before
constructing any quantum metric. The relational prescription of X-A
is retained, but no quantum conclusion is inferred from its classical
covariance.

A production plan cannot require both a background de Donder term
and an unbroken diffeomorphism identity for the resulting action.
It also cannot identify a quadratic gravitational action with the
nonlinear Einstein action at the next two coupling orders.
Accordingly, the model below has one declared gauge-fixed action,
together with its explicitly identified invariant part:
\[
 S_{\mathrm{tot}}=S_{\inv}+S_{\gf},
 \qquad
 S_{\inv}=S_{\EH}+S_g^{\mathrm{sc}}.
\]
The identities for these two functionals are different and are printed
separately.

Five corrections determine the release.

\begin{enumerate}
\item
The Einstein--Hilbert action is specified before expansion. Its
quadratic coefficient is Fierz--Pauli. Its cubic and quartic
coefficients are retained in the classical sheet when required by
coupling order. They are not evaluated as graphs.

\item
The scalar action contains no metric derivatives. Its metric variation
therefore contains $g$ but no $\partial g$, including at second order.
Adding such terms would change the stated minimally coupled model.

\item
The nonlinear Noether identity uses the formal adjoint of the
metric pullback action. A bare partial divergence is only its
leading flat contribution.

\item
Covariance of a metric Euler derivative means covariance as a
rank-two tensor density. Gauge fixing contributes a noncovariant
background-dependent term.

\item
An empty jet must have a specified input. The zero scalar and flat
metric provide an immediate counterexample to universal emptiness.
\end{enumerate}

The broader perturbative treatment of gravity distinguishes classical
gauge symmetry, gauge fixing, and subsequent BV quantization
\cite{BFR}. Here those distinctions are enforced without opening a
BV complex. Effective-field-theory discussions likewise do not justify
deleting the classical Einstein self-interactions
\cite{Donoghue}.

%=====================================================================
```

```
%%% [chunk II-A-section-L225 kind=section lines 225-270 labels=[]]


Use the X-A conventions:
\[
 \MT=[0,T]\times\R^3,\qquad
 \MTo=(0,T)\times\R^3,\qquad
 \Sigma=\{t=0\},
\]
\[
 \eta_{\mu\nu}=\operatorname{diag}(1,-1,-1,-1),
 \qquad
 \Box=\partial_t^2-\Delta,
 \qquad
 \Pfree=\Box+m^2,
 \qquad m\geq0.
\]
The mass is fixed for each instance of the model. All coefficient
fields are real and smooth on a neighbourhood of the slab.

Fix $g\in C_c^\infty(\MTo;[0,1])$ with nonempty plateau
$P_g=\Int\{g=1\}$. Fix $0<\eps<T$ and
\[
 \sig(t)=\chi(t/\eps),
 \qquad
 \chi(s)=
 \begin{cases}
 0,&s\leq0,\\[2pt]
 \displaystyle\frac{e^{-1/s}}{e^{-1/s}+e^{-1/(1-s)}},&0<s<1,\\[7pt]
 1,&s\geq1.
 \end{cases}
\]
Choose an open relatively compact $U$ such that
\[
 \overline U\subset P_g\cap\{\eps<t<T\},
 \qquad
 \ga=C_c^\infty(U;TM).
\]
The group $\Gp$ is generated by the corresponding compactly
supported flows. Every admitted transformation is the identity near
$\Sigma$ and preserves $g$ and $\sig$.

For coefficient calculations, genuine flows of
$\kappa\xi+\kappa^2\zeta$, $\xi,\zeta\in\ga$, provide all Taylor
coefficients needed below. They lift X-A's second-order formal
transformations without enlarging its support class.


%%% [chunk II-A-section-eq-q kind=section lines 271-308 labels=['eq:q', 'eq:b', 'eq:a1', 'eq:a2']]


The field variable is one classical symmetric tensor $\gamma$:
\begin{equation}
 q_{\mu\nu}=\eta_{\mu\nu}+\kappa\gamma_{\mu\nu}.
 \label{eq:q}
\end{equation}
There are no independent quantum fields $\widehat h,\widehat k$.
Write $\tau=\eta^{\mu\nu}\gamma_{\mu\nu}$ and set
\begin{align}
 b_1[\gamma]&=\tfrac12\tau,
 &
 b_2[\gamma]&=\tfrac18\tau^2
              -\tfrac14\gamma_{\mu\nu}\gamma^{\mu\nu},
 \label{eq:b}\\
 a_1^{\mu\nu}[\gamma]&=\tfrac12\tau\eta^{\mu\nu}
                       -\gamma^{\mu\nu},
 \label{eq:a1}\\
 a_2^{\mu\nu}[\gamma]&=
 \gamma^\mu{}_\rho\gamma^{\rho\nu}
 -\tfrac12\tau\gamma^{\mu\nu}+b_2[\gamma]\eta^{\mu\nu}.
 \label{eq:a2}
\end{align}
Thus
\[
 \sqrt{-q}=1+\kappa b_1+\kappa^2b_2+O(\kappa^3),
 \qquad
 \Adens_q^{\mu\nu}
 =\eta^{\mu\nu}+\kappa a_1^{\mu\nu}
       +\kappa^2a_2^{\mu\nu}+O(\kappa^3).
\]
```

```
These symbols are not coefficients of $\sqrt{-q}R[q]$.

The action is defined as a full formal generating expression before
its Euler derivative is projected through $\kappa^2$. In particular,
$\gamma$ may later be evaluated on a formal solution family.
This does not introduce another fundamental metric field.


%%% [chunk II-A-section-L309 kind=section lines 309-333 labels=[]]


X-A supplies the coordinate prescription
\[
 \partial_\alpha\left(
  \bigl[\eta^{\alpha\beta}
  +\sig(\Adens_q^{\alpha\beta}-\eta^{\alpha\beta})\bigr]
       \partial_\beta\Coord_q^\mu\right)=0,
\]
with the fixed data
$\Coord_q^\mu|_\Sigma=x^\mu|_\Sigma$ and
$\partial_t\Coord_q^\mu|_\Sigma=\delta^\mu_0$.
The observable is the fixed label-space smearing of
$\phi\circ\Coord_q^{-1}$.

Its plateau covariance and X-A's Route II are premises
\cite{XA}. They are not reproved here. The ramp does not enter the
gravitational action.

X-B, X-C, and X-D are not inputs. VI and VII-A are comparison
manuscripts with different signature and metric prescriptions.
VII-C, X-Enter, and the cone papers supply no premise. No quantum
cancellation from another relational prescription is imported
\cite{Frob}.


%%% [chunk II-A-section-L334 kind=section lines 334-347 labels=[]]


The factor $\kappa\Coord_1[\gamma]$ has one metric leg.
This observation alone does not eliminate a cubic Einstein vertex:
that vertex has coefficient $\kappa$, so the two factors have total
coupling degree two before any additional graph requirements are
specified. No rooted-tree topology is selected or calculated here.

More importantly, an action identity is not a graph selection rule.
The nonlinear classical identities require the relevant Einstein
coefficients. The Fierz--Pauli-only alternative is explicitly audited
in Section~\ref{sec:quadraticfailure}.

%=====================================================================

%%% [chunk II-A-section-eq-ricci kind=section lines 352-401 labels=['eq:ricci', 'eq:EH', 'eq:scalar-action', 'eq:gf']]


Fix the curvature convention
\begin{equation}
 R_{\mu\nu}
 =\partial_\rho\Gamma^\rho_{\mu\nu}
  -\partial_\nu\Gamma^\rho_{\mu\rho}
  +\Gamma^\rho_{\rho\lambda}\Gamma^\lambda_{\mu\nu}
  -\Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\rho},
 \label{eq:ricci}
\end{equation}
with the Levi-Civita Christoffel symbols and
$G_{\mu\nu}=R_{\mu\nu}-\frac12q_{\mu\nu}R$.
Use
\begin{equation}
 S_{\EH}[q]=-\frac{2}{\kappa^2}
                 \int_{\MT}\sqrt{-q}\,R[q]\dd^4x.
 \label{eq:EH}
\end{equation}
There is no cosmological constant and no gravitational switch.

The unchanged X-A scalar density is
\begin{align}
 \Lag_q[\phi]&=\tfrac12\sqrt{-q}
       \left(q^{\mu\nu}\partial_\mu\phi\partial_\nu\phi
                  -m^2\phi^2\right),\\
 \Lag_\eta[\phi]&=\tfrac12
       \left(\partial_\mu\phi\partial^\mu\phi-m^2\phi^2\right),\\
 S_g^{\mathrm{sc}}[q,\phi]
```

```
 &=\int_{\MT}\bigl((1-g)\Lag_\eta[\phi]+g\Lag_q[\phi]\bigr)\dd^4x.
 \label{eq:scalar-action}
\end{align}
Actions are interpreted through compact interior variations and compact
differences. Numerical finiteness on every smooth configuration is not
assumed. No metric boundary variation is taken.

Define
\[
 \bar\gamma_{\mu\nu}
 =\gamma_{\mu\nu}-\tfrac12\eta_{\mu\nu}\tau,
 \qquad
 D^\nu(\gamma)=\partial_\mu\bar\gamma^{\mu\nu}.
\]
The single frozen gauge-fixing term is
\begin{equation}
 S_{\gf}[\gamma]=\int_{\MT}D_\nu(\gamma)D^\nu(\gamma)\dd^4x.
 \label{eq:gf}
\end{equation}
Its coefficient has been fixed; no gauge-parameter family is used.


%%% [chunk II-4-proposition-prop-FP kind=proposition lines 404-445 labels=['prop:FP', 'eq:FP', 'eq:Glin']]
\begin{proposition}[Quadratic gravitational coefficient]
\label{prop:FP}
Modulo a total divergence, the quadratic bulk coefficient of
\eqref{eq:EH} is
\begin{align}
 S_{\FP}[\gamma]=\frac12\int_{\MT}\Bigl(
 &\partial_\lambda\gamma_{\mu\nu}
          \partial^\lambda\gamma^{\mu\nu}
 -2(\partial_\mu\gamma^{\mu\nu})
          (\partial^\rho\gamma_{\rho\nu})
 \nonumber\\
 &+2(\partial_\mu\gamma^{\mu\nu})\partial_\nu\tau
 -\partial_\lambda\tau\,\partial^\lambda\tau
 \Bigr)\dd^4x.
 \label{eq:FP}
\end{align}
Its Euler derivative is $2G_{\mathrm{lin}}^{\mu\nu}(\gamma)$, where
\begin{equation}
 G_{\mathrm{lin}}^{\mu\nu}(\gamma)
 =-\tfrac12\Box\bar\gamma^{\mu\nu}
  +\partial^{(\mu}D^{\nu)}
  -\tfrac12\eta^{\mu\nu}\partial_\rho D^\rho.
 \label{eq:Glin}
\end{equation}
\end{proposition}

\begin{proof}
Expanding \eqref{eq:ricci} to first order gives
\eqref{eq:Glin}. Variation with respect to the covariant metric gives
\[
 \delta S_{\EH}
 =\frac{2}{\kappa^2}\int\sqrt{-q}\,
             G^{\mu\nu}(q)\delta q_{\mu\nu}\dd^4x.
\]
Consequently the quadratic coefficient has derivative
$2G_{\mathrm{lin}}(\gamma)$. Integration by parts in \eqref{eq:FP}
gives that same derivative. Both quadratic functionals vanish at
$\gamma=0$; integrating their equal directional derivatives along
$s\gamma$, $0\leq s\leq1$, proves equality as bulk variational
functionals. The linear Einstein--Hilbert coefficient is a
divergence and has no bulk Euler derivative.
\end{proof}
%%% [chunk II-4-lemma-lem-TT kind=lemma lines 447-464 labels=['lem:TT']]
\begin{lemma}[Kinetic sign]
\label{lem:TT}
For $\gamma_{0\mu}=0$, $\partial_i\gamma_{ij}=0$, and
$\delta^{ij}\gamma_{ij}=0$, the quadratic density becomes
\[
 \Lag_{\FP}^{\TT}
 =\tfrac12\dot\gamma_{ij}\dot\gamma_{ij}
  -\tfrac12\partial_k\gamma_{ij}\partial_k\gamma_{ij}.
\]
Its time-derivative term has the same sign as
$\frac12\dot\phi^2$.
\end{lemma}

\begin{proof}
The trace and divergence terms in \eqref{eq:FP} vanish.
```

```
Raising both spatial indices contributes two minus signs, so
$\gamma^{ij}=\gamma_{ij}$. The stated expression follows.
\end{proof}
%%% [chunk II-A-lemma-lem-Newton kind=lemma lines 466-494 labels=['lem:Newton', 'eq:Newton']]
\begin{lemma}[Newtonian sign comparison]
\label{lem:Newton}
Let $H=q-\eta$ denote the physical metric perturbation. For an
auxiliary static conserved source
$T_{00}=\rho\geq0$, $T_{0i}=T_{ij}=0$, $\rho$ smooth compactly
supported and nonzero, the linear unswitched equation is solved by
\[
 H_{00}=2\Phi_N,\qquad
 H_{ij}=2\Phi_N\delta_{ij},\qquad H_{0i}=0,
\]
where
\begin{equation}
 \Delta\Phi_N=\frac{\kappa^2}{8}\rho,\qquad
 \Phi_N(x)=-\frac{\kappa^2}{32\pi}
       \int_{\R^3}\frac{\rho(y)}{|x-y|}\dd^3y<0.
 \label{eq:Newton}
\end{equation}
Thus $\kappa^2=32\pi G_N$ gives the attractive Newtonian sign.
\end{lemma}

\begin{proof}
The scalar metric variation computed below is negative, so the
unswitched metric equation is $G_{\mu\nu}=\kappa^2T_{\mu\nu}/4$.
For the displayed perturbation,
$\bar H_{00}=4\Phi_N$, $\bar H_{ij}=0$, and $D(H)=0$.
Equation \eqref{eq:Glin} gives
$G_{\mathrm{lin},00}(H)=2\Delta\Phi_N$ and zero spatial components.
The Newton inverse of $\Delta$ gives \eqref{eq:Newton}.
\end{proof}
%%% [chunk II-A-section-L494 kind=section lines 494-501 labels=[]]


This static calculation is a sign diagnostic, not a solution of the
compact-time switched model and not an adiabatic-limit construction.
The canonically normalized field is $\gamma=H/\kappa$; confusing
$\gamma_{00}$ with $H_{00}$ would change the normalization check.

%=====================================================================

%%% [chunk II-A-section-L506 kind=section lines 506-516 labels=[]]


Put $p_\mu=\partial_\mu\phi$ and define
\[
 \ell_q=\tfrac12(q^{\alpha\beta}p_\alpha p_\beta-m^2\phi^2),
 \qquad
 T_q^{\mu\nu}
 =q^{\mu\alpha}q^{\nu\beta}p_\alpha p_\beta-q^{\mu\nu}\ell_q.
\]
Write $T_0=T_\eta$ and $\ell_0=\Lag_\eta$.


%%% [chunk II-A-lemma-lem-vertex kind=lemma lines 517-544 labels=['lem:vertex', 'eq:vertex']]
\begin{lemma}[Stress sign and inherited vertex]
\label{lem:vertex}
For a covariant metric variation,
\[
 \delta_q\Lag_q=-\tfrac12\sqrt{-q}\,
                    T_q^{\mu\nu}\delta q_{\mu\nu}.
\]
In particular,
\begin{equation}
 V_{1,g}
 =\tfrac12\int g(a_1^{\mu\nu}p_\mu p_\nu-m^2b_1\phi^2)\dd^4x
 =-\tfrac12\int g\,\gamma_{\mu\nu}T_0^{\mu\nu}\dd^4x.
 \label{eq:vertex}
\end{equation}
\end{lemma}

\begin{proof}
Use
$\delta\sqrt{-q}=\frac12\sqrt{-q}\,q^{\mu\nu}\delta q_{\mu\nu}$
and
$\delta q^{\alpha\beta}=-q^{\alpha\mu}q^{\beta\nu}\delta q_{\mu\nu}$.
For the second assertion,
\[
 a_1^{\mu\nu}p_\mu p_\nu-m^2b_1\phi^2
```

```
 =-\gamma^{\mu\nu}p_\mu p_\nu+\tau\ell_0
 =-\gamma_{\mu\nu}T_0^{\mu\nu}.
\]
\end{proof}
%%% [chunk II-A-section-eq-V2 kind=section lines 544-554 labels=['eq:V2']]


The second matter coefficient is fixed, not guessed:
\begin{equation}
 V_{2,g}=\tfrac12\int
     g(a_2^{\mu\nu}p_\mu p_\nu-m^2b_2\phi^2)\dd^4x.
 \label{eq:V2}
\end{equation}
One switch multiplies this coefficient; replacing it by $g^2$
would change X-A's prescription.


%%% [chunk II-A-section-eq-Hdef kind=section lines 555-585 labels=['eq:Hdef', 'eq:J1']]


Define the Einstein tensor-density coefficients by
\begin{equation}
 \sqrt{-q}\,G^{\mu\nu}(q)
 =\kappa\Hcal_1^{\mu\nu}
  +\kappa^2\Hcal_2^{\mu\nu}
  +\kappa^3\Hcal_3^{\mu\nu}+O(\kappa^4).
 \label{eq:Hdef}
\end{equation}
Here $\Hcal_1=G_{\mathrm{lin}}(\gamma)$.
Appendix~\ref{app:coefficients} gives finite explicit formulas for
all three coefficients, without an external curvature convention.

Let
\[
 \sqrt{-q}\,T_q^{\mu\nu}
 =T_0^{\mu\nu}+\kappa J_1^{\mu\nu}+O(\kappa^2).
\]
Direct expansion gives
\begin{align}
 J_1^{\mu\nu}[\gamma,\phi]
 ={}&b_1T_0^{\mu\nu}
 -\gamma^{\mu\alpha}p_\alpha p^\nu
 -\gamma^{\nu\alpha}p_\alpha p^\mu
 \nonumber\\
 &+\gamma^{\mu\nu}\ell_0
 +\tfrac12\eta^{\mu\nu}\gamma^{\alpha\beta}p_\alpha p_\beta.
 \label{eq:J1}
\end{align}


%%% [chunk II-A-theorem-thm-euler kind=theorem lines 586-627 labels=['thm:euler', 'eq:eexact', 'eq:eexp', 'eq:egf']]
\begin{theorem}[Metric Euler coefficients]
\label{thm:euler}
With
$\Egamma^{\mu\nu}=\delta S_{\inv}/\delta\gamma_{\mu\nu}$,
the exact formal derivative and its order-two expansion are
\begin{align}
 \Egamma^{\mu\nu}
 &=\frac{2}{\kappa}\sqrt{-q}\,G^{\mu\nu}(q)
       -\frac{\kappa}{2}g\sqrt{-q}\,T_q^{\mu\nu},
 \label{eq:eexact}\\
 &=2\Hcal_1^{\mu\nu}
   +\kappa\left(2\Hcal_2^{\mu\nu}
                        -\tfrac12gT_0^{\mu\nu}\right)
   +\kappa^2\left(2\Hcal_3^{\mu\nu}
                        -\tfrac12gJ_1^{\mu\nu}\right)
   +O(\kappa^3).
 \label{eq:eexp}
\end{align}
The gauge-fixing contribution is separately
\begin{equation}
 \Egamma_{\gf}^{\mu\nu}
 =-2\partial^{(\mu}D^{\nu)}
       +\eta^{\mu\nu}\partial_\rho D^\rho,
 \qquad
 \partial_\mu\Egamma_{\gf}^{\mu\nu}=-\Box D^\nu.
 \label{eq:egf}
\end{equation}
Thus $\delta S_{\mathrm{tot}}/\delta\gamma=\Egamma+\Egamma_{\gf}$.
There are no $\partial g$ terms in \eqref{eq:eexact} or
\eqref{eq:eexp}.
```

```
\end{theorem}

\begin{proof}
Combine the Einstein variation in Proposition~\ref{prop:FP},
Lemma~\ref{lem:vertex}, and $\delta q=\kappa\delta\gamma$.
The flat scalar density is independent of $\gamma$.
Equations \eqref{eq:Hdef} and \eqref{eq:J1} give the expansion.
Varying $D_\nu D^\nu$ and integrating by parts gives
\eqref{eq:egf}; taking its divergence gives the last identity.
No integration by parts in a metric variation of the scalar density
is needed because that density has no metric derivatives.
\end{proof}
%%% [chunk II-A-section-L627 kind=section lines 627-644 labels=[]]


For $\phi=\phi_0+\kappa\phi_1+\cdots$, the order-one scalar source
is exactly $-\frac12gT_0[\phi_0]$. At order two, replace
$J_1[\gamma,\phi]$ in \eqref{eq:eexp} by
\[
 J_1[\gamma,\phi_0]+\dot T_0[\phi_0;\phi_1],
\]
where
\[
 \dot T_0^{\mu\nu}
 =p_0^\mu p_1^\nu+p_1^\mu p_0^\nu
  -\eta^{\mu\nu}
       (p_{0\alpha}p_1^\alpha-m^2\phi_0\phi_1).
\]
This distinguishes the metric variation of the stress density from
the correction due to the scalar solution.


%%% [chunk II-A-section-eq-Ephi kind=section lines 645-673 labels=['eq:Ephi', 'eq:scalar-switch']]


The scalar Euler derivative is
\begin{equation}
 \Ephi=\frac{\delta S_{\inv}}{\delta\phi}
 =-\left(\Pfree+\kappa\mathsf V_{1,g}
                  +\kappa^2\mathsf V_{2,g}\right)\phi+O(\kappa^3),
 \label{eq:Ephi}
\end{equation}
with
\begin{align}
 \mathsf V_{1,g}\phi
 &=g\left[\partial_\mu(a_1^{\mu\nu}p_\nu)+m^2b_1\phi\right]
       +(\partial_\mu g)a_1^{\mu\nu}p_\nu,\\
 \mathsf V_{2,g}\phi
 &=g\left[\partial_\mu(a_2^{\mu\nu}p_\nu)+m^2b_2\phi\right]
       +(\partial_\mu g)a_2^{\mu\nu}p_\nu.
 \label{eq:scalar-switch}
\end{align}
Also,
\[
 \partial_\mu(gT_0^{\mu\nu})
 =g(\Pfree\phi)\partial^\nu\phi
        +(\partial_\mu g)T_0^{\mu\nu}.
\]
These are the switch-derivative terms required by the stated model.
They must not be moved into its metric Euler derivative.

%=====================================================================

%%% [chunk II-A-section-eq-RouteII kind=section lines 678-708 labels=['eq:RouteII']]


For a compactly supported diffeomorphism $F$ in the interior,
X-A's Route II gives
\begin{align}
 \delta_F S_g^{\mathrm{sc}}
 ={}&\int(1-g)
          \bigl(\Lag_\eta[F^*\phi]-\Lag_\eta[\phi]\bigr)\dd^4x
 \nonumber\\
 &-\int(\partial_\mu g)
                Z_F^\mu[\Lag_q[\phi]]\dd^4x,
 \label{eq:RouteII}
\end{align}
where
$F_{\dens}^*r-r=\partial_\mu Z_F^\mu[r]$.
For the flow of $\kappa\xi+\kappa^2\zeta$,
\[
```

```
 Z_F^\mu[r]
 =\kappa\xi^\mu r
   +\kappa^2\left(\zeta^\mu r
          +\tfrac12\xi^\mu\partial_\rho(\xi^\rho r)\right)
   +O(\kappa^3).
\]
The proof of \eqref{eq:RouteII} is not repeated.

The unswitched Einstein--Hilbert density is natural. The same
divergence lemma therefore shows
$\delta_FS_{\EH}=0$ for compact interior $F$.
Consequently \eqref{eq:RouteII} is also the finite variation of
$S_{\inv}$. There is no analogous assertion for $S_{\gf}$.


%%% [chunk II-A-section-eq-N kind=section lines 709-735 labels=['eq:N', 'eq:B']]


For a symmetric contravariant density $u^{\mu\nu}$ define
\begin{equation}
 \Ncal_\nu(u)
 =2\partial_\mu(u^{\mu\rho}q_{\rho\nu})
            -u^{\mu\rho}\partial_\nu q_{\mu\rho}.
 \label{eq:N}
\end{equation}
For an arbitrary interior compact vector field $\xi$, the scaled
infinitesimal transformation is
\[
 \delta_\xi q=\kappa\Lie_\xi q,\qquad
 \delta_\xi\gamma=\Lie_\xi q,\qquad
 \delta_\xi\phi=\kappa X_\xi\phi.
\]
Thus $\delta_\xi\gamma=K\xi+\kappa\Lie_\xi\gamma$.

Introduce the explicit background-breaking density
\begin{equation}
 \Bcal_\nu
 =(\partial_\mu g)T_0^\mu{}_\nu
       -(1-g)(\Pfree\phi)\partial_\nu\phi
       +(\partial_\nu g)(\Lag_\eta[\phi]-\Lag_q[\phi]).
 \label{eq:B}
\end{equation}


%%% [chunk II-A-theorem-thm-noether kind=theorem lines 736-778 labels=['thm:noether', 'eq:noether', 'eq:noether-gf']]
\begin{theorem}[Fixed-background Noether identity]
\label{thm:noether}
Off shell, the invariant action satisfies
\begin{equation}
 \boxed{\quad
 \Ncal_\nu(\Egamma)
 =\kappa\left(\Ephi\,\partial_\nu\phi-\Bcal_\nu\right).
 \quad}
 \label{eq:noether}
\end{equation}
The gauge-fixed action instead satisfies
\begin{equation}
 \Ncal_\nu(\Egamma+\Egamma_{\gf})
 =\kappa\left(\Ephi\,\partial_\nu\phi-\Bcal_\nu\right)
       +\Ncal_\nu(\Egamma_{\gf}).
 \label{eq:noether-gf}
\end{equation}
On $U$, $\Bcal_\nu=0$. This removes the switching breaking,
not the gauge-fixing contribution.
\end{theorem}

\begin{proof}
By symmetry of $\Egamma$ and integration by parts,
\[
 \delta_\xi S_{\inv}
 =\int\xi^\nu
       \left[-\Ncal_\nu(\Egamma)
                   +\kappa\Ephi\,\partial_\nu\phi\right]\dd^4x.
\]
Take the infinitesimal coefficient of the imported
\eqref{eq:RouteII}. The flat scalar variation is
\[
 \delta_\xi\Lag_\eta
 =\kappa\left(
   T_0^\mu{}_\nu\partial_\mu\xi^\nu
       +\partial_\mu(\xi^\mu\Lag_\eta)\right).
```

```
\]
After integrating these two terms, the coefficient of
$\kappa\xi^\nu$ is exactly \eqref{eq:B}.
The fundamental lemma for arbitrary compact interior $\xi$ proves
\eqref{eq:noether}. Adding the separate gauge-fixing variation proves
\eqref{eq:noether-gf}. On $U$, both $1-g$ and $\partial g$ vanish.
\end{proof}
%%% [chunk II-A-corollary-L780 kind=corollary lines 780-790 labels=[]]
\begin{corollary}[Plateau contraction and Route II]
For $\xi\in\ga$,
\[
 \int\xi^\nu
       \left[-\Ncal_\nu(\Egamma)
                  +\kappa\Ephi\,\partial_\nu\phi\right]\dd^4x=0.
\]
This is the contraction of \eqref{eq:noether} and the infinitesimal
form of X-A's Route II. It is not a new covariance theorem for the
ramp observable.
\end{corollary}
%%% [chunk II-A-section-eq-N1 kind=section lines 792-819 labels=['eq:N1', 'eq:N2']]


For a $\kappa$-independent scalar write
$\Egamma=e_0+\kappa e_1+\kappa^2e_2+\cdots$ and
\[
 Q_\nu(u)
 =2\partial_\mu(u^{\mu\rho}\gamma_{\rho\nu})
                    -u^{\mu\rho}\partial_\nu\gamma_{\mu\rho}.
\]
Then \eqref{eq:noether} gives
\begin{align}
 2\partial_\mu(e_0^{\mu\rho}\eta_{\rho\nu})&=0,\\
 2\partial_\mu(e_1^{\mu\rho}\eta_{\rho\nu})+Q_\nu(e_0)
 &=-g(\Pfree\phi)\partial_\nu\phi
                    -(\partial_\mu g)T_0^\mu{}_\nu,
 \label{eq:N1}\\
 2\partial_\mu(e_2^{\mu\rho}\eta_{\rho\nu})+Q_\nu(e_1)
 &=-(\mathsf V_{1,g}\phi)\partial_\nu\phi
                    +(\partial_\nu g)\Lag^{(1)}[\gamma,\phi],
 \label{eq:N2}
\end{align}
where
$\Lag^{(1)}
 =\frac12(a_1^{\alpha\beta}p_\alpha p_\beta-m^2b_1\phi^2)$.
These formulas expose the missing coupling and nonlinear metric terms
in a bare partial-divergence identity. The source sign in
\eqref{eq:N1} agrees with $-\frac12gT_0$ in the metric derivative.


%%% [chunk II-A-proposition-L823 kind=proposition lines 823-846 labels=[]]
\begin{proposition}[Fierz--Pauli alone is insufficient]
Replacing $S_{\EH}$ by $S_{\FP}$ does not preserve the nonlinear
Noether identity even on the plateau.
\end{proposition}

\begin{proof}
Set $\phi=0$. For the complete Einstein action,
$\Ncal_\nu(\Egamma_{\EH})=0$.
Its order-one identity requires
\[
 2\partial_\mu(2\Hcal_2^{\mu\rho}\eta_{\rho\nu})
                  +Q_\nu(2\Hcal_1)=0.
\]
Deleting the cubic action deletes $2\Hcal_2$ but does not delete
$Q_\nu(2\Hcal_1)$. The latter is not identically zero.
For example, where $\gamma=\psi(t)\eta$,
\[
 Q_0(2G_{\mathrm{lin}}(\gamma))
                  =-6\dot\psi\,\ddot\psi.
\]
Choose a smooth compact tensor supported in $U$ which has this form
near a point where $\dot\psi\ddot\psi\neq0$.
The proposed quadratic nonlinear identity fails there.
\end{proof}
%%% [chunk II-A-proposition-prop-gfbad kind=proposition lines 848-870 labels=['prop:gfbad']]
\begin{proposition}[Gauge fixing is not plateau invariant]
\label{prop:gfbad}
The fixed term \eqref{eq:gf} is not invariant under $\Gp$.
At the linear level its breaking is $-\Box D^\nu$, as in
\eqref{eq:egf}.
\end{proposition}
```

```
\begin{proof}
Take $\xi=f\partial_1$, with nonzero $f\in C_c^\infty(U)$.
Directly,
\[
 D(K\xi)^\nu=\Box\xi^\nu,
 \qquad
 S_{\gf}[K\xi]=-\int(\Box f)^2\dd^4x\neq0.
\]
A nonzero compact interior $f$ cannot satisfy $\Box f=0$, by
Cauchy uniqueness with zero past data.
Starting from $\gamma=0$, a genuine pure-gauge metric has
$\gamma'=K\xi+O(\kappa)$, so the nonzero displayed term is the
leading gauge-fixing action. The Einstein action on that flat
pullback is unchanged. Hence the discrepancy is gauge fixing,
not a switching defect.
\end{proof}
%%% [chunk II-A-section-L870 kind=section lines 870-876 labels=[]]


No compact plateau gauge algebra is recovered by simply demanding
$\Box\xi=0$: compact support to the future of $\Sigma$ and Cauchy
uniqueness then force $\xi=0$.

%=====================================================================

%%% [chunk II-A-section-eq-tensorpull kind=section lines 878-893 labels=['eq:tensorpull']]

%=====================================================================

Let $F^*_{(2,0),\dens}$ denote pullback of a contravariant
rank-two density of weight one:
\begin{align}
 (F^*_{(2,0),\dens}u)^{\mu\nu}(x)
 ={}&\det DF(x)\,
 (DF(x)^{-1})^\mu{}_\alpha
 (DF(x)^{-1})^\nu{}_\beta
 \nonumber\\
 &\hspace{18mm}\times u^{\alpha\beta}(F(x)).
 \label{eq:tensorpull}
\end{align}
All admitted flows preserve orientation.


%%% [chunk II-A-theorem-thm-covariance kind=theorem lines 894-917 labels=['thm:covariance', 'eq:covariance']]
\begin{theorem}[Plateau metric Euler covariance]
\label{thm:covariance}
For $F\in\Gp$, put
$q'=F^*q$, $\phi'=F^*\phi$, and
$\gamma'=(q'-\eta)/\kappa$. Then
\begin{equation}
 \Egamma^{\mu\nu}[q',\phi']
   =(F^*_{(2,0),\dens}\Egamma[q,\phi])^{\mu\nu}.
 \label{eq:covariance}
\end{equation}
The identity is formal and holds after expansion through
$\kappa^2$, provided all coefficients needed before projection are
retained. It does not hold with $\Egamma$ replaced by
$\Egamma+\Egamma_{\gf}$ in general.
\end{theorem}

\begin{proof}
The Einstein tensor-density in \eqref{eq:eexact} is natural.
The scalar stress density $\sqrt{-q}T_q^{\mu\nu}$ is natural as
well. Since $F^*g=g$, its switched multiple transforms by
\eqref{eq:tensorpull}. This proves \eqref{eq:covariance}.
The counterexample in Proposition~\ref{prop:gfbad}, or the explicit
formula \eqref{eq:egf}, excludes the gauge-fixed assertion.
\end{proof}
%%% [chunk II-A-section-eq-covariance-defect kind=section lines 917-949 labels=['eq:covariance-defect']]


If covariant indices are preferred, define
\[
 \mathsf E_{\mu\nu}^{\mathrm{gr}}
       =q_{\mu\alpha}q_{\nu\beta}\Egamma^{\alpha\beta}.
\]
It transforms as a \emph{covariant rank-two density}; its two indices
also acquire Jacobian factors. Treating it as four-by-four scalar
densities would be incorrect.

```

```
For a general compact $F$ which need not preserve $g$, the exact
metric covariance defect is
\begin{align}
 &\Egamma[q',\phi']-F^*_{(2,0),\dens}\Egamma[q,\phi]
 \nonumber\\
 &\qquad
 =-\frac{\kappa}{2}(g-F^*g)\,
                  \sqrt{-q'}\,T_{q'}[\phi'].
 \label{eq:covariance-defect}
\end{align}
Its infinitesimal form involves $X_\xi g$.
The divergence defect involves $(\partial_\mu g)T^{\mu\nu}$,
as in \eqref{eq:N1}; these two defects are related but not identical
formulas.

In particular, $\partial g$ is a source-conservation issue, not an
extra term in the minimally coupled metric Euler derivative.
The restriction to $\Gp$ removes the covariance defect. It does
not imply global solvability of the metric equation in the
switching region.

%=====================================================================

%%% [chunk II-A-definition-def-jet kind=definition lines 956-973 labels=['def:jet']]
\begin{definition}[Fixed-input Einstein jet]
\label{def:jet}
For a fixed free scalar $\phi_0$, let $\Jcal_g(\phi_0)$ consist of
smooth formal coefficient families
\[
 \gamma(\kappa)=\gamma_0+\kappa\gamma_1+\kappa^2\gamma_2,
 \qquad
 \phi(\kappa)=\phi_0+\kappa\phi_1+\kappa^2\phi_2,
\]
such that the coefficients through $\kappa^2$ of
$\Egamma=0$ and $\Ephi=0$ vanish. Positive-order scalar
coefficients have the X-A zero Cauchy data.
No condition is imposed on metric Cauchy data.

This is the jet of the invariant Einstein equation, not of the
gauge-fixed Euler equation. Requiring $D=0$ in addition is a
possible further restriction, not a replacement of that equation.
\end{definition}
%%% [chunk II-A-lemma-lem-compatibility kind=lemma lines 975-1009 labels=['lem:compatibility', 'eq:compatibility', 'eq:source-
    jet']]
\begin{lemma}[Necessary source conservation]
\label{lem:compatibility}
If $\Jcal_g(\phi_0)$ is nonempty, then
\begin{equation}
 (\partial_\mu g)T_0^{\mu\nu}[\phi_0]=0.
 \label{eq:compatibility}
\end{equation}
For the source-response subclass $\gamma_0=0$, this is already the
compatibility condition for
\begin{equation}
 G_{\mathrm{lin}}^{\mu\nu}(\gamma_1)
                      =\tfrac14gT_0^{\mu\nu}[\phi_0].
 \label{eq:source-jet}
\end{equation}
\end{lemma}

\begin{proof}
The linearized Bianchi identity
$\partial_\mu G_{\mathrm{lin}}^{\mu\nu}=0$ follows directly from
\eqref{eq:Glin}. Also
$\partial_\mu T_0^{\mu\nu}
  =(\Pfree\phi_0)\partial^\nu\phi_0=0$.
Taking the divergence of \eqref{eq:source-jet} proves the assertion
in that subclass.

For arbitrary $\gamma_0$, evaluate the order-one coefficient of
the exact Noether identity on a purported jet.
Since the metric Euler coefficients vanish, its left side
vanishes. Since $\Pfree\phi_0=0$, the order-zero scalar Euler
term vanishes. The order-zero value of $\Bcal_\nu$ is precisely
$(\partial_\mu g)T_0^\mu{}_\nu[\phi_0]$.
Equation \eqref{eq:noether} therefore forces
\eqref{eq:compatibility}. This argument permits arbitrary vacuum
leading metric data and does not presume a harmonic section.
\end{proof}
%%% [chunk II-A-section-eq-input kind=section lines 1011-1023 labels=['eq:input']]

```

```
For the chosen mass define
\begin{equation}
 \phi_m(t,x)=
 \begin{cases}
  \sin(mt)/m,&m>0,\\
  t,&m=0.
 \end{cases}
 \label{eq:input}
\end{equation}
It is a smooth free solution for every $m\geq0$.


%%% [chunk II-A-theorem-thm-empty kind=theorem lines 1024-1059 labels=['thm:empty', 'eq:empty']]
\begin{theorem}[Empty fixed-input Einstein jet]
\label{thm:empty}
For every nonzero compact switch of the frozen class,
\begin{equation}
 \boxed{\qquad \Jcal_g(\phi_m)=\varnothing. \qquad}
 \label{eq:empty}
\end{equation}
The obstruction occurs in the first sourced metric coefficient,
before the second-order scalar or metric coefficients can repair it.
\end{theorem}

\begin{proof}
The input has no spatial derivative. A direct computation gives
\[
 T_0^{00}[\phi_m]=\tfrac12,\qquad
 T_0^{0i}[\phi_m]=0,
\]
and
\[
 T_0^{ij}[\phi_m]=
 \begin{cases}
 \tfrac12\cos(2mt)\delta^{ij},&m>0,\\
 \tfrac12\delta^{ij},&m=0.
 \end{cases}
\]
Consequently the $\nu=0$ component of
\eqref{eq:compatibility} is
\[
 \tfrac12\partial_tg=0.
\]
A nonzero function compactly supported in the slab interior cannot
be independent of time: for each fixed spatial point it vanishes
near both time boundaries. Since the switch has a nonempty plateau,
$\partial_tg$ is not identically zero. The necessary compatibility
condition fails.
\end{proof}
%%% [chunk II-A-section-L1059 kind=section lines 1059-1069 labels=[]]


The same obstruction can use compact scalar Cauchy data.
Choose a spatial cutoff equal to one on a ball containing the
initial causal shadows of $\supp g$, and prescribe
$\phi_0(0)=0$, $\dot\phi_0(0)$ equal to that cutoff.
Finite propagation and uniqueness imply that this compact-data
solution equals \eqref{eq:input} on a neighbourhood of $\supp g$.
Thus its stress gives the same witness. This uses the classical
Cauchy theory already admitted in X-A, not a gravitational propagator.


%%% [chunk II-A-section-eq-bump kind=section lines 1070-1095 labels=['eq:bump']]


Because $g$ rises from zero to one somewhere, there is an interior
open bump region on which $\partial_tg\geq c>0$.
For nonnegative $f$ supported there with $\int f\dd^4x=1$,
\begin{equation}
 \int f\,(\partial_\mu g)T_0^{\mu0}[\phi_m]\dd^4x
                  \geq\frac c2>0.
 \label{eq:bump}
\end{equation}
This is an evaluated conservation obstruction, not an Einstein
solution.

In the massless diagnostic $\phi_0=t$, take also
$\gamma=\psi\eta$, $\psi$ smooth. Then
\[
 T_0^{00}=\tfrac12,\qquad
```

```
 (\partial_\mu g)T_0^{\mu0}=\tfrac12\partial_tg,
 \qquad
 G_{\mathrm{lin}}[\psi\eta]_{\mu\nu}
 =-\partial_\mu\partial_\nu\psi+\eta_{\mu\nu}\Box\psi.
\]
Choosing a conformal test tensor does not alter the nonzero number
\eqref{eq:bump} and does not solve the coupled metric equation.
This is a check of the compatibility gate only.


%%% [chunk II-A-section-L1096 kind=section lines 1096-1118 labels=[]]


For every smooth vector field,
\[
 G_{\mathrm{lin}}[K\xi]=0,\qquad D(K\xi)=\Box\xi.
\]
Substitution into \eqref{eq:Glin} verifies the first identity.
For $\xi\in\ga$ this is the required pure-gauge diagnostic.
It checks the linearization, not the existence of a sourced jet.

The unrestricted family of Einstein jets is not empty:
$\phi_0=0$ and all metric and scalar coefficients zero solve the
equations. Other inputs satisfying \eqref{eq:compatibility} are not
classified here. Necessity is not being promoted to a general
sufficiency theorem.

No second-order correction can cancel the nonzero order-one
compatibility coefficient. The stress tensor is not modified, and
the Einstein--Hilbert density is not switched. A possible removal
or controlled comparison of compact cutoffs belongs to a successor
local-coupling or adiabatic problem; no such limit is taken.

%====================================================================

%%% [chunk II-A-section-eq-quadraticoperator kind=section lines 1120-1160 labels=['eq:quadraticoperator', 'eq:
    constraintbreaking']]

%====================================================================

At quadratic order the sum of \eqref{eq:FP} and \eqref{eq:gf} is
\[
 S_{\FP}+S_{\gf}
 =\int\left(
 \tfrac12\partial_\lambda\gamma_{\mu\nu}
              \partial^\lambda\gamma^{\mu\nu}
 -\tfrac14\partial_\lambda\tau\,\partial^\lambda\tau
 \right)\dd^4x,
\]
with Euler operator
\begin{equation}
 2G_{\mathrm{lin}}^{\mu\nu}+\Egamma_{\gf}^{\mu\nu}
                    =-\Box\bar\gamma^{\mu\nu}.
 \label{eq:quadraticoperator}
\end{equation}

In the source-response expansion, the gauge-fixed equation is
\[
 -\Box\bar\gamma_1^{\mu\nu}
                     =\tfrac12gT_0^{\mu\nu}[\phi_0].
\]
Componentwise smooth Cauchy theory permits such an equation with a
nonconserved smooth compact source. Taking its divergence gives
\begin{equation}
 \Box D^\nu(\gamma_1)
            =-\tfrac12(\partial_\mu g)T_0^{\mu\nu}[\phi_0].
 \label{eq:constraintbreaking}
\end{equation}
For the witness input, $D$ therefore cannot vanish identically.

Thus a gauge-fixed solution is not an Einstein solution in de Donder
gauge unless the source compatibility is restored. Gauge fixing
does not remove the Bianchi obstruction; it changes the equation
being solved. Equation \eqref{eq:quadraticoperator} specifies an
operator only. No graviton Green function, two-point function,
propagator normalization, or state is constructed.

%====================================================================

%%% [chunk II-A-section-L1162 kind=section lines 1162-1218 labels=[]]

```

```
%=====================================================================

\begin{center}
\small
\begin{tabularx}{\linewidth}{@{}p{.54\linewidth}X@{}}
\toprule
Gate & Result\\
\midrule
Signature, curvature, coupling normalization & Frozen\\
Positive TT kinetic sign & Proved\\
Attractive Newtonian comparison & Proved\\
Quadratic action and de Donder term & Explicit\\
Metric Euler derivative through $\kappa^2$ & Proved; no
$\partial g$ in the minimal metric variation\\
Scalar and source-divergence switch derivatives & Displayed\\
Invariant nonlinear Noether identity & Proved with the correct
adjoint gauge operator and coupling factors\\
Gauge-fixed unbroken Noether identity & Proved false; breaking
term retained\\
Invariant metric Euler covariance on $\ga$ & Proved as
tensor-density covariance\\
Nonlinear Fierz--Pauli-only replacement & Proved insufficient\\
Fixed-input Einstein jet $\Jcal_g(\phi_m)$ & Empty\\
Universal emptiness for all inputs & False; vacuum counterexample\\
Gauge-fixed wave equation & Different equation; not an
Einstein-jet repair\\
Quantum metric, propagator, tree, BRST class & Not constructed\\
\bottomrule
\end{tabularx}
\end{center}

The corrected sheet reaches \textbf{Constructed}. The specified
on-shell Einstein reading reaches \textbf{Proved obstructed}.
The literal demand for a nonlinear covariant, gauge-fixed
Fierz--Pauli-only sheet cannot receive a construction theorem.

XI-B receives:
\begin{enumerate}
\item the signed quadratic operator
$2G_{\mathrm{lin}}+\Egamma_{\gf}=-\Box\bar\gamma$;
\item the explicit de Donder term and its breaking;
\item the stress-tensor vertex \eqref{eq:vertex};
\item the unswitched Einstein action and the order bookkeeping
needed for its nonlinear identities;
\item the Noether identity \eqref{eq:noether}, including its
background-switch defect;
\item the empty fixed-input Einstein fiber and its distinction
from gauge-fixed wave solvability.
\end{enumerate}

It receives no propagator, ghost, antifield, tree, or quantum state.
The algebra $\ga$ acts on classical configurations; it has not
been shown to act on a quantum graviton. Establishing a BRST
description of gauge fixing would be a later construction, not a
consequence of naming the de Donder term \cite{BFR}.


%%% [chunk XI-A-section-L1219 kind=section lines 1219-1235 labels=[]]


A coherent classical sheet has been frozen without putting the
metric on shell. Its invariant and gauge-fixed derivatives are
distinguished, and its source obstruction has a named input:
\[
 \boxed{
 \text{signed classical action and identities constructed},
 \qquad
 \Jcal_g(\phi_m)=\varnothing.
 }
\]
The ramp relational prescription remains the inherited X-A object.
No quantum-gravitational observable has been constructed.

%=====================================================================
\appendix

%%% [chunk XI-A-section-eq-Gamma-n kind=section lines 1237-1298 labels=['eq:Gamma-n', 'eq:R-n', 'eq:r-n', 'eq:C-n', 'eq:H-n']]

%=====================================================================

This appendix makes \eqref{eq:Hdef} an explicit coefficient
```

```
prescription. All contractions use $\eta$ unless a coefficient
$u_i$ is written.

Set
\[
 u_0^{\mu\nu}=\eta^{\mu\nu},\qquad
 u_1^{\mu\nu}=-\gamma^{\mu\nu},\qquad
 u_2^{\mu\nu}=\gamma^\mu{}_\rho\gamma^{\rho\nu}.
\]
For $n=1,2,3$ define
\begin{equation}
 \Gamma_n^\rho{}_{\mu\nu}
 =\tfrac12 u_{n-1}^{\rho\sigma}
    \left(\partial_\mu\gamma_{\nu\sigma}
         +\partial_\nu\gamma_{\mu\sigma}
         -\partial_\sigma\gamma_{\mu\nu}\right).
 \label{eq:Gamma-n}
\end{equation}
These are precisely the first three coefficients of the
Christoffel symbols of $q=\eta+\kappa\gamma$.

Define, for $1\leq n\leq 3$,
\begin{align}
 R_{n,\mu\nu}
 ={}&\partial_\rho\Gamma_n^\rho{}_{\mu\nu}
       -\partial_\nu\Gamma_n^\rho{}_{\mu\rho}
 \nonumber\\
 &+\sum_{\substack{a+b=n\\a,b\geq 1}}
 \left(
 \Gamma_a^\rho{}_{\rho\lambda}\Gamma_b^\lambda{}_{\mu\nu}
 -\Gamma_a^\rho{}_{\nu\lambda}\Gamma_b^\lambda{}_{\mu\rho}
 \right),
 \label{eq:R-n}\\
 r_n&=\sum_{\substack{i+j=n\\i\geq 0,\ j\geq 1}}
                       u_i^{\mu\nu}R_{j,\mu\nu},
 \qquad r_0=0,
 \label{eq:r-n}\\
 C_{n,\mu\nu}
 &=R_{n,\mu\nu}
       -\tfrac12\eta_{\mu\nu}r_n
       -\tfrac12\gamma_{\mu\nu}r_{n-1}.
 \label{eq:C-n}
\end{align}
Only $u_0,u_1,u_2$ occur in these sums. Let $b_0=1$ and use
$b_1,b_2$ from \eqref{eq:b}. Then
\begin{equation}
 \Hcal_n^{\mu\nu}
 =
 \sum_{\substack{i+j+\ell+s=n\\
                 i,j,\ell\geq 0,\ s\geq 1}}
 b_i\,u_j^{\mu\alpha}u_\ell^{\nu\beta}C_{s,\alpha\beta},
 \qquad 1\leq n\leq 3.
 \label{eq:H-n}
\end{equation}
Again, no index above two is required for $b_i,u_j,u_\ell$.
Equations \eqref{eq:Gamma-n}--\eqref{eq:H-n} are finite local
differential formulas, not placeholders for unknown vertices.


%%% [chunk II-A-lemma-L1299 kind=lemma lines 1299-1314 labels=[]]
\begin{lemma}
The formulas in this appendix give the coefficients
$\Hcal_1,\Hcal_2,\Hcal_3$ of \eqref{eq:Hdef}.
\end{lemma}

\begin{proof}
Expand
$q^{-1}=u_0+\kappa u_1+\kappa^2u_2+\cdots$
in the exact Christoffel formula. Since $\partial q=\kappa\partial
\gamma$, the order-$n$ Christoffel coefficient is
\eqref{eq:Gamma-n}. Substitution into \eqref{eq:ricci} yields
\eqref{eq:R-n}. Contracting the Ricci expansion with the inverse
metric yields \eqref{eq:r-n}; multiplying $q_{\mu\nu}R$ gives
\eqref{eq:C-n}. Finally, multiplying by $\sqrt{-q}$ and by two
inverse metrics gives \eqref{eq:H-n}.
\end{proof}
%%% [chunk II-A-section-L1314 kind=section lines 1314-1333 labels=[]]


The Einstein action has bulk expansion
\[
```

```
 S_{\EH}=S_{\FP}+\kappa S_{\EH,3}
                    +\kappa^2S_{\EH,4}+O(\kappa^3),
\]
where the higher coefficients are fixed by \eqref{eq:EH}.
Their Euler derivatives are $2\Hcal_2$ and $2\Hcal_3$.
This identifies their classical role without constructing a tree.

Because the Einstein Euler derivative contains $1/\kappa$,
its order-two coefficient involves the order-three metric curvature
coefficient. A pullback covariance check must likewise retain
the corresponding metric Taylor coefficients before projection.
Truncating the metric pullback prematurely at second order would
lose required information. Genuine compact flows avoid that
bookkeeping ambiguity.

%=====================================================================

%%% [chunk II-A-section-L1335 kind=section lines 1335-1372 labels=[]]

%=====================================================================

The proofs use:
\begin{enumerate}
\item the explicit curvature convention and action;
\item algebraic density expansions and compact interior variations;
\item X-A's finite varied-action formula and divergence lemma;
\item the ordinary tensorial naturality of curvature and scalar stress;
\item the displayed linearized Bianchi identity;
\item classical Cauchy uniqueness and finite propagation for the
optional compact-data witness.
\end{enumerate}

They do not use an external graviton propagator, quantum Ward
identity, counterterm catalogue, or response matrix.

The following objects remain distinct:
\[
 \frac{\delta S_{\inv}}{\delta\gamma},
 \qquad
 \frac{\delta S_{\mathrm{tot}}}{\delta\gamma},
 \qquad
 G_{\mathrm{lin}}(\gamma),
 \qquad
 -\Box\bar\gamma.
\]
The first obeys the nonlinear fixed-background identity.
The second includes gauge-fixing breaking.
The third has the linearized Bianchi identity.
The fourth is the gauge-fixed quadratic operator.

Similarly, a conserved Einstein source, a switched nonconserved
source, and a source admitted by a gauge-fixed wave equation are not
interchangeable. The witness distinguishes them at a lower order
than any radiative or renormalization question.

%=====================================================================

%%% [chunk II-A-section-L1373 kind=section lines 1373-1400 labels=[]]

\addcontentsline{toc}{section}{Declarations}

\paragraph{Classical scope.}
All fields and products in this manuscript are smooth classical
coefficient fields. The metric is an independent classical
configuration variable, not a quantized field.

\paragraph{Scope correction.}
The title refers to an empty \emph{fixed-input} Einstein jet.
Gauge fixing is not claimed to preserve the invariant action's
Noether identity. The full Einstein action is specified because
nonlinear classical identities cannot be certified from the
quadratic coefficient alone.

\paragraph{Verification.}
No successful \texttt{pdflatex} execution, proof-assistant
verification, independently checked symbolic calculation, or
referee review is asserted. No empirical data or numerical graph
calculation is used.

\paragraph{Successor work.}
A free multiplet, gravitational propagator, ghost and antifield
```

```
sector, BRST identity, quantum state, relational product
renormalization, tree calculation, and sharp-ramp or adiabatic
limit remain separate tasks.

%=====================================================================

%%% [chunk XI-A-bibliography-L1401 kind=bibliography lines 1401-1441 labels=[]]
\begin{thebibliography}{9}
\addcontentsline{toc}{section}{References}

\bibitem{XA}
N.~Cheung,
\emph{Landmark X-A: Classical covariance of one fixed-regulator
switched relational scalar through order $\kappa^2$}.
Supplied companion manuscript.
Used only for the declared classical prescription, its Cauchy
facts, and the varied-action formula identified in the text.

\bibitem{BFR}
R.~Brunetti, K.~Fredenhagen, and K.~Rejzner,
``Quantum gravity from the point of view of locally covariant
quantum field theory,''
\emph{Communications in Mathematical Physics}
\textbf{345} (2016), 741--779.
\href{https://arxiv.org/abs/1306.1058}{arXiv:1306.1058}.
Comparison framework only; no quantum construction is imported.

\bibitem{Donoghue}
J.~F.~Donoghue,
``General relativity as an effective field theory:
The leading quantum corrections,''
\emph{Physical Review D}
\textbf{50} (1994), 3874--3888.
\href{https://arxiv.org/abs/gr-qc/9405057}{arXiv:gr-qc/9405057}.
Comparison background only; the sign checks in this manuscript
are derived in the declared signature.

\bibitem{Frob}
M.~B.~Fr\"ob,
``Gauge-invariant quantum gravitational corrections to
correlation functions,''
\emph{Classical and Quantum Gravity}
\textbf{35} (2018), 055006.
\href{https://arxiv.org/abs/1710.00839}{arXiv:1710.00839}.
Comparison background only; no quantum cancellation or
coordinate Green-function prescription is used as a premise.

\end{thebibliography}
```

3.3 Landmark XI-B (full source)

Listing 57: XI-B.tex

```
% Reconstructed from live /browse/XI-B chunks (fileName=XI-B.tex, contentHash=862627cd, version=1); gaps between chunks
     omitted
%%% [chunk XI-B-abstract-L129 kind=abstract lines 129-154 labels=[]]
\begin{abstract}
Starting from the signed quadratic operator supplied by Landmark
XI-A, we construct a free Einstein--scalar gauge-fixed multiplet on
a finite Minkowski slab. One gauge fermion reproduces exactly the
frozen de Donder term, fixes the ghost kinetic normalization, and
introduces an independent Nakanishi--Lautrup field. The auxiliary
field is retained for off-shell nilpotence.

We derive retarded and advanced Green operators for the graviton,
scalar, ghost, and auxiliary blocks. The graviton inverse is minus
trace reversal times the zero-data wave inverse; its coefficient is
fixed by the handed-off action, with the Newtonian sign providing
an independent check. A state-independent graded polynomial field
algebra is constructed from the resulting causal kernels. The free
differential is shown to preserve its equations and commutation
relations.

The plateau algebra has trivial intersection with residual de Donder
gauge. For classical de Donder fields, the ramp coordinate coefficient
is the retarded response to a layer source, not a field supported
only inside the layer. We give an explicit nonzero example and retain
```

```
the fixed regulator width. The empty fixed-input Einstein jet of
XI-A is not repaired: its gauge-fixed retarded response exists,
but has nonzero divergence. No Hadamard state, interacting
differential, tree, mixed loop, or relational BRST class is constructed.
\end{abstract}
%%% [chunk XI-B-scope-L156 kind=scope lines 156-169 labels=[]]
\scopebox{Release: the free gate is constructed; the Einstein jet is
not repaired}{%
The free graded multiplet, one gauge fermion, normalized causal
Green operators, a nilpotent free differential, the trivial
plateau residual algebra, and the classical layer formula are
constructed.

The operator inverted here is the \emph{gauge-fixed} operator.
The empty fixed-input Einstein jet remains empty.

No interaction is contracted with the dressing. No claim is made
that the plateau algebra acts by nontrivial residual transformations
on the free gauge-fixed graviton, or that the ramp scalar defines
an interacting BRST class.}
%%% [chunk XI-B-section-L175 kind=section lines 175-204 labels=[]]

%=====================================================================

This manuscript begins where XI-A stopped: with a signed quadratic
operator rather than a graviton two-point function. The construction
is performed at the free background $(\eta,0)$. A prescribed nonzero
free scalar used in a diagnostic is an input to a coupling expansion,
not a replacement background for the free Hessian.

There are three different objects:
\[
 \text{classical background pullbacks},\qquad
 \text{residual gauge transformations},\qquad
 \text{the free graded differential}.
\]
They must not be identified merely because they involve derivatives
of vector fields.

The polynomial algebra constructed below uses compactly smeared
linear fields and their finite products. It has causal commutation
relations, but no chosen Hadamard covariance or state. In particular,
it is not the Wick algebra used in VII-C. This restricted algebraic
construction suffices for the free multiplet and does not presume
admission of a nonlocal relational composite.

The standard distinction between free gauge fixing and interacting
BV quantization is respected \cite{BFR}. We construct the former,
including its free differential, without supplying an interacting
master equation.


%%% [chunk XI-B-theorem-thm-main kind=theorem lines 205-225 labels=['thm:main']]
\begin{theorem}[Free-multiplet release]
\label{thm:main}
For the XI-A quadratic normalization and the conventions of
Section~\ref{sec:handoff}, the following are established.
\begin{enumerate}[label=\textup{(\roman*)}]
\item One gauge fermion yields the frozen term
$S_{\gf}=\int D_\nu D^\nu$ after auxiliary-field elimination.
\item The full free operator has unique retarded and advanced
Green operators on compact interior sources, with the stated
causal support.
\item The free differential is nilpotent off shell with the
auxiliary field retained, and descends to the causal polynomial
field algebra.
\item The plateau algebra has zero intersection with residual
de Donder gauge.
\item The regulated first coordinate coefficient of a classical
de Donder field is a propagated layer functional.
\end{enumerate}
These statements do not construct an interacting or relational
BRST cohomology class.
\end{theorem}
%%% [chunk XI-B-section-L225 kind=section lines 225-230 labels=[]]


The proofs are given in the successive sections. No tree calculation
is part of the theorem.

%=====================================================================
```

```
%%% [chunk XI-B-section-L235 kind=section lines 235-277 labels=[]]


Retain
\[
 \MT=[0,T]\times\R^3,\qquad
 \MTo=(0,T)\times\R^3,\qquad
 \Sigma_0=\{t=0\},
\]
\[
 \eta=\operatorname{diag}(1,-1,-1,-1),\qquad
 \Box=\partial_t^2-\Delta,\qquad
 \Pfree=\Box+m^2,\qquad m\geq0.
\]
Fields are smooth on a neighbourhood of the closed slab.
Indices are raised and lowered with $\eta$; parentheses contain
the factor $1/2$.

The switch satisfies
\[
 g\in C_c^\infty(\MTo;[0,1]),\qquad
 P_g=\Int\{g=1\}\neq\varnothing.
\]
The fixed ramp is
\[
 \sig(t)=\chi(t/\eps),\qquad 0<\eps<T,
\]
where
\[
 \chi(s)=
 \begin{cases}
 0,&s\leq0,\\[2pt]
 \displaystyle
 \frac{e^{-1/s}}{e^{-1/s}+e^{-1/(1-s)}},&0<s<1,\\[7pt]
 1,&s\geq1.
 \end{cases}
\]
Choose $U$ open and relatively compact with
\[
 \overline U\subset P_g\cap\{\eps<t<T\},\qquad
 \ga=C_c^\infty(U;TM).
\]
Neither $g$ nor $\sig$ occurs in the free kinetic operator.


%%% [chunk XI-B-section-eq-basic kind=section lines 278-331 labels=['eq:basic', 'eq:handoff', 'eq:vertex']]


For a symmetric tensor $\gamma$ define
\[
 \bar\gamma_{\mu\nu}
 =\gamma_{\mu\nu}
       -\tfrac12\eta_{\mu\nu}\tr_\eta\gamma,\qquad
 D^\nu\gamma=\partial_\mu\bar\gamma^{\mu\nu},
\]
\[
 (K\xi)_{\mu\nu}
       =\partial_\mu\xi_\nu+\partial_\nu\xi_\mu.
\]
Let $\PP$ denote trace reversal, so $\bar\gamma=\PP\gamma$.
In four dimensions,
\begin{equation}
 \PP^2=\id,\qquad
 D(K\xi)=\Box\xi,\qquad
 G_{\mathrm{lin}}(K\xi)=0.
 \label{eq:basic}
\end{equation}

XI-A gives
\begin{equation}
 2G_{\mathrm{lin}}(\gamma)+E_{\gf}(\gamma)
                        =-\Box\PP\gamma,
 \qquad
 S_{\gf}=\int_{\MT}D_\nu\gamma D^\nu\gamma\dd^4x.
 \label{eq:handoff}
\end{equation}
This is not the normalization $-\frac12\Box\PP$.

The inherited interaction is
\begin{equation}
```

```
 V_{1,g}[\gamma,\phi]
       =-\tfrac12\int_{\MT}
             g\,\gamma_{\mu\nu}T_0^{\mu\nu}[\phi]\dd^4x,
 \label{eq:vertex}
\end{equation}
with
\[
 T_0^{\mu\nu}[\phi]
 =\partial^\mu\phi\partial^\nu\phi
  -\tfrac12\eta^{\mu\nu}
      \left(\partial_\rho\phi\partial^\rho\phi-m^2\phi^2\right).
\]
It is not inserted into the free action or into the free differential.

The invariant XI-A action has a fixed-background Noether identity;
its gauge-fixed action has an additional breaking term. The
switching contribution includes $(\partial_\mu g)T_0^{\mu\nu}$.
None of these facts is replaced by free gauge-fixed solvability
\cite{XIA}.


%%% [chunk XI-B-section-L332 kind=section lines 332-353 labels=[]]


VII-C's Hadamard graviton matrix is not an input. X-D is not used.
The text of X-B-v3 is not supplied here; no externally numbered
result from it is presumed. The scalar sector is constructed
directly below, with its off-shell and free-shell presentations
kept separate.

The independent fields and their gradings are:
\[
\begin{array}{c|ccccc}
 \text{field}&\gamma_{\mu\nu}&\phi&c^\mu&\bar c_\mu&b_\mu\\
 \hline
 \text{parity}&0&0&1&1&0\\
 \text{ghost number}&0&0&1&-1&0
\end{array}
\]
The algebra is complexified to accommodate the factor $i$ in
the antighost transformation. No positivity or ghost reality
condition is needed for this release.

%=====================================================================

%%% [chunk XI-B-section-eq-R kind=section lines 358-382 labels=['eq:R', 'eq:s']]


Define
\begin{equation}
 \RR c=\tfrac12Kc,\qquad
 (\RR c)_{\mu\nu}=\partial_{(\mu}c_{\nu)}.
 \label{eq:R}
\end{equation}
Thus the requested parenthesis convention is retained.
XI-A's variation $K\xi$ is recovered by setting $c=2\xi$.
This fixes a ghost normalization; it introduces neither a new
metric gauge condition nor a gauge-parameter family.

The free odd derivation is specified on independent fields by
\begin{equation}
 \sfree\gamma=\RR c,\qquad
 \sfree\phi=0,\qquad
 \sfree c=0,\qquad
 \sfree\bar c_\mu=ib_\mu,\qquad
 \sfree b_\mu=0.
 \label{eq:s}
\end{equation}
It commutes with spacetime derivatives and raises ghost number
by one. Its action on products is the graded Leibniz rule.


%%% [chunk XI-B-lemma-lem-nilpotence kind=lemma lines 383-396 labels=['lem:nilpotence']]
\begin{lemma}[Off-shell nilpotence]
\label{lem:nilpotence}
The derivation \eqref{eq:s} satisfies $\sfree^2=0$ on the
independent fields and on their polynomial differential algebra.
\end{lemma}

\begin{proof}
The second variation of $\gamma$ is $\RR(\sfree c)=0$.
```

```
The second variation of $\bar c$ is $i\sfree b=0$.
The remaining fields already have zero first variation.
The square of an odd derivation is an even derivation; hence
vanishing on the generators proves vanishing on their polynomial
algebra.
\end{proof}
%%% [chunk II-B-section-eq-FP kind=section lines 398-427 labels=['eq:FP', 'eq:Psi']]


The quadratic Fierz--Pauli action is copied in its signed form:
\begin{align}
 S_{\FP}[\gamma]=\frac12\int_{\MT}\Bigl(
 &\partial_\lambda\gamma_{\mu\nu}
       \partial^\lambda\gamma^{\mu\nu}
 -2(\partial_\mu\gamma^{\mu\nu})
       (\partial^\rho\gamma_{\rho\nu})
 \nonumber\\
 &+2(\partial_\mu\gamma^{\mu\nu})
       \partial_\nu\tr_\eta\gamma
 -\partial_\lambda\tr_\eta\gamma\,
       \partial^\lambda\tr_\eta\gamma
 \Bigr)\dd^4x.
 \label{eq:FP}
\end{align}
The free scalar action is
\[
 S_\phi=\tfrac12\int_{\MT}
       \left(\partial_\mu\phi\partial^\mu\phi-m^2\phi^2\right)\dd^4x.
\]

Choose exactly one fermion:
\begin{equation}
 \Psi=-i\int_{\MT}
           \bar c_\mu\left(2D^\mu\gamma-b^\mu\right)\dd^4x.
 \label{eq:Psi}
\end{equation}


%%% [chunk II-B-proposition-prop-fermion kind=proposition lines 428-460 labels=['prop:fermion', 'eq:S0', 'eq:sPsi']]
\begin{proposition}[Recovery of the frozen gauge term]
\label{prop:fermion}
The free auxiliary action
\begin{equation}
 S_0=S_{\FP}+S_\phi+\sfree\Psi
 \label{eq:S0}
\end{equation}
has
\begin{equation}
 \sfree\Psi
 =\int_{\MT}
       \left(2b_\mu D^\mu\gamma-b_\mu b^\mu
                       +i\bar c_\mu\Box c^\mu\right)\dd^4x.
 \label{eq:sPsi}
\end{equation}
Its auxiliary equation is $b_\mu=D_\mu\gamma$.
Eliminating $b$ gives precisely $S_{\gf}$ and the ghost term
$i\int\bar c_\mu\Box c^\mu$.
\end{proposition}

\begin{proof}
Equation \eqref{eq:basic} gives
$\sfree D^\mu\gamma=\frac12\Box c^\mu$.
The minus sign from applying an odd derivation to the second
factor of $\bar c(2D-b)$ yields \eqref{eq:sPsi}.
The auxiliary variation is $2(D^\mu\gamma-b^\mu)$.
Finally,
\[
 2b_\mu D^\mu\gamma-b_\mu b^\mu
 =D_\mu\gamma D^\mu\gamma
       -(b-D\gamma)_\mu(b-D\gamma)^\mu.
\]
\end{proof}
%%% [chunk II-B-section-L460 kind=section lines 460-473 labels=[]]


All action identities are bulk variational identities, modulo
divergences and tested with compact interior variations. No
BRST boundary charge or boundary variational principle is asserted.

The Fierz--Pauli part is invariant under its linear gauge map,
and the scalar is unchanged by $\sfree$. Consequently
```

```
$\sfree S_0=0$ as a free bulk variational functional, since
$\sfree^2\Psi=0$. This does not contradict the XI-A breaking:
the bosonic gauge-fixing term alone is not gauge invariant.
The independent auxiliary and ghost terms produce the free
BRST-invariant completion.


%%% [chunk XI-B-section-L474 kind=section lines 474-495 labels=[]]


After elimination,
$\sfree\bar c=iD\gamma$, and therefore
\[
 \sfree^2\bar c=\frac i2\Box c.
\]
This vanishes on the free ghost shell, not on unrestricted ghost
configurations. We retain $b$ in the off-shell multiplet.

For bookkeeping, name the antifields
\[
 \gamma^\ddagger,\quad \phi^\ddagger,\quad c^\ddagger,\quad
 \bar c^\ddagger,\quad b^\ddagger.
\]
An antifield has opposite parity and ghost number
$-1-\ghnum(\text{field})$. These names denote external source
slots only. They are not additional propagating generators here;
no antifield differential, BV antibracket, or interacting master
action is claimed.

%=====================================================================

%%% [chunk XI-B-section-eq-ER kind=section lines 500-518 labels=['eq:ER', 'eq:EA']]


For $\mu\geq0$, let
$\Omega_\mu=\sqrt{-\Delta+\mu^2}$. For a smooth source $F$ define
\begin{align}
 (\Egen^{\mathrm R}F)(t)
 &=\int_0^t
       \frac{\sin((t-s)\Omega_\mu)}{\Omega_\mu}F(s)\dd s,
 \label{eq:ER}\\
 (\Egen^{\mathrm A}F)(t)
 &=\int_t^T
       \frac{\sin((s-t)\Omega_\mu)}{\Omega_\mu}F(s)\dd s.
 \label{eq:EA}
\end{align}
The quotient has its continuous zero-frequency value. For
nondecaying smooth sources these formulas are interpreted locally
by finite propagation. On compact interior sources they are the
restrictions of the full Minkowski retarded and advanced operators.


%%% [chunk XI-B-lemma-lem-Green kind=lemma lines 519-551 labels=['lem:Green']]
\begin{lemma}[Slab Green identities]
\label{lem:Green}
The operators \eqref{eq:ER}--\eqref{eq:EA} invert
$\Box+\mu^2$ with, respectively, zero initial and zero final
Cauchy data. For $F\in C_c^\infty(\MTo)$,
\[
 (\Box+\mu^2)\Egen^\pm F=F,\qquad
 \Egen^\pm(\Box+\mu^2)F=F,
\]
and
\[
 \supp\Egen^{\mathrm R}F\subset J^+(\supp F)\cap\MT,\qquad
 \supp\Egen^{\mathrm A}F\subset J^-(\supp F)\cap\MT.
\]
These support bounds are compact in the \emph{closed} slab.
Moreover $(\Egen^{\mathrm R})^*=\Egen^{\mathrm A}$ on interior
test functions.
\end{lemma}

\begin{proof}
Differentiating the sine formulas twice verifies the equation
and the prescribed endpoint data. Uniqueness follows from
the homogeneous Cauchy problem. Applying this uniqueness to
$\Egen^\pm(\Box+\mu^2)F-F$ gives the left inverse on compact
interior tests. Finite propagation gives the support bounds.
The intersection of a causal future or past of a compact set
with a bounded-time closed slab is compact.
Green's identity proves transposition: the retarded solution
```

```
has zero data at $0$, the advanced solution has zero data at
$T$, and there is no spatial boundary contribution.
These are the standard smooth Cauchy properties
\cite[Chapter~3]{BGP}.
\end{proof}
%%% [chunk XI-B-section-L551 kind=section lines 551-555 labels=[]]


The endpoint prescriptions select the inverses. They do not
require all free solutions or ghosts to have zero Cauchy data.


%%% [chunk XI-B-section-L556 kind=section lines 556-562 labels=[]]


Under the flat Lorentz pairings, the graviton operator is
\[
 L_\gamma=-\Box\PP.
\]


%%% [chunk XI-B-theorem-thm-graviton kind=theorem lines 563-586 labels=['thm:graviton', 'eq:Ggr']]
\begin{theorem}[Normalized graviton inverse]
\label{thm:graviton}
Its retarded and advanced inverses are
\begin{equation}
 \boxed{\qquad
 \Ggr^{\mathrm R}=-\PP\Ezero^{\mathrm R},\qquad
 \Ggr^{\mathrm A}=-\PP\Ezero^{\mathrm A}.
 \qquad}
 \label{eq:Ggr}
\end{equation}
Thus the coefficient in
$c\,\PP\Ezero^{\mathrm R}$ is $c=-1$.
\end{theorem}

\begin{proof}
Trace reversal commutes with the wave operator and satisfies
$\PP^2=\id$. Hence
\[
 (-\Box\PP)(-\PP\Ezero^\pm)=\id
\]
and the left inverse follows on compact tests. Any coefficient
$c$ instead gives $-c\,\id$, forcing $c=-1$.
The support statement follows from Lemma~\ref{lem:Green}.
\end{proof}
%%% [chunk XI-B-section-L586 kind=section lines 586-603 labels=[]]


This fixes the normalization from XI-A's Hessian. The same sign
reproduces its Newtonian comparison: for
$H=q-\eta$ and a static positive source,
\[
 H_{00}=2\Phi_N,\qquad
 \Phi_N(x)=-\frac{\kappa^2}{32\pi}
                  \int\frac{\rho(y)}{|x-y|}\dd^3y<0.
\]
A coefficient $-2$ would invert a different Hessian. The Newtonian
check is not permission to change the quadratic normalization.

The scalar \emph{equation} operator is $\Pfree$, with inverse
$\Em^\pm$. Its \emph{action Hessian} is $-\Pfree$, with inverse
$-\Em^\pm$. This distinction determines the scalar causal bracket
below.


%%% [chunk XI-B-section-eq-Dstar kind=section lines 604-629 labels=['eq:Dstar', 'eq:Laux']]


Let $D^*$ be the formal adjoint of $D$. With flat index
identifications,
\begin{equation}
 (D^*v)_{\mu\nu}
 =-\partial_{(\mu}v_{\nu)}
        +\tfrac12\eta_{\mu\nu}\partial_\rho v^\rho.
 \label{eq:Dstar}
\end{equation}
Write $A=2G_{\mathrm{lin}}$. The even metric--auxiliary operator is
\begin{equation}
 \mathcal L_{\gamma b}
 =
```

```
 \begin{pmatrix}
 A&2D^*\\
 2D&-2\id
 \end{pmatrix}.
 \label{eq:Laux}
\end{equation}
Its source equation is
\[
 A\gamma+2D^*b=J,\qquad
 2D\gamma-2b=j.
\]


%%% [chunk II-B-proposition-prop-aux kind=proposition lines 630-671 labels=['prop:aux', 'eq:Gaux']]
\begin{proposition}[Full auxiliary inverse]
\label{prop:aux}
The unique retarded and advanced inverses are
\begin{equation}
 \Gaux^\pm=
 \begin{pmatrix}
 \Ggr^\pm&\Ggr^\pm D^*\\
 D\Ggr^\pm&D\Ggr^\pm D^*-\frac12\id
 \end{pmatrix}
 =
 \begin{pmatrix}
 -\PP\Ezero^\pm&\RR\Ezero^\pm\\
 D(-\PP\Ezero^\pm)&0
 \end{pmatrix}.
 \label{eq:Gaux}
\end{equation}
The identities in the second matrix hold on compact interior
sources.
\end{proposition}

\begin{proof}
The auxiliary equation gives $b=D\gamma-j/2$. Substitution
reduces the metric equation to
\[
 (A+2D^*D)\gamma=J+D^*j.
\]
The handed-off identity is $A+2D^*D=-\Box\PP$.
Equation \eqref{eq:Ggr} therefore gives the first matrix and
uniqueness.

For the simplification, direct calculation gives
\[
 \PP D^*=-\RR,\qquad
 D\RR=\tfrac12\Box.
\]
Consequently
$\Ggr^\pm D^*=\RR\Ezero^\pm$ and
$D\Ggr^\pm D^*=\frac12\id$.
The last block cancels the algebraic contact term.
The source elimination proves both Green identities; derivatives
do not enlarge causal support.
\end{proof}
%%% [chunk II-B-section-L671 kind=section lines 671-676 labels=[]]


The vanishing $bb$ block does not make $b$ a discarded field:
the $\gamma b$ and $b\gamma$ blocks are nonzero. Nor is an
independent wave inverse for $b$ postulated.


%%% [chunk II-B-section-eq-Gghost kind=section lines 677-698 labels=['eq:Gghost']]


Use left Grassmann variations. The ghost kinetic term yields
\[
 \frac{\delta^L S_0}{\delta c}=-i\Box\bar c,\qquad
 \frac{\delta^L S_0}{\delta\bar c}=i\Box c.
\]
In the order $(c,\bar c)$, with dual vector indices understood,
\begin{equation}
 \mathcal L_{\gh}
 =
 \begin{pmatrix}0&-i\Box\\ i\Box&0\end{pmatrix},
 \qquad
 \Ggh^\pm=
 \begin{pmatrix}0&-i\Ezero^\pm\\ i\Ezero^\pm&0\end{pmatrix}.
 \label{eq:Gghost}
```

```
\end{equation}
Matrix multiplication proves the two Green identities.
The scalar, auxiliary, and ghost blocks together give the full
free multiplet Green operator. No Feynman inverse is selected.

%=====================================================================

%%% [chunk II-B-section-eq-causal-even kind=section lines 703-742 labels=['eq:causal-even', 'eq:causal-aux', 'eq:causal-ghost']]


Let $\Aoff$ be the complex graded commutative polynomial algebra
generated by compact interior smearings of the five independent
fields. Products are finite. They have smooth finite-rank
multivariable kernels, so no coincident-field composite is
needed to define them.

This is an off-shell algebra: no field equation is used in
Lemma~\ref{lem:nilpotence}. A second, free-shell algebra is obtained
by imposing the linear equations and causal commutation relations.
Making that second presentation does not retroactively impose
free equations on classical relational functionals.

Put
\[
 \delta_\mu=\mathsf E_\mu^{\mathrm R}
                       -\mathsf E_\mu^{\mathrm A}.
\]
The full causal matrix is the retarded-minus-advanced inverse
of the action Hessian. Its nonzero blocks are
\begin{align}
 \Delta_{\gamma\gamma}&=-\PP\delta_0,
 &
 \Delta_{\phi\phi}&=-\delta_m,
 \label{eq:causal-even}\\
 \Delta_{\gamma b}&=\RR\delta_0,
 &
 \Delta_{b\gamma}&=D(-\PP\delta_0),
 \qquad \Delta_{bb}=0,
 \label{eq:causal-aux}\\
 \Delta_{c\bar c}&=-i\delta_0,
 &
 \Delta_{\bar c c}&=i\delta_0.
 \label{eq:causal-ghost}
\end{align}
The appropriate vector Kronecker symbols accompany the ghost
blocks. These are causal commutator kernels, not Hadamard
two-point functions.


%%% [chunk II-B-definition-def-algebra kind=definition lines 743-762 labels=['def:algebra', 'eq:CCR']]
\begin{definition}[Free causal polynomial algebra]
\label{def:algebra}
The algebra $\Afree$ is generated over $\C[[\hbar]]$ by the
compact interior smeared fields, subject to:
\begin{enumerate}[label=\textup{(\alph*)}]
\item the auxiliary free equations
\[
 A\gamma+2D^*b=0,\qquad b=D\gamma,\qquad
 \Pfree\phi=0,\qquad \Box c=\Box\bar c=0;
\]
\item the graded commutation relations
\begin{equation}
 [\Phi_A(f),\Phi_B(k)]_{\mathrm{gr}}
       =i\hbar\,\pair{f}{\Delta_{AB}k}\,\mathbf1.
 \label{eq:CCR}
\end{equation}
\end{enumerate}
Only finite polynomial degree is used at each formal
$\hbar$ coefficient.
\end{definition}
%%% [chunk II-B-proposition-prop-algebra kind=proposition lines 764-788 labels=['prop:algebra']]
\begin{proposition}[Existence of the algebra]
\label{prop:algebra}
Definition~\ref{def:algebra} gives a nonzero associative graded
polynomial algebra. All pairings in \eqref{eq:CCR} exist.
\end{proposition}

\begin{proof}
Smooth compact tests can be paired with the displayed causal
distributions. Transposition of the Green operators implies
```

```
graded antisymmetry:
the even blocks are antisymmetric, while the odd--odd block is
symmetric under simultaneous exchange of its slots.

The equations annihilate the causal matrix on both sides, so
the pairing descends to the test space modulo equation tests.
It is a constant even graded antisymmetric form on that quotient.
The associated central graded Lie algebra satisfies the Jacobi
identity because every bracket is central. Its enveloping
algebra, with the central generator set to the unit and scaled
by $i\hbar$, supplies the required algebra.
Equivalently, symmetrization gives the constant graded Moyal
product on polynomials. This establishes associativity and the
polynomial vector-space presentation without choosing a state
\cite{FR}.
\end{proof}
%%% [chunk II-B-section-L788 kind=section lines 788-793 labels=[]]


No extension to all local or microcausal polynomial functionals
is asserted. In particular, \eqref{eq:vertex} is not being
multiplied in $\Afree$.


%%% [chunk II-B-lemma-lem-equations kind=lemma lines 796-819 labels=['lem:equations']]
\begin{lemma}[Stability of the equation ideal]
\label{lem:equations}
The derivation $\sfree$ preserves the free equation ideal.
\end{lemma}

\begin{proof}
Since $A\RR=0$ and $\sfree b=0$,
\[
 \sfree(A\gamma+2D^*b)=0.
\]
Also,
\[
 \sfree(2D\gamma-2b)=\Box c.
\]
The ghost equation for $c$ is unchanged, while
\[
 \sfree(\Box\bar c)=i\Box b.
\]
Taking the divergence of the metric--auxiliary equation gives
$\Box b=0$: the divergence of $A\gamma$ is zero and the
divergence of $2D^*b$ is $-\Box b$.
The scalar equation is unchanged. Thus every transformed
equation belongs to the same linear equation ideal.
\end{proof}
%%% [chunk II-B-section-L819 kind=section lines 819-824 labels=[]]


The eliminated relations then give
$\sfree^2\bar c=\frac i2\Box c=0$ in the free-shell algebra,
while independent $b$ already provided off-shell nilpotence.


%%% [chunk II-B-theorem-thm-differential kind=theorem lines 827-865 labels=['thm:differential']]
\begin{theorem}[Free differential on the causal algebra]
\label{thm:differential}
The field rule \eqref{eq:s} induces a ghost-number-one
nilpotent graded derivation of $\Afree$.
\end{theorem}

\begin{proof}
Equation stability is Lemma~\ref{lem:equations}.
It remains to check the commutation relations, not merely to
postulate a Leibniz rule.

The only nontrivial mixed check is the initially zero relation
between $\gamma$ and $\bar c$. The ghost block gives
\[
 \{c^\mu(x),\bar c_\nu(y)\}
      =\hbar\,\delta^\mu_\nu\delta_0(x,y).
\]
The auxiliary block gives
\[
 [\gamma(x),b(y)]
      =i\hbar\,\RR_x\delta_0(x,y),
\]
with its tensor indices understood. Therefore
```

```
\[
 \sfree[\gamma(x),\bar c(y)]
 =\{\RR c(x),\bar c(y)\}
           +[\gamma(x),ib(y)]
 =\hbar\RR_x\delta_0-\hbar\RR_x\delta_0=0.
\]
These are distributional identities tested with compact
smearings; no pointwise field multiplication is involved.

All other transformed relations contain only zero cross-blocks,
or $cc$ or $bb$ brackets, which vanish. The nonzero
$\gamma\gamma$, $\phi\phi$, and $c\bar c$ brackets have constant
right-hand sides and zero transformed left-hand sides.
Hence $\sfree$ preserves the relation ideal.
Nilpotence descends from Lemma~\ref{lem:nilpotence}.
\end{proof}
%%% [chunk XI-B-section-L865 kind=section lines 865-872 labels=[]]


Thus the free algebra is a differential graded algebra. No
cohomology classification or interacting quantum differential
$\widehat s_{\mathrm{int}}$ follows from this statement.
The construction contains no stress-tensor vertex.

%=====================================================================

%%% [chunk XI-B-section-L877 kind=section lines 877-885 labels=[]]


A classical de Donder field has $D\gamma=0$.
Under $\gamma\mapsto\gamma+K\xi$,
\[
 D(\gamma+K\xi)=D\gamma+\Box\xi.
\]
Its residual de Donder parameters therefore satisfy $\Box\xi=0$.


%%% [chunk XI-B-theorem-thm-residual kind=theorem lines 886-901 labels=['thm:residual', 'eq:residual']]
\begin{theorem}[Trivial plateau residual algebra]
\label{thm:residual}
\begin{equation}
 \boxed{\qquad
 \ga\cap\{\xi:\Box\xi=0\}=\{0\}.
 \qquad}
 \label{eq:residual}
\end{equation}
\end{theorem}

\begin{proof}
A vector field in $\ga$ vanishes on a neighbourhood of
$\Sigma_0$. Every component has zero initial Cauchy data.
If it also solves the homogeneous wave equation, Cauchy
uniqueness forces it to vanish everywhere on the slab.
\end{proof}
%%% [chunk XI-B-section-L901 kind=section lines 901-921 labels=[]]


This does not say that all residual parameters on Minkowski
space vanish. Smooth noncompact wave solutions exist.
It says that the frozen plateau support class contains none
except zero.

Consequently $\ga$ is not a nontrivial residual action on
classical de Donder free solutions. More generally, a compact
shift $K\xi$ preserving the gauge-fixed free equation would
require $K\Box\xi=0$; the compactly supported vector
$\Box\xi$ would then be a Killing vector and hence zero,
again giving $\xi=0$.

The admitted plateau algebra remains an algebra acting on the
classical background prescription. It is not thereby promoted
to an interacting action on the free graviton algebra.
The graded transformation involving the ghost is a different
construction. In particular, ghosts are not required to be
compactly supported plateau wave solutions.


%%% [chunk XI-B-section-eq-response kind=section lines 922-971 labels=['eq:response', 'eq:divergence', 'eq:witness']]


For XI-A's source-response coefficient put
```

```
\[
 J^{\mu\nu}=\tfrac12gT_0^{\mu\nu}[\phi_0].
\]
The gauge-fixed retarded solution is
\begin{equation}
 \gamma_1^{\mathrm R}
     =-\tfrac12\PP\Ezero^{\mathrm R}
                          (gT_0[\phi_0]).
 \label{eq:response}
\end{equation}
Taking its divergence on the free scalar shell gives
\begin{equation}
 D^\nu\gamma_1^{\mathrm R}
 =-\tfrac12\Ezero^{\mathrm R}
               \bigl((\partial_\mu g)T_0^{\mu\nu}[\phi_0]\bigr),
 \qquad
 \Box D^\nu\gamma_1^{\mathrm R}
 =-\tfrac12(\partial_\mu g)T_0^{\mu\nu}[\phi_0].
 \label{eq:divergence}
\end{equation}

For the named input
\[
 \phi_m(t)=
 \begin{cases}\sin(mt)/m,&m>0,\\ t,&m=0,\end{cases}
\]
one has $T_0^{00}=1/2$ and $T_0^{0i}=0$. Thus
\begin{equation}
 D^0\gamma_1^{\mathrm R}
     =-\tfrac14\Ezero^{\mathrm R}(\partial_tg),
 \qquad
 \Box D^0\gamma_1^{\mathrm R}=-\tfrac14\partial_tg.
 \label{eq:witness}
\end{equation}
A nonzero compact interior switch has
$\partial_tg\not\equiv0$, so this divergence is nonzero.

Both \eqref{eq:response} and its divergence exist as smooth
retarded responses. The failed requirement is
$D\gamma_1=0$, not existence of the divergence.
The ungauge-fixed Einstein equation would require conservation
of $gT_0$ by its Bianchi identity. XI-A's empty
$\Jcal_g(\phi_m)$ is therefore unchanged. This diagnostic is
not a failure of the Green construction and uses no contraction
with the interaction.

%=====================================================================

%%% [chunk XI-B-section-eq-Igeneral kind=section lines 976-999 labels=['eq:Igeneral']]


The density coefficient inherited from X-A is
\[
 a_1^{\mu\nu}[\gamma]
       =\tfrac12\eta^{\mu\nu}\tr_\eta\gamma-\gamma^{\mu\nu}.
\]
Its divergence gives exactly the de Donder combination:
\[
 C^\mu[\gamma]
       =-\partial_\alpha a_1^{\alpha\mu}[\gamma]
       =D^\mu\gamma.
\]
The regulated first coordinate coefficient, with zero initial
data, is therefore
\begin{equation}
 \Coord_1^\mu[\gamma]
 =\Ezero^{\mathrm R}
       \left(\sig D^\mu\gamma-\sig'a_1^{0\mu}[\gamma]\right).
 \label{eq:Xgeneral}
\end{equation}
This is the divergence-form ramp prescription, not a
multiplied-equation replacement.


%%% [chunk XI-B-theorem-thm-layer kind=theorem lines 1000-1030 labels=['thm:layer', 'eq:layer']]
\begin{theorem}[Classical layer formula]
\label{thm:layer}
For every smooth classical tensor satisfying $D\gamma=0$,
\begin{equation}
 \boxed{\qquad
 \Coord_1^\mu[\gamma]
```

```
       =-\Ezero^{\mathrm R}
                    \left(\sig'a_1^{0\mu}[\gamma]\right).
 \qquad}
 \label{eq:layer}
\end{equation}
Its source vanishes outside $0<t<\eps$.
If that source has spatially compact support, then
\[
 \supp\Coord_1
 \subset J^+\!\left(
       \supp(\sig'a_1^{0\cdot}[\gamma])\right)\cap\MT.
\]
The propagated coefficient need not vanish for $t>\eps$.
It is nonzero precisely when the source vector in
\eqref{eq:layer} is nonzero.
\end{theorem}

\begin{proof}
Set $D\gamma=0$ in \eqref{eq:Xgeneral}.
The derivative $\sig'$ is smooth, vanishes off the ramp,
and is flat at both endpoints. Finite propagation proves the
support inclusion. Since
$\Box\Coord_1=-\sig'a_1^{0\cdot}$, zero coefficient implies
zero source. The converse follows from zero-data uniqueness.
\end{proof}
%%% [chunk II-B-section-L1030 kind=section lines 1030-1042 labels=[]]


Strictly, the closed support of $\sig'$ can meet both $0$ and
$\eps$, even though the function vanishes at those endpoints.
The support statement is about a smooth layer source, not a
hypersurface distribution.

A nonzero tensor in the layer can have
$a_1^{0\mu}=0$ and hence zero dressing. For example, a constant
purely spatial trace-free tensor has this property.
Thus ``nonzero layer data'' must refer to the projected source
in \eqref{eq:layer}, not to arbitrary nonzero metric components.


%%% [chunk II-B-proposition-prop-layerexample kind=proposition lines 1045-1075 labels=['prop:layerexample', 'eq:layerexample']]
\begin{proposition}[Nonzero de Donder diagnostic]
\label{prop:layerexample}
Let $\lambda\neq0$ and $\gamma_{\mu\nu}=\lambda\eta_{\mu\nu}$.
It solves the homogeneous gauge-fixed free equation and
$D\gamma=0$. Its coordinate coefficient is
\[
 \Coord_1^i=0,\qquad
 \Coord_1^0(t)
       =-\lambda\int_0^t\sig(s)\dd s.
\]
In particular, for $t\geq\eps$,
\begin{equation}
 \Coord_1^0(t)=-\lambda(t-\eps/2)\neq0.
 \label{eq:layerexample}
\end{equation}
\end{proposition}

\begin{proof}
The constant tensor has zero derivatives, and
$a_1^{\mu\nu}=\lambda\eta^{\mu\nu}$.
For its spatially constant source, the zero-frequency wave
formula gives
\[
 \Coord_1^0(t)
 =-\lambda\int_0^t(t-s)\sig'(s)\dd s
 =-\lambda\int_0^t\sig(s)\dd s.
\]
The ramp satisfies $\chi(1-s)=1-\chi(s)$, so
$\int_0^\eps\sig(s)\dd s=\eps/2$.
This yields \eqref{eq:layerexample}.
\end{proof}
%%% [chunk II-B-section-L1075 kind=section lines 1075-1081 labels=[]]


No spatial decay was imposed on classical configurations.
This example is legitimate under the local smooth Cauchy
prescription and shows explicitly why the output cannot be
labelled layer-supported.

```

```
%%% [chunk XI-B-section-eq-layerkernel kind=section lines 1082-1114 labels=['eq:layerkernel']]


For a compact interior covector test $a_\mu$,
transposition gives the de Donder-layer identity
\begin{equation}
 \int a_\mu\Coord_1^\mu[\gamma]\dd^4x
 =-\int_{\MT}\sig'(z)
       (\Ezero^{\mathrm A}a)_\mu(z)
          a_1^{0\mu}[\gamma](z)\dd^4z.
 \label{eq:layerkernel}
\end{equation}
The resulting linear kernel is smooth at fixed $\eps$.
Its support is contained in the closed layer intersected with
the causal past of $\supp a$.

Equation \eqref{eq:layerkernel} is a classical restriction to
de Donder tensors. The full auxiliary multiplet is not restricted
by $D\gamma=0$: its equation is $b=D\gamma$. Imposing $D=0$
as an additional algebraic quotient would require a separate
analysis and is not done here.

A boundary-supported trace is suggested by formally shrinking
the layer. No convergence theorem, sharp limit, or identification
with a sharp Cauchy kernel is asserted. In particular, a smooth
closed-slab kernel need not be a compactly supported interior
test. This classical kernel fact is not an admission theorem
for a quantum relational composite.

Neither $\Coord_1$ nor its scalar dressing is declared
$\sfree$-closed. The plateau covariance of X-A does not prove
such a claim for unrestricted ghost Cauchy data.

%=====================================================================

%%% [chunk XI-B-section-L1116 kind=section lines 1116-1177 labels=[]]

%=====================================================================

VII-C concerns a sharp zero-Cauchy insertion, a fixed Hadamard
Wick matrix in the opposite signature, and a specified
normal-topology extension. Those are not the data of the
causal polynomial algebra constructed here \cite{VIIC}.

For precision, its four extension requirements are recorded
below. ``Unmet'' means not established or not instantiated by
this manuscript, not disproved for every possible successor
construction.

\begin{center}
\small
\begin{tabularx}{\linewidth}{@{}p{.08\linewidth}X p{.27\linewidth}@{}}
\toprule
 & VII-C requirement & Status in XI-B\\
\midrule
E1 &
The old microcausal Wick algebra embeds into a Hausdorff locally
convex $\C[[\hbar]]$-algebra, with its fixed Hadamard Wick product
unchanged. &
Unmet. We construct a causal polynomial algebra, not that
Hadamard Wick embedding.\\[0.5em]

E2 &
Sharp insertions $Q_\Sigma(B_f)$ are admitted, with
$[Q_\Sigma(B_f)]_{\hbar^0}=B_f$ on all smooth configurations. &
Unmet. No sharp insertion or quantum relational composite is
admitted.\\[0.5em]

E3 &
Admission on the stated degree-$(1,1)$ kernel stages is
sequentially continuous in the normal topology and agrees with
the old regular inclusion; the regularized insertions converge
to the sharp element. &
Unmet. No such admission map or topology is constructed.\\[0.5em]

E4 &
Multiplication is hypocontinuous on bounded sets, and the
field-independent $\hbar^3$ coefficient map is continuous and
agrees with old coefficient extraction. &
Unmet as a topological package. Our product is algebraic;
```

```
no specified locally convex continuity result is proved.\\
\bottomrule
\end{tabularx}
\end{center}

Formal coefficient extraction is available algebraically.
That does not establish E4. Likewise the tensor $\PP$ derived
from the XI-A operator is not an imported Hadamard block.
A causal inverse does not select a positive-frequency
bisolution.

We do not form the word
$B_f\star W_g\star B_{f'}$, do not evaluate a reflected
light-ray residue, and do not subtract one. There is no
conclusion here that the ramp resolves VII-C's obstruction,
nor that VII-C obstructs the present restricted free algebra.

%=====================================================================

%%% [chunk XI-B-section-L1179 kind=section lines 1179-1231 labels=[]]

%=====================================================================

\begin{center}
\small
\begin{tabularx}{\linewidth}{@{}p{.55\linewidth}X@{}}
\toprule
Gate & Result\\
\midrule
One gauge fermion and frozen normalization & Constructed\\
Independent Nakanishi--Lautrup field & Retained\\
Ghost kinetic coefficient & $i\bar c_\mu\Box c^\mu$\\
Graviton causal inverse & $-\PP\Ezero^{\mathrm R/A}$\\
Scalar equation versus Hessian inverse & Distinguished\\
Full auxiliary and ghost Green blocks & Constructed\\
Free causal polynomial algebra & Constructed, without a state\\
Off-shell free nilpotence & Proved with independent $b$\\
Free product compatibility & Proved on the graded relations\\
Plateau residual algebra & $\{0\}$\\
De Donder ramp coefficient & Propagated layer functional\\
Fixed-input Einstein jet & Not repaired\\
Stress-tensor vertex versus dressing & Not contracted\\
Hadamard Wick extension or VII-C hypotheses & Not established\\
Tree, mixed loop, interacting differential & Not constructed\\
Sharp-ramp limit & Not taken\\
\bottomrule
\end{tabularx}
\end{center}

The ending is \textbf{Constructed} for the free gate.
The existence of its causal Green operators is not evidence
against XI-A's on-shell obstruction. Different equations are
being solved.

XI-C receives:
\begin{enumerate}
\item the signed operator $-\Box\PP$ and its causal inverses;
\item the single gauge fermion, auxiliary equations, ghost
normalization, and full free causal matrix;
\item the free graded polynomial algebra and its differential;
\item the residual-gauge result \eqref{eq:residual};
\item the classical layer formula, with its support qualifications;
\item the inherited vertex \eqref{eq:vertex}, still uncontracted;
\item the distinction between the classical plateau action and
a residual action on the gauge-fixed free graviton.
\end{enumerate}

It does not receive a Hadamard state, Feynman prescription,
tree, mixed loop, renormalized relational product, or
interacting BRST class. The first future contraction must name
its domain and its covariance; neither is supplied by the
word ``propagator.''


%%% [chunk XI-B-section-L1232 kind=section lines 1232-1256 labels=[]]


The free construction closes:
\[
 \boxed{
 \text{XI-A signed operator}
```

```
 \ \longrightarrow\
 \text{one auxiliary graded multiplet}
 \ \longrightarrow\
 \text{causal inverses and a free differential}.
 }
\]
At the same time,
\[
 \ga\cap\{\Box\xi=0\}=\{0\},
 \qquad
 \Jcal_g(\phi_m)=\varnothing
 \quad\text{for the XI-A Einstein reading}.
\]
The former is a residual-gauge statement, the latter a coupled
source obstruction. Neither is a failure of the constructed
gauge-fixed Green operator.

%=====================================================================
\appendix

%%% [chunk XI-B-section-L1261 kind=section lines 1261-1279 labels=[]]


The reduced graviton Hessian is $-\Box\PP$, so its inverse is
$-\PP\Ezero^\pm$.

The scalar action Hessian is $-(\Box+m^2)$, so its inverse is
$-\Em^\pm$, even though the equation inverse is $+\Em^\pm$.

The retarded-minus-advanced scalar Hessian kernel is therefore
$-\delta_m$. With the convention \eqref{eq:CCR},
\[
 [\phi(x),\phi(y)]=-i\hbar\delta_m(x,y).
\]
Since $\partial_{x^0}\delta_m$ at equal times is the spatial
delta, this gives
$[\phi,\partial_t\phi]=i\hbar\delta_{\R^3}$
in the canonical ordering. Thus the action and commutator signs
agree.


%%% [chunk XI-B-section-L1280 kind=section lines 1280-1301 labels=[]]


The half-normalized metric differential is $\sfree\gamma=\RR c$.
The gauge fermion produces $i\bar c\Box c$, not
$2i\bar c\Box c$.

Its causal anticommutator is
\[
 \{c,\bar c\}=\hbar\delta_0,
\]
while
$[\gamma,b]=i\hbar\RR\delta_0$.
These exact coefficients produce the cancellation in
Theorem~\ref{thm:differential}. Replacing one coefficient
without rescaling the ghost would destroy that compatibility.

One could instead use the full ghost normalization
$\sfree\gamma=K\widetilde c$, with $c=2\widetilde c$.
Then the corresponding ghost kinetic coefficient changes.
That is an equivalent reparametrization, not a second gauge
chosen in this paper.


%%% [chunk XI-B-section-L1302 kind=section lines 1302-1314 labels=[]]


For an unrestricted tensor,
\[
 \Box\Coord_1^\mu
       =\sig D^\mu\gamma-\sig'a_1^{0\mu}.
\]
Only the classical condition $D\gamma=0$ removes the first
term. The gauge-fixed equation $-\Box\bar\gamma=0$
does not itself impose $D\gamma=0$.
The auxiliary field records that distinction.

%=====================================================================

%%% [chunk XI-B-section-L1316 kind=section lines 1316-1329 labels=[]]
```

```
%====================================================================

The proofs use the explicit quadratic action, compact interior
variation, ordinary smooth wave Cauchy theory, algebraic trace
reversal, and graded central commutation relations. They do
not use a graviton Hadamard matrix from another manuscript.

The supplied XI-A is used for its signed action and the
Einstein-jet obstruction. The ramp coefficient is derived
from the displayed divergence-form prescription. VII-C is
used only to identify the separate hypotheses that are not
constructed here.


%%% [chunk XI-B-section-L1330 kind=section lines 1330-1353 labels=[]]

\addcontentsline{toc}{section}{Declarations}

\paragraph{Construction level.}
The free algebra is a formal algebra of compactly smeared
linear fields and finite products. No Hilbert-space
representation or state is selected.

\paragraph{Boundary level.}
Initial and final data specify causal inverses. No independent
boundary algebra, BRST boundary charge, or admission topology
is constructed.

\paragraph{Interaction level.}
The switched stress-tensor vertex remains a named classical
interaction. It is not used in any contraction with the ramp
dressing and is not included in $\sfree$.

\paragraph{Verification.}
No successful \texttt{pdflatex} execution, proof-assistant
verification, executed symbolic computation, independent
referee review, or journal acceptance is asserted.

%====================================================================

%%% [chunk XI-B-bibliography-L1354 kind=bibliography lines 1354-1402 labels=[]]
\begin{thebibliography}{9}
\addcontentsline{toc}{section}{References}

\bibitem{XIA}
N.~Cheung,
\emph{Landmark XI-A: A switched Einstein--scalar action for the
ramp relational scalar: sign conventions, Noether identities,
and an empty fixed-input Einstein jet}.
Supplied companion manuscript.
Used for the signed quadratic operator, the switching
conventions, and the fixed-input Einstein-jet obstruction.

\bibitem{VIIC}
\emph{Landmark VII-C: A mixed-Wick obstruction to a
normal-topology boundary extension for sharp Cauchy harmonic
insertions}.
Supplied comparison manuscript.
Used only for the extension checklist; its free matrix and
products are not inputs to the present construction.

\bibitem{BGP}
C.~B\"ar, N.~Ginoux, and F.~Pf\"affle,
\emph{Wave Equations on Lorentzian Manifolds and Quantization},
ESI Lectures in Mathematics and Physics,
European Mathematical Society, 2007.
\href{https://arxiv.org/abs/0806.1036}{arXiv:0806.1036}.
Used for smooth Cauchy theory, causal Green operators, and
finite propagation.

\bibitem{FR}
K.~Fredenhagen and K.~Rejzner,
\emph{Perturbative algebraic quantum field theory},
arXiv:1208.1428v2, 2013.
\href{https://arxiv.org/abs/1208.1428}{arXiv:1208.1428}.
Background for the state-independent polynomial deformation;
all block normalizations and compatibility checks are derived
in this manuscript.

\bibitem{BFR}
```

```
R.~Brunetti, K.~Fredenhagen, and K.~Rejzner,
``Quantum gravity from the point of view of locally covariant
quantum field theory,''
\emph{Communications in Mathematical Physics}
\textbf{345} (2016), 741--779.
\href{https://arxiv.org/abs/1306.1058}{arXiv:1306.1058}.
Comparison framework for the distinction between free gauge
fixing and an interacting BV construction.

\end{thebibliography}
```

3.4 Landmark XII-C (full source)

Listing 58: XII-C.tex

```
% !TeX program = pdflatex
%=====================================================================
% XII-C.tex
%
% Standalone source for pdfLaTeX.
% Build twice:
%   pdflatex -interaction=nonstopmode -halt-on-error XII-C.tex
%   pdflatex -interaction=nonstopmode -halt-on-error XII-C.tex
%
% Explicit correction of the production specification:
%   Multiplying the entire Einstein--Hilbert density by sigma
%   does NOT preserve the XI-A/XI-B lower-order coefficients.
%   It introduces a flat-background Euler term and changes the
%   quadratic operator.
%
% The single model selected here is therefore:
%   density weighting before variation, at CUBIC ORDER ONLY.
%
% Its gravitational action is
%   S_EH + kappa int (sigma-1) ell_3,
% where ell_3 is the specified, unintegrated cubic coefficient.
% The scalar action and gauge-fixing term remain unchanged.
%
% This correction is stated in the abstract, release box, and C0.
% It is not attributed retroactively to XII-A or XII-B.
%
% Proved here:
%   - a finite prescription for the cubic density representative;
%   - the actual weighted cubic Euler source;
%   - all first- and second-ramp-derivative source terms;
%   - the changed fixed-background Noether identity;
%   - cancellation of apparent third ramp derivatives;
%   - emptiness of the named fixed-input coupled jet;
%   - the vacuum counterexample to universal jet emptiness.
%
% Not constructed:
%   kernel admission, a metric retarded response, wavefront bounds,
%   a response plateau identity, a quantum retarded product,
%   an interacting differential, a state, or a sharp-width limit.
%
% No successful compilation, mechanized verification,
% independent referee review, or journal acceptance is asserted.
%=====================================================================

\documentclass[11pt,a4paper]{article}

\usepackage[T1]{fontenc}
\usepackage[utf8]{inputenc}
\usepackage{lmodern}
\usepackage[margin=27mm]{geometry}
\usepackage{microtype}
\usepackage{amsmath,amssymb,amsthm,mathtools,mathrsfs}
\usepackage{array,booktabs,tabularx}
\usepackage{enumitem}
\usepackage[hidelinks]{hyperref}

\allowdisplaybreaks
\setlength{\emergencystretch}{3em}
\setlist{leftmargin=*,itemsep=3pt,topsep=5pt}
\numberwithin{equation}{section}
\setcounter{tocdepth}{1}

```

```
\newtheorem{theorem}{Theorem}[section]
\newtheorem{proposition}[theorem]{Proposition}
\newtheorem{lemma}[theorem]{Lemma}
\newtheorem{corollary}[theorem]{Corollary}
\theoremstyle{definition}
\newtheorem{definition}[theorem]{Definition}
\theoremstyle{remark}
\newtheorem{remark}[theorem]{Remark}
\newtheorem{inputstatement}[theorem]{Input statement}

\newcommand{\R}{\mathbb R}
\newcommand{\C}{\mathbb C}
\newcommand{\dd}{\,\mathrm d}
\newcommand{\eps}{\varepsilon}
\newcommand{\sig}{\sigma_\eps}
\newcommand{\MT}{M_T}
\newcommand{\MTo}{M_T^\circ}
\newcommand{\PP}{\mathcal P}
\newcommand{\RR}{\mathcal R}
\newcommand{\Lie}{\mathcal L}
\newcommand{\ga}{\mathfrak g_U}
\newcommand{\Pmass}{P_m}
\newcommand{\Ggr}{G_{\mathrm{gr}}^{\mathrm R}}
\newcommand{\ER}{\mathsf E_0^{\mathrm R}}
\newcommand{\Csol}{\mathscr C_{\mathrm{sol}}}
\newcommand{\Fpol}{\mathscr F_{\mathrm{mc,pol}}}
\newcommand{\Nsol}{\mathscr N_{\mathrm{sol}}}
\newcommand{\Ealg}{\mathscr E_H}
\newcommand{\Afree}{\mathscr A_0}
\newcommand{\Aone}{\mathcal A_1}
\newcommand{\Cthree}{\mathcal C_3}
\newcommand{\Jcal}{\mathcal J}
\newcommand{\Hcal}{\mathcal H}
\newcommand{\Lag}{\mathfrak L}
\newcommand{\Euler}{\mathsf{EL}}
\newcommand{\Ncal}{\mathcal N}
\newcommand{\Bcal}{\mathcal B}
\newcommand{\Rampdef}{\mathfrak R}
\newcommand{\Jdensity}{J_{3,\mathrm D}}
\newcommand{\Dsource}{\mathcal D_\sig}
\newcommand{\Zsource}{Z_\sig}
\newcommand{\pair}[2]{\left\langle #1,#2\right\rangle}
\newcommand{\coeff}[2]{\left[#1^{#2}\right]}

\DeclareMathOperator{\supp}{supp}
\DeclareMathOperator{\tr}{tr}
\DeclareMathOperator{\Int}{int}
\DeclareMathOperator{\id}{id}

\newcommand{\scopebox}[2]{%
  \par\medskip\noindent
  \fbox{\begin{minipage}{%
    \dimexpr\linewidth-2\fboxsep-2\fboxrule\relax}
    \textbf{#1}\par\smallskip #2
  \end{minipage}}
  \par\medskip
}

\title{\textbf{Landmark XII-C}\\[0.4em]
One declared cubic weighting:\\
actual Euler source, changed Noether identity,\\
and the coupled-jet question}
\author{Ngo Cheung}
\date{}

\begin{document}
\maketitle

\begin{abstract}
We freeze one density-before-variation prescription at fixed
regulator width. A necessary correction is made explicit:
weighting the entire Einstein--Hilbert density would change the
lower-order sheet and would not retain the XI-B free operator.
To preserve that handoff, only the specified cubic density
coefficient is multiplied by the existing ramp. All other
gravitational coefficients, the switched scalar action, and
the gauge-fixing term remain those of XI-A and XI-B.

A finite, unintegrated prescription fixes the cubic density
representative. Its actual Euler source is computed as
$\sigma_\varepsilon J_3+\mathcal D_{\sigma_\varepsilon}$.
```

```
The complete tensor formula displays all first and second
derivatives of the ramp and reproduces XII-B's pure-trace
layer term. The changed fixed-background Noether identity
contains both the scalar-switch defect and the additional
breaking caused by weighting only one gravitational
coefficient. Its apparent third ramp derivatives cancel
algebraically.

For the named free inputs $\phi_m$, the coupled invariant
metric--scalar jet is empty for every switch of the declared
plateau class. This is proved anew: the compatibility condition
has a layer correction, but a nonzero late switching defect
survives where the ramp is already one. Arbitrary leading
vacuum metric data are allowed. The zero-field vacuum remains
a counterexample to universal jet emptiness.

No Green operator is applied. No completed insertion is admitted,
no wavefront theorem or response plateau law is proved, and no
interacting differential or sharp-width limit is introduced.
\end{abstract}

\scopebox{Release and explicit specification correction}{%
\textbf{Selected model:}
density before variation, at \emph{cubic order only}.
This is the unique weighting used in the manuscript.

\smallskip
\textbf{Correction at CO:}
the production formula weighting the entire Einstein--Hilbert
density cannot also satisfy the requirement that the lower-order
XI-A/XI-B sheet remain unchanged. We therefore replace that
formula by the explicitly declared cubic-only action in
Section~\ref{sec:freeze}. The literal full-density model is not
certified here.

\smallskip
\textbf{Constructed:}
the action freeze, actual cubic source, changed Noether identity,
and successor data.

\smallskip
\textbf{Proved obstructed:}
the fixed-input coupled jet for the family $\phi_m$.
This is an obstruction of that solution fiber, not of an
observable or of XII-A's carrier.

\smallskip
\textbf{Not opened:}
B1--B4, kernel admission, any new metric retarded response,
a response plateau cancellation, a state, or
$\widehat s_{\mathrm{int}}$.}

\tableofcontents

%=====================================================================
\section{Release criteria and the inherited prerequisite sheet}
\label{sec:prerequisites}
%=====================================================================

The three endings remain:
\begin{enumerate}
\item \textbf{Constructed}: the weighting and density representative
are fixed, their actual source and Noether identity are proved,
and the jet question has a named verdict.
\item \textbf{Proved obstructed}: an explicit incompatibility is
proved for the chosen specification or for the named input family.
\item \textbf{Unresolved}: a necessary definition or calculation
is absent.
\end{enumerate}
The action freeze and the solution-fiber verdict are reported
separately. A constructed action need not have a solution for
every prescribed scalar input.

\[
 \boxed{\text{XII-C freezes an action. It does not cross B1.}}
\]

\subsection{Copied prerequisite statements}

The following statements are used with their predecessor scope.
They are not conclusions about the new weighting.
```

```
\begin{inputstatement}[XII-A-v2]
The supplied carrier is
\[
 \Ealg
 =
 (\Fpol[[\hbar,\kappa]],\star_H)/
             \Nsol[[\hbar,\kappa]],
\]
with equal equation and action kernels $\Csol$ and a regular
quotient recovering the causal free algebra.
Ordering transport is an isomorphism on the named composite
sector. For detected observations, the original unramped
$\Cthree|_{\Csol}$ is excluded.
Undetected observations, the Einstein subshell, global
hypocontinuity, and enlargement remain undecided.
No quantum retarded product is supplied \cite{XIIA}.
\end{inputstatement}

\begin{inputstatement}[XII-B]
The requested changed-model handoff was incomplete, so B0--B4
were left Unresolved. The proved variational distinction is
\begin{equation}
 \Euler(\sigma\ell_3)
   =\sigma J_3+\mathcal D_\sigma[\ell_3],
 \label{eq:input-weighted}
\end{equation}
with the pure-trace evaluation
\begin{equation}
 \eta_{\mu\nu}\mathcal D_\sigma^{\mu\nu}
       \big|_{\gamma=\psi\eta}
 =-6(\partial_\mu\sigma)\psi\partial^\mu\psi
 \not\equiv0.
 \label{eq:input-trace}
\end{equation}
The simply weighted source is not an Euler derivative on the
unrestricted off-shell class. These results distinguish source
weighting from density weighting; they do not obstruct an action
with its actual layer terms retained \cite{XIIB}.
\end{inputstatement}

\begin{inputstatement}[XI-B]
The inherited inverse normalization is
$\Ggr=-\PP\ER$.
The free rule is $s_0\gamma=\RR c$, with $\RR=K/2$ and
$c=2\xi$ recovering $K\xi$.
The compact plateau residual intersection is zero.
The de Donder layer formula is conditional on $D\gamma=0$,
and propagated support is not layer support \cite{XIB}.
\end{inputstatement}

\begin{inputstatement}[XI-A and XI-G]
XI-A's empty fixed-input Einstein jet concerns its original
unswitched gravitational density.
XI-G's detected unramped shell class remains excluded; its
source-ramping remark is not a completed-kernel admission theorem
\cite{XIA,XIG}.
\end{inputstatement}

Nothing below searches for a representative of the old
$\Cthree$. X-E/X-F's normalization, X-G's reciprocity pairing,
VII-C's word, the IX sharp commutator, and any cone or Enter
readout are not inputs.

The general variational treatment of curvature with a scalar
prefactor and of nondynamical background breaking provides
useful comparison literature \cite{SF,BS}. The model-specific
formulas below are derived in the displayed conventions.

%=====================================================================
\section{CO: one choice and the lower-order consistency correction}
\label{sec:freeze}
%=====================================================================

\[
 \boxed{\text{Choice D: density before variation, at cubic order only.}}
\]

The nonvariational forcing option is not selected.

\subsection{Geometry and fixed external functions}
```

```

Use
\[
 M=\R^4,\quad
 \eta=\operatorname{diag}(1,-1,-1,-1),\quad
 \Box=\partial_t^2-\Delta_{\mathbf x},\quad
 \Pmass=\Box+m^2,
\]
\[
 \MT=[0,T]\times\R^3,\qquad
 \MTo=(0,T)\times\R^3,\qquad 0<\eps<T.
\]
All fields are smooth on a neighborhood of the closed slab.
The ramp is exactly the inherited function
\[
 \sig(t)=\chi(t/\eps),\qquad
 \chi(s)=
 \begin{cases}
 0,&s\le0,\\[2pt]
 \displaystyle
 \frac{e^{-1/s}}{e^{-1/s}+e^{-1/(1-s)}},&0<s<1,\\[7pt]
 1,&s\ge1.
 \end{cases}
\]
The scalar switch satisfies
\[
 g\in C_c^\infty(\MTo;[0,1]),\qquad
 P_g=\Int\{g=1\}\ne\varnothing,
\]
and
\[
 \overline U\subset P_g\cap\{\eps<t<T\},\qquad
 \ga=C_c^\infty(U;TM).
\]
Neither $g$ nor $\sig$ is varied. The class $\ga$ is not
subsequently reduced.

Put
\[
 q_{\mu\nu}=\eta_{\mu\nu}+\kappa\gamma_{\mu\nu},\qquad
 \PP r=r-\tfrac12\eta\,\tr_\eta r,
\]
\[
 D^\nu\gamma=\partial_\mu(\PP\gamma)^{\mu\nu},
 \qquad
 (K\xi)_{\mu\nu}=\partial_\mu\xi_\nu+\partial_\nu\xi_\mu.
\]
Parentheses in indexed formulas include $\tfrac12$.
The curvature convention is
\begin{equation}
 R_{\mu\nu}
 =
 \partial_\rho\Gamma^\rho_{\mu\nu}
 -\partial_\nu\Gamma^\rho_{\mu\rho}
 +\Gamma^\rho_{\rho\lambda}\Gamma^\lambda_{\mu\nu}
 -\Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\rho}.
 \label{eq:curvature}
\end{equation}

\subsection{Why the full-density wording must be corrected}

\begin{proposition}[Full weighting changes the free sheet]
\label{prop:fullweight}
Multiplication of the entire signed Einstein--Hilbert density
by $\sig$ does not retain the XI-A/XI-B lower-order model.
Its metric Euler derivative at the flat background contains
\[
 \frac2\kappa
 \left(\eta^{\mu\nu}\Box\sig-\partial^\mu\partial^\nu\sig\right).
\]
This expression is nonzero for the declared ramp.
\end{proposition}

\begin{proof}
The compact-variation calculation proved in
Lemma~\ref{lem:weighted-curvature} below gives the displayed
expression at $q=\eta$.
Its spatial components are
$-2\delta^{ij}\sig''/\kappa$.
The ramp has derivative zero at both endpoints but changes
from zero to one, so $\sig''$ cannot vanish identically.
```

```
Thus flat space is not stationary even with zero scalar input.
In particular, the original zero-background free sheet is
not preserved.
\end{proof}

This is a consistency check on a rejected formula, not a second
selected model. Keeping the entire weighted density would require
a different lower-order freeze and a different background audit.

\subsection{The selected formal action}

Let $\ell_3$ be the precise cubic density fixed in
Section~\ref{sec:density}. Define
\begin{equation}
 S_{\mathrm{grav,D}}
 =
 S_{\mathrm{EH}}
 +\kappa\int_{\MT}(\sig-1)\ell_3[\gamma]\dd^4x,
 \qquad
 S_{\mathrm{EH}}
 =-\frac2{\kappa^2}
       \int_{\MT}\sqrt{-q}\,R[q]\dd^4x.
 \label{eq:selected-action}
\end{equation}
The unchanged scalar action is
\begin{equation}
 S_g^{\mathrm{sc}}
 =
 \int_{\MT}
 \bigl((1-g)\Lag_\eta[\phi]+g\Lag_q[\phi]\bigr)\dd^4x,
 \qquad
 \Lag_q[\phi]
 =\tfrac12\sqrt{-q}
       \left(q^{\mu\nu}\partial_\mu\phi\partial_\nu\phi
                          -m^2\phi^2\right).
 \label{eq:scalar}
\end{equation}
Write
\[
 S_{\mathrm D}=S_{\mathrm{grav,D}}+S_g^{\mathrm{sc}},
 \qquad
 S_{\mathrm{D,gf}}=S_{\mathrm D}+S_{\mathrm{gf}},
 \qquad
 S_{\mathrm{gf}}=\int_{\MT}D_\nu\gamma D^\nu\gamma\dd^4x.
\]

Actions mean formal bulk variational functionals, tested with
compact interior variations. Numerical finiteness on all smooth
configurations is not required. The unweighted linear
Einstein--Hilbert divergence is treated as in XI-A.
The selected cubic representative is not changed by a later
integration by parts.

Equation~\eqref{eq:selected-action} weights only the cubic
coefficient. The quadratic coefficient and all quartic and higher
coefficients remain unweighted. This fixes the higher-order
completion rather than leaving it unspecified.

\begin{proposition}[Unchanged free handoff]
The quadratic gravitational and gauge-fixing coefficients of
$S_{\mathrm{D,gf}}$ are the XI-A/XI-B coefficients. Their metric
operator is
\[
 A+2D^*D=-\Box\PP,\qquad A=2G_{\mathrm{lin}}.
\]
The auxiliary and ghost normalization of XI-B is unchanged.
\end{proposition}

\begin{proof}
The added functional in \eqref{eq:selected-action} is cubic in
$\gamma$. Its first and second derivatives vanish at $\gamma=0$.
It therefore changes neither the free metric Hessian nor the
independent quadratic auxiliary and ghost terms.
\end{proof}

This cubic-only correction is a definition made in XII-C.
It is not attributed retrospectively to XII-A or XII-B.

%=====================================================================
\section{C1: the density representative and actual cubic source}
\label{sec:density}
```

```
%=================================================================

\subsection{A finite unintegrated density prescription}

Write $\tau=\tr_\eta\gamma$ and set
\[
 u_0^{\mu\nu}=\eta^{\mu\nu},\qquad
 u_1^{\mu\nu}=-\gamma^{\mu\nu},\qquad
 u_2^{\mu\nu}=\gamma^\mu{}_\lambda\gamma^{\lambda\nu},
\]
\[
 b_0=1,\qquad b_1=\tfrac12\tau,\qquad
 b_2=\tfrac18\tau^2-\tfrac14\gamma_{\mu\nu}\gamma^{\mu\nu}.
\]
For $n=1,2,3$, define
\begin{align}
 \Gamma_n^\rho{}_{\mu\nu}
 &=
 \tfrac12u_{n-1}^{\rho\lambda}
 \left(\partial_\mu\gamma_{\nu\lambda}
      +\partial_\nu\gamma_{\mu\lambda}
      -\partial_\lambda\gamma_{\mu\nu}\right),\\
 R_{n,\mu\nu}
 &=
 \partial_\rho\Gamma_n^\rho{}_{\mu\nu}
 -\partial_\nu\Gamma_n^\rho{}_{\mu\rho}
 \nonumber\\
 &\quad+
 \sum_{\substack{a+b=n\\a,b\ge1}}
 \left(
 \Gamma_a^\rho{}_{\rho\lambda}\Gamma_b^\lambda{}_{\mu\nu}
 -\Gamma_a^\rho{}_{\nu\lambda}\Gamma_b^\lambda{}_{\mu\rho}
 \right),\\
 r_n&=\sum_{i=0}^{n-1}u_i^{\mu\nu}R_{n-i,\mu\nu}.
 \label{eq:finite-curvature}
\end{align}
The declared representative is
\begin{equation}
 \boxed{\ell_3=-2(r_3+b_1r_2+b_2r_1).}
 \label{eq:ell3}
\end{equation}
It is the coefficient of $\kappa$ in the unintegrated signed
Einstein--Hilbert density. Equivalently, it is $-2$ times the
coefficient of $\kappa^3$ in $\sqrt{-q}R[q]$.

Every term is local, cubic, and of jet order at most two.
The formulas specify the density itself, not only its Euler
equivalence class.

For clarity, the original source can also be evaluated finitely.
Put
\[
 V_{1,\mu\nu}=R_{1,\mu\nu}-\tfrac12\eta_{\mu\nu}r_1,\qquad
 V_{2,\mu\nu}=R_{2,\mu\nu}-\tfrac12\eta_{\mu\nu}r_2
                           -\tfrac12\gamma_{\mu\nu}r_1.
\]
Then
\begin{equation}
 J_3^{\mu\nu}
 =
 2\left(
 V_2^{\mu\nu}+b_1V_1^{\mu\nu}
 -\gamma^{\mu\alpha}V_{1,\alpha}{}^\nu
 -\gamma^{\nu\alpha}V_{1,\alpha}{}^\mu
 \right).
 \label{eq:J3}
\end{equation}
Compact variation of \eqref{eq:ell3} gives
$\Euler_\gamma(\ell_3)=J_3$ with the XI-A sign.

\subsection{The curvature variation needed for the expansion}

\begin{lemma}[Weighted curvature variation]
\label{lem:weighted-curvature}
For a fixed smooth scalar $\sigma$, compact covariant metric
variation gives
\begin{equation}
 \frac{\delta}{\delta\gamma_{\mu\nu}}
 \left(-\frac2{\kappa^2}
        \int\sigma\sqrt{-q}R[q]\dd^4x\right)
 =
```

```
 \frac2\kappa\sqrt{-q}
 \left[
 \sigma G^{\mu\nu}
 +q^{\mu\nu}\Box_q\sigma-\nabla^\mu\nabla^\nu\sigma
 \right].
 \label{eq:weighted-full-variation}
\end{equation}
This identity is used for coefficient extraction, not to select
the full-density model.
\end{lemma}

\begin{proof}
For $h_{\mu\nu}=\delta q_{\mu\nu}$, write
$h=q^{\mu\nu}h_{\mu\nu}$ and
$h^{\mu\nu}=q^{\mu\alpha}q^{\nu\beta}h_{\alpha\beta}$.
The Christoffel variation gives
\[
 \delta R=-R^{\mu\nu}h_{\mu\nu}
              +\nabla_\mu\nabla_\nu h^{\mu\nu}-\Box_q h.
\]
Together with $\delta\sqrt{-q}=\frac12\sqrt{-q}\,h$,
two compact integrations by parts give
\[
 \delta\int\sigma\sqrt{-q}R
 =
 \int\sqrt{-q}
 \left[-\sigma G^{\mu\nu}
       +\nabla^\mu\nabla^\nu\sigma-q^{\mu\nu}\Box_q\sigma\right]
 h_{\mu\nu}\dd^4x.
\]
Multiply by $-2/\kappa^2$ and use
$h_{\mu\nu}=\kappa\,\delta\gamma_{\mu\nu}$.
\end{proof}

The coefficient of $\kappa$ on the left is the Euler derivative
of $\sigma\ell_3$. Thus coefficient extraction supplies its
complete layer correction without changing the chosen density.

\subsection{All first and second ramp derivatives}

Define
\[
 C^{\mu\nu\alpha\beta}(q)
 =
 \sqrt{-q}
 \left(q^{\mu\nu}q^{\alpha\beta}
                   -q^{\mu(\alpha}q^{\beta)\nu}\right)
 =C_0+\kappa C_1+\kappa^2C_2+O(\kappa^3).
\]
The three coefficients are explicitly
\begin{align}
 C_0^{\mu\nu\alpha\beta}
 &=\eta^{\mu\nu}\eta^{\alpha\beta}
                          -\eta^{\mu(\alpha}\eta^{\beta)\nu},\\
 C_1^{\mu\nu\alpha\beta}
 &=b_1C_0^{\mu\nu\alpha\beta}
   +u_1^{\mu\nu}\eta^{\alpha\beta}
   +\eta^{\mu\nu}u_1^{\alpha\beta}
 \nonumber\\
 &\quad-u_1^{\mu(\alpha}\eta^{\beta)\nu}
          -\eta^{\mu(\alpha}u_1^{\beta)\nu},\\
 C_2^{\mu\nu\alpha\beta}
 &=b_2C_0^{\mu\nu\alpha\beta}
   +b_1\left(
    u_1^{\mu\nu}\eta^{\alpha\beta}
   +\eta^{\mu\nu}u_1^{\alpha\beta}
   -u_1^{\mu(\alpha}\eta^{\beta)\nu}
   -\eta^{\mu(\alpha}u_1^{\beta)\nu}\right)
 \nonumber\\
 &\quad+u_2^{\mu\nu}\eta^{\alpha\beta}
       +\eta^{\mu\nu}u_2^{\alpha\beta}
       +u_1^{\mu\nu}u_1^{\alpha\beta}
 \nonumber\\
 &\quad-u_2^{\mu(\alpha}\eta^{\beta)\nu}
       -\eta^{\mu(\alpha}u_2^{\beta)\nu}
       -u_1^{\mu(\alpha}u_1^{\beta)\nu}.
 \label{eq:Ccoefficients}
\end{align}
Also define the quadratic tensor
\begin{equation}
 M_2^{\mu\nu\rho}
```

```
 =
 C_0^{\mu\nu\alpha\beta}\Gamma_2^\rho{}_{\alpha\beta}
 +C_1^{\mu\nu\alpha\beta}\Gamma_1^\rho{}_{\alpha\beta}.
 \label{eq:M2}
\end{equation}

\begin{theorem}[Actual density-weighted cubic source]
\label{thm:actualsource}
The selected cubic vertex
\[
 V_{3,\mathrm D}[\gamma]
       =\int_{\MT}\sig\ell_3[\gamma]\dd^4x
\]
has Euler source
\begin{equation}
 \boxed{
 \Jdensity^{\mu\nu}
 =\sig J_3^{\mu\nu}+\Dsource^{\mu\nu},\qquad
 \Dsource^{\mu\nu}
 =2C_2^{\mu\nu\alpha\beta}\partial_\alpha\partial_\beta\sig
             -2M_2^{\mu\nu\rho}\partial_\rho\sig .
 }
 \label{eq:actualsource}
\end{equation}
For the time-only ramp,
\begin{equation}
 \Dsource^{\mu\nu}
 =2C_2^{\mu\nu00}\sig''
                  -2M_2^{\mu\nu0}\sig'.
 \label{eq:time-layer}
\end{equation}
These are all ramp-derivative terms in the cubic source.
\end{theorem}

\begin{proof}
The extra term in \eqref{eq:weighted-full-variation} is
\[
 \frac2\kappa C^{\mu\nu\alpha\beta}(q)
 \left(
 \partial_\alpha\partial_\beta\sig
            -\Gamma^\rho_{\alpha\beta}(q)\partial_\rho\sig
 \right).
\]
Its coefficient of $\kappa$ is
\eqref{eq:actualsource}, using \eqref{eq:Ccoefficients}
and \eqref{eq:M2}. The Einstein term at that coefficient is
$\sig J_3$. No derivative of the fixed weight beyond order
two occurs.
\end{proof}

The correction is quadratic in $\gamma$ and its derivatives.
It vanishes outside the closed layer $0\le t\le\eps$ and is
smooth and flat at its endpoints. Its nonzero values occur
inside the open layer; its closed support need not avoid the
endpoints. There is no hypersurface distribution.

\subsection{The copied pure-trace check}

XII-B's identity \eqref{eq:input-trace} is a consistency
requirement, not the general source formula.
For $\gamma=\psi\eta$, algebraic substitution gives
$C_1=C_2=0$ and $\Gamma_2=-\psi\Gamma_1$.
Consequently \eqref{eq:actualsource} gives
\begin{equation}
 \Dsource^{\mu\nu}\big|_{\gamma=\psi\eta}
 =
 -\psi\left[
 \eta^{\mu\nu}(\partial_\rho\sig)\partial^\rho\psi
 +(\partial^\mu\sig)\partial^\nu\psi
 +(\partial^\nu\sig)\partial^\mu\psi
 \right].
 \label{eq:full-trace-sector}
\end{equation}
Contracting with $\eta_{\mu\nu}$ reproduces
\eqref{eq:input-trace}. Thus the predecessor's nonzero layer
term is retained.

Density representatives cannot be changed silently.
If $\ell_3$ is replaced by $\ell_3+\partial_\mu B^\mu$,
the weighted source changes by
$-\Euler_\gamma((\partial_\mu\sig)B^\mu)$, as proved in
```

```
XII-B. Equation~\eqref{eq:ell3} eliminates that ambiguity.

%=====================================================================
\section{C2: the changed fixed-background Noether identity}
\label{sec:noether}
%=====================================================================

\subsection{Euler operators and the unchanged matter defect}

Define
\[
 e_{\mathrm D}^{\mu\nu}
       =\frac{\delta S_{\mathrm D}}{\delta\gamma_{\mu\nu}},
 \qquad
 E_\phi=\frac{\delta S_{\mathrm D}}{\delta\phi}.
\]
The scalar equation is unchanged. Explicitly,
\begin{equation}
 E_\phi
 =
 -\partial_\mu\left(
 [(1-g)\eta^{\mu\nu}+g\sqrt{-q}\,q^{\mu\nu}]
                          \partial_\nu\phi\right)
 -m^2[(1-g)+g\sqrt{-q}]\,\phi.
 \label{eq:scalar-euler}
\end{equation}

Put $w=\sig-1$ and
\begin{equation}
 \Zsource^{\mu\nu}
   =wJ_3^{\mu\nu}+\Dsource^{\mu\nu}.
 \label{eq:Z}
\end{equation}
The exact formal metric derivative is
\begin{equation}
 e_{\mathrm D}^{\mu\nu}
 =
 \frac2\kappa\sqrt{-q}\,G^{\mu\nu}
 -\frac{\kappa}{2}g\sqrt{-q}\,T_q^{\mu\nu}
 +\kappa\Zsource^{\mu\nu},
 \label{eq:eD}
\end{equation}
where
\[
 T_q^{\mu\nu}
 =q^{\mu\alpha}q^{\nu\beta}\partial_\alpha\phi\partial_\beta\phi
 -\tfrac12q^{\mu\nu}
       (q^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi
                         -m^2\phi^2).
\]
In particular, for coupling-independent fields,
\[
 e_{\mathrm D}
 =A\gamma
   +\kappa\left(\Jdensity-\tfrac12gT_0\right)
   +O(\kappa^2).
\]

For a symmetric contravariant density $v$, set
\begin{align}
 \Ncal_\nu(v)
 &=2\partial_\mu(v^{\mu\rho}q_{\rho\nu})
                          -v^{\mu\rho}\partial_\nu q_{\mu\rho},\\
 N_{0,\nu}(v)
 &=2\eta_{\rho\nu}\partial_\mu v^{\mu\rho},\\
 Q_\nu(v)
 &=2\partial_\mu(v^{\mu\rho}\gamma_{\rho\nu})
                          -v^{\mu\rho}\partial_\nu\gamma_{\mu\rho}.
 \label{eq:Noperators}
\end{align}
Thus $\Ncal_\nu(v)=N_{0,\nu}(v)+\kappa Q_\nu(v)$.

The unchanged scalar-background defect is
\begin{equation}
 \Bcal_{g,\nu}
 =
 (\partial_\mu g)T_0^\mu{}_\nu
 -(1-g)(\Pmass\phi)\partial_\nu\phi
 +(\partial_\nu g)(\Lag_\eta[\phi]-\Lag_q[\phi]).
 \label{eq:matterdefect}
\end{equation}
```

```
The derivative-of-switch term remains visible.
There are no derivatives of $g$ in the metric variation of
the minimally coupled scalar density itself.

\subsection{An explicit ramp defect with no third derivatives}

Define
\begin{equation}
 \begin{split}
 \Rampdef_{\sig,\nu}
 ={}&
 2\eta_{\rho\nu}(\partial_\mu\sig)J_3^{\mu\rho}\\
 &+4\eta_{\rho\nu}
 \left[
  (\partial_\mu C_2^{\mu\rho\alpha\beta})
                    \partial_\alpha\partial_\beta\sig
  -(\partial_\mu M_2^{\mu\rho\lambda})\partial_\lambda\sig
  -M_2^{\mu\rho\lambda}
                    \partial_\mu\partial_\lambda\sig
 \right].
 \end{split}
 \label{eq:Rampdef}
\end{equation}
For the declared ramp this is
\begin{equation}
 \begin{split}
 \Rampdef_{\sig,\nu}
 ={}&2\eta_{\rho\nu}J_3^{0\rho}\sig'\\
 &+4\eta_{\rho\nu}
 \left[
   (\partial_\mu C_2^{\mu\rho00})\sig''
   -(\partial_\mu M_2^{\mu\rho0})\sig'
   -M_2^{0\rho0}\sig''
 \right].
 \end{split}
 \label{eq:Rampdef-time}
\end{equation}

\begin{lemma}[Derivative cancellation and cubic divergence]
\label{lem:divergence}
The following identities hold off shell:
\begin{align}
 N_{0,\nu}(J_3)+Q_\nu(A\gamma)&=0,
 \label{eq:original-cubic-noether}\\
 N_{0,\nu}(\Zsource)
 &=-wQ_\nu(A\gamma)+\Rampdef_{\sig,\nu}.
 \label{eq:Zdivergence}
\end{align}
In particular, differentiating the actual source introduces
no surviving third derivative of $\sig$.
\end{lemma}

\begin{proof}
The contracted Bianchi identity, written for
$\sqrt{-q}G^{\mu\nu}$, gives
$\Ncal_\nu(2\sqrt{-q}G/\kappa)=0$.
Its coefficient of $\kappa$ is
\eqref{eq:original-cubic-noether}.

Now differentiate \eqref{eq:Z}. The terms from $wJ_3$ are
\[
 wN_{0,\nu}(J_3)
       +2\eta_{\rho\nu}(\partial_\mu\sig)J_3^{\mu\rho}.
\]
Differentiation of \eqref{eq:actualsource} gives the remaining
terms of \eqref{eq:Rampdef} and an apparent term
\[
 4\eta_{\rho\nu}
 C_2^{\mu\rho\alpha\beta}
               \partial_\mu\partial_\alpha\partial_\beta\sig.
\]
For every symmetric three-covector tensor
$S_{\mu\alpha\beta}$,
\[
 C^{\mu\rho\alpha\beta}(q)S_{\mu\alpha\beta}=0:
\]
the two products of inverse metrics become equal after
permuting the three contracted indices.
Every coefficient $C_r$ has the same property.
The apparent third derivative therefore vanishes.
Substitute \eqref{eq:original-cubic-noether}.
```

```
\end{proof}

\subsection{The actual changed identity}

\begin{theorem}[Changed Noether identity]
\label{thm:noether}
For the selected action, off shell,
\begin{equation}
 \boxed{
 \begin{split}
 \Ncal_\nu(e_{\mathrm D})
 =\kappa\bigl[
 &E_\phi\partial_\nu\phi-\Bcal_{g,\nu}
 -wQ_\nu(A\gamma)+\Rampdef_{\sig,\nu}
 +\kappa Q_\nu(\Zsource)
 \bigr].
 \end{split}}
 \label{eq:changed-noether}
\end{equation}
All ramp derivatives are displayed in
\eqref{eq:actualsource} and \eqref{eq:Rampdef}.
The $Q_\nu(\Zsource)$ term is the explicitly defined first-order
operator \eqref{eq:Noperators} applied to the displayed
quadratic source.
\end{theorem}

\begin{proof}
For an arbitrary compact interior $\xi$, use the scaled
metric--scalar variation
\[
 \delta_\xi\gamma=K\xi+\kappa\Lie_\xi\gamma=\Lie_\xi q,
 \qquad
 \delta_\xi\phi=\kappa\xi^\rho\partial_\rho\phi.
\]
Compact integration by parts gives
\[
 \delta_\xi S_{\mathrm D}
 =
 \int\xi^\nu
 \left[-\Ncal_\nu(e_{\mathrm D})
                        +\kappa E_\phi\partial_\nu\phi\right]\dd^4x.
\]

For the unweighted Einstein--Hilbert part, naturality gives
zero variation. Variation of the unchanged switched scalar
action gives
$\kappa\int\xi^\nu\Bcal_{g,\nu}\dd^4x$.
This follows by expanding the flat scalar variation as
\[
 \delta_\xi\Lag_\eta
 =\kappa\left[
 T_0^\mu{}_\nu\partial_\mu\xi^\nu
                   +\partial_\mu(\xi^\mu\Lag_\eta)\right]
\]
and the curved scalar variation as a density divergence,
then integrating the factors $1-g$ and $g$.

The added cubic functional in \eqref{eq:selected-action}
varies as
\[
 \kappa\int
 \Zsource^{\mu\nu}
      (K\xi+\kappa\Lie_\xi\gamma)_{\mu\nu}\dd^4x
 =
 -\kappa\int\xi^\nu
       [N_{0,\nu}(\Zsource)+\kappa Q_\nu(\Zsource)]\dd^4x.
\]
Comparison of the two variation expressions, followed by
Lemma~\ref{lem:divergence}, proves
\eqref{eq:changed-noether}.
\end{proof}

This is not XI-A's identity with a relabeled source.
The additional term $-wQ_\nu(A\gamma)$ is important:
weighting only the cubic coefficient, while leaving the
quadratic coefficient unchanged, produces breaking even
where the weight is a constant different from one.
Thus the new breaking is not exclusively a ramp-derivative
term. Omitting this bulk coefficient mismatch would falsely
treat a cubic-only modification as weighting the complete
natural gravitational density.
```

```

For the gauge-fixed action, add the separate contribution
$\Ncal_\nu(e_{\mathrm{gf}})$ to the right, where
\[
 e_{\mathrm{gf}}^{\mu\nu}
 =-2\partial^{(\mu}D^{\nu)}
                      +\eta^{\mu\nu}\partial_\rho D^\rho.
\]
Gauge fixing is not made invariant by the ramp.

\begin{corollary}[Plateau contraction, not a response law]
For $\xi\in\ga$, the new action correction and scalar-switch
breaking vanish in the contracted variational identity.
Thus
\[
 \int\xi^\nu
 \left[-\Ncal_\nu(e_{\mathrm D})
                       +\kappa E_\phi\partial_\nu\phi\right]\dd^4x=0.
\]
\end{corollary}

\begin{proof}
On a neighborhood of $\supp\xi$, one has
$g=\sig=1$ and all their derivatives vanish.
Hence $\Bcal_g$, $w$, $\Dsource$, $\Zsource$, and
$\Rampdef_\sig$ vanish there.
\end{proof}

This local action statement supplies no propagated cancellation.
No source has been evolved, and no plateau ledger is computed.

%=====================================================================
\section{C3: the coupled jet for the selected action}
\label{sec:jet}
%=====================================================================

\begin{definition}[New fixed-input invariant jet]
For a fixed free scalar $\phi_0$, let
$\Jcal_{g,\sig}^{\mathrm D}(\phi_0)$ consist of smooth formal
families
\[
 \gamma(\kappa)=\gamma_0+\kappa\gamma_1+\kappa^2\gamma_2,
 \qquad
 \phi(\kappa)=\phi_0+\kappa\phi_1+\kappa^2\phi_2,
\]
for which the coefficients through $\kappa^2$ of
$e_{\mathrm D}=0$ and $E_\phi=0$ vanish.
Positive-order scalar coefficients have the inherited zero
initial data. Metric Cauchy data are unrestricted.

This is a jet of $S_{\mathrm D}$, not of the gauge-fixed
action. In particular, $\gamma_0$ is not required to vanish.
\end{definition}

\subsection{The recomputed compatibility condition}

\begin{proposition}[Order-one compatibility]
\label{prop:compatibility}
Every element of
$\Jcal_{g,\sig}^{\mathrm D}(\phi_0)$ satisfies
\begin{equation}
 (\partial_\mu g)T_0^\mu{}_\nu[\phi_0]
       =\Rampdef_{\sig,\nu}[\gamma_0].
 \label{eq:new-compatibility}
\end{equation}
For $t>\eps$, this reduces to
\begin{equation}
 (\partial_\mu g)T_0^\mu{}_\nu[\phi_0]=0.
 \label{eq:late-compatibility}
\end{equation}
\end{proposition}

\begin{proof}
The leading metric equation is $A\gamma_0=0$.
The next metric equation is
\[
 A\gamma_1+\Jdensity[\gamma_0]
                           -\tfrac12gT_0[\phi_0]=0.
\]
Apply $N_0$. The linearized Bianchi identity gives
$N_0(A\gamma_1)=0$.
```

```
Lemma~\ref{lem:divergence} also gives
\[
 N_{0,\nu}(\Jdensity)
 =-\sig Q_\nu(A\gamma_0)+\Rampdef_{\sig,\nu}
 =\Rampdef_{\sig,\nu}.
\]
Since $\Pmass\phi_0=0$,
$\partial_\mu T_0^\mu{}_\nu[\phi_0]=0$.
The resulting equation is
\eqref{eq:new-compatibility}.
Above the ramp, \eqref{eq:Rampdef-time} vanishes identically.
\end{proof}

The new compatibility condition permits a layer contribution
from leading metric data. It is therefore not obtained by
transferring XI-A's unswitched condition to the whole slab.

\subsection{The named family and a late switching witness}

For every $m\ge0$, set
\[
 \phi_m(t,\mathbf x)=
 \begin{cases}
 \sin(mt)/m,&m>0,\\
 t,&m=0.
 \end{cases}
\]
It is a free scalar and
\[
 T_0^{00}[\phi_m]=\tfrac12,\qquad
 T_0^{0i}[\phi_m]=0.
\]

\begin{lemma}[A falling switch region above the ramp]
\label{lem:late-switch}
There are $c>0$ and a nonempty interior open set
$O\subset\{t>\eps\}$ on which $\partial_tg\le-c$.
\end{lemma}

\begin{proof}
Choose $(t_*,\mathbf x_*)\in U$.
Then $t_*>\eps$ and $g(t_*,\mathbf x_*)=1$.
Compact interior support gives a time
$t_1\in(t_*,T)$ with $g(t_1,\mathbf x_*)=0$.
The mean value theorem yields a point between them where
$\partial_tg<0$. Smoothness gives the claimed open
neighborhood and a strictly positive $c$.
\end{proof}

\begin{theorem}[Empty fixed-input jet for the new action]
\label{thm:empty}
For every mass $m\ge0$ and every switch of the declared
plateau class,
\begin{equation}
 \boxed{\Jcal_{g,\sig}^{\mathrm D}(\phi_m)=\varnothing.}
 \label{eq:empty}
\end{equation}
The obstruction already occurs in the order-one compatibility
condition, independently of the allowed leading metric data.
\end{theorem}

\begin{proof}
For $\nu=0$, equation~\eqref{eq:late-compatibility} becomes
$\frac12\partial_tg=0$ above the ramp.
This contradicts Lemma~\ref{lem:late-switch}.

More explicitly, choose $f\ge0$ supported in $O$ with
$\int f\dd^4x=1$. Then
\[
 \int f\left[
 (\partial_\mu g)T_0^\mu{}_0[\phi_m]
                  -\Rampdef_{\sig,0}[\gamma_0]\right]\dd^4x
 \le-\frac c2<0,
\]
because the ramp defect is zero on $O$.
No leading vacuum metric configuration can alter this
pairing. Higher-order coefficients cannot cancel a nonzero
lower-order compatibility coefficient.
\end{proof}

This proof is new to the selected action.
```

```
The layer correction is present in
\eqref{eq:new-compatibility}; it simply cannot cancel the
late switch defect. The witness is in the switching region,
not an asserted nonzero plateau variation of an observable.
The plateau algebra has not been shrunk.

\begin{proposition}[Vacuum and the limits of emptiness]
The unrestricted jet family of the selected action is not
empty. The family with $\gamma=0$ and $\phi=0$ is a solution.
\end{proposition}

\begin{proof}
The flat Einstein tensor and scalar stress vanish.
Both $J_3$ and $\Dsource$ are quadratic in the metric field,
so their values at $\gamma=0$ vanish.
The scalar Euler expression also vanishes.
\end{proof}

This vacuum statement depends on the cubic-only freeze.
It would be false for the flat background of the rejected
full-density formula in Proposition~\ref{prop:fullweight}.

Other scalar inputs are not classified. The necessary
condition \eqref{eq:new-compatibility} is not promoted to a
general sufficiency theorem. Gauge-fixed solvability is a
different equation because $e_{\mathrm{gf}}$ changes the
divergence constraint. No gauge-fixed solution is constructed
here, and none is declared an Einstein-jet repair.

%=====================================================================
\section{C4: successor data without a response}
\label{sec:handoff}
%=====================================================================

The action supplies a local quadratic source and its polarized
form
\[
 \Jdensity(\gamma_1,\gamma_2)
 =
 \tfrac12\left[
 \Jdensity[\gamma_1+\gamma_2]
 -\Jdensity[\gamma_1]-\Jdensity[\gamma_2]\right].
\]
This definition applies equally to the displayed layer terms.
It contains no propagator.

\subsection{An honest classical source-pairing functional}

A successor may prescribe a compact root datum:
a continuous linear map
\[
 a_{f,g}:C^\infty(M)\longrightarrow
 C_c^\infty(M;\operatorname{Sym}^2 T^*M)
\]
whose range is supported in a fixed compact set
\[
 K_a\subset J^-(\supp f)\cap\{t\ge0\}.
\]
Here the range support and the map are \emph{given data}.
They are not claimed to arise from a retarded construction
already performed in this paper.

For such a datum, the classical trilinear source pairing is
\begin{equation}
 \mathcal I_{\mathrm D,a}
       (f,g;\gamma_1,\gamma_2,\phi)
 =
 \int a_{f,g}[\phi]_{\mu\nu}(y)
       \Jdensity^{\mu\nu}(\gamma_1,\gamma_2)(y)\dd^4y.
 \label{eq:source-pairing}
\end{equation}
It is well defined on smooth configurations: the test has
compact support and the source is a local smooth polynomial.

Equation~\eqref{eq:source-pairing} is an actual classical
functional with an independently supplied root datum.
It is not asserted to be the completed insertion of XI-E.
No claim about its distributional kernel or admission is made.

\subsection{The unconstructed completion slot}
```

```
The notation $\mathcal C_{3,\mathrm D}(f,g)$ is reserved for
the eventual scalar-completed insertion.
Its intended signature is the inherited completion
$\Aone[f,g](r;\phi)$ with a new metric-response slot $r$.
That slot has not been constructed.

A successor must determine the completed root datum associated
with both the coordinate and scalar-response legs, establish
its support, and then identify the resulting functional.
We do not turn this unconstructed slot into a response formula.

This is a necessary type distinction.
A source pairing is scalar valued; an intermediate compact
test is tensor valued; a metric response is a smooth metric
configuration. Naming one does not construct either of the
others. The action freeze alone cannot supply the compact
root test generated by an unperformed evolution.

\subsection{The inherited inverse is recorded, not applied}

The successor receives the normalization
\[
 \Ggr=-\PP\ER.
\]
No occurrence of this identity in the manuscript is an
application to $\Jdensity$, $\sig J_3$, or any other source.

The successor also receives XII-A's existing carrier and
the unchanged exclusion of the original detected unramped
classes. It does not receive membership of a new insertion
in that carrier.

%=====================================================================
\section{C5: non-identifications and final decision}
\label{sec:decision}
%=====================================================================

The selected source is not XI-F's $J_3$: it is
$\sig J_3+\Dsource$.
It is not the simply source-weighted expression $\sig J_3$.
It is not XII-A's excluded original $\Cthree$ and is not
identified with X-F's mixed map or VII-D's triangle.

No X-track normalization is introduced.
The scalar switch and coordinate ramp are unchanged; neither
has been adjusted to remove a defect.
The original XI-F ledger is not recomputed for this model.

\begin{center}
\small
\begin{tabularx}{\linewidth}{@{}p{.08\linewidth}
                              p{.24\linewidth}X@{}}
\toprule
Gate & Ending & Exact scope\\
\midrule
C0 & Constructed, with explicit correction &
One cubic-only density weighting; the full-density wording
is rejected because it changes the lower-order sheet.\\[0.4em]
C1 & Constructed &
A fixed density representative and its actual Euler source;
all first and second ramp derivatives retained.\\[0.4em]
C2 & Constructed &
Changed Noether identity, scalar-switch defect, layer defect,
and cubic-only bulk breaking all displayed.\\[0.4em]
C3 & Proved obstructed for the named fiber &
The new fixed-input jet is empty for every $\phi_m$ and
every declared switch; vacuum prevents universal emptiness.\\[0.4em]
C4 & Specified as successor data &
Local source pairing and typed completion slot;
the completed insertion and generated root are not constructed.\\[0.4em]
C5 & Recorded &
No old-class repair, X-track identification, admission,
retarded response, or sharp limit.\\
\bottomrule
\end{tabularx}
\end{center}

\begin{theorem}[Freeze and solution-fiber release]
The selected action is a coherent density-before-variation
cubic modification preserving the XI-A/XI-B free handoff.
Its actual source and changed Noether identity are those of
```

```
Theorems~\ref{thm:actualsource} and \ref{thm:noether}.
Its fixed-input family $\phi_m$ has no coupled invariant jet,
by Theorem~\ref{thm:empty}.

The freeze is Constructed.
If a single ending is assigned by the worst-gate convention,
the named fixed-input release is Proved obstructed at C3.
Neither statement is a kernel-admission or observable no-go.
\end{theorem}

\begin{proof}
The action and density are fixed by
\eqref{eq:selected-action} and \eqref{eq:ell3}.
The cubic modification preserves the free Hessian.
The cited source and identity theorems prove its variational
and Noether consistency, while the jet theorem supplies the
specified incompatibility.
No admission or response premise is used in these arguments.
\end{proof}

\subsection*{Exact handoff}

The next paper receives:
\begin{enumerate}
\item the explicit cubic-only action, including the unweighted
higher gravitational coefficients;
\item the density representative \eqref{eq:ell3};
\item the actual source
$\Jdensity=\sig J_3+\Dsource$;
\item the changed identity \eqref{eq:changed-noether};
\item the new jet verdict and its late-switch witness;
\item the unchanged XI-track inverse normalization;
\item XII-A's carrier and the unreopened old exclusion;
\item the classical source-pairing signature and the
unconstructed scalar-completion slot.
\end{enumerate}

It receives no completed-kernel theorem, wavefront estimate,
admission map for this source, response plateau cancellation,
or quantum retarded product.
\textbf{B1--B4 were not opened.}

\subsection*{Conclusion}

\[
 \boxed{
 \begin{gathered}
 S_{\mathrm{grav,D}}
 =S_{\mathrm{EH}}+\kappa\int(\sig-1)\ell_3,\qquad
 \Jdensity=\Euler_\gamma(\sig\ell_3)
                    =\sig J_3+\Dsource,\\[2pt]
 \eta_{\mu\nu}\Dsource^{\mu\nu}\big|_{\gamma=\psi\eta}
 =-6(\partial_\mu\sig)\psi\partial^\mu\psi,\\[2pt]
 \Jcal_{g,\sig}^{\mathrm D}(\phi_m)=\varnothing,
 \qquad
 \text{no metric Green operator has been applied}.
 \end{gathered}}
\]

The nonzero jet defect is not a failure of the polynomial
carrier. Retaining the layer source is not admission.
The changed action is not a repaired representative of the
old cubic class.

%=====================================================================
\section*{Declarations}
\addcontentsline{toc}{section}{Declarations}

\paragraph{Specification correction.}
This manuscript selects density weighting at cubic order only.
It explicitly amends the full-density formula in the production
brief because that formula is incompatible with the required
unchanged lower-order handoff. No claim is made that the two
actions are equivalent.

\paragraph{Classical scope.}
All new products and variations are local classical expressions
on smooth configurations. No new quantum product is defined.

\paragraph{Dependency scope.}
XII-B's weighted Euler distinction and pure-trace layer identity
```

```
are copied as inputs and used as checks. The new general tensor
source, changed identity, and coupled-jet verdict are proved here.
XI-A's original jet theorem is not used as the new jet verdict.

\paragraph{Boundary scope.}
The cubic representative is fixed before weighting.
All variational proofs use compact interior variations.
No boundary action, boundary charge, or additional variational
principle is constructed.

\paragraph{Unopened work.}
No metric retarded map, kernel admission, wavefront test,
response ledger, interacting differential, antibracket,
ghost loop, state, enlargement, or $\eps\to0$ limit is included.

\paragraph{Verification.}
No successful \texttt{pdflatex} execution, proof-assistant
verification, independently executed symbolic computation,
independent referee review, or journal acceptance is asserted.

%====================================================================
\begin{thebibliography}{99}
\addcontentsline{toc}{section}{References}

\bibitem{XIIA}
N.~Cheung,
\emph{Landmark XII-A: One graded polynomial carrier for the
finite-ramp multiplet, with ordering-stable composites and
a robust observation exclusion}.
Supplied manuscript \texttt{XII-A-v2.tex}.
Carrier, ordering transport, and original-class exclusion input.

\bibitem{XIIB}
N.~Cheung,
\emph{Landmark XII-B: Admission and one retarded response of the
ramped cubic Einstein source, at fixed regulator width:
handoff audit, variational consistency, and the unresolved release}.
Supplied manuscript \texttt{XII-B.tex}.
Weighted Euler identity, pure-trace layer term, and
off-shell integrability distinction.

\bibitem{XIA}
N.~Cheung,
\emph{Landmark XI-A: A switched Einstein--scalar action for the
ramp relational scalar: sign conventions, Noether identities,
and an empty fixed-input Einstein jet}.
Supplied manuscript \texttt{XI-A.tex}.
Original signed action, scalar switch, curvature convention,
and original jet quantifier.

\bibitem{XIB}
N.~Cheung,
\emph{Landmark XI-B: The free de Donder multiplet for the
ramp Einstein--scalar sheet: causal propagators, residual
gauge, and the layer dressing}.
Supplied manuscript \texttt{XI-B.tex}.
Free normalization, plateau class, and support distinctions.

\bibitem{XIE}
N.~Cheung,
\emph{Landmark XI-E: On-shell admission of the fixed-width tree,
a localized first-order identity, and the held-fixed defect}.
Supplied manuscript \texttt{XI-E.tex}.
Inherited scalar-completion signature only;
no new completion is attributed to that manuscript.

\bibitem{XIG}
N.~Cheung,
\emph{Landmark XI-G: The free-shell class of the cubic Einstein
kernel: the onset covector is inherited by every representative}.
Supplied manuscript \texttt{XI-G.tex}.
Closed original obstruction and non-reopening boundary.

\bibitem{SF}
T.~P.~Sotiriou and V.~Faraoni,
``$f(R)$ theories of gravity,''
\emph{Reviews of Modern Physics}
\textbf{82} (2010), 451--497,
arXiv:0805.1726.
Comparison for metric curvature variation and scalar-prefactor
derivative terms; the present coefficient calculation is explicit.
```

```
\bibitem{BS}
R.~Bluhm and A.~Sehic,
``Noether identities in gravity theories with nondynamical
backgrounds and explicit spacetime symmetry breaking,''
\emph{Physical Review D}
\textbf{94} (2016), 104034,
arXiv:1610.02892.
Comparison for fixed-background Noether defects;
the selected cubic-only identity is derived here.

\end{thebibliography}

\end{document}
```

3.5 Landmark XVI-M (full source)

Listing 59: XVI-M.tex

```
\documentclass{article}
\usepackage[utf8]{inputenc}
\usepackage{amsmath,amssymb,amsthm}
\usepackage[margin=1in]{geometry}
\usepackage{longtable}
\usepackage{listings}
\lstset{basicstyle=\ttfamily\scriptsize,breaklines=true,columns=fullflexible,keepspaces=true,literate={?}{{\textrm{\S}}}1}

\newtheorem{proposition}{Proposition}
\newtheorem{theorem}[proposition]{Theorem}
\newtheorem{lemma}[proposition]{Lemma}
\newtheorem{corollary}[proposition]{Corollary}
\newtheorem{definition}[proposition]{Definition}
\newtheorem{inputstatement}[proposition]{Input}
\newtheorem{gate}[proposition]{Gate}
\theoremstyle{remark}
\newtheorem{remark}[proposition]{Remark}

\newcommand{\scopebox}[2]{%
  \par\medskip\noindent
  \fbox{\begin{minipage}{0.92\textwidth}
  \textbf{#1}\par\smallskip #2
  \end{minipage}}
  \par\medskip
}
\newcommand{\RL}{\mathfrak R_{\mathrm L}}
\newcommand{\RK}{\mathfrak R_{\mathrm K}}
\newcommand{\RM}{\mathfrak R_{\mathrm M}}
\newcommand{\Pc}{\mathcal P}
\newcommand{\JD}{\mathcal J_{\mathrm D}}
\newcommand{\sigeps}{\sigma_\varepsilon}
\newcommand{\chiI}{\chi_{\mathrm I}}
\newcommand{\Ccoh}{\mathcal C_{\mathrm{coh}}}
\newcommand{\Creg}{\mathcal C_{\mathrm{reg}}}
\newcommand{\Tcol}{\mathcal T_{\mathrm{collision}}}
\newcommand{\Coll}{\mathrm{Collapse}}

\title{Landmark XVI-M\\
\large From coherent graph renormalization to local counterterm functionals and symmetry-compatible effective quantum gravity \\
\large Draft: integration of work packages A--C (final verification); Gate~S-bv checked; M-C$_{\mathrm{curv}}$ proved obstructed; broad M-C not achieved ($\varpi$ unspecified); M-C$_{\mathrm{data}}$ satisfied}
\author{}
\date{}

\begin{document}
\setlength{\emergencystretch}{3em}
\maketitle

\begin{abstract}
XVI-M begins the passage from the coherent graph renormalization of XVI-L (a frozen dependency) toward local counterterm functionals.
(M.1, unconditional) The complete-source two-edge coefficients are assembled with independent residual-metric jets.
The general table specializes entry by entry to the XVI-L background table (all $1065$ grouped entries), and the formal label support is all $255$ admissible labels ($235$ on the XVI-L background).
(M.2) Gate~S-bv is certified (Theorem~\ref{thm:wpA-Sbv}): on $\mathbb R^4\setminus\{0\}$ the predecessor edge kernel is $S=h_0+iE_0^{\mathrm A}=-W_0+iE_0^{\mathrm A}=\frac1{4\pi^2}(Q-i0)^{-1}$, including across the nonzero null cone, so $c=1/(4\pi^2)$ with boundary side $Q-i0$. This formula is already printed in XVI-J (eq:rational, with $S=-d$ by eq:DF);
```

```
      XVI-M supplies a written real-time check of it, including a local boundary-value argument at nonzero null points (Lemma ~\ref{lem:wpA-varreg}). The predecessor $S$ is globally defined (XVI-J); what is not established here is its position-space identification at the origin.
With a boundary-value calculus lemma this gives a two-direction equivalence between numerator identities and distributional relations, and hence, unconditionally on the punctured space, the graded presentation $d_N=(1,4,10,20,35)$, $\dim\mathcal V_{\mathrm{adm}}=70$, with $185$ certified relations (only $c\neq0$ and the single boundary side are used).
(M.3) The coherent contact space has dimension $3321$, against $14321$ raw slots.
The graph-contact map is injective in the formal universal-jet reading, while $1750$ coherent directions are invisible on the XVI-L background.
(M.4) Under a stated variational contract (Gate~V), the contact-collapse map is computed exactly.
It is injective in the formal reading, and its formally self-adjoint (variational) subspace is $26$-dimensional there and consists only of ultralocal potentials $\chiI\sigeps K(h,h)$; the formal universal-jet collapse image of this selected graph contains no nonzero variational tensor of total derivative degree $D\ge1$.
The same holds in the universal time-only reading $\sigeps=\sigma(t)$; if in addition $\chiI\equiv1$, there are $546$ further spatial-only, non-Lorentz variational directions.
In the double-plateau reading ($\sigeps\equiv1$, $\chiI\equiv1$) the variational image has dimension $1325$; it contains the linearized Euler tensors of $\int R^2$ and $\int R_{\mu\nu}R^{\mu\nu}$, with certified preimages, while the specified quadratic Einstein--Hilbert Euler tensor is not in that selected image.
These curvature-square matchings do not survive the time-only reading once $\sigeps'$, $\sigeps''$ are retained.
Stage~2: fixed-data criteria, universal region comparisons and the time-only classification are constructed; the transfer to the actual XII-C ramp on $(0,\varepsilon)\times\mathbb R^3$ is a proposition with explicit hypotheses; regions where the actual $\chiI$ is non-constant, and transitions, remain Unresolved.
M.5 records the named matching theorem \texttt{PlateauCurvatureSquareMatching} with explicit domain, order, pairing, and equivalence; ledgers~(3)--(4) remain Unresolved (EOM/redefs undefined).
(M.6) The $S^2$ scaling residue is $r=-2i\pi^2c^2=-i/(8\pi^2)$.
For the XVI-L prescription the scale derivative equals the dilation defect at every grade $N\le4$ ($A_\mu=\mathcal A_\mu$); top-order residues are $\varpi$-independent, while lower orders depend on the cutoff $\varpi$, which XVI-L does not fix.
In the double plateau, the cutoff-independent $D=4$ component is variational and Lorentz covariant, but not linearized-gauge invariant. The full scale tensor has cutoff-dependent lower-degree components and need not be variational. The $D=4$ component lies outside $\mathrm{span}(E_{R^2},E_{\mathrm{Ric}^2})$, certified by an exact separating functional ($\ell(v_{\mathrm{scale}})=-\tfrac1{16}$, $\ell(v_{R^2})=\ell(v_{\mathrm{Ric}^2})=0$). Hence the curvature-ledger scaling criterion M-C$_{\mathrm{curv}}$ is \emph{Proved obstructed} for this graph, while broad M-C (Definition~\ref{def:MCcurv}: matching plus an explicit cutoff and a functional representation of the entire collapsed scale variation) is \emph{not achieved} and remains Unresolved, because XVI-L leaves $\varpi$ unspecified (thm:L6); M-C$_{\mathrm{data}}$ is satisfied (not a claim about gravity generally).
M.7 (audited): with the contact distributions $\partial_\gamma\delta$ transported by the metric intertwiner, the Lorentz-equivariant contact assignments have dimension $14=5+9$; their exact intersection $V$ with the variational subspace has $\dim V=1$ (formal, $\chi$-plateau; $V\subset\Creg$) and $\dim V=12$ in the double plateau ($\dim V\cap\mathcal C_{\mathrm K}=3$, $\dim V\cap\Creg=7$); the curvature-square Euler tensors satisfy the linear divergence identity $\partial_\mu E^{\mu\nu}=0$ (this is not a Ward identity).
No Ward or BV statement is claimed; scale-variation coefficients are not physical beta functions.
\end{abstract}

\scopebox{Status (XVI-M, this draft).}{%
M.0--M.3: Constructed as stated (M.1 unconditional; M.2/M.3 use Input~(S-bv), now certified; Lemma~\ref{lem:bv}(c) by same-sign BV product).
Gate~S-bv: \textbf{certified} (Theorem~\ref{thm:wpA-Sbv}: $c=1/(4\pi^2)$, boundary side $Q-iO$, including across the nonzero null cone); a written check of XVI-J, eq:rational; the Euclidean motivation (Remark~\ref{rem:Sbv-motivation}) is not the certification; the predecessor $S$ is globally defined (XVI-J), and its position-space identification at the origin is not established here; not discharged by matching.
M.4: Constructed (Gate~V) in the formal, $\chi$-plateau, preparation-plateau, double-plateau and time-only readings (the formal, preparation-plateau and time-only readings are universal formal-jet readings). Stage~2: fixed-data criteria, universal region comparisons and the time-only classification constructed; actual-data transfer on $(0,\varepsilon)\times\mathbb R^3$ only as a proposition with explicit hypotheses; non-constant-$\chiI$ regions and transitions Unresolved.
M.5: Functional ledger with named theorem \texttt{PlateauCurvatureSquareMatching}; ledgers (1)--(2) filled; (3)--(4) Unresolved (EOM/redefs undefined).
M.6: dilation identity and $A_\mu=\mathcal A_\mu$ for all $N\le4$ Constructed; $S^2$ residue $r=-i/(8\pi^2)$ Constructed; top-order residues exact and $\varpi$-independent; lower-order values Unresolved ($\varpi$ not fixed by XVI-L). In the double plateau, the cutoff-independent $D=4$ component is variational and Lorentz covariant, but not linearized-gauge invariant. The full scale tensor has cutoff-dependent lower-degree components and need not be variational (Section~\ref{sec:wpB}).
M.7: \textbf{Constructed after audit} (Gate~V for the SA items; (S-bv) certified) for the exactly verified items: Lorentz action on $B_N$ and on $\partial_\gamma\delta$ with explicit $\eta$ intertwiner and boost check; equivariant dims $1,1,3,3,6$; exact $\mathrm{Equiv}\cap\mathrm{SA}$ ($1/1/12$) with projections and intersections; linear divergence identities. Obs$_L(R_L)$ Unresolved. See the audit log, Remark~\ref{rem:M7-audit}.
M.8--M.9: light stubs (Target II / BV Unresolved; M.8 should wait for XVI-N).
Release ladder: \textbf{M-A} Constructed (presentation unconditional on the punctured space); \textbf{M-B} Constructed in the formal reading (Gate~V convention); \textbf{M-C$_{\mathrm{curv}}$} (curvature-ledger scaling criterion, Definition~\ref{def:MCcurv}) Proved obstructed, graph-specific (Corollary~\ref{cor:wpB-MC}); broad \textbf{M-C} not achieved (Unresolved: $\varpi$ unspecified by XVI-L, thm:L6); \textbf{M-C$_{\mathrm{data}}$} Satisfied; \textbf{M-D/M-E} not claimed.
XVI-L is a frozen dependency.
No $\varepsilon\to0$, no loop-as-result, no UV claim, no interacting BRST/Ward identity as proved.
}

\scopebox{Programme arrow (domains/images/kernels/intersections/compatibility).}{%
\[
\boxed{\begin{aligned}
&\text{coherent numerical contact assignments}\\
&\quad\longrightarrow
\text{unrooted local collision tensors}\\
```

```
&\quad\longleftarrow
\text{variations of local counterterm functionals}\\
&\quad\longleftarrow
\text{symmetry-compatible EFT counterterms}.
\end{aligned}}
\]
The arrows need not be surjective or injective.
This draft computes the first arrow (M.3) and the intersection of its collapsed image with the variational tensors (M.4).
The symmetry arrow is reserved for M.7--M.9.
}

\noindent
Conventions: ambient $M=\mathbb R^4$, $\eta=\mathrm{diag}(1,-1,-1,-1)$, $\Pc r=r-\tfrac12\eta\,\mathrm{tr}_\eta r$, Gram $G=\mathrm{diag}(1,-2,-2,-2,$ $1,2,2,1,2,1)$, dual basis $B^i=(G^{-1})^{ii}B_i$; Hessian on field directions $B_i$; propagator/source contraction on dual slots $-\Pc B^i$.
The XI inverse $G_{\mathrm{gr}}^{\mathrm R}=-\Pc E_0^{\mathrm R}$ is not equated with the X-sheet $\Delta_R^{\mathrm{gr}}=-4\Pc E_0^{\mathrm R}$.
Series voice: Constructed / Proved obstructed / Unresolved / Non-claims.
Predecessor quotations are exact.

%=====================================================================
\section{M.0 Dependency, normalization, gates, and target ledger}
\label{sec:m0}

\subsection{Frozen XVI-L dependency}

XVI-L.tex and \texttt{XVI-L-artifacts/} are a frozen dependency: they are cited and read, not edited.
XVI-K-artifacts/\texttt{hessian\_full\_out.txt} is likewise frozen.
The following items are imported \emph{as stated in XVI-L}.
\begin{enumerate}
\item Printed cubic density $\ell_3$ (XVI-L, eq:L-ell3; XII-C, eq:ell3).
\item Complete weighted source $\JD=\sigeps J_3+2C_2\partial\partial\sigeps-2M_2\partial\sigeps$ (XVI-L, eq:L-JD).
\item Full cubic Hessian $L^{pq}_{ij}$ (XVI-K, prop:hess, eq:Lpq).
\item Metric-dual contraction on $-\Pc B^i$.
\item Ambient numerical span $\mathcal V_{\mathrm{adm}}$, $|\Lambda_{\mathrm{adm}}|=255$ (XVI-L, def:VL).
\item Selected extension $\RL$ with $\RL|_{\mathcal V_{\mathrm K}}=\RK$ (XVI-L, def:RL, prop:RL).
\item Coherent contact freedom (XVI-L, thm:L5); raw counts $14321/321/14000$ ``are not dimensions''.
\item Subtraction-scale family $\mathfrak R_{\mathrm L,\mu}$ and generator $A_\mu$ (XVI-L, thm:L6); explicit nonzero scaling residue Unresolved there.
\item Conditional root attachment under (R1)--(R3) (XVI-L, thm:L7).
\end{enumerate}

\paragraph{Preservation versus new results.}
XVI-M does not alter any release claim of XVI-L: L-A/L-C remain as constructed there, L-B remains as stated there, and L-D remains not claimed.
The new statements below are results \emph{about the objects of XVI-L} (its admissible span, its contact freedom, its graph coefficients).
They strengthen knowledge of the span without retroactively rewriting XVI-L.
In particular, the graded presentation of M.2 is a new result resting on an explicitly checked kernel identification (Gate~S-bv, now certified by Theorem~\ref{thm:wpA-Sbv}); it is not a correction of XVI-L, Theorem thm:L5.

\subsection{Two separate targets}

\begin{definition}[Target I and Target II]
\label{def:targets}
\textbf{Target I (unrooted counterterm matching).} Determine which coherent contact assignments of $\RL$ produce collision tensors that arise as variations of local metric or insertion functionals, without using the root hypotheses (R1)--(R3).
\textbf{Target II (gravitational perturbative-sector interpretation).} Interpret those matched functionals inside a specified gravitational action, insertion, gauge choice, ghost content, and diagram inventory.
\end{definition}
The M.4 results serve Target I.
Target II begins only in skeleton Section~\ref{sec:m8}.

\begin{definition}[Scaling criteria M-C, M-C$_{\mathrm{data}}$ and M-C$_{\mathrm{curv}}$ (final verification)]
\label{def:MCcurv}
\textbf{M-C (broad)} is the release-ladder \emph{matched-scale} criterion: it requires
(i)~at least one proved functional matching (Target~I), and
(ii)~an explicitly specified cutoff $\varpi$ together with a functional representation of the \emph{entire} collapsed scale variation in a stated reading.
\textbf{M-C$_{\mathrm{data}}$} is the weaker \emph{data} criterion: at least one proved functional matching together with explicitly specified, prescription-dependent scaling formulas (residues, $\nu_k[\varpi]$, top-order coefficients) for the collapsed scale variation; it does not require a fixed numerical cutoff.
\textbf{M-C$_{\mathrm{curv}}$} is the \emph{curvature-ledger scaling criterion}: the scale variation of the graph is matched inside the span of the M.5 curvature-ledger targets (the Euler tensors $E_{R^2},E_{\mathrm{Ric}^2}$ at $D=4$, and the lower-degree curvature targets of the ledger), with the matched-scale data of (ii).
M-C$_{\mathrm{curv}}$ implies broad M-C, not conversely; M-C$_{\mathrm{data}}$ is satisfied in this draft (Theorem~\ref{thm:wpB-B0}, Theorem~\ref{thm:wpB-dp}, Section~\ref{sec:wpB}), while broad M-C remains Unresolved because XVI-L leaves $\varpi$ unspecified (XVI-L, thm:L6; Remark rem:Lressretch).
Enlarging the target ledger by non-gauge-invariant quadratic representatives (for example $I^{\mathrm{top}}_{\mathrm{scale}}$ of Theorem~\ref{thm:wpB-dp}) changes the matching domain of M-C$_{\mathrm{curv}}$ but may remain within broad Target~I; gauge-fixing/ghost or further graph contributions are a different, independent enlargement and are not the only one.
```

```
\end{definition}

\subsection{Normalization table}
\label{sec:m0norm}

XVI-L determines the combinatorial graph factor $(1/2!)\cdot2=1$ and leaves coupling and loop weights undetermined (XVI-L, prop:Lnorm).
No number is invented below.

\begin{center}
\begin{longtable}{p{0.20\textwidth}p{0.44\textwidth}p{0.26\textwidth}}
\hline
Item & Value / source & Status\\
\hline
Metric fluctuation & $q_{\mu\nu}=\eta_{\mu\nu}+\kappa\gamma_{\mu\nu}$ (XII-C) & Imported\\
Residual $h$ vs $\gamma$ & Dummy of the same species; no rescaling (XVI-I, prop:Ft) & Imported\\
$\ell_3$ vs action & $\ell_3$ is the coefficient of $\kappa$ in the unintegrated signed EH density; $S_{\mathrm{EH}}=-\frac2{\kappa^2}\int\sqrt{-q}\,R$; $S_{\mathrm{grav,D}}=S_{\mathrm{EH}}+\kappa\int(\sigma-1)\ell_3$ (XII-C) & Imported\\
Graph vertex & $V(h)=\int\chiI\ell_3(h)$, $\chiI$ independent of $\sigeps,f,g$ (XVI-I, prop:second) & Imported; relation of $\chiI$ to $\kappa\int\sigma\ell_3$ Unresolved\\
Kinetic operator & $K=-\square\Pc=A+2D^*D$, $A=2G_{\mathrm{lin}}$ (XII-C; XVI-I) & Imported\\
Propagator & $D_F=H+iG^{\mathrm A}=-\Pc(h_0+iE_0^{\mathrm A})$, $KD_F=i\,\mathrm{id}$ (XVI-J, eq:DF, quoted in XVI-K, eq:J-DF); ``with $h_0=-W_0$ (XI-E.tex, eq:h)'' (XVI-K, K.0); $S=h_0+iE_0^{\mathrm A}=-\Delta_F^{\mathrm{FR}}$ (XVI-J, lem:identify) & Imported\\
Position-space form of $S$ on $\mathbb R^4\setminus\{0\}$ & $S=c\,(Q-i0)^{-1}$ with $c=\frac1{4\pi^2}$, boundary side $Q-i0$ (Theorem~\ref{thm:wpA-Sbv}); $W_0=-\frac1{4\pi^2}(Q-i0\,\mathrm{sgn}\,z^0)^{-1}$, $E_0^{\mathrm A}=\frac1{2\pi}\theta(-z^0)\delta(Q)$ & Certified (Gate~S-bv; written check of XVI-J, eq:rational); Euclidean motivation is not the certification; $S$ is global (XVI-J), its position-space identification at $z=0$ is not established\\
$S^2$ residue and scale units & $r=-2i\pi^2c^2=-\frac{i}{8\pi^2}$ (Theorem~\ref{thm:M6-S2}); top-order scale residues in units $i\pi^2c^2=\frac{i}{16\pi^2}$ (Section~\ref{sec:wpB}) & Constructed; not a beta function\\
Powers of $\kappa$ & Action cubic carries $\kappa$; $a_\lambda$ and the graph factor carry none & Overall order Unresolved\\
Powers of $\hbar$ & ``No power of $\hbar$ is assigned'' (XVI-I, prop:sector) & Unresolved\\
Factors of $i$ & $i$ in $D_F$ fixed by $KD_F=i\,\mathrm{id}$; vertex factors not assigned & Partial; remainder Unresolved\\
Insertion & $F_t(h,\phi)=\langle t_{\mathrm D},h\otimes h\otimes\phi\rangle$; $D_h^2F_t[a,c]=2\langle t_{\mathrm D},a\otimes c\otimes\phi\rangle$ (XVI-I) & Imported\\
Loop order & ``A count of one cycle is not a loop order of an interacting observable'' (XVI-I) & Unresolved\\
Variational contract & Gate~V (Section~\ref{sec:gates}) & Chosen Phase II convention; not certified by this table\\
\hline
\end{longtable}
\end{center}

\subsection{Gates}
\label{sec:gates}

\begin{gate}[Gate S-bv]
\label{gate:Sbv}
Input~(S-bv) (Input~\ref{inp:Sbv}) was introduced as an explicit input of XVI-M.
Certification had to establish the representation for the \emph{actual predecessor kernel} $S=h_0+iE_0^{\mathrm A}$ on the punctured space $\Omega$, including across the nonzero null cone---not only for a model kernel with the same wave equation, and not only off the cone.
(The alternative route, a direct certification of the relations of M.2 and of their completeness, is no longer needed.)
Earlier drafts proceeded conditionally on (S-bv); no downstream matching result was used to remove that condition.
The unconditional general-$h$ assembly (M.1) and the wave-operator bound (Proposition~\ref{prop:wave}) are stated separately and never depended on this gate.
\textbf{Gate~S-bv certified (Theorem~\ref{thm:wpA-Sbv}):} $S=\frac1{4\pi^2}(Q-i0)^{-1}$ in $\mathcal D'(\Omega)$, i.e.\ $c=1/(4\pi^2)$ with boundary side $Q-i0$ (Section~\ref{sec:wpA-Sbv}, Remark~\ref{rem:Sbv-status}).
\end{gate}

\begin{gate}[Gate V: variational contract for Phase II]
\label{gate:V}
\emph{Chosen Phase II convention; it is not certified by the normalization ledger.}
\begin{enumerate}
\item Field variable: the symmetric lowered residual field $h_{\mu\nu}$, with no rescaling and no trace reversal.
\item Variation: $h\mapsto h+t$ with $t$ symmetric and compactly supported.
\item Pairing: $\delta I[h](t)=\int d^4y\,t_{\mu\nu}E_I^{\mu\nu}[h]$ with the full symmetric-index contraction over all $16$ pairs $(\mu,\nu)$.
In the ten-component coordinates $h_{\mu\nu}$, $\mu\le\nu$:
\[
t_{\mu\nu}E^{\mu\nu}=\sum_{\mu\le\nu}w_{\mu\nu}\,t_{\mu\nu}E^{\mu\nu},\qquad w_{\mu\mu}=1,\quad w_{\mu\nu}=2\ (\mu<\nu),
\]
so the Euler tensor of a density $\mathcal L$ in the independent components is
\[
E_I^{\mu\nu}=\frac1{w_{\mu\nu}}\sum_\beta(-D)^\beta\frac{\partial\mathcal L}{\partial(\partial^\beta h_{\mu\nu})}.
\]
\item External data: $\sigeps$ and $\chiI$ are fixed under field variation, but their derivatives are kept in every integration by parts.
\item Order: $h$-linear collapsed tensors; quadratic functional ansatz modulo field-independent terms.
\item Helmholtz test: $\int r\cdot(DE_C[h]\cdot t)=\int t\cdot(DE_C[h]\cdot r)$; for homogeneous linear $E_C$ this is $\int\langle r,E_C[t]\rangle=\int\langle t,E_C[r]\rangle$.
\end{enumerate}
\end{gate}
```

```
\subsection{Verification boundary}
\label{sec:vb}

The following are \emph{reported computational results}, with exact certificates in the archive (Appendix~\ref{app:script}):
\begin{itemize}
\item the $1065$-entry specialization of M.1 and the support counts;
\item the numerator ranks $d_N$, reconstruction certificates and relation vectors of M.2;
\item the general-$h$ ranks $\dim U_N$ and the background/plateau kernels of M.3;
\item the collapse ranks, self-adjoint subspaces, Helmholtz certificates, and matching preimages of M.4;
\item the symbolic/numerical cross-checks of work package A (Lemma~\ref{lem:wpA-comp}); the top-order residues, coefficient files and scale-tensor collapse certificates of work package B; the time-only kernels, route~1/route~2 agreement and matching probes of work package C (Lemma~\ref{lem:to-comp});
\item the audit-3 checks: the separating functional on the archived target vectors, its pure-gauge sympy verification and the recollapse of $C_{\mathrm{top}}$ (Proposition~\ref{prop:wpB-sep}); the numerical variable-regulator pairing (Lemma~\ref{lem:wpA-comp}); the Beta-integral convolution of Lemma~\ref{lem:bv}(c).
\end{itemize}
The analytic statements (Lemma~\ref{lem:bv}, Lemma~\ref{lem:wpA-varreg}, Theorem~\ref{thm:wpA-Sbv}, Proposition~\ref{prop:wpB-sep}, Propositions~\ref{prop:relations}, \ref{prop:Gsplit}, \ref{prop:collinj}, Lemma~\ref{lem:helm}, Lemma~\ref{lem:wpB-dw}, Theorem~\ref{thm:wpB-BO}, Lemmas~\ref{lem:to-intertwine} and~\ref{lem:to-indep}, Corollaries~\ref{cor:wpB-timeonly} and~\ref{cor:wpB-MC}) are written proofs.
The outputs are reproduced verbatim in Appendix~\ref{app:out}.

\subsection{Release ladder}
\label{sec:ladder}

\begin{center}
\begin{longtable}{p{0.08\textwidth}p{0.32\textwidth}p{0.50\textwidth}}
\hline
Level & Target claim & Status in this draft\\
\hline
M-A & General-$h$ coefficients and an explicit presentation of the admissible span & \textbf{Constructed} (M.1 unconditional; the presentation depended only on (S-bv) and is now unconditional on the punctured space, Theorem~\ref{thm:wpA-Sbv})\\
M-B & Contact freedom $\to$ local collision tensors, with variational-integrability conditions & \textbf{Constructed in the formal reading} (M.3, M.4) under the Gate~V convention; no longer conditional on (S-bv). Time-only classification constructed (Section~\ref{sec:to-timeonly}); actual fixed-data dimensions unresolved (Stage~2)\\
M-C & Matched-scale criterion (Definition~\ref{def:MCcurv}): (i)~proved functional matching; (ii)~explicit $\varpi$ and a functional representation of the entire collapsed scale variation & \textbf{Not achieved (Unresolved).} Criterion~(i) holds (e.g.\ $I^{\mathrm{top}}_{\mathrm{scale}}$, Theorem~\ref{thm:wpB-dp}); criterion~(ii) fails because XVI-L leaves $\varpi$ unspecified (thm:L6). Their numerical values and the variationality of the resulting full tensor are not determined for the unspecified cutoff. \textbf{M-C$_{\mathrm{data}}$} (matching + explicit prescription-dependent scaling formulas) is \textbf{Satisfied}.\newline \textbf{M-C$_{\mathrm{curv}}$} (curvature-ledger scaling criterion, Definition~\ref{def:MCcurv}): \textbf{Proved obstructed (graph-specific).} For the specified two-edge graph, the XVI-L prescription and any admissible $\varpi$, the full collapsed scale variation does not lie in the span of the M.5 curvature targets, because its fixed $D=4$ component already lies outside (Theorem~\ref{thm:wpB-dp}, Proposition~\ref{prop:wpB-sep}, Corollary~\ref{cor:wpB-MC}); not a claim about gravity generally\\
M-D & Ward/BV compatibility for a diagrammatically closed sector & \textbf{Not claimed}\\
M-E & Full renormalization for a specified EFT truncation and order & \textbf{Not claimed}\\
\hline
\end{longtable}
\end{center}

%=====================================================================
\section{M.1 General-$h$ coefficient assembly}
\label{sec:m1}

XVI-L assembles all-sector coefficients on the background $h_{\mu\nu}=(w^1)^2\eta_{\mu\nu}$ (XVI-L, lem:L2comp: ``NOT a general-$h$ symbolic assembly'').
This section retains the independent residual-metric jets of $L^{pq}_{ij}$.

\begin{definition}[General-$h$ graph coefficients]
\label{def:aGeneral}
With $T_i^r:=(-\Pc B^i)^r$,
\[
\widetilde P_{A,ij}^{ab,\mu\nu}=\sum_{r,s}P_{A,rs}^{ab,\mu\nu}T_i^rT_j^s,\qquad
\hat a^{A\,,\mu\nu}_{ab,pq}=(-1)^{|p|+|q|}\sum_{i,j}L^{pq}_{ij}(w;h)\widetilde P_{A,ij}^{ab,\mu\nu}(y),
\]
and $\hat a^{A\,,\mu\nu}_\lambda$ is the accumulation over $\{a+p,b+q\}=\lambda$, exactly as in XVI-L, def:La / eq:L-apq, \emph{with $\chiI$ suppressed}.
The physical (switched) coefficient---XVI-L style---is the single product
\[
a^{A\,,\mu\nu}_\lambda=\chiI(w)\,\hat a^{A\,,\mu\nu}_\lambda.
\]
The hat marks the switch-free factor: $a_\lambda$ and $\hat a_\lambda$ are not the same object.
Here $L^{pq}_{ij}$ is kept as the symbolic linear forms of XVI-K in independent $h$-jet symbols $h_{ab}$, $\partial_ch_{ab}$, $\partial_c\partial_dh_{ab}$.
The recorded table \texttt{graph\_general\_table.txt} stores the switch-free $\hat a$ ($\chiI$ suppressed), as in XVI-L.
\end{definition}

\begin{proposition}[General-field coefficient construction and grading]
\label{prop:M1}
```

```
For every assembled switch-free coefficient $\hat a^{A\,\mu\nu}_\lambda(y,w;h)$:
\begin{enumerate}
\item it is linear in the residual-metric jet of order $2-v$, where $v=|p|+|q|$;
\item it is linear in the preparation jet of order $2-s$, where $s=|a|+|b|$ ($s=0,1,2$ for $C_2,M_2,J_3$);
\item $N=s+v$, and every monomial is one preparation-jet symbol times one $h$-jet symbol, with a rational coefficient;
\item it is symmetric in $(\mu,\nu)$;
\item it is consistent under edge interchange, $(A;a,b;p,q)\leftrightarrow(A;b,a;q,p)$;
\item no unintended generators occur, and the arithmetic is exact rational.
\end{enumerate}
The supports satisfy $\Lambda_{\mathrm{graph}}^{\mathrm{background}}\subseteq\Lambda_{\mathrm{graph}}^{\mathrm{formal}}\
     subseteq\Lambda_{\mathrm{adm}}$ with cardinalities $235\le255\le255$ (sectorwise $C_2$ $25/25/25$, $M_2$ $84/84/84$, $J
     _3$ $202/250/250$).
\end{proposition}

\begin{proof}
(1)--(3) and (6) are inherited from XVI-K, prop:hess, and XVI-L, prop:jet, through Definition~\ref{def:aGeneral}.
(4) holds because $P_A$ is symmetric in $(\mu,\nu)$.
(5) holds because the defining sum is symmetric under simultaneous interchange of the two edges.
The supports, and a coefficientwise confirmation of (1)--(6), are the reported computation of Lemma~\ref{lem:M1comp}.
\end{proof}

\begin{theorem}[Exact specialization to the XVI-L table]
\label{thm:M1spec}
After the specialization $h_{\mu\nu}=(w^1)^2\eta_{\mu\nu}$ (with $\chiI$ suppressed as in XVI-L), the general-$h$ table equals
      \texttt{two\_edge\_coeff\_table.txt} entry by entry.
All $1065$ nonzero grouped coefficients $(A,\lambda,(\mu,\nu))$ agree as polynomials in the preparation jet and $w^1$, and the
      nonzero key sets coincide.
\end{theorem}

\begin{lemma}[Reported computation for M.1]
\label{lem:M1comp}
\texttt{graph\_general.py} loads $P_A$ from the frozen XVI-L source table and $L^{pq}_{ij}$ from the frozen XVI-K Hessian list
     .
It assembles the switch-free $\hat a$ of Definition~\ref{def:aGeneral} in exact $\mathbb Q$ arithmetic and exits with status
     $0$, recording the following.
\begin{enumerate}
\item[(G1)] Proposition~\ref{prop:M1}(1)--(6). The edge-interchange check is \emph{exhaustive}: it runs over every enumerated
     ordered placement whose swap is distinct, which gives $1000$ unordered pairs.
\item[(G2)] The supports of Proposition~\ref{prop:M1}. Twenty labels lie in $\Lambda^{\mathrm{formal}}_{\mathrm{graph}}\
     setminus\Lambda^{\mathrm{background}}_{\mathrm{graph}}$.
\item[(G3)] Theorem~\ref{thm:M1spec} ($1065/1065$ entries).
\item[(G4)] An independent $C_2$ route via \texttt{xvi\_common.polar\_s} ($250$ comparisons).
\item[(G5)] $3282$ nonzero $(A,\lambda,(\mu,\nu))$ lines. The sector-summed table has $2322$ entries with $18646$ bilinear
     terms.
\end{enumerate}
\end{lemma}

%=====================================================================
\section{M.2 Numerical-kernel relations and Gate S-bv}
\label{sec:m2}

XVI-L does not compute the full relation space of its $255$ admissible labels (XVI-L, thm:L5).
This section constructs the presentation $\pi:\mathbb C^{\Lambda_{\mathrm{adm}}}\to\mathcal V_{\mathrm{adm}}$, $e_\lambda\
     mapsto q^\circ_\lambda$, degree by degree.

\begin{inputstatement}[Input (S-bv); certified by Theorem~\ref{thm:wpA-Sbv}]
\label{inp:Sbv}
On $\Omega:=\mathbb R^4\setminus\{0\}$,
\[
S=c\,(Q-i0)^{-1},\qquad Q(z)=\eta(z,z)=z_0^2-z_1^2-z_2^2-z_3^2,\qquad c\neq0,
\]
where $S=h_0+iE_0^{\mathrm A}$ is the predecessor edge kernel (XVI-K, eq:S) and $(Q-i0)^{-1}$ is the boundary-value
     distribution $T_1$ of Lemma~\ref{lem:bv}.
In its original formulation the constant $c$ was left unspecified; the graded results of M.2--M.4 and M.7 use only $c\neq0$
     and the side $Q-i0$.
Theorem~\ref{thm:wpA-Sbv} certifies the input with $c=1/(4\pi^2)$.
\end{inputstatement}

\begin{remark}[Motivation for Input (S-bv)]
\label{rem:Sbv-motivation}
XVI-K quotes $D_F=-\Pc(h_0+iE_0^{\mathrm A})$ and states ``with $h_0=-W_0$ (XI-E.tex, eq:h)''.
XI-E, eq:h, reads $h_\mu=-W_\mu$, $h_\mu-h_\mu^{\mathsf T}=i\delta_\mu$.
Hence $S=h_0+iE_0^{\mathrm A}=-W_0+iE_0^{\mathrm A}$, and XVI-J, lem:identify, identifies this with $-\Delta_F^{\mathrm{FR}}$.
A Euclidean calculation motivates the candidate form.
Substituting $x^0-y^0=-i\tau_{\mathrm E}$ ($\tau_{\mathrm E}>0$) into the $W_0$ integral and using $\int_0^\infty e^{-p\tau_{\
     mathrm E}}\sin(pr)\,dp=r/(\tau_{\mathrm E}^2+r^2)$ (checked symbolically in \texttt{kernel\_relations.py}) gives $1/(4\
     pi^2(\tau_{\mathrm E}^2+r^2))$, which suggests the motivational value $c=1/(4\pi^2)$.
\emph{This Euclidean calculation motivates the candidate; it does not certify the real-time identification of $S$ on $\Omega$
     as a single boundary value.}
The certification is the real-time argument of Theorem~\ref{thm:wpA-Sbv}, which uses the predecessor Fourier family of $W_0$
     and the step representation of $E_0^{\mathrm A}$, not this Euclidean slice.
```

```
XVI-K states: ‘‘No explicit position-space formula for \(S\) is used.’’
\textbf{Boundary value versus $\varepsilon$.} The symbol $i0$ denotes a distributional boundary value.
It is unrelated to the fixed preparation parameter $\varepsilon$.
The series rule ‘‘no $\varepsilon\to0$’’ (XVI-I, prop:signs(5): ‘‘There is no \(i\varepsilon\) prescription and no limit \(\varepsilon\to0\)’’) does not forbid the $i0$ notation, and Input~(S-bv) introduces no prescription and changes no definition.
\end{remark}

\begin{remark}[Status of Gate S-bv]
\label{rem:Sbv-status}
\emph{What predecessors supply.} XVI-K, eq:S, defines $S:=h_0+iE_0^{\mathrm A}=-W_0+iE_0^{\mathrm A}=-\Delta_F^{\mathrm{FR}}$ through two globally defined distributions, with Fourier $z^0$-family formulae (XVI-K, lem:families / eq:W-fam, eq:E-fam) quoted from XVI-J and XI-E, eq:h ($h_0=-W_0$).
Homogeneity of degree $-2$ is proved (XVI-K, lem:homog); $\square S=0$ on $\Omega$ (XVI-K, lem:grading(3)); the H\"ormander product of two copies exists on $y\neq w$ (XVI-J / XVI-K, lem:prod).
XVI-K states explicitly: ‘‘No explicit position-space formula for \(S\) is used.’’
XI-E, eq:W and eq:h, likewise give the Fourier definition of $W_\mu$ and the normalization $h_\mu=-W_\mu$, with no $(Q-i0)^{-1}$ formula.
XVI-J, however, already prints the position-space formula: eq:DF sets $S=h_0+iE_0^{\mathrm A}=-d$ with $d=W_0-iE_0^{\mathrm A}$, and eq:rational states $d(z)=-\frac1{4\pi^2}(Q_{\mathrm{geom}}(z)-i0)^{-1}$ on $z\neq0$, including nonzero null points; XVI-J, prop:q, uses it (quotations in Section~\ref{sec:wpA-identify}).
Its printed justification is ‘‘direct Fourier evaluation’’.

\emph{What was missing.} A written real-time derivation of XVI-J, eq:rational, including the boundary-value argument at nonzero null points. The Euclidean Wick calculation of Remark~\ref{rem:Sbv-motivation} motivates the candidate $c=1/(4\pi^2)$; it is \emph{not} a real-time identification of $S$ with $c\,T_1$ on $\Omega$, and it does not settle equality across the nonzero null cone.
Lemma~\ref{lem:bv} establishes the model calculus for $T_m$; it does not identify $S$ with $c\,T_1$.
No matching probe, collapse rank, or Helmholtz certificate discharges (S-bv).

\emph{Certification (this cycle): a written check of XVI-J, eq:rational, not a new extension.} Theorem~\ref{thm:wpA-Sbv} establishes, from the predecessor Fourier families, all four required items on $\Omega$: (i) the distributional representation $S=c\,(Q-i0)^{-1}$; (ii) the boundary-value side, the same as for $T_1$ (namely $Q-i0$, not $Q+i0$); (iii) the normalization $c=1/(4\pi^2)\neq0$; (iv) equality across the nonzero null cone, not merely on $\{Q\neq0\}$.
The alternative route (certifying the $185$ relations and their completeness directly) is therefore not needed.
Consequently every graded M.2--M.4 (and M.7) conclusion that uses the presentation of Theorem~\ref{thm:M2} holds unconditionally on $\Omega$; only $c\neq0$ and the single side were ever used.
Statements that also use Gate~V remain relative to that chosen convention.
\end{remark}

\begin{lemma}[Boundary-value calculus on $\Omega$]
\label{lem:bv}
This lemma concerns the model family $T_m$ itself; it does not yet identify $T_m$ with derivatives of the predecessor kernel $S$.
For $m\ge1$ let $T_m:=(Q-i0)^{-m}:=\lim_{\tau\downarrow0}(Q-i\tau)^{-m}$ in $\mathcal D'(\Omega)$.
The limit exists, and:
\begin{enumerate}
\item[(a)] $Q\,T_{m+1}=T_m$;
\item[(b)] $\partial_\rho T_m=-m\,(\partial_\rho Q)\,T_{m+1}$;
\item[(c)] $T_mT_n=T_{m+n}$, where the product is the existing H\"ormander product on $\Omega$ (wavefront criterion).
\end{enumerate}
\end{lemma}

\begin{proof}
\emph{Points with $Q(z_0)\neq0$.} Near such a point $Q\neq0$ and $(Q-i\tau)^{-m}\to Q^{-m}$ in $C^\infty$, so $T_m=Q^{-m}$ there and (a)--(c) are identities of smooth functions.

\emph{Nonzero null points.} Let $Q(z_0)=0$ with $z_0\ne0$.
Then $dQ(z_0)=2\eta(z_0,\cdot)\neq0$, because $\eta$ is nondegenerate and $z_0\neq0$.
Hence $Q$ is a local coordinate: there is a neighbourhood $U$ and a diffeomorphism $\Phi:U\to(-\delta,\delta)\times V'$ with $\Phi_1=Q$.
On $(-\delta,\delta)$, $(s-i\tau)^{-m}\to(s-i0)^{-m}$ in $\mathcal D'$.
This is the boundary value from $\{\mathrm{Im}\,\zeta<0\}$ of the holomorphic function $\zeta^{-m}$, and $(s-i0)^{-m}=\frac{(-1)^{m-1}}{(m-1)!}\partial_s^{m-1}(s-i0)^{-1}$.
Pullback by the submersion $\Phi$ (H\"ormander, Thm.~6.1.2) is continuous, so $T_m|_U=\Phi^*((s-i0)^{-m}\otimes1)$ and the limit exists on $U$.

Its Fourier transform satisfies $\widehat{(s-i0)^{-m}}(\sigma)=\frac{2\pi i\,(-i\sigma)^{m-1}}{(m-1)!}\,\theta(-\sigma)$ for $\hat u(\sigma)=\int e^{-is\sigma}u(s)\,ds$.
This transform is supported in the single half-line $\sigma\le0$, the same for every $m$.
Hence $\mathrm{WF}((s-i0)^{-m})\subset\{(0,\sigma):\sigma<0\}$, and by the pullback theorem (H\"ormander, Thm.~8.2.4)
\[
\mathrm{WF}(T_m)\subset\Gamma_-:=\{(z,\sigma\,dQ(z)):Q(z)=0,\ z\ne0,\ \sigma<0\},
\]
with one sign for all $m$.

\emph{Compatibility.} All $T_m$ are defined by the same limit $\tau\downarrow0$ from the same side, globally on $\Omega$.
The local descriptions therefore agree on overlaps, and with $Q^{-m}$ off the cone.

(a) $Q(Q-i\tau)^{-m-1}=(Q-i\tau)^{-m}+i\tau(Q-i\tau)^{-m-1}$.
```

In the local coordinate, $\tau(s-i\tau)^{-m-1}$ is a constant times $\partial_s^{m}\bigl(\tau(s-i\tau)^{-1}\bigr)$, and $\tau(s-i\tau)^{-1}=\partial_s(\tau\log(s-i\tau))$ with $\tau\log(s-i\tau)\to0$ in L^1_{loc}; so the remainder tends to 0 in $\mathcal D'$.

(b) Derivatives commute with limits in $\mathcal D'$, and $\partial_\rho(Q-i\tau)^{-m}=-m(\partial_\rho Q)(Q-i\tau)^{-m-1}$.

(c) \emph{Same-sign boundary-value product.} Since $\mathrm{WF}(T_m),\mathrm{WF}(T_n)\subset\Gamma_-$ and Γ_- contains no opposite covectors $(z,\xi),(z,-\xi)$, H\"ormander's criterion (Thm.~8.2.10) defines the product T_mT_n in $\mathcal D'(\Omega)$.
It remains to identify that product with T_{m+n}.

All T_m are boundary values from the \emph{same} lower half-plane.
Locally near a nonzero null point, in the coordinate $s=Q$ of the first paragraph,
\[
T_m|_U=\Phi^*\bigl((s-i0)^{-m}\otimes 1\bigr),\qquad
(s-i0)^{-m}=\mathrm{bv}_{\mathrm{Im}\,\zeta<0}(\zeta^{-m}).
\]
The one-variable distributions $(s-i0)^{-m}$ and $(s-i0)^{-n}$ therefore arise as boundary values of the holomorphic functions $\zeta\mapsto\zeta^{-m}$ and $\zeta\mapsto\zeta^{-n}$ from $\{\mathrm{Im}\,\zeta<0\}$.
We identify their H\"ormander product by an explicit Fourier computation.
Write $u_m:=(s-i0)^{-m}$ and use the one-variable convention $\hat u(\sigma)=\int e^{-is\sigma}u(s)\,\mathrm ds$, with inverse $u(s)=\frac1{2\pi}\int e^{is\sigma}\hat u(\sigma)\,\mathrm d\sigma$; then $\widehat{uv}=\frac1{2\pi}\,\hat u*\hat v$ (H\"ormander's $(2\pi)^{-n}$ with $n=1$).
As above, $\hat u_m(\sigma)=\frac{2\pi i\,(-i\sigma)^{m-1}}{(m-1)!}\theta(-\sigma)$ (for $m=1$: $\widehat{\mathrm{PV}\frac1s}=-i\pi\,\mathrm{sgn}\,\sigma$ and $\widehat{i\pi\delta}=i\pi$; then $\widehat{\partial_su}=i\sigma\hat u$).

\emph{Beta integral.} For $\sigma<0$ the convolution runs over the finite interval $[\sigma,0]$, where both factors are supported; with $\sigma'=\sigma x$, $x\in[0,1]$, $\mathrm d\sigma'$ over $[\sigma,0]$ becomes $|\sigma|\,\mathrm dx$, and
\[
\frac1{2\pi}(\hat u_m*\hat u_n)(\sigma)
=\frac{(2\pi i)^2}{2\pi\,(m-1)!(n-1)!}\int_\sigma^0\bigl(-i(\sigma-\sigma')\bigr)^{m-1}(-i\sigma')^{n-1}\,\mathrm d\sigma'
=\frac{2\pi i\cdot i|\sigma|\,(-i\sigma)^{m+n-2}}{(m-1)!(n-1)!}\,B(m,n).
\]
Since $B(m,n)=\frac{(m-1)!(n-1)!}{(m+n-1)!}$ and $i|\sigma|=-i\sigma$ for $\sigma<0$, this equals $\frac{2\pi i(-i\sigma)^{m+n-1}}{(m+n-1)!}=\hat u_{m+n}(\sigma)$; for $\sigma>0$ the integrand vanishes.
So $\frac1{2\pi}\hat u_m*\hat u_n=\hat u_{m+n}$, the integral being absolutely convergent.

\emph{Identification with the H\"ormander product.} For $\varphi\in C^\infty_c(\mathbb R)$ the H\"ormander product is characterized locally by $\widehat{\varphi^2uv}=\frac1{2\pi}\,\widehat{\varphi u}*\widehat{\varphi v}$, the integral converging absolutely under the wavefront condition (H\"ormander, proof of Thm.~8.2.10).
Here $\widehat{\varphi u_m}=\frac1{2\pi}\hat\varphi*\hat u_m$.
Because $\hat\varphi$ is Schwartz while $\hat u_m,\hat u_n$ are polynomially bounded and supported in $\sigma\le0$, the iterated integral $\int\!\!\int\!\!\int|\hat\varphi(a)\hat\varphi(b)\hat u_m(\sigma-a-b-c)\hat u_n(c)|$ runs in c over an interval of length at most $|\sigma-a-b|$ and converges absolutely; Fubini allows regrouping:
\[
\tfrac1{2\pi}\widehat{\varphi u_m}*\widehat{\varphi u_n}
=\tfrac1{2\pi}\,\widehat{\varphi^2}*\Bigl(\tfrac1{2\pi}\hat u_m*\hat u_n\Bigr)
=\tfrac1{2\pi}\,\widehat{\varphi^2}*\hat u_{m+n}
=\widehat{\varphi^2u_{m+n}} .
\]
Hence $\varphi^2(u_mu_n)=\varphi^2u_{m+n}$ for every φ, i.e.
\[
(s-i0)^{-m}\,(s-i0)^{-n}=(s-i0)^{-(m+n)}
\quad\text{in }\mathcal D'(\mathbb R_s).
\]
\emph{Transfer to the chart.} With $p(s,y')=s$, $u_m\otimes1=p^*u_m$, and pullback is multiplicative on distributions satisfying the product condition (H\"ormander, Thms.~8.2.4 and~8.2.10: the product is the pullback of the tensor product by the diagonal map, and pullbacks compose); thus $p^*u_m\cdot p^*u_n=p^*u_{m+n}$, and, pulling back further by the diffeomorphism Φ, $T_mT_n=T_{m+n}$ on U.
Off the cone the identity is the smooth identity $Q^{-m}Q^{-n}=Q^{-(m+n)}$.
On overlaps the two local descriptions agree by the global same-side limit $\tau\downarrow0$.
Thus $T_mT_n=T_{m+n}$ on Ω.
\end{proof}

\begin{proposition}[Numerator representation]
\label{prop:num}
By Theorem~\ref{thm:wpA-Sbv}, Input~(S-bv) holds.
Then, using Lemma~\ref{lem:bv}(a),(b), $\partial^\alpha S=c\,p_\alpha\,T_{1+|\alpha|}$ on Ω.
Here $p_0=1$ and $p_{\alpha+e_\rho}=Q\,\partial_\rho p_\alpha-(1+|\alpha|)\,p_\alpha\,\partial_\rho Q$, so p_α is homogeneous of degree $|\alpha|$.
For $\lambda=\{\alpha,\beta\}$ of order $N=|\alpha|+|\beta|$, Lemma~\ref{lem:bv}(c) gives
\[
q^\circ_\lambda=(\partial^\alpha S)(\partial^\beta S)=c^2\,n_\lambda\,T_{2+N},\qquad n_\lambda:=p_\alpha p_\beta\in\mathrm{Poly}_N(\mathbb R^4).
\]
\end{proposition}

\begin{proof}
By induction, $\partial_\rho(p_\alpha T_k)=(\partial_\rho p_\alpha)T_k-k\,p_\alpha(\partial_\rho Q)T_{k+1}$ by (b), and $T_k=QT_{k+1}$ by (a); this gives the recursion.

```
The product is the H\"ormander product used for $q^\circ$ on $y\neq w$ (XVI-J: ‘‘the pointwise product of two copies of \(\Delta_F^{\mathrm{FR}}\) exists’’), and (c) applies.
\texttt{kernel\_relations.py} computes $p_\alpha$ from the same recursion, realised as $\partial^\alpha(1/Q)=p_\alpha/Q^{1+|\alpha|}$, for all $|\alpha|\le4$.
\end{proof}

\begin{proposition}[Two-direction equivalence of relations]
\label{prop:relations}
Use Theorem~\ref{thm:wpA-Sbv} and fix $N$.
For constants $r_\lambda$:
\begin{enumerate}
\item if $\sum_\lambda r_\lambda n_\lambda=0$ as a polynomial, then $\sum_\lambda r_\lambda q^\circ_\lambda=0$ in $\mathcal D’(\Omega)$;
\item if $\sum_\lambda r_\lambda q^\circ_\lambda=0$ in $\mathcal D’(\Omega)$, then $\sum_\lambda r_\lambda n_\lambda=0$ as a polynomial.
\end{enumerate}
Relations mixing different $N$ split into homogeneous components, because $q^\circ_\lambda$ is homogeneous of degree $-4-N$.
Hence the nullspace of the numerator map is exactly the relation space of $\pi$.
\end{proposition}

\begin{proof}
(1) By Proposition~\ref{prop:num}, $\sum r_\lambda q^\circ_\lambda=c^2(\sum r_\lambda n_\lambda)T_{2+N}=0$.
(2) Restrict to the nonempty open set $\{Q<0\}\subset\Omega$ (it contains $(0,1,0,0)$).
There $T_{2+N}=Q^{-2-N}$ is smooth and nowhere zero, so $c^2(\sum r_\lambda n_\lambda)Q^{-2-N}=0$ pointwise.
Hence the polynomial $\sum r_\lambda n_\lambda$ vanishes on a nonempty open set and is therefore zero.
For mixed $N$, dilations $z\mapsto tz$ act on $\Omega$, and distinct homogeneity degrees separate.
\end{proof}

\begin{theorem}[Certified graded presentation]
\label{thm:M2}
With Input~(S-bv) certified (Theorem~\ref{thm:wpA-Sbv}):
\begin{center}
\begin{tabular}{ccccc}
\hline
$N$ & labels & $\dim\mathrm{Poly}_N$ & $d_N=\dim\mathcal H_{4+N}$ & $\dim\ker\pi_N$\\
\hline
$0$ & $1$ & $1$ & $1$ & $0$\\
$1$ & $4$ & $4$ & $4$ & $0$\\
$2$ & $20$ & $10$ & $10$ & $10$\\
$3$ & $60$ & $20$ & $20$ & $40$\\
$4$ & $170$ & $35$ & $35$ & $135$\\
\hline
Total & $255$ & & $70$ & $185$\\
\hline
\end{tabular}
\end{center}
The numerator map is onto $\mathrm{Poly}_N$.
A basis $\mathcal B_N$ extends a basis of $\mathcal V_{\mathrm K}=\mathcal H_4\oplus\mathcal H_5\oplus\mathcal H_6$ ($\dim15$).
Every label reconstructs as $q^\circ_\lambda=\sum_{b\in\mathcal B_N}R_{\lambda b}q^\circ_b$ with exact rational $R$.
\end{theorem}

\begin{proof}
Propositions~\ref{prop:num} and~\ref{prop:relations}, together with the exact $\mathbb Q$-linear algebra reported in Lemma~\ref{lem:M2comp}.
\end{proof}

\begin{proposition}[Unconditional wave-operator family]
\label{prop:wave}
Without using Input~(S-bv), $\sum_{\rho,\sigma}\eta^{\rho\sigma}q^\circ_{\{\alpha,\beta+e_\rho+e_\sigma\}}=0$ on $\Omega$, since $\square S=0$ there (XVI-K, lem:grading(3)).
The family spans a subspace of dimensions $(0,0,1,8,36)$, so $\dim\mathcal V_{\mathrm{adm}}\le210$ without any kernel identification.
With (S-bv) (certified), $140$ further independent relations appear.
\end{proposition}

\begin{remark}[A relation outside the wave family; sign check]
With $\eta=\mathrm{diag}(1,-1,-1,-1)$, $\partial_0Q=2z_0$ and $\partial_1Q=-2z_1$, so $p_{e_0}=-2z_0$, $p_{e_1}=2z_1$, $p_{e_0}p_{e_1}=-4z_0z_1$, and $p_{e_0+e_1}=-8z_0z_1$.
Hence $(\partial_0S)(\partial_1S)=\tfrac12\,S\,\partial_0\partial_1S$ on $\Omega$ (Theorem~\ref{thm:wpA-Sbv}); this is the relation \texttt{q\{1, 0\} - (1/2) q\{(), 01\} = 0}.
The script checks exactly this example (\texttt{require}: ‘‘sign example’’).
The relation is not generated by $\square S=0$.
\end{remark}

\begin{lemma}[Reported computation for M.2]
\label{lem:M2comp}
\texttt{kernel\_relations.py} exits with status $0$ and records:
\begin{itemize}
\item the table of Theorem~\ref{thm:M2};
```

```
\item reconstruction certificates for all labels (\texttt{kernel\_basis\_table.txt}, which also lists every numerator row $n_\lambda$);
\item $185$ certified relation vectors and $45$ wave-operator generators (\texttt{kernel\_relation\_matrix.txt});
\item the sign example.
\end{itemize}
\texttt{regression\_M.py} independently spot-checks every relation (on $(\partial^\alpha S)(\partial^\beta S)$ at three rational points).
\end{lemma}

\begin{remark}[Analytic boundary and dependency on (S-bv)]
\label{rem:M2boundary}
An identity of rational functions away from the null cone is not automatically a distributional identity on $\Omega$.
Lemma~\ref{lem:bv} is about the model family $T_m$ and does not by itself identify $S$ with $c\,T_1$.
The identification of the numerical kernels $q^\circ_\lambda$ with the corresponding products of $T_m$'s---and hence the numerical conclusions of M.2--M.4 that use the graded presentation---rests on Input~(S-bv), which is certified by Theorem~\ref{thm:wpA-Sbv}; these conclusions therefore hold unconditionally on $\Omega$.
Proposition~\ref{prop:wave} remains the part that needs no kernel identification.
The linear Helmholtz lemma (Lemma~\ref{lem:helm}) is pure variational calculus under Gate~V and never depended on (S-bv).
\end{remark}
%-----------------------------------------------------------------------
\subsection{Kernel identification: Gate S-bv certified (work package A)}
\label{sec:wpA-Sbv}
\scopebox{Status (work package A).}{%
\textbf{Constructed (proved):} on $\Omega:=\mathbb R^4\setminus\{0\}$,
\[
S=h_0+iE_0^{\mathrm A}=-W_0+iE_0^{\mathrm A}
=\frac1{4\pi^2}\,(Q-i0)^{-1},
\]
with $Q(z)=\eta(z,z)$, $\eta=\mathrm{diag}(1,-1,-1,-1)$, including across the nonzero null cone (Theorem~\ref{thm:wpA-Sbv}).
\textbf{Constructed:} convention ledger (\texttt{wpA/kernel\_conventions.json}); symbolic and numerical cross-checks (\texttt{wpA/kernel\_identification\_checks.py}); local boundary-value lemma for the variable regulator (Lemma~\ref{lem:wpA-varreg}).
\textbf{Predecessor status:} the formula is printed in XVI-J, eq:rational (with $S=-d$, XVI-J, eq:DF); this subsection is a written check of it, not a new extension.\newline\textbf{Non-claims:} the predecessor $S$ is globally defined (XVI-J); what is not established here is its position-space identification at the origin; no $i\varepsilon$ Feynman contour is introduced as a definition; the massive case is not needed for this edge; nothing about M-C, Ward or BV.
Gate~S-bv / Input~(S-bv) is discharged by the theorem below, with the explicit constant $c=1/(4\pi^2)$ and boundary side $Q-i0$.
The Euclidean calculation of Remark~\ref{rem:Sbv-motivation} remains a motivation only; it is not the certification.
}

\paragraph{Convention ledger.}
\label{sec:wpA-conventions}
\begin{center}
\small
\begin{longtable}{p{0.20\textwidth}p{0.46\textwidth}p{0.25\textwidth}}
\hline
Item & Value (predecessor-authoritative) & Source\\
\hline
\endhead
Difference variable & $z=x-y$; $S$ is a translation-invariant distribution of $z$ & XVI-K, after eq:S\\
Caveat & XVI-K uses $z=y-w$ for the \emph{bubble extension} only; this is not a redefinition of the edge kernel $S$ & XVI-K (extension step)\\
Signature / $Q$ & $\eta=\mathrm{diag}(1,-1,-1,-1)$, $Q(z)=z_0^2-|\mathbf z|^2$ & XI-B; Input~\ref{inp:Sbv}\\
Wave operator & $\square=\partial_t^2-\Delta$; $\square E_0^{\mathrm{R/A}}=\mathrm{id}$ & XI-B, lem:Green; XVI-J, lem:signs\\
Spatial Fourier in $W_0$ & $W_0(x,y)=\int\frac{e^{-i|\mathbf p|(x^0-y^0)+i\mathbf p\cdot(\mathbf x-\mathbf y)}}{(2\pi)^3\,2|\mathbf p|}\,\mathrm d^3\mathbf p$ (massless) & XI-E, eq:W; XVI-K, eq:W-fam\\
Test-function FT & $\tilde\varphi(z^0,\mathbf p)=\int e^{i\mathbf p\cdot\mathbf z}\varphi\,\mathrm d^3\mathbf z$ & XVI-K, lem:families\\
$h_0$ & $h_0=-W_0$, $h_0-h_0^{\mathsf T}=i\delta_0$ & XI-E, eq:h\\
$E_0^{\mathrm A}$ (sine) & $(E_0^{\mathrm A}F)(t)=\int_t^\infty\frac{\sin((s-t)\Omega_0)}{\Omega_0}F(s)\,\mathrm ds$ & XVI-J, eq:sineA; XVI-K, eq:E-fam\\
$E_0^{\mathrm A}$ (step) & $E_0^{\mathrm A}=-\theta(y^0-x^0)\,\delta_0=-\theta(-z^0)\,\delta_0$ & XVI-J, eq:step\\
$S$ & $S:=h_0+iE_0^{\mathrm A}=-W_0+iE_0^{\mathrm A}=-\Delta_F^{\mathrm{FR}}$ & XVI-K, eq:S; XVI-J, lem:identify\\
Reconciliation & ``$S=h_0+iE_0^{\mathrm A}$'' and ``$S=-W_0+iE_0^{\mathrm A}$'' are equivalent via $h_0=-W_0$ & XI-E, eq:h; XVI-K, eq:S\\
Transposition & $\langle T^{\mathsf T},\phi(x,y)\rangle=\langle T,\phi(y,x)\rangle$; for $u(z)$, $u^{\mathsf T}(z)=u(-z)$ & XVI-J: $(E_0^{\mathrm A})^{\mathsf T}=E_0^{\mathrm R}$, $W_0^{\mathsf T}=W_0+i\delta_0$\\
$(Q-i0)^{-1}$ & $\lim_{\tau\downarrow0}(Q-i\tau)^{-1}$ in $\mathcal D'(\Omega)$; equivalently $\Phi^*((s-i0)^{-1}\otimes1)$ near nonzero null points & Lemma~\ref{lem:bv}\\
One-variable identity & $(s-i0)^{-1}=\mathrm{PV}(1/s)+i\pi\delta(s)$ (sign $+i\pi$) & classical; checked in \texttt{wpA/kernel\_\allowbreak identification\_\allowbreak checks.py}\\
Constant / side & $c=1/(4\pi^2)$; boundary side $Q-i0$ (the side of $T_1$) & Theorem~\ref{thm:wpA-Sbv}\\
\hline
\end{longtable}
\end{center}
No abstract $\mathbb R^4$ Fourier transform of $S$ is fixed by the predecessors---only the spatial Fourier formulae for $W_0$ and the sine kernel.
Signs are \emph{not} inferred from the word ``Feynman''; $\Delta_F^{\mathrm{FR}}$ enters only through XVI-J, lem:identify, as an identification of the already-defined combination $\theta_+W_0+\theta_-W_0^{\mathsf T}$.

```

\paragraph{Position-space identification.}
\label{sec:wpA-identify}
\emph{What the predecessors already print (audit~3).}
XI-E, eq:h (box copy \texttt{enter-companion/raw/tex/XI-E.tex}, lines 366--371, verified) only normalizes the Wightman factor and records the bisolution jump:
\begin{quote}\small
``To match the proposed block notation put
\(h_\mu=-W_\mu,\ h_\mu-h_\mu^{\mathsf T}=i\delta_\mu.\)''
\end{quote}
It gives no position-space boundary value.
XVI-J (version~1, \texttt{content/manuscripts/XVI-J.tex} of \texttt{cheungngo/enter-companion}), eq:W, eq:EA and eq:DF read (macros expanded):
\begin{quote}\small
``Define the massless positive-frequency distribution by
\(W_0(z)=\int_{\mathbb R^3}\frac{e^{-i|\mathbf p|z^0+i\mathbf p\cdot\mathbf z}}{(2\pi)^3\,2|\mathbf p|}\,\mathrm d^3\mathbf p.\)
The integral is distributional. Set \(h_0=-W_0\).''
\ldots\ ``\(\widehat E_0^{\mathrm A}(t,\mathbf p)=-\theta(-t)\frac{\sin(t|\mathbf p|)}{|\mathbf p|}\)'' \ldots\
``Put
\(d=W_0-iE_0^{\mathrm A},\quad S=h_0+iE_0^{\mathrm A}=-d,\quad D_F=-\mathcal PS=\mathcal Pd.\)''
\end{quote}
XVI-J, eq:rational:
\begin{quote}\small
``Let \(Q_{\mathrm{geom}}(z)=(z^0)^2-|\mathbf z|^2.\)
On \(z\ne0\), direct Fourier evaluation gives
\(d(z)=-\frac1{4\pi^2}\bigl(Q_{\mathrm{geom}}(z)-i0\bigr)^{-1}.\)
At a nonzero null point, \(Q_{\mathrm{geom}}\) is a submersion.
The boundary value in [eq:rational] has one, rather than both, conormal orientations.''
\end{quote}
XVI-J, prop:q:
\begin{quote}\small
``The distribution \(q^\circ(z)=S(z)^2=d(z)^2,\ z\ne0,\) exists on \(\mathbb R^4\setminus\{0\}\), including nonzero null points.
It is homogeneous of degree \(-4\), has scaling degree four at the origin, and is nonzero.''
\end{quote}
Thus $S=-d=\frac1{4\pi^2}(Q-i0)^{-1}$ on $z\neq0$, including nonzero null points, is the XVI-J formula ($Q=Q_{\mathrm{geom}}$), and XVI-J, prop:q, uses it.
Theorem~\ref{thm:wpA-Sbv} is a \emph{written check} of XVI-J, eq:rational, whose printed justification is ``direct Fourier evaluation''; it is not a new extension.
XVI-M's contribution is (i) the written real-time derivation $W_0=-\frac1{4\pi^2}\bigl(\mathrm{PV}\frac1Q+i\pi\,\mathrm{sgn}(z^0)\delta(Q)\bigr)$, $E_0^{\mathrm A}=\frac1{2\pi}\theta(-z^0)\delta(Q)$ and the cancellation giving $\frac1{4\pi^2}(Q-i0)^{-1}$; and (ii) the local justification at nonzero null points of the boundary value of the variable regulator $Q-2i\tau z^0-\tau^2$ (Lemma~\ref{lem:wpA-varreg}).
\emph{Predecessor provenance.} Each predecessor version used below is pinned by SHA-256 (local files) or by an archived corpus excerpt under \texttt{XVI-M-artifacts/\allowbreak provenance/} (regression re-checks those hashes). Citations are by label only.
\begin{itemize}\setlength{\itemsep}{0.15em}\small
\item \textbf{Local} \texttt{XVI-J.tex} (\texttt{dcba2602ab4a\ldots}): labels eq:W0, eq:sineA, eq:step, lem:signs, lem:identify. Role: supplementary (earlier box version).
\item \textbf{Repository fetch} of XVI-J (\texttt{d5558cd419c8\ldots}, \texttt{provenance/XVI-J_github_\allowbreak fetch.tex}): HTTP~404 stub. Role: failed retrieval; does not remove the corpus file.
\item \textbf{Corpus excerpt} of XVI-J (\texttt{eda87cb7400d\ldots}, \texttt{provenance/XVI-J_corpus_\allowbreak eqW_DF_\allowbreak rational_propq.txt}): labels eq:W, eq:EA, eq:DF, eq:rational, prop:q. Role: \textbf{supplementary corroboration}; eq:rational is \emph{not} part of the frozen XVI-L dependency; Theorem~\ref{thm:wpA-Sbv} and Lemma~\ref{lem:wpA-varreg} stand independently.
\item \textbf{Frozen} \texttt{XVI-L.tex} (\texttt{db5f027ef08e\ldots}): thm:L5, thm:L6, def:RL,\ldots\ Role: \textbf{frozen dependency}. Corpus excerpt of thm:L6 (\texttt{1ef32e999bfc\ldots}).
\item \textbf{Local} \texttt{XVI-K.tex}/artifacts (\texttt{6264bc8ec685\ldots}): eq:S, prop:hess,\ldots\ Role: frozen interface. Corpus excerpt (\texttt{f0ebf560b5d6\ldots}): prop:hess, eq:Lpq.
\item \textbf{Corpus/local} XI-E (\texttt{ea40d58d5912\ldots}/\texttt{d686b6ec595e\ldots}): eq:W, eq:h. Role: supplementary corroboration.
\item \textbf{Corpus/local} XII-C (\texttt{79c1765ef7d8\ldots}/\texttt{83c6d0383b0a\ldots}): lem:weighted-curvature. Role: supplementary corroboration.
\end{itemize}
A failed repository retrieval does not make eq:rational a premise of XVI-L.

\begin{lemma}[Variable regulator near a nonzero null point (audit~3)]
\label{lem:wpA-varreg}
Let $z_\ast\in\Omega$ with $Q(z_\ast)=0$ and put $s_\ast:=\mathrm{sgn}(z_\ast^0)$.
Then $z^0_\ast\neq0$, and there are an open neighbourhood $U\Subset\Omega$ of $z_\ast$ and a chart $\Phi=(Q,y'):U\to(-\delta,\delta)\times V'$ such that, with $a_0:=|z^0_\ast|$, $s_\ast z^0\ge a_0/2>0$ on U.
For $0<\tau\le1$ put $g_\tau(z):=Q(z)-2i\tau z^0-\tau^2$.
Then g_τ has no zero on $\mathbb R^4$, and
\[
\lim_{\tau\downarrow0}\frac1{g_\tau}=\Phi^*\bigl((s-i0\,,s_\ast)^{-1}\otimes1\bigr)
\quad\text{in }\mathcal D'(U),
\qquad
(s-i0\,,s_\ast)^{-1}:=\lim_{\eta\downarrow0}(s-i\eta\, s_\ast)^{-1}=\mathrm{PV}\frac1{s+i\pi\, s_\ast\,\delta(s).
\]
For $s_\ast=+1$ the right-hand side is $T_1|_U$ of Lemma~\ref{lem:bv}.
\end{lemma}

```
\begin{proof}
If $z^0_\ast=0$ then $Q(z_\ast)=-|\mathbf z_\ast|^2=0$ forces $z_\ast=0\notin\Omega$.
The chart with first coordinate $Q$ exists by the proof of Lemma~\ref{lem:bv}; shrinking it, we may assume $\overline U$
     compact in $\Omega$ and $s_\ast z^0\ge a_0/2$ on $\overline U$, with $a_0:=|z^0_\ast|$.
A zero of $g_\tau$ needs $\mathrm{Im}\,g_\tau=-2\tau z^0=0$ and then $Q=-|\mathbf z|^2=\tau^2>0$, which is impossible.
In the chart, $g_\tau\circ\Phi^{-1}=w_\tau:=s-\tau^2-i\tau a$ with $a:=2z^0\circ\Phi^{-1}$ smooth, $s_\ast a\ge a_0$, $|a|\le
     a_1$, $|\partial_sa|\le a_2$ on the chart (after a further shrinking).
Pullback by the diffeomorphism $\Phi$ is continuous on $\mathcal D'$ and the Jacobian is absorbed in the test function, so it
     suffices to show, for $\psi\in C^\infty_c((-\delta,\delta)\times V')$,
$\int\psi\,w_\tau^{-1}\,\mathrm ds\,\mathrm dy'\to\langle(s-i0\,,s_\ast)^{-1}\otimes1,\psi\rangle$.

Since $\mathrm{Im}\,w_\tau=-\tau a$ has the fixed sign $-s_\ast$, $w_\tau$ stays in one open half-plane, on which the
     principal logarithm $L_\tau:=\mathrm{Log}\,w_\tau$ is smooth, and $\partial_sL_\tau=(1-i\tau\partial_sa)/w_\tau$.
Hence
\[
\frac1{w_\tau}=\partial_sL_\tau+\frac{i\tau\,\partial_sa}{w_\tau}.
\]
(i) \emph{Remainder.} $|\tau\,\partial_sa/w_\tau|\le\tau a_2/|\mathrm{Im}\,w_\tau|\le a_2/a_0$, and for $s\neq0$ the term
     tends to $0$ (since $w_\tau\to s\ne0$).
By dominated convergence on the compact support of $\psi$, $\int\psi\,i\tau\partial_sa/w_\tau\to0$.

(ii) \emph{Main term.} Integrating by parts in $s$, $\int\psi\,\partial_sL_\tau=-\int(\partial_s\psi)L_\tau$.
For $s\ne0$, $L_\tau\to\mathrm{Log}_{s_\ast}(s):=\log|s|-i\pi s_\ast\theta(-s)$ (for $s<0$ the point $w_\tau$ approaches the
     negative axis from the half-plane $\mathrm{sgn}\,\mathrm{Im}=-s_\ast$).
Moreover $|\mathrm{Im}\,L_\tau|\le\pi$ and $|s-\tau^2|\le|w_\tau|\le C:=\delta+1+a_1$, so $|\mathrm{Re}\,L_\tau|\le|\log C|+\
     bigl|\log|s-\tau^2|\bigr|$.
The family $\{\log|s-\tau^2|\}_{0<\tau\le1}$ is bounded in $L^2(-\delta,\delta)$ (each is a translate of $\log|u|$ restricted
     to $(-\delta-1,\delta+1)$), hence uniformly integrable on $\mathrm{supp}\,\psi$; by Vitali's theorem $\int(\partial_s\
     psi)L_\tau\to\int(\partial_s\psi)\,\mathrm{Log}_{s_\ast}(s)$.

(iii) \emph{Identification.} The case $a\equiv2s_\ast$ without the shift $\tau^2$ (i.e.\ $w=s-i\eta s_\ast$, remainder $\equiv
     0$) gives by the same argument $(s-i\eta s_\ast)^{-1}\to\partial_s\mathrm{Log}_{s_\ast}(s)$ in $\mathcal D'$; hence both
      limits equal $\partial_s\mathrm{Log}_{s_\ast}$.
Finally $\partial_s\log|s|=\mathrm{PV}\frac1s$ and $\partial_s\bigl(-i\pi s_\ast\theta(-s)\bigr)=i\pi s_\ast\delta(s)$.
For $s_\ast=+1$, $(s-i\eta)^{-1}\otimes1$ is $(Q-i\eta)^{-1}$ in the chart, whose limit defines $T_1|_U$.
\end{proof}

\begin{theorem}[Certified identification for Gate S-bv]
\label{thm:wpA-Sbv}
Let $W_0$, $h_0=-W_0$ and $E_0^{\mathrm A}$ be as in XVI-K, eq:S / lem:families (quoting XI-E, eq:W, eq:h and XVI-J, eq:sineA,
      eq:step), massless case.
Set $S:=h_0+iE_0^{\mathrm A}$ and $Q(z)=\eta(z,z)$.
Then, in $\mathcal D'(\Omega)$,
\begin{equation}
\label{eq:wpA-id}
S=\frac1{4\pi^2}\,(Q-i0)^{-1}
=\frac1{4\pi^2}\Bigl(\mathrm{PV}\frac1Q+i\pi\,\delta(Q)\Bigr),
\end{equation}
where $(Q-i0)^{-1}$ is the boundary-value distribution $T_1$ of Lemma~\ref{lem:bv}, and $\delta(Q)$ denotes the pullback
     surface distribution $\Phi^*(\delta(s)\otimes1)$ in any local chart with first coordinate $Q$.
In particular Input~(S-bv) holds with
\[
c=\frac1{4\pi^2}\neq0
\]
and with the same boundary side $Q-i0$ as $T_1$.
Since $S=-d$ (XVI-J, eq:DF), this is XVI-J, eq:rational; the theorem is a written check of it, not a new extension.
\end{theorem}

\begin{proof}
\emph{(1) Position-space formulae from the Fourier definitions.}
\textit{Wightman factor.}
From XVI-K, eq:W-fam / XI-E, eq:W, with $z=x-y$ and $\omega_0=|\mathbf p|$,
\[
W_0(z)
=\int_{\mathbb R^3}\frac{e^{-i|\mathbf p|z^0+i\mathbf p\cdot\mathbf z}}{(2\pi)^3\,2|\mathbf p|}\,\mathrm d^3\mathbf p\ .
\]
Angular integration yields the radial formula
\[
W_0(z)=\frac1{4\pi^2\,r}\int_0^\infty\sin(pr)\,e^{-ipz^0}\,\mathrm dp,
\qquad r=|\mathbf z|.
\]
Insert the positive-frequency damping $z^0\mapsto z^0-i\tau$ ($\tau\downarrow0$), i.e.\ $\alpha=\tau+iz^0$ with $\mathrm{Re}
     }\,\alpha>0$, and use
$\int_0^\infty e^{-\alpha p}\sin(pr)\,\mathrm dp=r/(\alpha^2+r^2)$.
(The damping parameter is called $\tau$ throughout; it is unrelated to the preparation width $\varepsilon$. The factor $e^{-\
     tau|\mathbf p|}$ only realises the oscillatory integral as a limit in $\mathcal D'(\mathbb R^4)$: for a test function $\
     varphi$, $\langle W_0^{(\tau)},\varphi\rangle=\int\mathrm dz^0\int\frac{e^{-(\tau+iz^0)|\mathbf p|}}{(2\pi)^32|\mathbf p
     |}\tilde\varphi(z^0,-\mathbf p)\,\mathrm d^3\mathbf p$, and the integrand is dominated by $|\tilde\varphi(z^0,-\mathbf p
     )|/((2\pi)^32|\mathbf p|)$, which is integrable; it changes no definition and introduces no $i\varepsilon$ prescription
     .)
```

One obtains the smooth function
\[
W_0^{(\tau)}(z)=\frac1{4\pi^2(\alpha^2+r^2)}
=-\frac1{4\pi^2\bigl(Q-2i\tau z^0-\tau^2\bigr)}=-\frac1{4\pi^2\,g_\tau(z)} .
\]
Off the cone, $g_\tau^{-1}\to Q^{-1}$ in C^∞; near each nonzero null point, where z^0 has a fixed sign, Lemma~\ref{lem:wpA-varreg} gives $g_\tau^{-1}\to(Q-i0\,\mathrm{sgn}(z^0))^{-1}$.
Passing to $\tau\downarrow0$ therefore gives the distributional identity on Ω
\begin{equation}
\label{eq:wpA-W}
W_0(z)
=-\frac1{4\pi^2}\,\frac1{Q-i0\,\mathrm{sgn}(z^0)}
=-\frac1{4\pi^2}\Bigl(\mathrm{PV}\frac1Q+i\pi\,\mathrm{sgn}(z^0)\,\delta(Q)\Bigr).
\end{equation}
(The Euclidean slice $z^0=-i\tau_{\mathrm E}$, $\tau_{\mathrm E}>0$, recovers $W_0=1/(4\pi^2(\tau_{\mathrm E}^2+r^2))$, matching Remark~\ref{rem:Sbv-motivation} and the check in \texttt{kernel_relations.py}.)

\textit{Causal factor.}
From $W_0-W_0^{\mathsf T}=-i\delta_0$ (XI-E, after eq:W; XVI-J) and \eqref{eq:wpA-W},
$\delta_0=\frac1{2\pi}\,\mathrm{sgn}(z^0)\,\delta(Q)$ on Ω.
XVI-J, eq:step, gives $E_0^{\mathrm A}=-\theta(-z^0)\,\delta_0$, hence
\begin{equation}
\label{eq:wpA-EA}
E_0^{\mathrm A}(z)=\frac1{2\pi}\,\theta(-z^0)\,\delta(Q)
\quad\text{on }\Omega .
\end{equation}
(Equivalently, $E_0^{\mathrm R}=\theta(z^0)\,\delta_0=\frac1{2\pi}\theta(z^0)\delta(Q)$ is the standard massless retarded kernel for $\square E=\delta$.)

\emph{(2) Combination: cancellation of $\mathrm{sgn}(z^0)$.}
Using $h_0=-W_0$,
\[
-W_0=\frac1{4\pi^2}\mathrm{PV}\frac1Q+\frac{i}{4\pi}\,\mathrm{sgn}(z^0)\,\delta(Q),
\qquad
iE_0^{\mathrm A}=\frac{i}{2\pi}\,\theta(-z^0)\,\delta(Q).
\]
On $\{z^0>0\}$ one has $\mathrm{sgn}=+1$, $\theta(-z^0)=0$; on $\{z^0<0\}$, $\mathrm{sgn}=-1$, $\theta(-z^0)=1$.
The coefficient of $\delta(Q)$ becomes
\[
\frac{i}{4\pi}\mathrm{sgn}(z^0)+\frac{i}{2\pi}\theta(-z^0)
=\frac{i}{4\pi}\bigl(\theta(z^0)+\theta(-z^0)\bigr)
=\frac{i}{4\pi}
=\frac1{4\pi^2}\cdot i\pi
\]
on both sheets.
(The identity $\theta(z^0)+\theta(-z^0)=1$ holds on $\Omega\cap\{Q=0\}$, because $Q=0$ and $z^0=0$ force $\mathbf z=0$, hence $z=0\notin\Omega$.)
Therefore
\[
S=-W_0+iE_0^{\mathrm A}
=\frac1{4\pi^2}\Bigl(\mathrm{PV}\frac1Q+i\pi\,\delta(Q)\Bigr)
=\frac1{4\pi^2}\,(Q-i0)^{-1},
\]
where the last step is the pullback of $(s-i0)^{-1}=\mathrm{PV}(1/s)+i\pi\delta(s)$.
The $\mathrm{sgn}(z^0)$-dependence of W_0 is exactly cancelled by $iE_0^{\mathrm A}$, which yields the \emph{single} boundary value $Q-i0$ (not $Q+i0$).

\emph{(3) Normalization off the cone.}
On the open sets $\{Q>0\}$ and $\{Q<0\}$, $\delta(Q)=0$ and $E_0^{\mathrm A}=0$, so $S=c/Q$ with $c=1/(4\pi^2)$; e.g.\ $S=-c$ at $(0,1,0,0)$ and $S=c$ at $(1,0,0,0)$.

\emph{(4) Equality across the nonzero null cone.}
Let $z_\ast\in\Omega$ with $Q(z_\ast)=0$.
Then $\mathrm dQ(z_\ast)=2\eta(z_\ast,\cdot)\neq0$, so Q is a local coordinate: there is a chart Φ with $\Phi_1=Q$ (proof of Lemma~\ref{lem:bv}).
In that chart $(Q-i0)^{-1}=\Phi^*((s-i0)^{-1}\otimes1)$, and step~(2) shows that $-W_0+iE_0^{\mathrm A}$ equals c times this same pullback on both time sheets.
The surface distribution $\delta(Q)$ is well defined on the cone minus the origin by the same submersion, and the sheet-wise boundary values $(Q-i0\,\mathrm{sgn}\,z^0)^{-1}$ used in~\eqref{eq:wpA-W} are those of Lemma~\ref{lem:wpA-varreg}.
Thus \eqref{eq:wpA-id} holds near every nonzero null point, and therefore on all of Ω.

\emph{(5) Cross-checks.}
(a) $\square W_0=0$ and $\square E_0^{\mathrm A}=\delta$ (XI-B, lem:Green), so $\square S=i\delta$ on $\mathbb R^4$ and $\square S=0$ on Ω, matching XVI-K, lem:grading(3); off the cone $\square(c/Q)=0$ as a rational identity in four dimensions.
(b) Both sides are homogeneous of degree -2 (XVI-K, lem:homog).
(c) $W_0-W_0^{\mathsf T}=-i\delta_0$ and $E_0^{\mathrm R}-E_0^{\mathrm A}=\delta_0$ imply $S^{\mathsf T}=S$, consistently with $D_F=-\Pc S$ (XVI-J, lem:signs).
(d) $KD_F=i\,\mathrm{id}$ with $K=-\square\Pc$ and $D_F=-\Pc S$ is equivalent to $\square S=i\delta$.
\end{proof}

```
\begin{remark}[Downstream impact]
\label{rem:wpA-M6}
\emph{M.2--M.4, M.7.} The numerator relations, the graded presentation and everything built on it use only $c\neq0$ and the single side $T_1$; they are now unconditional on $\Omega$ (statements that also use Gate~V stay relative to that convention).
\emph{M.6.} Theorem~\ref{thm:M6-S2} gives $r=-2i\pi^2c^2$; with $c=1/(4\pi^2)$,
$r=-2i\pi^2\cdot\frac1{16\pi^4}=-\frac{i}{8\pi^2}$.
The boundary side is $Q-i0$, so the angular integral $\int_{S^3}(Q-i0)^{-2}\,\mathrm d\sigma=-2i\pi^2$ of \texttt{scaling\_anomaly.py} is unchanged in sign; the counterfactual $Q+i0$ would conjugate that integral and flip $r$, and it is excluded by Theorem~\ref{thm:wpA-Sbv}.
The top-order scale-residue unit of Section~\ref{sec:wpB} is $i\pi^2c^2=i/(16\pi^2)$.
\end{remark}

\begin{remark}[What remains open after work package A]
\label{rem:wpA-open}
The predecessor $S$ is globally defined (XVI-J); what is not established here is its position-space identification at the origin. (Gate~S-bv concerns $\Omega$ only.)
The massive formulae ($\mu>0$ in XI-E, eq:W) are not needed for the massless edge kernel of XVI-K/J.
Quoted predecessor displays (XVI-K, eq:S and eq:W-fam; XVI-J, eq:step; XI-E, eq:h) are reproduced in \texttt{wpA/kernel\_identification.tex} and \texttt{wpA/kernel\_conventions.json}.
\end{remark}

\begin{lemma}[Reported computation for work package A]
\label{lem:wpA-comp}
\texttt{wpA/kernel\_identification\_checks.py} exits with status $0$ (about $2$\,s) and prints \texttt{ALL CHECKS PASSED}.
It checks: the radial Laplace integral and the damped closed form of $W_0$ (symbolic, with numerical spot checks); the sign of $(s\mp i0)^{-1}$; the cancellation of $\mathrm{sgn}(z^0)$ and the coefficients $c$ and $c\,i\pi$; $\delta_0$ and $E_0^{\mathrm A}$ on $\Omega$; off-cone normalization at $12$ points; numerical spatial Fourier integrals against the closed form; the null-cone chart $s=Q$ on both sheets; the variable regulator of Lemma~\ref{lem:wpA-varreg} (numerical pairing for $a=2+\sin s$ and $a=-(1.5+0.5\cos s)$ at $\tau=10^{-2},10^{-3},10^{-4}$, error decreasing linearly in $\tau$; audit~3); $\square(c/Q)=0$, homogeneity, transposition, propagator normalization; and $r=-i/(8\pi^2)$.
The output is deterministic (fixed seeds) and is re-generated and compared byte-for-byte by \texttt{regression\_M.py}.
\end{lemma}

%=====================================================================
\section{M.3 Contact space, graph maps, and invisible directions}
\label{sec:m3}

\begin{definition}[Coherent contact space]
\label{def:Ccoh}
Let $\mathcal D_{\le N}=\mathrm{span}\{\partial^\gamma\delta:|\gamma|\le N\}$.
After Theorem~\ref{thm:M2}, and using XVI-L, thm:L5,
\[
\Ccoh=\bigoplus_{N=0}^4\mathrm{Hom}\bigl(\mathcal H_{4+N},\mathcal D_{\le N}\bigr),\qquad\dim\Ccoh=3321
\]
(using Theorem~\ref{thm:M2}, hence Theorem~\ref{thm:wpA-Sbv}; raw labelled slots $14321$).
Coordinates are $C_{N,b,\gamma}$ with $b\in\mathcal B_N$ and $|\gamma|\le N$.
The regression-preserving subspace $\Creg=\{\Delta\RL|_{\mathcal V_{\mathrm K}}=0\}$ consists of the coordinates with $N\ge3$; it has dimension $3150$.
\end{definition}

\begin{definition}[Three graph maps and two named variants]
\label{def:Gmaps}
Write $\bar a_b:=\sum_\lambda R_{\lambda b}a_\lambda$ for the reduced coefficient of a basis kernel $b$, using the physical (switched) $a_\lambda=\chiI\hat a_\lambda$ of Definition~\ref{def:aGeneral}.
Equivalently $\bar a_b=\chiI(w)\,\hat{\bar a}_b$ with the switch-free factor $\hat{\bar a}_b:=\sum_\lambda R_{\lambda b}\hat a_\lambda$ (built from the table $\hat a$).
For $C\in\Ccoh$ put
\[
\mathcal G(C)=\sum_{N,b,\gamma}C_{N,b,\gamma}\,\bar a_b^{\mu\nu}(y,w)\,(\partial^\gamma\delta)(y-w)
\]
(equivalently $\sum C_{N,b,\gamma}\,\chiI(w)\,\hat{\bar a}_b^{\mu\nu}(y,w)\,(\partial^\gamma\delta)(y-w)$; the preferred form matches $E_C$ below).
\begin{enumerate}
\item $\mathcal G_{\mathrm{formal}}$, the \emph{formal universal-jet reading}.
External data $\sigma$ (preparation), $\chi$ (switch) and the field $h$ are arbitrary smooth functions.
$\Tcol^{\mathrm{formal}}$ is the space of collision tensors $\sum_\gamma F_\gamma(y,w)\,\partial^\gamma\delta(y-w)$ whose coefficients are polynomials in the jet symbols: $F_\gamma\in\mathbb Q[\sigma\text{-jet}(y)]\otimes\mathbb Q[h\text{-jet}(w),\chi(w)]\otimes\mathbb Q^{10}$ (output components).
$\mathcal G_{\mathrm{formal}}(C)=0$ means vanishing for \emph{all} smooth data.
Since every finite jet at a point is realised by a polynomial function, this is equivalent to vanishing as a polynomial identity in independent jet symbols.
\item $\mathcal G_{\mathrm{background}}$ (the XVI-L specialization): $h=(w^1)^2\eta$, with $\sigma,\chi$ arbitrary; $\Tcol^{\mathrm{background}}$ has coefficients in $\mathbb Q[\sigma\text{-jet}(y)]\otimes\mathbb Q[w^1]\,\chi(w)\otimes\mathbb Q^{10}$.
\item $\mathcal G_{\mathrm{fixed}}$: the actual fixed preparation $\sigeps$ and switch $\chiI$, with $h$ arbitrary; $\Tcol^{\mathrm{fixed}}$ is the space of $h$-linear maps into $\mathcal D'(\mathbb R^4\times\mathbb R^4)$.
\end{enumerate}
Named variants of the formal reading are \emph{time-only}, where $\sigma$ depends on time only and only $\sigma,\partial_0\sigma,\partial_0^2\sigma$ survive, and \emph{plateau}, on $\operatorname{supp}\chiI\cap\{\sigeps=1\}$, where only the $J_3$ sector survives.
```

```
\end{definition}

\paragraph{Reduced coefficient expansion.}
Because $\bar a_b=\chiI\,\hat{\bar a}_b$, expand the switch-free factor $\hat{\bar a}_b^{\mu\nu}=\sum_m c_{m,b}\,m$ over coefficient monomials $m=(\mu\nu)\otimes(\sigma\text{-symbol})\otimes(h\text{-symbol})$; the output component is part of the monomial.
(The overall factor $\chiI(w)$ in $\bar a_b$ factors out of the linear algebra on coefficients.)
Then
\[
\sum_{b\in\mathcal B_N}\hat{\bar a}_b\otimes b=\sum_m m\otimes u_m,\qquad u_m:=\sum_bc_{m,b}\,b\in\mathcal H_{4+N},\qquad U_N:=\mathrm{span}\{u_m\}.
\]
\texttt{graph\_contact\_map.txt} lists a basis of each $U_N$ as vectors $u_m$, labelled by their pivot monomials $m$.

\begin{proposition}[Reduction of $\mathcal G$]
\label{prop:Gsplit}
In the formal universal-jet reading of Definition~\ref{def:Gmaps} (with the presentation of Theorem~\ref{thm:M2}):
Then $\mathcal G_{\mathrm{formal}}(C)=0$ if and only if $C_N|_{U_N}=0$ for every $N$.
Hence
\[
\ker\mathcal G_{\mathrm{formal}}=\bigoplus_N\mathrm{Hom}(\mathcal H_{4+N}/U_N,\mathcal D_{\le N}),\qquad
\dim\operatorname{im}\mathcal G_{\mathrm{formal}}=\sum_N\dim U_N\cdot\dim\mathcal D_{\le N}.
\]
\end{proposition}

\begin{proof}
Coherence gives $\mathcal G(C)=\sum_\gamma F_\gamma(y,w)\,\partial^\gamma\delta(y-w)$ with $F_\gamma=\sum_{N,b}C_{N,b,\gamma}\bar a_b=\chiI(w)\sum_{N,b}C_{N,b,\gamma}\hat{\bar a}_b$.
Taylor expansion of $F_{\gamma'}(w+z,w)$ in $z$ gives the unique canonical form $\sum_\gamma g_\gamma(w)\partial^\gamma\delta(y-w)$, with
$g_\gamma=\sum_{\gamma'\ge\gamma}(-1)^{|\gamma'-\gamma|}\binom{\gamma'}{\gamma'-\gamma}\partial_y^{\gamma'-\gamma}F_{\gamma'}|_{y=w}$.
Proceed by descending induction on $|\gamma|$: the top coefficient is $F_\gamma(w,w)$.
In the formal reading the $\sigma$-jet symbols (at $y$) and the $h$-jet and $\chi$ symbols (at $w$) are independent, and evaluation on the diagonal does not identify any two of them.
So $F_\gamma(w,w)=0$ for all data if and only if $\sum_bC_{N,b,\gamma}\hat{\bar a}_b\equiv0$ as a polynomial (the factor $\chiI(w)$ in $\bar a_b$ is independent), and then all its $y$-derivatives vanish too.
Thus $\mathcal G(C)=0$ if and only if $\sum_bC_{N,b,\gamma}\hat{\bar a}_b\equiv0$ for every $\gamma$.
By the reduced expansion this means $\sum_bC_{N,b,\gamma}c_{m,b}=0$ for all $m$, i.e.\ $C_N(\cdot)_\gamma$ annihilates $U_N$.
Monomials of different $N$ are disjoint (preparation order $2-s$, $h$-order $2-N+s$, checked in \texttt{contact\_space.py}), so the conditions split by $N$.
\end{proof}

\begin{remark}[Faithfulness for the specializations]
For $\mathcal G_{\mathrm{background}}$ the coefficient class is $\mathbb Q[\sigma\text{-jet}(y)]\otimes\mathbb Q[w^1]\chi(w)$.
Diagonal evaluation $F\mapsto F(w,w)$ is injective on this class: at a point $w$ with $\chi(w)\ne0$, the $\sigma$-jet at $w$ ranges over all values independently of $w^1$, because $\sigma$ is arbitrary.
The induction above therefore applies with $U_N^{\mathrm{background}}$ computed from the specialized expansion.
The same holds for the time-only class, whose remaining symbols are independent, and for the plateau class, which is a sector restriction.
For $\mathcal G_{\mathrm{fixed}}$ the coefficients are fixed functions (the imported $\sigeps$, $\chiI$).
In general one only has $\ker\mathcal G_{\mathrm{formal}}\subseteq\ker\mathcal G_{\mathrm{fixed}}$.
Equality with the formal kernel is not automatic from formal $D\ge1$ vanishing; it requires a separate injectivity analysis.
Stage~2 records sufficient criteria (Proposition~\ref{prop:fixed-crit}) and a region ledger (Proposition~\ref{prop:fixed-regions}); the time-only graph map being injective does not by itself settle the time-only collapse/Helmholtz problem; that problem is settled in Section~\ref{sec:to-timeonly}, the transfer to the concrete XII-C ramp is Proposition~\ref{prop:to-actual} (explicit hypotheses), and regions where the actual $\chiI$ is non-constant remain Unresolved.
\end{remark}

\begin{theorem}[Ambient freedom versus graph-visible collision freedom]
\label{thm:M3}
These are reported exact ranks, cross-checked mod $p$ (Lemma~\ref{lem:M3comp}):
\begin{center}
\begin{tabular}{lcccc}
\hline
Map / variant & $\dim U_N$ & $\dim\operatorname{im}$ & $\dim\ker$ & on $\Creg$: im / ker\\
\hline
$\mathcal G_{\mathrm{formal}}$ & $(1,4,10,20,35)$ & $3321$ & $0$ & $3150/0$\\
time-only (formal) & $(1,4,10,20,35)$ & $3321$ & $0$ & $3150/0$\\
$\mathcal G_{\mathrm{background}}$ & $(1,4,10,20,10)$ & $1571$ & $1750$ & $1400/1750$\\
time-only + background & $(1,4,10,16,10)$ & $1431$ & $1890$ & $1260/1890$\\
plateau ($J_3$ only) & $(0,0,10,20,35)$ & $3300$ & $21$ & $3150/0$\\
$\mathcal G_{\mathrm{fixed}}$ & --- & --- & --- & Criteria only; Unresolved (Stage~2)\\
\hline
\end{tabular}
\end{center}
Injectivity is claimed only for $\mathcal G_{\mathrm{formal}}$ (and its time-only variant).
\end{theorem}

\begin{lemma}[Reported computation for M.3]
\label{lem:M3comp}
```

\texttt{contact_space.py} exits with status 0.
It records the table of Theorem~\ref{thm:M3}, the disjointness of monomials across N, the vectors u_m spanning U_N, and certified annihilators: for every annihilator vector x of U_N, $\sum_bx_b\hat{\bar a}_b=0$ exactly (\texttt{graph_contact_map.txt}).
\end{lemma}

%==
\section{M.4 Contact collapse and Helmholtz conditions}
\label{sec:m4}

\begin{definition}[Contact collapse]
\label{def:collapse}
For $C\in\Ccoh$, with the physical reduced coefficient $\bar a_b$ of Definition~\ref{def:Gmaps},
\[
E_C^{\mu\nu}(y)=\sum_{N,b,\gamma}C_{N,b,\gamma}\,\partial_w^\gamma\bigl(\bar a_b^{\mu\nu}(y,w;h)\bigr)\Big|_{w=y}
=\int dw\;\mathcal G(C)^{\mu\nu}(y,w).
\]
There is no second χI outside $\bar a_b$: the single switch factor already sits inside $\bar a_b=\chiI\,\hat{\bar a}_b$.
The second equality holds because $\int dw\,F(y,w)(\partial^\gamma\delta)(y-w)=(\partial_w^\gamma F)(y,y)$ and $\mathcal G(C)=\sum C\,\bar a\,\partial^\gamma\delta$.
∂_w^γ distributes by Leibniz over the factors of $\bar a_b=\chiI(w)\,\hat{\bar a}_b(y,w;h)$ (the h-jet and χI); the $\sigeps$-jet sits at y and is not hit (XVI-L, prop:Ljet(2)).
$\Coll:C\mapsto E_C$ maps into local tensors that are linear in the h-jet, with coefficients that are monomials in the external jets.
Four computational readings are used, together with the Stage~2 fixed-data region ledger:
\begin{itemize}
\item \emph{formal}: $\sigeps$- and χI-jets are independent symbols;
\item \emph{χ-plateau}: $\chiI\equiv1$ locally and $\sigeps$ formal;
\item \emph{preparation plateau}: $\sigeps\equiv1$ locally and χI formal (variable switch; only J_3 survives);
\item \emph{double plateau}: $\chiI\equiv1$ and $\sigeps\equiv1$ locally, so only J_3 terms survive and the coefficients are constant.
\end{itemize}
\end{definition}

\begin{remark}[Single switch; preferred presentation]
\label{rem:singleswitch}
The preferred three-equation form keeps the physical coefficient $a_\lambda=\chiI\hat a_\lambda$ (XVI-L style) and the reduced $\bar a_b=\sum_\lambda R_{\lambda b}a_\lambda$, and collapses by $\partial^\gamma\bar a_b$ with no further χI outside $\bar a_b$.
Equivalently one may write everything in switch-free hats with a single explicit χI in $\mathcal G$ and in $E_C=\partial(\chiI\hat{\bar a})$; both presentations are the same if and only if exactly one overall switch factor is used.
Earlier drafts that wrote $a_\lambda=\chiI\hat a_\lambda$ \emph{and} a second χI in the collapse $\partial(\chiI\,\bar a)$ were double-counting in prose.
The recorded computation uses a \emph{single} switch factor: \texttt{graph_general_table.txt} stores $\hat a$ with χI suppressed, and \texttt{m4_common.collapse_column} applies Leibniz once to $\partial^\gamma(\chiI\hat{\bar a})=\partial^\gamma\bar a$.
Certificates A/B/B2 already used one switch factor; the numerical collapse matrices are unchanged by this notation alignment.
\end{remark}

\begin{lemma}[Joint derivative budget]
\label{lem:joint}
If r_h of the $|\gamma|$ collapse derivatives hit the metric coefficient and r_χ hit the switch, then $(2-v+r_h)+(2-s)+r_\chi\le4-N+|\gamma|\le4$.
Consequently $D:=4-N+|\gamma|$, the total derivative count on $(h,\sigeps,\chiI)$, grades $\Coll$.
\end{lemma}

\begin{proof}
$r_h+r_\chi=|\gamma|\le N$; add $4-s-v=4-N$.
\end{proof}

\begin{proposition}[Collapse rank]
\label{prop:collinj}
\begin{enumerate}
\item In the formal reading, $\Coll$ is injective on $\Ccoh$.
\item In the χ-plateau reading, $\Coll$ is injective (reported computation).
\item In the double-plateau reading, $\dim\ker\Coll=22$ (with 4 at $D=3$ and 18 at $D=4$), $\dim\operatorname{im}\Coll=3299$, and $\ker\Coll\cap\Creg=0$ (reported computation).
\end{enumerate}
Injectivity of $\mathcal G$ does not by itself imply injectivity after collapse.
In the plateau reading, $\ker\mathcal G\subseteq\ker\Coll$ grows from 21 to 22.
\end{proposition}

\begin{proof}
(1) Factor $\bar a_b=\chiI\,\hat{\bar a}_b$ and expand the Leibniz rule for $\partial_w^\gamma\bar a_b$:
write $E_C=\sum_\rho\chi_\rho F_\rho$ with $\chi_\rho=\partial^\rho\chiI$ and
$F_\rho=\sum_{\gamma\ge\rho}\binom{\gamma}{\rho}\sum_{N,b}C_{N,b,\gamma}\,\partial_h^{\gamma-\rho}\hat{\bar a}_b$, where ∂_h acts on the h-jet only (the displayed F_ρ use the switch-free $\hat{\bar a}_b$).
In the formal reading the χ_ρ are independent, so $E_C=0$ forces $F_\rho=0$ for all ρ.
Proceed by descending induction on k.
If $C_{\cdot,\cdot,\gamma}=0$ for $|\gamma|>k$, then for $|\rho|=k$ one has $F_\rho=\sum_{N,b}C_{N,b,\rho}\hat{\bar a}_b=0$.
Since $U_N=\mathcal H_{4+N}$ (Theorem~\ref{thm:M3}) and monomials of different N are disjoint, $C_{\cdot,\cdot,\rho}=0$.

```
(2), (3): Lemma~\ref{lem:M4comp}.
\end{proof}

\begin{definition}[Formal adjoint and self-adjoint subspace]
\label{def:adjoint}
Write $E^{\mu\nu}[h]=\sum_{a\le b,\beta}K^{\mu\nu}_{ab,\beta}\,\partial^\beta h_{ab}$, where $K$ depends on the external jets according to the reading.
Under Gate~V the formal adjoint is
\[
E^{\dagger\,ab}[t]=\frac1{w_{ab}}\sum_{\mu\le\nu,\beta}(-\partial)^\beta\bigl(w_{\mu\nu}K^{\mu\nu}_{ab,\beta}\,t_{\mu\nu}\bigr),
\]
with derivatives hitting the external coefficients retained (Leibniz on $\sigeps$- and $\chiI$-jets, unless these are constant in the reading).
Then $\int\langle t,E[h]\rangle=\int\langle E^\dagger[t],h\rangle$ for compactly supported $t$.
The \emph{self-adjoint subspace} is $\mathrm{SA}:=\{C\in\Ccoh:E_C=E_C^\dagger\}$, as an identity in formal jets.
By Lemma~\ref{lem:joint}, $\mathrm{SA}=\bigoplus_D\mathrm{SA}_D$.
\end{definition}

\begin{lemma}[Helmholtz lemma for linear operators]
\label{lem:helm}
Independent of Input~(S-bv).
Let $E$ be a linear differential operator $h\mapsto E[h]$ with fixed smooth (or formal-jet) coefficients.
Let $\mathcal L_E:=\tfrac12\langle h,E[h]\rangle$ with Gate~V weights, and let $\mathrm{EL}$ denote the Gate~V Euler operator.
Then:
\begin{enumerate}
\item $\mathrm{EL}(\mathcal L_E)=\tfrac12(E+E^\dagger)$;
\item $E$ is the Euler derivative of \emph{some} local Lagrangian if and only if $E=E^\dagger$, and then $E=\mathrm{EL}(\mathcal L_E)$, i.e.\ $E=\delta I_E/\delta h$ with $I_E[h]=\tfrac12\int\langle h,E[h]\rangle$, modulo divergences and field-independent terms.
\end{enumerate}
\end{lemma}

\begin{proof}
(1) For compactly supported $t$: $\frac{d}{d\epsilon}\big|_0\int\mathcal L_E[h+\epsilon t]=\tfrac12\int\langle t,E[h]\rangle+\tfrac12\int\langle h,E[t]\rangle$.
Integrating by parts, with the boundary terms vanishing because $t$ has compact support and derivatives hitting the external coefficients retained, the second term equals $\tfrac12\int\langle t,E^\dagger[h]\rangle$.
Hence $\int\langle t,\mathrm{EL}(\mathcal L_E)\rangle=\int\langle t,\tfrac12(E+E^\dagger)[h]\rangle$ for all $t$.
(2) If $E=E^\dagger$, then (1) gives $E=\mathrm{EL}(\mathcal L_E)$.
Conversely, if $E=\mathrm{EL}(\mathcal L)$ for some local $\mathcal L$, the Fr\'echet derivative $D_E$ is formally self-adjoint (Helmholtz conditions; Olver, Thm.~5.92).
Since $E$ is linear, $D_E=E$, so $E=E^\dagger$.
\end{proof}

Thus $\mathrm{SA}$ is the preimage of the variational tensors, and $\Coll(\mathrm{SA})=\mathcal T_{\mathrm{variational}}\cap\operatorname{im}\Coll$ at this order.

\begin{theorem}[Variational subspace of the contact collapse]
\label{thm:M4}
Assume Gate~V. These are reported exact computations (Lemma~\ref{lem:M4comp}):
\begin{center}
\small
\begin{tabular}{lccccc}
\hline
Reading & $\ker\Coll$ & $\mathrm{SA}$ & $\mathrm{SA}_D$, $D=0..4$ & var.\ image$^\ast$ & $\mathrm{SA}\cap\Creg$\\
\hline
formal & $0$ & $26$ & $(26,0,0,0,0)$ & $26$ & $26$\\
$\chi$-plateau & $0$ & $26$ & $(26,0,0,0,0)$ & $26$ & $26$\\
prep.\ plateau$^\dagger$ & $21$ & $47$ & $(26,0,0,4,17)$ & $26$ & $=(26,0,0,0,0)$ & (D=0: $26$)\\
double plateau & $22$ & $1347$ & $(26,0,271,12,1038)$ & $1325$ & $=(26,0,271,8,1020)$ & $1208$\\
time-only$^\ddagger$ & $0$ & $26$ & $(26,0,0,0,0)$ & $26$ & $26$\\
time-only, $\chiI\equiv1$$^\ddagger$ & $0$ & $572$ & $(26,0,156,0,390)$ & $572$ & $572$\\
fixed data & \multicolumn{5}{c}{Criteria only; actual dimensions Unresolved (Stage~2; Prop.~\ref{prop:fixed-regions})}\\
\hline
\end{tabular}
\normalsize

$^\ast$var.\ image $=\dim(\mathcal T_{\mathrm{variational}}\cap\operatorname{im}\Coll)=\dim\mathrm{SA}-\dim\ker\Coll$; all entries are dimensions.
$^\dagger$universal reading with formal $\chiI$-jets (exact for that reading; not a statement about the actual fixed $\chiI$).
$^\ddagger$universal time-only readings, $\sigeps=\sigma(t)$ with formal $\sigma^{(k)}$ (SA $=\ker H_t$; Section~\ref{sec:to-timeonly}).
\end{center}
In particular:
\begin{enumerate}
\item[(i)] \textbf{Formal-reading vanishing (not a global obstruction).} Under Gate~V, the formal universal-jet collapse image of this selected graph contains no nonzero variational tensor of total derivative degree $D\ge1$.
Every variational collapsed tensor in that image is an ultralocal multiplication operator $E^{\mu\nu}=\chiI\sigeps\sum_{a\le b}K^{\mu\nu}_{ab}h_{ab}$ with the Gate~V weighted symmetry
$w_{\mu\nu}K^{\mu\nu}_{ab}=w_{ab}K^{ab}_{\mu\nu}$
```

```
(a $26$-dimensional family, all in $\Creg$, all from $N=4$, $\gamma=0$).
(Certificate~A checks $w_{00}K^{00}_{02}=-56/3=w_{02}K^{02}_{00}$; the index-reversed form fails.)
The matched functionals are $I_C=\tfrac12\int\chiI\sigeps\langle h,Kh\rangle$.
This is a statement about the selected two-edge graph in the formal reading; it does not claim absence of derivative counterterms in a completed gravitational sector, failure for every fixed preparation, or impossibility after adding other diagrams.
\item[(ii)] The $\chi$-plateau reading has the same $26$-dimensional SA basis.
\item[(iii)] In the double-plateau reading, derivative counterterms do occur: $271$ second-order, $8$ third-order, and $1020$ fourth-order variational directions in the image (variational per-$D$ dimensions $(26,0,271,8,1020)$).
\end{enumerate}
\end{theorem}

\begin{remark}[Eight third-order double-plateau variational directions]
\label{rem:thirdorder}
In the double-plateau reading the SA subspace has dimension $12$ at total derivative degree $D=3$, of which $4$ lie in $\ker\Coll$ (hence contribute $0$ to the variational image) and $8$ survive in $\mathcal T_{\mathrm{variational}}\cap\operatorname{im}\Coll$.
Each of the eight image directions collapses to a pure third-order operator on the residual field $h$ (every monomial in the collapsed tensor has field multi-index of order exactly $3$; no undifferentiated or second-order $h$ terms).
They arise from $N=2$ coherent contact assignments with $|\gamma|=1$ (and linear combinations thereof), not from $N=4$ ultralocal potentials.
\textbf{They are not gravitational scalar counterterms.}
In particular they are not Euler tensors of $\int R$, $\int R^2$, $\int\mathrm{Ric}^2$, or of any other Lorentz-scalar curvature polynomial at this order: those live in the $D=0$ ultralocal block or the $D=2$/$D=4$ matched curvature-squared sector of Proposition~\ref{prop:match}.
The eight directions are variational collision tensors of total derivative degree $3$ inside the selected double-plateau image; their functional interpretation (if any) is deferred to the M.5 ledger and is not claimed here.
Exact sparse SA supports are recorded in \texttt{helmholtz\_certificates.txt} (double\_plateau, $D=3$).
\end{remark}

\begin{remark}[Explicit matched functional; certificates]
\label{rem:cert}
Certificate A, in the formal reading, is the single coordinate $C[N=4,\,b=\{(),0222\},\,\gamma=()]=1$. It gives
\[
E^{00}=-\tfrac{56}3\chiI\sigeps h_{02},\quad E^{02}=-\tfrac{28}3\chiI\sigeps(h_{00}+h_{22}),\quad E^{22}=-\tfrac{56}3\chiI\sigeps h_{02},
\]
with all other components zero, and
\[
I_C=-\tfrac{56}3\int d^4y\;\chiI\sigeps\,h_{02}\,(h_{00}+h_{22}).
\]
The Gate~V Euler derivative of $I_C$ is computed by an independent sympy total-derivative route (\texttt{jet\_euler.py}) and equals $E_C$ exactly.
Certificates B ($D=4$) and B2 ($D=2$), in the double plateau, are $C[N=4,\{(),0222\},\gamma=0002]$ and $C[\ldots,\gamma=02]$.
For example, B gives $E^{00}=-\tfrac{56}3\partial_0^3\partial_2h_{02}$, $E^{02}=-\tfrac{28}3\partial_0^3\partial_2(h_{00}+h_{22})$, $E^{22}=-\tfrac{56}3\partial_0^3\partial_2h_{02}$, and $I_C=-\tfrac{28}3\int(h_{00}\partial_0^3\partial_2h_{02}+\partial_0^3\partial_2h_{00}\,h_{02}+h_{02}\partial_0^3\partial_2h_{22}+\partial_0^3\partial_2h_{02}\,h_{22})$.
For a non-self-adjoint unit vector (negative control C) the Euler derivative of $I_C$ differs from $E_C$.
All certificates are in \texttt{helmholtz\_certificates.txt}.
\end{remark}

\begin{proposition}[Matching probes]
\label{prop:match}
Assume Gate~V.
Membership in a linear space does not depend on normalization, so the target normalizations below only fix the preimage coefficients, which are not Wilson coefficients.
Targets, at quadratic order on the flat background, with $R_{\mathrm{lin}}=\partial^\mu\partial^\nu h_{\mu\nu}-\square h$ and $R^{\mathrm{lin}}_{\mu\nu}=\tfrac12(\partial^\rho\partial_\mu h_{\rho\nu}+\partial^\rho\partial_\nu h_{\rho\mu}-\square h_{\mu\nu}-\partial_\mu\partial_\nu h)$:
\[
P_1=\int w\,h_{\mu\nu}h^{\mu\nu},\ \ P_2=\int w\,h^2,\ \ \mathrm{EH}=\int h^{\mu\nu}G^{\mathrm{lin}}_{\mu\nu},\ \ R2=\int R_{\mathrm{lin}}^2,\ \ RC2=\int R^{\mathrm{lin}}_{\mu\nu}R_{\mathrm{lin}}^{\mu\nu}.
\]
Here the scalar weight is
$w=\chiI\sigeps$ in the formal reading,
$w=\sigeps$ in the $\chi$-plateau reading, and
$w=1$ in the double-plateau reading.
$\int\mathrm{Riem}_{\mathrm{lin}}^2$ is not separate: at quadratic order $\int(\mathrm{Riem}^2-4\mathrm{Ric}^2+R^2)_{\mathrm{lin}}$ is a total derivative.
\begin{enumerate}
\item In all three readings at $D=0$: $\mathrm{span}\{E_{P_1},E_{P_2}\}\cap\operatorname{im}\Coll$ is one-dimensional, spanned by the Euler tensor of $P_1-\tfrac{89}{140}P_2$.
The preimage is the single coordinate $C[N=4,\{33,33\},()]=3/70\in\Creg$.
Neither $P_1$ nor $P_2$ alone, nor the Fierz--Pauli combination $P_1-P_2$, is an image.
\item \textbf{Non-membership (double plateau).} Under Gate~V, the specified quadratic Einstein--Hilbert Euler tensor $E_{\mathrm{EH}}\propto G^{\mathrm{lin}}$ is not in the selected double-plateau collapse image ($D=2$), either in $\Ccoh$ or in $\Creg$.
\item \textbf{Constructed (double plateau).} $E_{R2}$ and $E_{RC2}$ are each collapse images ($D=4$), with explicit exact preimages in $\Ccoh$ ($286$ nonzero coefficients each).
In $\Creg$ only the single combination $R2-\tfrac{188813}{88456}RC2$ is an image.
```

```
Matching $R^2$ or $\mathrm{Ric}^2$ individually therefore requires coherent changes on the $\mathcal V_{\mathrm K}$ block, which do not preserve regression.
\item In the formal reading no target with $D\ge1$ is an image of a variational collapse (Theorem~\ref{thm:M4}(i)).
The double-plateau matchings hold locally on the open set where $\sigeps\equiv1$ and $\chiI\equiv1$; they fail once switch and preparation derivatives are retained.
This is a separation of plateau matching from full switched matching.
In the time-only reading the weighted $R^2$ and $\mathrm{Ric}^2$ Euler tensors are not images either, even with $\chiI\equiv1$ (Proposition~\ref{prop:to-match}(3)): the double-plateau matching does not survive the time-only reading.
\end{enumerate}
\end{proposition}

\begin{proof}
For each target $T$: $T$ is self-adjoint (checked).
Membership is decided by the exact verified kernel of the matrix whose columns are the $\Coll$ columns of the $(D,\text{parity }0)$ block, augmented by $T$.
Each preimage is reassembled and compared with $T$ exactly; \texttt{regression\_M.py} re-verifies the $R2$ preimage with an independently coded $R_{\mathrm{lin}}$.
Targets are Lorentz scalars, hence of reflection parity $0$, so only parity-$0$ blocks contribute.
The target construction is checked by the linearized Bianchi identity in jets.
Audit (Proposition~\ref{prop:M7-id}): $E_{R2}$ and $E_{RC2}$ are rebuilt independently from the metric linearization and coincide exactly with the targets; both targets and the certified preimages (after $\Coll$) satisfy $\partial_\mu E^{\mu\nu}=0$.
\end{proof}

\begin{lemma}[Reported computation for M.4]
\label{lem:M4comp}
\texttt{contact\_collapse.py}, \texttt{helmholtz\_check.py} and \texttt{matching\_probe.py} exit with status $0$.
Kernels are computed exactly over $\mathbb Q$ in each block of $(D,\text{reflection parity})$; there are $40$ blocks, of maximal size $292$.
The procedure has three steps: rows independent mod $p$ are selected; the exact nullspace of the selected rows is computed; and every candidate kernel vector is verified exactly against \emph{all} rows.
Since $\ker(\text{all rows})\subseteq\ker(\text{selected})$, verified equality is exact.
The adjoint implementation is unit-tested against Euler derivatives of random quadratic Lagrangians computed by the independent route.
Archives: \texttt{collapse\_kernel\_certificates.txt}, \texttt{helmholtz\_certificates.txt}, \texttt{matching\_certificates.txt}.
The full collapse matrices are regenerated deterministically by the scripts and are not stored; they have about $6.7\times10^5$ (formal) and $1.5\times10^6$ (adjoint-defect) nonzero entries.
\end{lemma}

\paragraph{M.4 status.}
Constructed in the formal, $\chi$-plateau, preparation-plateau, double-plateau and time-only readings (Gate~S-bv certified): the collapse ranks, the SA subspaces, the variational images, the Helmholtz certificates, and the matching probes.
Under Gate~V: the formal collapse image has no variational $D\ge1$ tensors; the preparation-plateau variational image is likewise only the $26$-dimensional ultralocal family; the specified quadratic Einstein--Hilbert Euler tensor is not in the selected double-plateau collapse image.
Fixed-data: criteria, universal region comparisons and the time-only classification in Stage~2 (Propositions~\ref{prop:fixed-crit},~\ref{prop:fixed-regions}, Theorem~\ref{thm:to-timeonly}); actual fixed-data dimensions unresolved where $\chiI$ is non-constant; not assumed settled by formal $D\ge1$ vanishing.
Unresolved: physical normalization ($\kappa$, $\hbar$, $i$); actual fixed-data dimensions where $\chiI$ is non-constant; whether transition-region contributions near plateau edges can be absorbed by insertion or switch counterterms.

%=====================================================================
\subsection{Stage~2: fixed-data and region-wise readings}
\label{sec:m4-fixed}

Gate~V is unchanged.
The imported fixed external data are those of the XVI-L Scope box: the preparation $\sigeps$ is the fixed time-only ramp of XII-C (Section~sec:freeze), never varied and with no $\varepsilon\to0$; the interaction switch $\chiI\in C_c^\infty(M)$ is independent and compact.
The closed form of $\sigeps$ is the XII-C profile recalled in Lemma~\ref{lem:to-indep}, with $\varepsilon$ fixed but not numerically specified; the predecessors give no closed form of $\chiI$, and none is selected inside XVI-M.

\begin{proposition}[Sufficient criteria for fixed data]
\label{prop:fixed-crit}
Assume Gate~V.
Always $\ker\mathcal G_{\mathrm{formal}}\subseteq\ker\mathcal G_{\mathrm{fixed}}$ and $\ker\Coll_{\mathrm{formal}}\subseteq\ker\Coll_{\mathrm{fixed}}$ (specialization).
Let $U$ be an open set meeting the relevant support, and let $\mathrm{ev}_U$ be evaluation of formal external-jet monomials (the $\sigeps$- and $\chiI$-jets occurring) on the actual fixed data over $U$.
Consider the finite-dimensional coefficient spaces spanned by the external-jet coefficients occurring in
\begin{enumerate}
\item[(i)] the canonical graph map $\mathcal G$;
\item[(ii)] the collapse map $\Coll$;
\item[(iii)] the adjoint-defect map $C\mapsto E_C^\dagger-E_C$ (whose coefficients include the external-jet derivatives produced by integration by parts).
\end{enumerate}
If $\mathrm{ev}_U$ is injective on the space in (i), then $\ker\mathcal G_{\mathrm{fixed}}|_U=\ker\mathcal G_{\mathrm{formal}}=0$; if it is injective on the space in (ii), then $\ker\Coll_{\mathrm{fixed}}|_U=\ker\Coll_{\mathrm{formal}}=0$; if it is injective on the space in (iii), then $\mathrm{SA}_{\mathrm{fixed}}|_U=\mathrm{SA}_{\mathrm{formal}}$, the $26$-dimensional ultralocal family.
```

```
Condition (iii) is needed for the SA statement: injectivity in (i) or (ii) alone does not control the defect map.
\begin{itemize}
\item \emph{Time-only caveat.} The actual $\sigeps$ is a time-only ramp, so every spatial $\sigeps$-derivative vanishes identically and the full spacetime preparation jets are not independent; conditions (i)--(iii) cannot hold for the formal coefficient spaces.
A time-only comparison class (spatial $\sigeps$-jets set to zero, $\chiI$ formal) needs its own Helmholtz analysis, with its own collapse and adjoint-defect maps; injectivity of the time-only graph map alone is insufficient.
That analysis is carried out in Section~\ref{sec:to-timeonly} (Theorem~\ref{thm:to-timeonly}, Proposition~\ref{prop:to-chiplateau}); its transfer to the actual ramp is Proposition~\ref{prop:to-actual}, which has explicit hypotheses.
\item \emph{Universal versus actual.} Readings with formal $\chiI$-jets (formal; preparation plateau) are universal variable-switch readings: their dimensions are exact for that universal reading.
They coincide with the actual fixed $\chiI$ only under (i)--(iii) for the actual $\chiI$ on $U$, which are not verified here.
\end{itemize}
These are sufficient criteria, not a certification of the concrete XII-C ramp.
Formal universal-jet vanishing at $D\ge1$ does \emph{not} by itself settle any fixed-data case.
\end{proposition}

\begin{proposition}[Region ledger for fixed preparation/switch]
\label{prop:fixed-regions}
Assume Gate~V. Distinguish four regions.
\begin{enumerate}
\item \textbf{Full switched/prepared region.} Comparison classes: formal universal reading, and the time-only class, whose Helmholtz analysis is now exact (Theorem~\ref{thm:to-timeonly}). Status: sufficient criteria (Proposition~\ref{prop:fixed-crit}); on $(0,\varepsilon)\times\mathbb R^3$ with a universal switch the time-only table applies under the explicit hypotheses of Proposition~\ref{prop:to-actual}; actual dimensions where $\chiI$ is non-constant Unresolved.
\item \textbf{Preparation plateau with variable switch, universal reading.} On $\{\sigeps\equiv1\}$ with formal $\chiI$-jets (reading \texttt{prep\_plateau}): support kills $C_2$/$M_2$; $\dim\ker\Coll=21$ exactly; $\mathrm{SA}=(26,0,0,4,17)$ by $D$; the $D\ge1$ SA directions lie in $\ker\Coll$, so the variational image is exactly the $26$-dimensional ultralocal family. Exact for the universal reading; for the actual fixed $\chiI$ it holds under (ii)--(iii) only.
\item \textbf{Double plateau.} On $\{\sigeps\equiv1,\chiI\equiv1\}$: recover Theorem~\ref{thm:M4} exactly ($\dim\ker\Coll=22$, variational image $1325$).
\item \textbf{Transition regions} near plateau boundaries: exact dimensions Unresolved. Only the domain-size bound $0\le\dim\ker\Coll_{\mathrm{fixed}}\le3321$ is asserted. (The previously stated upper bound $22$ was unsupported: rank is only lower semicontinuous, so special points of the fixed data could have larger kernels. No sharper bound is proved.)
\end{enumerate}
Vanishing by support, additional relations among fixed coefficient functions, and genuine Gate~V self-adjointness are kept distinct: no concrete ramp identity enlarging $\ker\mathcal G_{\mathrm{fixed}}$ is certified.
\textbf{Status: fixed-data criteria, universal region comparisons and the time-only classification constructed; actual fixed-data dimensions unresolved where $\chiI$ is non-constant} (status table at the end of Section~\ref{sec:to-timeonly}).
Archives: \texttt{fixed\_data\_reading.py}, \texttt{fixed\_data\_reading.json}, \texttt{prep\_plateau\_sa.json}; \texttt{wpC/}.
\end{proposition}

%-----------------------------------------------------------------------
\subsection{Time-only universal Helmholtz classification (work package C)}
\label{sec:to-timeonly}

Everything in this subsection is relative to the current coherent presentation $\Ccoh$ of M.2 ($\dim\Ccoh=3321$, $\dim\Creg=3150$, K block $\mathcal C_{\mathrm K}=\{N\le2\}$ of dimension $171$); since Gate~S-bv is certified (Theorem~\ref{thm:wpA-Sbv}), no (S-bv) condition remains.
Gate~V is unchanged: $\eta=\mathrm{diag}(1,-1,-1,-1)$; $\int t_{\mu\nu}E^{\mu\nu}=\int\sum_{\mu\le\nu}w_{\mu\nu}t_{\mu\nu}E^{\mu\nu}$ with $w=1$ on the diagonal and $w=2$ off it; symmetric lowered $h$; $\sigeps$ and $\chiI$ are fixed under the variation, and their derivatives are retained in every integration by parts.
The collapse is the single-switch collapse of Definition~\ref{def:collapse}, $E_C=\partial_w^\gamma\bar a|_{w=y}$ with $a_\lambda=\chiI\hat a_\lambda$, and there is no second $\chiI$.
This subsection supplies the time-only analysis that Proposition~\ref{prop:fixed-crit} (time-only caveat) had left Unresolved.

\begin{definition}[Time-only formal reading and the specialization $S_t$]
\label{def:to-timeonly}
In the \emph{time-only formal} reading, $\sigeps=\sigma(t)$, $t=x^0$, and the symbols $\sigma,\sigma',\sigma'',\dots$ are independent.
The $\chiI$-jets $\chi_\rho$ are independent spacetime jets, and $h$ is a general spacetime field with independent jets.
The time-only total derivatives are as follows: $D_0^{(t)}$ acts by Leibniz on $h$- and $\chi$-jets and maps $\sigma^{(k)}\mapsto\sigma^{(k+1)}$; $D_i^{(t)}$ ($i=1,2,3$) acts by Leibniz on $h$- and $\chi$-jets and annihilates every $\sigma^{(k)}$.
The specialization $S_t$ from the formal reading sends $\sigma_\alpha\mapsto\sigma^{(\alpha_0)}$ if $\alpha=(\alpha_0,0,0,0)$, and $\sigma_\alpha\mapsto0$ if $\alpha$ has a spatial index; it leaves the $h$- and $\chi$-jets unchanged.
Put $\Coll_t:=S_t\circ\Coll_{\mathrm{formal}}$ and $H_t(C):=S_t(E_C^\dagger-E_C)$.
The supplementary reading \emph{time-only $\chi$-plateau} additionally sets $\chiI\equiv1$, so that all $\chi$-derivatives vanish.
\end{definition}

\begin{lemma}[Intertwining]
\label{lem:to-intertwine}
Independent of Input~(S-bv).
For every polynomial $F$ in the formal jets, $S_t(D_\mu F)=D_\mu^{(t)}S_t(F)$ for $\mu=0,\dots,3$.
Hence $S_t\circ\dagger_{\mathrm{formal}}=\dagger_t\circ S_t$, where $\dagger_t$ is the Gate~V formal adjoint built from $D^{(t)}$, and $H_t(C)=E_{C,t}^{\dagger_t}-E_{C,t}$ with $E_{C,t}=\Coll_t(C)$.
\end{lemma}

\begin{proof}
```

```
Both sides are derivations that agree on generators.
On $h$- and $\chi$-jets $S_t$ is the identity.
For $\sigma_\alpha$: $D_\mu\sigma_\alpha=\sigma_{\alpha+e_\mu}$.
If $\mu=0$, then $\alpha+e_0$ is time-only exactly when $\alpha$ is, and both sides equal $\sigma^{(\alpha_0+1)}$ or $0$.
If $\mu=i\ge1$, then $\alpha+e_i$ is spatial, so the left side is $0$, and the right side is $D_i^{(t)}S_t(\sigma_\alpha)=0$.
The adjoint is the composite of multiplications and iterated total derivatives, so it commutes with the ring homomorphism $S_t$.
\end{proof}

\paragraph{Tests (\texttt{wpC/time\_only\_specialization\_tests.txt}).}
The intertwining is tested exactly on $400$ random rational polynomials (seed $20261006$) in all four directions, and for random iterated multi-indices $|\beta|\le4$.
It is tested by an independent sympy route with $\sigma=\sigma(x^0)$, $\chi(x)$ and $h(x)$ as functions, and on $960$ actual collapse and formal adjoint-defect expressions ($120$ contact variables $\times$ $4$ directions).
A negative control, a $D_1^{(t)}$ that differentiates $\sigma$, violates the identity.
\textbf{Route~1 versus route~2.}
Route~1 is $S_t\circ\Coll_{\mathrm{formal}}$, and $S_t$ applied to the formal adjoint defect.
Route~2 applies $S_t$ to the graph table and then builds the collapse by iterated single $\partial_w$ (Leibniz on $\chiI$ and $h$ only) and the adjoint by iterated $D^{(t)}$ on (coefficient $\times$ test field), with no binomial closed forms.
The two routes agree exactly over $\mathbb Q$ on all $3321$ columns of both maps: $446292$ nonzero entries for $\Coll_t$ and $573582$ for $H_t$.
In the $\chi$-plateau variant they also agree, with $91568$ and $130110$ entries.

\begin{theorem}[Time-only classification]
\label{thm:to-timeonly}
Assume Gate~V.
In the time-only formal reading the following dimensions are exact (Lemma~\ref{lem:to-comp}):
\begin{center}
\small
\begin{tabular}{c|c|cc|cc|cc|c}
\hline
$D$ & size & $k_C$ & $i_C$ & $k_H$ & $v$ & $k_C^{\mathrm{reg}}$ & $k_C^{\mathrm K}$ & $k_H^{\mathrm{reg}}$ & $k_H^{\mathrm K}$ & $v^{\mathrm{reg}}$\\
\hline
0 & 35 & 0 & 35 & 26 & 26 & 0 & 0 & 26 & 0 & 26\\
1 & 160 & 0 & 160 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
2 & 440 & 0 & 440 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
3 & 944 & 0 & 944 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
4 & 1742 & 0 & 1742 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\hline
$\Sigma$ & 3321 & 0 & 3321 & 26 & 26 & 0 & 0 & 26 & 0 & 26\\
\hline
\end{tabular}
\normalsize
\end{center}
Here $k_C=\dim\ker\Coll_t$, $i_C=\dim\operatorname{im}\Coll_t$, $k_H=\dim\ker H_t$, $v=\dim\Coll_t(\ker H_t)$ (variational image), $k^{\mathrm{reg}}_\bullet$, $k^{\mathrm K}_\bullet$ are the dimensions of the intersections with $\Creg$ and $\mathcal C_{\mathrm K}$, and $v^{\mathrm{reg}}=\dim\Coll_t(\ker H_t\cap\Creg)$ (regression-preserving variational image).
Thus $\Coll_t$ is injective for every $D$, and the time-only variational image is exactly the $26$-dimensional ultralocal family $E^{\mu\nu}=\chiI\sigeps\sum_{a\le b}K^{\mu\nu}_{ab}h_{ab}$ with $w_{\mu\nu}K^{\mu\nu}_{ab}=w_{ab}K^{ab}_{\mu\nu}$.
The family is supported on $N=4$, $\gamma=0$, lies entirely in $\Creg$ (so it preserves regression), and coincides with the formal SA family: the $26$ formal SA vectors lie in $\ker H_t$, and $\dim\ker H_t=26$.
There is no time-only variational direction of total derivative degree $D\ge1$ for a variable (formal) switch.
\end{theorem}

\begin{proposition}[Time-only $\chi$-plateau: new spatial variational directions]
\label{prop:to-chiplateau}
Assume Gate~V.
With $\chiI\equiv1$ and $\sigma=\sigma(t)$ formal, $\Coll_t$ is injective, $\ker H_t$ has dimensions $(26,0,156,0,390)$ by $D$ ($572$ in total), all of it lies in $\Creg$, $\ker H_t\cap\mathcal C_{\mathrm K}=0$, and $\Coll_t(\ker H_t)$ has the same dimensions.
Exactly,
\[
\ker H_t\big|_{D}=
\begin{cases}
\operatorname{span}\Bigl\{\displaystyle\sum_b K_b\,C[N{=}4,b,\gamma]:\
K\in\mathcal K_{\mathrm{SA}},\ \gamma\text{ purely spatial},\ |\gamma|=D\Bigr\},
& D=0,2,4,\\[0.6em]
0,& D=1,3,
\end{cases}
\]
so that for $D=0,2,4$ one has $E_C=\sigma(t)\,K\,\partial^\gamma h$ with $\gamma$ purely spatial; here $26\times6=156$ and $26\times15=390$, and the dimensions $(26,0,156,0,390)$ are unchanged.
Reason: for $E[h]=\sigma(t)K\,\partial_1 h$ one has $E^\dagger=-E$, since $\partial_1\sigma=0$ (odd spatial order is anti-self-adjoint); likewise for every odd-order purely spatial multi-index. Even-order purely spatial derivatives are formally self-adjoint and commute with $\sigma(t)$, so the even-$D$ directions are self-adjoint; they are variational only by virtue of the time-only structure.
The $D=0$ ultralocal family is the formal self-adjoint family already recorded above; the sentence that none is self-adjoint in the spacetime-formal $\chi$-plateau reading refers to the \emph{extra} $D=2,4$ directions (checked), so those are time-only-specific.
```

```
All even-$D$ directions remain self-adjoint in the double plateau.
For all $546$ directions with $D=2$ or $D=4$, $\mathrm{EL}(\tfrac12\langle h,E_C h\rangle)=E_C$ is verified by an independent sympy time-only jet route, and for samples also by \texttt{euler\_equations} on functions.
They are spatial-only and not Lorentz covariant, and they are not Euler tensors of $\int\sigma R$, $\int\sigma R^2$ or $\int\sigma\mathrm{Ric}^2$ (Proposition~\ref{prop:to-match}).
\end{proposition}

\begin{proposition}[Time-only matching probes]
\label{prop:to-match}
Assume Gate~V.
Take the weight $w=\chiI(x)\sigeps(t)$, or $w=\sigeps(t)$ in the $\chi$-plateau variant, with all preparation and switch derivatives retained in the Euler tensors.
\begin{enumerate}
\item $D=0$: $\mathrm{span}\{E_{wP_1},E_{wP_2}\}\cap\operatorname{im}\Coll_t$ is one-dimensional, spanned by $P_1-\tfrac{89}{140}P_2$, with preimage $C[N=4,\{33,33\},()]=3/70\in\Creg$ (regression-preserving), in both $\Ccoh$ and $\Creg$.
Neither $P_1$ nor $P_2$ is an image by itself.
The Helmholtz check $\mathrm{EL}(\tfrac12\langle h,E_C h\rangle)=E_C$ is verified by two independent routes (time-only jets; \texttt{euler\_equations} with $\sigma(x^0)$, $\chi(x)$, $h(x)$).
\item $D=2$: the weighted Einstein--Hilbert Euler tensor is not in $\operatorname{im}\Coll_t$, in either $\Ccoh$ or $\Creg$.
\item $D=4$: $\mathrm{span}\{E_{wR^2},E_{w\mathrm{Ric}^2}\}\cap\operatorname{im}\Coll_t=0$, in both $\Ccoh$ and $\Creg$.
\textbf{The double-plateau matchings of Proposition~\ref{prop:match}(3) and Theorem~\ref{thm:M5-plateau-R2} therefore do not survive the time-only reading:} once $\sigeps'(t)$, $\sigeps''(t),\dots$ are retained, the weighted $R^2$ and $\mathrm{Ric}^2$ Euler tensors are not images, even with $\chiI\equiv1$.
\item Divergence identities (upper-index $E$, no metric factor; time-only jets): $\partial_\mu E^{\mu\nu}=0$ for $\int wR_{\mathrm{lin}}^2$ and $\int w\mathrm{Ric}_{\mathrm{lin}}^2$; $\partial_\mu E_{\mathrm{EH}}^{\mu\nu}=(\partial_\mu w)\,G_{\mathrm{lin}}^{\mu\nu}$; nonzero for the ultralocal $P_1$.
\end{enumerate}
Each weighted target is checked to be self-adjoint under $\dagger_t$.
Membership is decided by the exact verified kernel of the $(D,\text{parity }0)$ block augmented by the targets.
\end{proposition}

\begin{lemma}[Linear independence of the actual ramp derivatives]
\label{lem:to-indep}
Independent of Input~(S-bv) and of Gate~V (analytic argument).
The actual $\sigeps$ is explicit: XII-C, Section~sec:freeze (imported by XVI-L, Scope), defines $\sigeps(t)=\rho(t/\varepsilon)$, where
$\rho(s)=0$ for $s\le0$, $\rho(s)=e^{-1/s}/(e^{-1/s}+e^{-1/(1-s)})$ for $0<s<1$, and $\rho(s)=1$ for $s\ge1$; here $0<\varepsilon<T$ is fixed but not numerically specified.
For every open interval $J\subset(0,\varepsilon)$ and every $m$, the functions $\sigeps,\sigeps',\dots,\sigeps^{(m)}$ are linearly independent over $\mathbb R$ on $J$.
The actual $\chiI$ is \emph{not} explicit: XVI-J/K/L give only $\chiI\in C_c^\infty(M)$, an independent compact switch.
\end{lemma}

\begin{proof}
Since $\sigeps^{(k)}(t)=\varepsilon^{-k}\rho^{(k)}(t/\varepsilon)$, it suffices to treat $\rho$ on $J'\subset(0,1)$.
Suppose $\sum_{k\le m}c_k\rho^{(k)}=0$ on $J'$ with $c_m\ne0$.
If $m=0$, then $\rho=0$ on $J'$, which is impossible because $\rho>0$ on $(0,1)$.
Otherwise, $\rho$ is real-analytic on $(0,1)$ (a quotient of analytic functions with positive denominator), so the identity holds on all of $(0,1)$.
Then $\rho|_{(0,1)}$ solves a constant-coefficient linear ODE of order $m$.
Since $\rho$ is smooth on $\mathbb R$ and vanishes for $s\le0$, $\rho^{(j)}(s)\to0$ as $s\to0^+$ for all $j$, so the solution has zero Cauchy data at $0$ and vanishes identically, a contradiction.
\end{proof}
A supplementary floating-point (60-digit) evaluation of the Wronskian of $\rho,\dots,\rho^{(4)}$ at $s=\tfrac12,\tfrac13,\tfrac17$ is nonzero ($\approx8.8\times10^{13}$, $2.6\times10^{15}$, $4.1\times10^{17}$); it is not used in the proof and is re-evaluated independently by \texttt{regression\_M.py}.

\begin{proposition}[Actual preparation versus universal switch; explicit hypotheses]
\label{prop:to-actual}
\emph{Status: a proposition with explicit hypotheses; it is not counted as a Constructed fixed-data result.}
Assume Gate~V and:
\begin{enumerate}
\item[(H1)] $\sigeps$ is the actual XII-C ramp of Lemma~\ref{lem:to-indep}, and $U$ is a nonempty open set with $U\subset(0,\varepsilon)\times\mathbb R^3$;
\item[(H2)] either the switch is treated \emph{universally} on $U$ (the identities $E_C=0$, resp.\ $E_C=E_C^\dagger$, are required for all $\chiI$), or $U\subset\{\chiI=1\}^\circ$ for the actual $\chiI$;
\item[(H3)] on $U$ every coefficient of $\Coll_t(C)$ and of $H_t(C)$ is a finite sum $\sum c_{k,\rho,\beta}\,\sigma^{(k)}(t)\chi_\rho(x)\,\partial^\beta h$ with $k\le4$, linear in the $\sigma$-jet (each graph coefficient carries exactly one $\sigma$-jet factor, Proposition~\ref{prop:M1}(3), and collapse and adjoint preserve this; Lemma~\ref{lem:joint}).
\end{enumerate}
Then, with a universal switch, $\Coll$ and $\mathrm{SA}$ on $U$ are exactly those of Theorem~\ref{thm:to-timeonly}; and if $U\subset\{\chiI=1\}^\circ\cap(0,\varepsilon)\times\mathbb R^3$, with actual $\sigeps$ and actual $\chiI$, they are those of Proposition~\ref{prop:to-chiplateau} (vacuous if that set is empty).
\end{proposition}

\begin{proof}
By (H3), vanishing of a coefficient for all $h$, and (in the universal-switch case) for all $\chiI$, forces $\sum_kc_{k,\rho,\beta}\sigma^{(k)}=0$ on the time projection of $U$, hence $c=0$ by Lemma~\ref{lem:to-indep}.
In the plateau case the $\chi_\rho$ with $\rho\ne0$ vanish identically on $U$, and the same argument applies.
\end{proof}
```

\begin{lemma}[Reported computation for the time-only classification]
\label{lem:to-comp}
\texttt{wpC/time_only_helmholtz.py} exits with status 0 (98 checks; total runtime about 155\,s on the box).
Kernels are computed exactly over $\mathbb Q$ in each of the 40 $(D,\text{reflection parity})$ blocks, for $\Ccoh$, $\Creg$ and $\mathcal C_{\mathrm K}$.
The grading is checked: every key has total derivative count D and the block parity, so the blocks have disjoint rows.
Rows independent mod $p=2^{61}-1$ are selected, their exact nullspace is computed, and every kernel vector is verified exactly against \emph{all} rows and re-verified by route~2.
\emph{Completeness:} the rank mod p of the selected integer rows equals $n-\#\ker$, which gives $\operatorname{rank}_{\mathbb Q}\ge n-\#\ker$; the verified independent kernel vectors give the reverse inequality.
Ranks mod 10^9+7 and 998244353 of all rows agree in every block; they are supplementary only.
$\ker\Coll_t\subseteq\ker H_t$ is checked, and $\dim\Coll_t(\ker H_t)$ is certified as a mod-p lower bound meeting the upper bound $\dim\ker H_t-\dim\ker\Coll_t$.
Archives: \texttt{time_only_collapse_certificates.json}, \texttt{time_only_sa_certificates.json}, \texttt{time_only_matching_certificates.json}, \texttt{time_only_summary.json}, \texttt{time_only_specialization_tests.txt}, \texttt{time_only_helmholtz_out.txt} (all in \texttt{wpC/}); their status strings ``CONDITIONAL on (S-bv)'' predate Theorem~\ref{thm:wpA-Sbv} and are superseded by it.
\end{lemma}

\paragraph{Status of the time-only analysis.}
\begin{itemize}
\item \textbf{Constructed} (Gate~V; on the current coherent presentation):
the intertwining lemma and its tests; exact route~1 / route~2 agreement;
the time-only classification (Theorem~\ref{thm:to-timeonly}): $\Coll_t$ injective, variational image $=$ the 26 ultralocal directions, all regression-preserving, no $D\ge1$ direction;
the time-only χ-plateau classification $(26,0,156,0,390)$ with its exact spatial-operator identification;
the matching probes and divergence identities (Proposition~\ref{prop:to-match});
the linear independence of $\sigeps^{(k)}$ on open subintervals of $(0,\varepsilon)$ (Lemma~\ref{lem:to-indep}).
\item \textbf{Proposition with explicit hypotheses} (not Constructed): the transfer to the actual XII-C ramp with a universal switch on $(0,\varepsilon)\times\mathbb R^3$, and to open parts of $\{\chiI=1\}^\circ$ in the slab $(0,\varepsilon)\times\mathbb R^3$ (Proposition~\ref{prop:to-actual}, hypotheses (H1)--(H3)).
\item \textbf{Unresolved}:
the actual fixed pair $(\sigeps,\chiI)$ wherever χI is non-constant, because χI is not given explicitly;
transition neighbourhoods of $\partial\{\chiI=1\}$ and of $t\in\{0,\varepsilon\}$.
\emph{Sufficient test (recorded):} on an open U, a full-rank exact evaluation matrix, on U, of the external-data coefficient functions $\sigeps^{(k)}\partial^\rho\chiI$ that occur in (i) the graph map, (ii) $\Coll_t$ and (iii) the adjoint-defect map H_t.
Full rank for (iii) gives $\mathrm{SA}_{\mathrm{fixed}}|_U=\ker H_t$.
The condition concerns the \emph{joint} family $\{\sigeps^{(k)}(t)\,\partial^\rho\chiI(x)\}$; it is not verified for any concrete χI, and it does not follow from separate linear independence of the $\sigeps^{(k)}$ (Lemma~\ref{lem:to-indep}) and of the $\partial^\rho\chiI$ (Remark~\ref{rem:to-joint}).
\item \textbf{Non-claims}:
the 546 spatial χ-plateau directions are not gravitational (Lorentz-scalar) counterterms and are not matched to any curvature functional;
no statement is made after other diagrams are added, and none about a completed gravitational sector;
no $\varepsilon\to0$ limit is taken;
preimage coefficients are not Wilson coefficients;
the double-plateau R^2/Ric^2 matchings (Theorem~\ref{thm:M5-plateau-R2}) are unchanged as double-plateau statements, but they do not admit the stated universal time-only completion (Proposition~\ref{prop:to-match}(3)); the actual-ramp conclusion holds under Proposition~\ref{prop:to-actual}.
\end{itemize}

\begin{remark}[Separate independence does not give joint independence (audit~3)]
\label{rem:to-joint}
Linear independence of $\{\sigeps^{(k)}\}_k$ on an interval and of $\{\partial^\rho\chiI\}_\rho$ on an open set does not imply linear independence of the products $\{\sigeps^{(k)}(t)\,\partial^\rho\chiI(x)\}_{k,\rho}$ on U.
Counterexample: if locally in the slab $\chiI(x)=\sigeps(t)$ (a switch that is a function of time on U), then
\[
\sigeps'\,\chiI-\sigeps\,\partial_0\chiI=\sigeps'\sigeps-\sigeps\sigeps'=0
\]
on U: a nontrivial linear relation between the products $\sigeps^{(1)}\cdot\chiI$ ($k=1$, $\rho=0$) and $\sigeps^{(0)}\cdot\partial_0\chiI$ ($k=0$, $\rho=e_0$), although $\{\sigeps,\sigeps'\}$ is independent (Lemma~\ref{lem:to-indep}) and so is $\{\chiI,\partial_0\chiI\}$.
The sufficient test above is therefore stated for the joint family.
The actual-ramp transfer of Proposition~\ref{prop:to-actual} is unaffected: with a universal switch the jets of χI at a point can be prescribed independently of $\sigeps$, and on $\{\chiI=1\}^\circ$ only $\rho=0$ occurs.
\end{remark}

\paragraph{Stage~2 / fixed-data status after work package C.}
\begin{center}
\small
\begin{longtable}{p{0.30\textwidth}p{0.35\textwidth}p{0.26\textwidth}}
\hline
Region / reading & Result & Status\\
\hline
\endhead
Formal universal reading & $\ker\Coll=0$; $\mathrm{SA}=(26,0,0,0,0)$ & Constructed (Theorem~\ref{thm:M4})\\
Time-only universal reading ($\sigma(t)$ formal, χI formal) & $\ker\Coll_t=0$ for all D; $\ker H_t=(26,0,0,0,0)$; variational image $=26$ ultralocals, regression-preserving & Constructed (Theorem~\ref{thm:to-timeonly})\\
Time-only, $\chiI\equiv1$ & $\ker H_t=(26,0,156,0,390)$; spatial-only, non-Lorentz, variational only by the time-only structure & Constructed (Proposition~\ref{prop:to-chiplateau})\\

```
Preparation plateau, universal $\chiI$ & $\ker\Coll=21$; $\mathrm{SA}=(26,0,0,4,17)$; variational image $26$ & Constructed for the universal reading (Proposition~\ref{prop:fixed-regions})\\
Double plateau & $\ker\Coll=22$; variational image $1325$ & Constructed (Theorem~\ref{thm:M4})\\
Actual $\sigeps$ on $(0,\varepsilon)\times\mathbb R^3$, universal switch & time-only table exact & Proposition~\ref{prop:to-actual} under (H1)--(H3); not counted as Constructed\\
Actual $\sigeps$ and $\chiI$ on open parts of $\{\chiI=1\}^\circ$ in the slab & time-only $\chi$-plateau table exact (vacuous if empty) & Proposition~\ref{prop:to-actual} under (H1)--(H3)\\
Actual $\chiI$ where non-constant & --- & Unresolved ($\chiI$ not explicit); sufficient test recorded\\
Preparation plateau with actual $\chiI$ & universal numbers under criteria (ii)--(iii) only & Unresolved for the actual $\chiI$\\
Transitions ($\partial\{\chiI=1\}$; $t\in\{0,\varepsilon\}$) & only $0\le\dim\ker\Coll_{\mathrm{fixed}}\le3321$ & Unresolved\\
\hline
\end{longtable}
\end{center}


\section{M.5 Counterterm-functionals ledger}
\label{sec:m5}

All statements in this section use Gate~V; Input~(S-bv) enters through the presentation of M.2 and is certified (Theorem~\ref{thm:wpA-Sbv}).
Four ledgers are maintained separately:
\begin{enumerate}
\item literal local representatives;
\item equality modulo total derivatives;
\item EOM reduction;
\item field-redefinition equivalence.
\end{enumerate}
\textbf{EOM and permitted field redefinitions are not defined in Target~I.}
Consequently ledgers~(3) and~(4) are left empty/Unresolved; work proceeds in~(1)--(2).
(Defining EOM/redefs is a Target~II / M.8 obligation.)

\paragraph{Ultralocal family and Lorentz scalars.}
The $26$-dimensional ultralocal SA family ($N=4$, $\gamma=()$, all in $\Creg$) consists of Gate~V-symmetric multiplication operators
$E^{\mu\nu}=\chiI\sigeps\sum K^{\mu\nu}_{ab}h_{ab}$, with matched functionals $I=\tfrac12\int\chiI\sigeps\langle h,Kh\rangle$.
The SA basis realises $26$ distinct nonzero $K$-patterns (\texttt{counterterm\_functionals.py}).
Inside $\mathrm{span}\{E_{P_1},E_{P_2}\}\cap\operatorname{im}\Coll$ there is a single Lorentz-scalar line $P_1-\tfrac{89}{140}P_2$, with preimage $C[N=4,\{33,33\},()]=3/70$.

\paragraph{$R^2$/$\mathrm{Ric}^2$ preimages and K-block changes.}
In the double plateau, $E_{R2}$ and $E_{RC2}$ are collapse images in $\Ccoh$ ($286$ nonzero coefficients each).
Their supports use $N=2$ slots, hence change the $\mathcal V_{\mathrm K}$ block and are \emph{not} regression-preserving.
In $\Creg$ only the combination $R2-\tfrac{188813}{88456}RC2$ is an image.
At quadratic order $\int(\mathrm{Riem}^2-4\mathrm{Ric}^2+R^2)_{\mathrm{lin}}$ is a total derivative, so $\int\mathrm{Riem}_{\mathrm{lin}}^2$ is not a separate probe (ledger~(2)).

\paragraph{Eight third-order directions.}
The eight double-plateau variational directions at $D=3$ (Remark~\ref{rem:thirdorder}) are recorded in ledger~(1) as Gate~V variational collision tensors.
They are \textbf{not} gravitational scalar counterterms.

\paragraph{Plateau completion gate.}
\begin{proposition}[Completion obstruction inside the selected formal/$\chi$-plateau image]
\label{prop:completion}
Under Gate~V: the formal and $\chi$-plateau variational images have $\mathrm{SA}_D=0$ for $D\ge1$ (Theorem~\ref{thm:M4}).
The double-plateau Euler tensors of $\int R_{\mathrm{lin}}^2$ and $\int\mathrm{Ric}_{\mathrm{lin}}^2$ have total degree $D=4$, hence cannot lie in those images.
Therefore the plateau matchings do \emph{not} complete inside the already-stated selected formal/$\chi$-plateau image.
Any enlarged ansatz (retained switch/preparation derivatives, or additional diagrams) is a \textbf{new domain}, not a completion established by M.4.
\end{proposition}

\begin{theorem}[Plateau curvature-square matching]
\label{thm:M5-plateau-R2}
\textbf{(Named theorem \texttt{PlateauCurvatureSquareMatching}.)}
Assume Gate~V.
\begin{itemize}
\item \textbf{Domain:} double-plateau reading of the selected XVI-L complete-source two-edge graph (open set where $\sigeps\equiv1$ and $\chiI\equiv1$ locally); coherent contact space $\Ccoh$ and regression-preserving subspace $\Creg$.
\item \textbf{Order:} quadratic in the residual field $h$ on the flat background; total derivative degree $D=4$.
\item \textbf{Pairing:} Gate~V.
\item \textbf{Equivalence:} literal local Euler-tensor representatives (ledger~(1)), with the optional quadratic Gauss--Bonnet identification of $\int\mathrm{Riem}_{\mathrm{lin}}^2$ in ledger~(2); EOM and field-redefinition equivalences are not used.
\end{itemize}
Then the Gate~V Euler tensors of $\int R_{\mathrm{lin}}^2$ and of $\int\mathrm{Ric}_{\mathrm{lin}}^2$ each lie in $\Coll(\Ccoh)$, with explicit exact preimages.
In $\Coll(\Creg)$ only the single combination $R2-\tfrac{188813}{88456}RC2$ is an image.
The specified quadratic Einstein--Hilbert Euler tensor is not an image.
```

```
These matchings are local to the double-plateau domain and do not complete inside the formal or $\chi$-plateau selected images (Proposition~\ref{prop:completion}).
\emph{Audit status.} The theorem still holds after the M.7 audit: the matching code uses neither the corrected divergence routine nor the equivariant vectors, the coefficient $188813/88456$ is unchanged on rerun, the targets agree with an independent metric-based construction, and the targets and their certified preimages satisfy the linear divergence identity $\partial_\mu E^{\mu\nu}=0$ (Proposition~\ref{prop:M7-id}).
\end{theorem}

\begin{remark}[Scope of \texttt{PlateauCurvatureSquareMatching}: time-only reading and scale tensor]
\label{rem:M5-timeonly}
Theorem~\ref{thm:M5-plateau-R2} is a double-plateau statement and remains true as such.
\textbf{It does not survive the time-only reading:} with weight $w=\chiI\sigeps(t)$, or $w=\sigeps(t)$ when $\chiI\equiv1$, and with $\sigeps'$, $\sigeps''$ retained, $\mathrm{span}\{E_{wR^2},E_{w\mathrm{Ric}^2}\}\cap\operatorname{im}\Coll_t=0$ in $\Ccoh$ and in $\Creg$, and $E_{w\mathrm{EH}}$ is not an image; only the ultralocal line $P_1-\tfrac{89}{140}P_2$ ($D=0$) survives (Proposition~\ref{prop:to-match}).
Moreover the controlled scale tensor of Section~\ref{sec:wpB} is not in the span of the M.5 curvature targets in any reading, because its fixed $D=4$ component already lies outside (Theorem~\ref{thm:wpB-dp}, Corollary~\ref{cor:wpB-MC}); it is recorded as an unmatched entry relative to the curvature ledger, not projected.
\end{remark}

\paragraph{M.5 status.}
Ledgers~(1)--(2) constructed from the M.4 probes; named theorem recorded (double-plateau domain only; not surviving the time-only reading, Remark~\ref{rem:M5-timeonly}); completion-gate obstruction recorded; ledgers~(3)--(4) Unresolved (definitions missing).
Archives: \texttt{counterterm\_functionals.py}, \texttt{matching\_matrix.*}, \texttt{matching\_certificates\_M5.txt}, \texttt{ledger\_*.json}, \texttt{ledger\_tables.txt}.


%==================================================================
\section{M.6 Dilation identity, residues, and matched scale variation}
\label{sec:m6}

This section imports the XVI-L scale family $\varpi_\mu$, $\mathfrak R_{\mathrm L,\mu}$, $A_\mu=\mu\,\mathrm d/\mathrm d\mu\,\mathfrak R_{\mathrm L,\mu}$, and the scale-variation coefficients of XVI-L,~L.6, without modifying XVI-L.
The term \emph{scaling residue} means the anomaly form $\mathcal A(u)=(E+4+N)\mathfrak R(u)$ (or its contact coefficients), as reserved in XVI-L,~rem:Lressretch.
Scale-variation coefficients are \textbf{not} physical gravitational beta functions (XVI-L,~rem:Lscale; Target~II unfinished).

\paragraph{Dilation identity for the XVI-L prescription.}
\begin{proposition}[Dilation identity for $W^{(\mu)}_\rho$ / zero-moment extensions]
\label{prop:M6-dil}
Let $\varpi_\mu(z)=\varpi((\mu/\mu_0)z)$ and let $W^{(\mu)}_\rho$, $w^{(\rho,\mu)}_\alpha$, $\mathfrak R_{\mathrm L,\mu}$ be as in XVI-L, Definition~def:RLmu.
Write $E=z^\rho\partial_\rho$.
\begin{enumerate}
\item For every $\rho$, $\mu\,\partial_\mu\varpi_\mu=E\varpi_\mu$.
\item For $\rho=0$ (so $N=0$, $w^{(0,\mu)}_0=\varpi_\mu$):
\[
\bigl\langle A_\mu(u),\varphi\bigr\rangle
=
-\varphi(0)\,\bigl\langle u,\,E\varpi_\mu\bigr\rangle,
\qquad
A_\mu(u)=-\bigl\langle u,\,E\varpi_\mu\bigr\rangle\,\delta.
\]
\item For $u\in\mathcal H_4$ homogeneous of degree $-4$ on $\mathbb R^4\setminus\{0\}$, the dilation defect satisfies
\[
\mathcal A(u):=(E+4)\mathfrak R_{\mathrm L,\mu}(u)=A_\mu(u).
\]
\end{enumerate}
\end{proposition}

\begin{proof}
(1) Chain rule: $\partial_\mu\varpi((\mu/\mu_0)z)=(z/\mu_0)\cdot(\nabla\varpi)((\mu/\mu_0)z)$, hence $\mu\partial_\mu\varpi_\mu=E\varpi_\mu$.
(2) For $\rho=0$, $\langle\mathfrak R_{\mathrm L,\mu}(u),\varphi\rangle=\langle u,\varphi-\varphi(0)\varpi_\mu\rangle$; differentiate in $\mu$ and use (1).
(3) $\langle(E+4)\mathfrak R,\varphi\rangle=-\langle\mathfrak R,E\varphi\rangle$.
Since $(E\varphi)(0)=0$, $W_0(E\varphi)=E\varphi$, so $\langle\mathfrak R,E\varphi\rangle=\langle u,E\varphi\rangle$.
Write $\varphi=W_0\varphi+\varphi(0)\varpi_\mu$ and $E\varphi=E(W_0\varphi)+\varphi(0)E\varpi_\mu$.
The continuous extension of $u$ to $\{\psi:\psi(0)=0\}$ obeys $\langle u,E(W_0\varphi)\rangle=0$ (homogeneity of degree $-4$; polar identity on the Euclidean sphere), whence $\langle\mathcal A(u),\varphi\rangle=-\varphi(0)\langle u,E\varpi_\mu\rangle=\langle A_\mu(u),\varphi\rangle$.
\end{proof}

\begin{remark}[General $N$]
\label{rem:M6-N}
For $u\in\mathcal H_{4+N}$ and $\rho=N$, differentiating the zero-moment formula yields the contact expansion
\[
A_\mu(u)=\sum_{|\alpha|\le N}a_\mu(u,\alpha)\,\partial^\alpha\delta,
\qquad
a_\mu(u,\alpha)=(-1)^{|\alpha|+1}\bigl\langle u,\,\mu\partial_\mu w^{(\rho,\mu)}_\alpha\bigr\rangle
```

```
\]
(with the pairing understood via $W^{(\mu)}_\rho$).
Equality $A_\mu=(E+4+N)\mathfrak R_{\mathrm L,\mu}$ for $N\ge1$, previously Unresolved here, is now \textbf{resolved}: the $\mu$-dependence of $M^{(\rho,\mu)}$ is accounted for by $\mu\,\mathrm dM^{-1}=DM^{-1}-M^{-1}D$, and $B_\mu=0$ for all $N\le4$ (Lemma~\ref{lem:wpB-dw}, Theorem~\ref{thm:wpB-B0}).
\end{remark}

\paragraph{Explicit $S^2$ residue.}
\begin{theorem}[$S^2$ scaling residue]
\label{thm:M6-S2}
By Theorem~\ref{thm:wpA-Sbv}, $S=c(Q-i0)^{-1}$ on $\mathbb R^4\setminus\{0\}$ with $c=1/(4\pi^2)$ and $Q=\eta(z,z)$.
Let $u=S^2=c^2(Q-i0)^{-2}\in\mathcal H_4$.
Then, for every $\mu>0$ and every admissible $\varpi$ with $\varpi(0)=1$,
\[
A_\mu(S^2)=r\,\delta,
\qquad
r=-2i\pi^2 c^2=-\frac{i}{8\pi^2}\neq0,
\]
and $\mathcal A(S^2)=(E+4)\mathfrak R_{\mathrm L,\mu}(S^2)=r\delta$.
The value of $r$ is independent of $\mu$ and of the radial profile of $\varpi$.
\end{theorem}

\begin{proof}
By Proposition~\ref{prop:M6-dil}, $r=-\langle S^2,E\varpi_\mu\rangle$.
In Euclidean polar coordinates $z=r\omega$ on the unit sphere $S^3$, the obstruction identity for a homogeneous density of degree $-4$ reduces $r$ to the spherical integral of the angular part of $S^2$.
By Theorem~\ref{thm:wpA-Sbv} that angular part is $c^2(Q-i0)^{-2}$.
Parametrizing $S^3$ by $\omega_0=\cos\chi$ and spatial radius $\sin\chi$, one has $Q=\cos(2\chi)$ and
\[
\int_{S^3}(Q-i0)^{-2}\,\mathrm d\sigma
=
4\pi\int_0^\pi\frac{\sin^2\chi}{(\cos(2\chi)-i0)^2}\,\mathrm d\chi
=
\pi\int_0^{2\pi}\frac{1-\cos u}{(\cos u-i0)^2}\,\mathrm d u.
\]
The $u$-integral equals $-2i\pi$ by the residue at the interior double pole $z_m=i(e-\sqrt{e^2+1})$ of the unit-circle integrand $2i(z-1)^2/(z^2-2iez+1)^2$ after $e\downarrow0$ (certificate: \texttt{scaling\_anomaly.py}).
Hence $\int_{S^3}(Q-i0)^{-2}\,\mathrm d\sigma=-2i\pi^2$ and $r=c^2(-2i\pi^2)$.
Nonvanishing uses $c\neq0$; with $c=1/(4\pi^2)$, $r=-2i\pi^2/(16\pi^4)=-i/(8\pi^2)$.
The side $Q-i0$ fixes the sign of the angular integral; the counterfactual $Q+i0$ would conjugate it and flip $r$ (Remark~\ref{rem:wpA-M6}).
Independence of the radial profile of $\varpi$ is the polar telescope $\int_0^\infty\partial_r\varpi_\mu(r\omega)\,\mathrm d r=-\varpi_\mu(0)=-1$.
\end{proof}

\begin{remark}[Status of the residue]
\label{rem:M6-Sbv}
Theorem~\ref{thm:M6-S2} uses Input~(S-bv), which is certified by Theorem~\ref{thm:wpA-Sbv}; the logic does not run the other way (the residue does not certify the gate), and matching played no role in the certification (Remark~\ref{rem:Sbv-status}).
The explicit value $r=-i/(8\pi^2)$ fills, as a result about XVI-L objects, the Unresolved stretch of XVI-L,~rem:Lressretch, for the $S^2$ kernel; XVI-L itself is not edited.
Higher graded residues ($N\ge1$) are computed in Section~\ref{sec:wpB}: exact and $\varpi$-independent at top order, $\varpi$-dependent at lower orders.
\end{remark}

\paragraph{The $N=0$ anomaly component in the coherent basis.}
\begin{definition}[$N=0$ anomaly component]
\label{def:M6-Canom}
The \emph{$N=0$ anomaly component} is the coherent contact vector $C_{\mathrm{anom}}\in\Ccoh$ with
\[
C_{\mathrm{anom}}\bigl[N=0,\,\{(),()\},\,\gamma=()\bigr]=r=-2i\pi^2 c^2=-\frac{i}{8\pi^2}
\]
($\dim\mathcal H_4=1$), and with all higher-grade coordinates ($N\ge1$) \emph{explicitly set to zero for this probe}.
Coherence of the $N=0$ coordinate follows from linearity of $A_\mu$ and uniqueness of the $N=0$ basis element.
$C_{\mathrm{anom}}$ is not the full graph scale variation; the full $C_{\mathrm{scale}}$, including all higher-grade coordinates, is treated in Section~\ref{sec:wpB}.
\end{definition}

\paragraph{Collapse and variational compatibility with the M.5 ledger.}
\begin{theorem}[Collapse of $C_{\mathrm{anom}}$; unmatched remainder retained]
\label{thm:M6-collapse}
Assume Gate~V. Let $C_{\mathrm{anom}}$ be as in Definition~\ref{def:M6-Canom} (unit $r=1$ for the linear probe; overall factor $r$ restores the physical coefficient).
\begin{enumerate}
\item \textbf{Formal / $\chi$-plateau.} $\Coll(C_{\mathrm{anom}})$ is nonzero (744 nonzero keys) and \emph{not} self-adjoint ($\mathrm{SA}$-defect nonempty). It is therefore \textbf{not} a nonzero variational tensor.
\item \textbf{Preparation / double plateau.} $\Coll(C_{\mathrm{anom}})=0$.
Structural reason: the $S^2$ coefficient $\hat a$ is supported only on $|\partial\sigeps|=2$ ($C_2$ sector); both plateaus set $\sigeps\equiv1$ and kill those derivatives.
```

```
\item \textbf{Matched projection (double plateau).} The orthogonal projection of $\Coll(C_{\mathrm{anom}})$ onto $\mathrm{span}\{E_{R2},E_{RC2}\}$ (the M.5 matched Euler tensors) is $(0,0)$, since the vector itself is $0$.
\item \textbf{Unmatched remainder.} In the double plateau the remainder is $0$.
In the formal reading the \emph{entire} non-self-adjoint $\Coll(C_{\mathrm{anom}})$ is retained as unmatched remainder.
No unmatched component is discarded and renamed ``the anomaly.''
\end{enumerate}
\end{theorem}

\begin{proof}
Exact collapse columns and SA defects: \texttt{scaling\_anomaly.py} (reads the same $\bar a$ and Collapse as M.4).
Linear independence of $E_{R2},E_{RC2}$ and vanishing of $\Coll(C_{\mathrm{anom}})$ in the double plateau give the projection claim.
\end{proof}

\begin{corollary}[Scope of the $N=0$ component relative to the M.5 ledger]
\label{cor:M6-compat}
The $N=0$ anomaly component does \emph{not} furnish a nonzero Gate~V-variational matched scale variation inside the double-plateau domain of Theorem~\ref{thm:M5-plateau-R2}: the $N=0$ component alone contributes no nonzero plateau matching.
In the formal/$\chi$-plateau readings it collapses to a non-variational tensor.
The full graph scale variation, including all higher-grade residues, is treated in Section~\ref{sec:wpB}; it leads to the graph-specific obstruction of M-C$_{\mathrm{curv}}$, while broad M-C is not achieved (Corollary~\ref{cor:wpB-MC}).
\end{corollary}

%-----------------------------------------------------------------------
\subsection{Scale derivative versus dilation defect at every grade (work package B)}
\label{sec:wpB}

\scopebox{Status (work package B).}{%
\textbf{Constructed} (with the certified $c=1/(4\pi^2)$ of Theorem~\ref{thm:wpA-Sbv}; Gate~V for (iv)):
(i) the complete $\mu$-derivative of the XVI-L subtraction formula, including $\mu\,\mathrm d M^{-1}=-M^{-1}(\mu\,\mathrm dM)M^{-1}=DM^{-1}-M^{-1}D$, at every order $\rho=N\le4$;
(ii) the identity $A_\mu=\mathcal A_\mu$, i.e.\ $B_\mu=0$, on every $b\in\mathcal B_N$, $N=0,\dots,4$, for every admissible $\varpi$;
(iii) the exact, $\mu$- and $\varpi$-independent top-order ($|\gamma|=N$) coefficients of $A_\mu(b)=\mathcal A_\mu(b)$, consistent with all $185$ relations and all labels;
(iv) the collapse, full adjoint defect, variationality and ledger membership of the assembled scale tensor in the formal, $\chi$-plateau, preparation-plateau and double-plateau readings, and (Corollary~\ref{cor:wpB-timeonly}) its non-variationality in the time-only readings.
In the double plateau, the cutoff-independent $D=4$ component is variational and Lorentz covariant, but not linearized-gauge invariant. The full scale tensor has cutoff-dependent lower-degree components and need not be variational.
\textbf{Proved obstructed:} M-C$_{\mathrm{curv}}$ (Definition~\ref{def:MCcurv}), graph-specific (Corollary~\ref{cor:wpB-MC}). \textbf{Not achieved (Unresolved):} broad M-C ($\varpi$ unspecified by XVI-L, thm:L6). \textbf{Satisfied:} M-C$_{\mathrm{data}}$.
\textbf{Unresolved:} the numerical values of the lower-order ($|\gamma|<N$) coefficients; they depend on the cutoff $\varpi$ through the functionals $\nu_k[\varpi]$, $k=1,\dots,4$, of \eqref{eq:wpB-nu}, and XVI-L does not fix $\varpi$ (the radial comparison is a separately named specialization).
\textbf{Non-claims:} no physical beta function; no claim about gravity generally; no orthogonal projection of an unmatched tensor; $B_\mu=0$ does not remove prescription dependence.}

\paragraph{Prescription (imported unchanged).}
XVI-L, Definition~def:RLmu: $\varpi_\mu(z)=\varpi(\lambda z)$, $\lambda=\mu/\mu_0$, $M^{(\rho,\mu)}_{\beta\alpha}=\partial^\beta(z^\alpha\varpi_\mu/\alpha!)(0)$,
$w^{(\rho,\mu)}_\alpha=\sum_\gamma((M^{(\rho,\mu)})^{-1})_{\gamma\alpha}z^\gamma\varpi_\mu/\gamma!$, $\langle\mathfrak R_{\mathrm L,\mu}(u),\varphi\rangle=\langle u,W^{(\mu)}_\rho\varphi\rangle$, $\rho=N$.
$A_\mu(u)=\mu\,\mathrm d_\mu\mathfrak R_{\mathrm L,\mu}(u)$; $\mathcal A_\mu(u)=(E+4+N)\mathfrak R_{\mathrm L,\mu}(u)$ with $E=z^\rho\partial_\rho$ and $\langle ET,\varphi\rangle=-\langle T,(E+4)\varphi\rangle$, so $\langle\mathcal A_\mu(u),\varphi\rangle=\langle\mathfrak R_{\mathrm L,\mu}(u),(N-E)\varphi\rangle$; $B_\mu=A_\mu-\mathcal A_\mu$.
Contacts: $T=\sum_\gamma t_\gamma\partial_\gamma\delta$, $\langle\partial_\gamma\delta,\varphi\rangle=(-1)^{|\gamma|}\partial^\gamma\varphi(0)$, $t_\gamma=(-1)^{|\gamma|}\langle T,w_\gamma\rangle$; these are the coordinates $C_{N,b,\gamma}$ of $\Ccoh$.

\begin{lemma}[Complete scale derivative of the subtraction functions]
\label{lem:wpB-dw}
For every $\rho\le4$, every $\varpi$ with $\varpi(0)=1$, and $D=\mathrm{diag}(|\beta|)$:
\[
\mu\,\partial_\mu M^{(\rho,\mu)}=[D,M^{(\rho,\mu)}],\qquad
\mu\,\partial_\mu (M^{(\rho,\mu)})^{-1}=-M^{-1}(\mu\,\partial_\mu M)M^{-1}=D M^{-1}-M^{-1}D,
\]
\[
\mu\,\partial_\mu w^{(\rho,\mu)}_\alpha=(E-|\alpha|)\,w^{(\rho,\mu)}_\alpha,\qquad
w^{(\rho,\mu)}_\alpha=\frac{z^\alpha}{\alpha!}\,T_{\rho-|\alpha|}\!\bigl[1/\varpi_\mu\bigr]\,\varpi_\mu ,
\]
with $T_k$ the Taylor polynomial of order $k$ at $0$. Equivalently $w^{(\rho,\mu)}_\alpha(z)=\lambda^{-|\alpha|}w^{(\rho)}_\alpha(\lambda z)$.
\end{lemma}
\begin{proof}
$M_{\beta\alpha}=\binom\beta\alpha\lambda^{|\beta|-|\alpha|}\partial^{\beta-\alpha}\varpi(0)$ (Leibniz), so $\mu\partial_\mu M=[D,M]$; differentiate $MM^{-1}=1$.
For the basis functions $f_\gamma:=z^\gamma\varpi_\mu/\gamma!$, $\mu\partial_\mu\varpi_\mu=E\varpi_\mu$ (Proposition~\ref{prop:M6-dil}(1)) and $Ez^\gamma=|\gamma|z^\gamma$ give $\mu\partial_\mu f_\gamma=(z^\gamma/\gamma!)E\varpi_\mu=(E-|\gamma|)f_\gamma$.
```

With $\mu\partial_\mu(M^{-1})_{\gamma\alpha}=(|\gamma|-|\alpha|)(M^{-1})_{\gamma\alpha}$ (the entries of $DM^{-1}-M^{-1}D$),
\[
\mu\partial_\mu w_\alpha=\sum_\gamma(M^{-1})_{\gamma\alpha}\bigl[(|\gamma|-|\alpha|)+(E-|\gamma|)\bigr]f_\gamma=\sum_\gamma(M^{-1})_{\gamma\alpha}(E-|\alpha|)f_\gamma=(E-|\alpha|)w_\alpha .
\]
The closed form holds because $z^\alpha T_k[1/\varpi_\mu]\varpi_\mu/\alpha!=z^\alpha/\alpha!+O(|z|^{\rho+1})$ and M is invertible.
Exact certificates (\texttt{wpB/higher_grade_scaling.py}, Section~A): symbolic jets for $\rho\le3$, two random rational jets for $\rho=3,4$, and an explicit non-radial $\varpi=e^{q}$ as functions, all $\rho\le4$; \texttt{regression_M.py} rechecks the matrix identities independently for $\rho\le2$ (symbolic) and $\rho=3$ (rational jets).
\end{proof}

\begin{theorem}[Scale derivative equals dilation defect at every grade]
\label{thm:wpB-B0}
For every $N\in\{0,\dots,4\}$, $b\in\mathcal B_N$, $\mu>0$ and admissible ϖ,
\[
A_\mu(b)=\mathcal A_\mu(b)=\sum_{|\gamma|\le N}a_{N,b,\gamma}(\mu)\,\partial_\gamma\delta,\qquad B_\mu(b)=0,
\qquad
a_{N,b,\gamma}(\mu)=(-1)^{|\gamma|+1}\bigl\langle b,(E-|\gamma|)w^{(N,\mu)}\bigr\rangle_\gamma .
\]
\begin{enumerate}
\item \emph{Top order} ($|\gamma|=N$): $a_{N,b,\gamma}=\frac{(-1)^N}{\gamma!}\int_{S^3}b\,\omega^\gamma\,\mathrm d\sigma$, independent of μ and ϖ; it is the intrinsic residue of b (common to all extensions of degree $4+N$).
\item \emph{Lower orders} ($k=N-|\gamma|\ge1$):
\begin{equation}
a_{N,b,\gamma}(\mu)=\frac{(-1)^{|\gamma|+1}}{\gamma!}\Bigl(\frac\mu{\mu_0}\Bigr)^{k}\bigl\langle b|_{S^3}\,\omega^\gamma,\nu_k[\varpi]\bigr\rangle_{S^3},
\qquad
\nu_k[\varpi](\omega)=k\!\int_0^\infty\! r^{-k-1}\bigl(T_k[1/\varpi]\varpi\,(r\omega)-1\bigr)\mathrm dr .
\label{eq:wpB-nu}
\end{equation}
\end{enumerate}
The identity $B_\mu=0$ is pure prescription calculus; the explicit top-order values below use Theorem~\ref{thm:wpA-Sbv}.
\end{theorem}
\begin{proof}
A_μ: differentiate $\langle b,W^{(\mu)}_N\varphi\rangle$; $\mu\partial_\mu w_\gamma$ vanishes to order N at 0 (since $\partial^\beta w_\gamma(0)=\delta^\beta_\gamma$ for all μ), so the pairing is the unique extension; Lemma~\ref{lem:wpB-dw}.
$\mathcal A_\mu$: by homogeneity $\mathcal A_\mu(b)$ is supported at 0 of order $\le N$; its coefficients are $(-1)^{|\gamma|}\langle\mathfrak R_{\mathrm L,\mu}b,(N-E)w_\gamma\rangle=(-1)^{|\gamma|}\langle\mathfrak R_{\mathrm L,\mu}b,(N-|\gamma|)w_\gamma-(E-|\gamma|)w_\gamma\rangle$; the first term is a zero moment, the second vanishes to order N. Hence the coefficients agree; $B_\mu=0$.
Polar coordinates, $r^{-|\gamma|}w_\gamma(r\omega)=\frac{\omega^\gamma}{\gamma!}T_k[1/\varpi_\mu]\varpi_\mu(r\omega)$ and one integration by parts give (1)--(2); for $k=0$ the radial integral telescopes to $-\varpi(0)=-1$.
Certificates: top order by two independent exact routes (contour residues; Hadamard finite part plus $i\pi\delta^{(m-1)}$ in $x=Q|_{S^3}$), equal on every slot; finite exact end-to-end tests with an explicit compactly supported non-radial ϖ for $N=0,1,2$ (Euclidean toy kernels, Section~B of the script).
\end{proof}

\paragraph{Exact top-order data.}
All top-order values are $i\pi^2c^2\times\mathbb Q$, and with the certified c the unit is $i\pi^2c^2=i/(16\pi^2)$ (the real, finite-part contributions vanish identically).
$N=0$: $a=-2i\pi^2c^2=-i/(8\pi^2)$, the M.6 residue r of Theorem~\ref{thm:M6-S2}. $N=1$: $a_{1,\{(),\mu\},\mu}=-i\pi^2c^2=-i/(16\pi^2)$, others 0.
Nonzero top slots per grade: $1,4,22,68,194$ of $1,16,100,400,1225$ (289 in total).
In the audited equivariant coordinates of M.7 ($C_{N,b,\gamma}=(-1)^{\gamma_1+\gamma_2+\gamma_3}[Q^k\square^k n_b]_\gamma$), the residue map is, in units $i\pi^2c^2=i/(16\pi^2)$,
$-2\phi_{00}$; $\tfrac12\phi_{00}$; $-\tfrac1{12}\phi_{00}-\tfrac1{48}\phi_{11}$; $\tfrac1{96}\phi_{00}+\tfrac1{384}\phi_{11}$; $-\tfrac1{960}\phi_{00}-\tfrac1{3840}\phi_{11}-\tfrac1{61440}\phi_{22}$ for $N=0,\dots,4$.
Tests where the hypotheses hold: Lorentz equivariance on all six generators (pull-back action, intertwiner $z^\mu\leftrightarrow\eta^{\mu\nu}\partial_\nu\delta$); $\mathrm{Res}(\partial_\alpha u)=\partial_\alpha\mathrm{Res}(u)$ for all 340 label/direction pairs with $N\le3$; every one of the 185 relation vectors annihilates the top-order, the radial-specialization and the general-ϖ coefficient vectors; all 255 labels reconstructed and checked against direct evaluation.
The derivative and Lorentz identities are \emph{not} used for, and do not hold for, the lower-order coefficients (ϖ is neither covariant nor Leibniz-compatible).

\paragraph{Prescription dependence.}
$B_\mu=0$ does not remove prescription dependence: it resides in the lower-order coefficients, common to A_μ and $\mathcal A_\mu$, which scale as $(\mu/\mu_0)^{N-|\gamma|}$ and depend on $\nu_k[\varpi]$, $k=1,\dots,4$.
\emph{Missing datum:} XVI-L fixes $\varpi\in C_c^\infty$, $\varpi(0)=1$, but does not specify it; the values of $\nu_k[\varpi]$ (equivalently the functionals $\Lambda_k(P)=\langle P(Q-i0)^{-6}|_{S^3},\nu_k\rangle$ on Poly_{8-k}) are what is missing.
For a reflection-even ϖ the odd-k coefficients vanish.
The named specialization $\mathfrak R_{\mathrm L,\mu}[\varpi_{\mathrm{rad}}]$ (Euclidean-radial ϖ; \emph{not} asserted by XVI-L) has constant ν_k^{rad}, only $k=2,4$ survive, and its contact difference from $\mathfrak R_{\mathrm L,\mu}[\varpi]$ has coefficients $-(-1)^{|\gamma|}\langle b,w_\gamma[\varpi]-w_\gamma[\varpi_{\mathrm{rad}}]\rangle_\gamma$, i.e.\ $\frac{(-1)^{|\gamma|+1}}{\gamma!}\langle b\,\omega^\gamma,\nu_k[\varpi]-\nu_k^{\mathrm{rad}}\rangle$ for $k\ge1$ and $\frac{(-1)^{N+1}}{\gamma!}\langle b\,\omega^\gamma,\int_0^\infty r^{-1}(\varpi-\varpi_{\mathrm{rad}})(r\omega)\,\mathrm dr\rangle$ for $k=0$.

```
\paragraph{Collapse of the full scale tensor.}
$C_{\mathrm{scale}}(\mu)=(a_{N,b,\gamma}(\mu))$ for all $N,b,\gamma$.
Collapse and adjoint defect are homogeneous in the total derivative degree $D=4-N+|\gamma|$, so the top part ($D=4$) and the lower parts ($D=4-k$) cannot cancel each other; cross-grade cancellations at fixed $D$ are included.
\begin{center}
\small
\begin{tabular}{lcccc}
\hline
reading & $\Coll$ nnz ($D=4$) & adjoint-defect nnz & variational & per-$N$ nnz ($N=0..4$)\\
\hline
formal & 14112 & 16608 & no & 744, 4440, 10278, 10040, 4604\\
$\chi$-plateau & 2522 & 4090 & no & 744, 1728, 2386, 1550, 514\\
preparation plateau & 4580 & 4090 & no & 0, 0, 2274, 4072, 4604\\
double plateau & 490 & 0 & \textbf{yes} & 0, 0, 474, 514, 514\\
\hline
\end{tabular}
\normalsize
\end{center}
Lower-order envelope (all $\varpi$): $\Coll$ is injective on each $\Lambda_k$-block; self-adjoint parameter subspaces have dimensions $0,0,0,26$ ($k=1,2,3,4$) in the formal, $\chi$- and preparation readings and $0,49,0,26$ in the double plateau; there $E_{\mathrm{EH}}\notin$ image ($k=2$) and $\mathrm{span}(E_{P_1},E_{P_2})\cap$ image is one-dimensional ($k=4$).
The radial specialization is non-self-adjoint at $k=2$ and $k=4$ in every reading.
Time-only readings: Corollary~\ref{cor:wpB-timeonly}.

\begin{theorem}[Double-plateau scale tensor: top component and full tensor]
\label{thm:wpB-dp}
Assume Gate~V. In the double plateau, the cutoff-independent $D=4$ component is variational and Lorentz covariant, but not linearized-gauge invariant. The full scale tensor has cutoff-dependent lower-degree components and need not be variational. In detail:
\begin{enumerate}
\item[(i)] \emph{Top component.} The $D=4$ component $\Coll(C_{\mathrm{top}})$ of $\Coll(C_{\mathrm{scale}})$ (the slots with $|\gamma|=N$; independent of $\mu$ and $\varpi$) is nonzero and variational, with
\[
I^{\mathrm{top}}_{\mathrm{scale}}=\frac{i}{16\pi^2}\!\int\Bigl(-\tfrac{71}{120}h_{\mu\nu}\square^2h^{\mu\nu}-\tfrac{27}{80}h\square^2h-\tfrac{11}{10}(\partial^\mu h_{\mu\nu})\square(\partial_\rho h^{\rho\nu})+\tfrac{19}{30}h\square\partial^\mu\partial^\nu h_{\mu\nu}-\tfrac{13}{15}(\partial^\mu\partial^\nu h_{\mu\nu})^2\Bigr)
\]
(mod total derivatives; the prefactor is $i\pi^2c^2$ with $c=1/(4\pi^2)$).
It is Lorentz covariant (a combination of Lorentz scalars), but $\partial_\mu E^{\mu\nu}\neq0$ (not linearized-gauge invariant), and it is \emph{not} in $\mathrm{span}(E_{R^2},E_{\mathrm{Ric}^2})$: in the basis above $R_{\mathrm{lin}}^2\mapsto(0,1,0,-2,1)$ and $\mathrm{Ric}_{\mathrm{lin}}^2\mapsto(\frac14,\frac14,\frac12,-\frac12,\frac12)$, the rank goes from $2$ to $3$ when the scale vector is adjoined, and Proposition~\ref{prop:wpB-sep} gives an exact separating functional.
In particular it is not a multiple of $E_{R^2-(188813/88456)\mathrm{Ric}^2}$, of $E_{\mathrm{EH}}$, or of an ultralocal.
\item[(ii)] \emph{Full tensor.} The lower-degree components ($D=4-k$, $k=1,\dots,4$) depend on the cutoff through $\Lambda_k$, i.e.\ through $\nu_k[\varpi]$ of~\eqref{eq:wpB-nu}.
The full tensor is variational iff $\Lambda_1=\Lambda_3=0$, $\Lambda_2$ lies in the $49$-dimensional and $\Lambda_4$ in the $26$-dimensional self-adjoint subspaces; so it need not be variational.
Lorentz covariance of the full tensor is \emph{not} established: the lower-order terms depend on the possibly noncovariant $\varpi$, and the Lorentz tests apply to the top order only.
\item[(iii)] \emph{Curvature span.} For \emph{every} $\mu$ and every admissible $\varpi$ the full collapsed scale tensor lies outside the span of the M.5 curvature targets: collapse is homogeneous in $D$, and its fixed $D=4$ component already lies outside $\mathrm{span}(E_{R^2},E_{\mathrm{Ric}^2})$, while the targets of degree $D=0,2$ lie in other blocks.
\end{enumerate}
The whole tensor is retained as unmatched relative to the curvature ledger.
\end{theorem}
\begin{proof}
(i) Exact double-plateau collapse and adjoint defect of $C_{\mathrm{top}}$ (\texttt{wpB/higher\_grade\_scaling.py}, Section~D; \texttt{full\_scale\_collapse\_certificates.json}, \texttt{readings.double\_plateau.top\_D4}): adjoint defect $0$, so Lemma~\ref{lem:helm} gives the potential; its coordinates in the five-term Lorentz-scalar basis are the displayed coefficients; the divergence is nonzero exactly; the rank statement is exact over $\mathbb Q$.
\texttt{regression\_M.py} independently recollapses $C_{\mathrm{top}}$ from \texttt{\_wpB\_state.json} and checks the pure-gauge coefficient of Proposition~\ref{prop:wpB-sep}.
(ii) Theorem~\ref{thm:wpB-B0}(2) and the lower-order envelope above; the Lorentz and derivative identities hold for the top order only (paragraph ``Exact top-order data'').
(iii) Homogeneity in $D$ (Lemma~\ref{lem:to-comp} and the paragraph above the table) and (i).
\end{proof}

\begin{proposition}[Exact separating functional for the displayed scale functional (audit~3)]
\label{prop:wpB-sep}
In the ordered basis $(L_1,\dots,L_5)=\bigl(h_{\mu\nu}\square^2h^{\mu\nu},\ h\square^2h,\ (\partial^\mu h_{\mu\nu})\square(\partial_\rho h^{\rho\nu}),\ h\square\partial^\mu\partial^\nu h_{\mu\nu},\ (\partial^\mu\partial^\nu h_{\mu\nu})^2\bigr)$ of Theorem~\ref{thm:wpB-dp} (the order of \texttt{C0V4} in \texttt{higher\_grade\_scaling.py} and of \texttt{conventions.json}), let
\[
\ell(a,b,c,d,e):=a+b-c+d+e\ .
\]
Then $\ell(v_{R^2})=0+1-0-2+1=0$, $\ell(v_{\mathrm{Ric}^2})=\frac14+\frac14-\frac12-\frac12+\frac12=0$, and
\[
\ell(v_{\mathrm{scale}})=-\tfrac{71}{120}-\tfrac{27}{80}+\tfrac{11}{10}+\tfrac{19}{30}-\tfrac{13}{15}=-\tfrac1{16}\neq0,
\]
```

```
so $v_{\mathrm{scale}}\notin\mathrm{span}(v_{R^2},v_{\mathrm{Ric}^2})$ exactly.
\emph{Interpretation.} On the pure-gauge configuration $h_{00}=f(t)$, all other components zero ($h_{\mu\nu}=\partial_\mu\xi_\nu+\partial_\nu\xi_\mu$ with $\xi_0=\frac12\int f$), one has $h=h^{00}=f$, $\partial^\mu h_{\mu0}=f'$, $\partial^\mu\partial^\nu h_{\mu\nu}=f''$, $\square=\partial_t^2$, so modulo total derivatives
\[
(L_1,L_2,L_3,L_4,L_5)\equiv(1,1,-1,1,1)\,f''''\,,
\]
($L_3=f'f'''\equiv-ff''''$, $L_5=(f'')^2\equiv ff''''$). Hence $\int\sum_ja_jL_j\equiv\ell(a)\int ff''''$: the curvature squares vanish on pure gauge, as they must, while the scale functional leaves $-\frac1{16}\,ff''''$ times the unit $i/(16\pi^2)$; its Euler--Lagrange expression in $f$ is $2\ell(a)f''''$.
This is verified symbolically with \texttt{sympy} in \texttt{regression\_M.py} (audit-3 section).
\emph{Scope.} The proposition certifies the displayed functional only; the identification of the graph's collapsed top scale tensor with $E[I^{\mathrm{top}}_{\mathrm{scale}}]$ (the graph$\to$functional obligation) is carried by the collapse certificate of Theorem~\ref{thm:wpB-dp}(i); \texttt{regression\_M.py} re-verifies it exactly (the recollapse of the archived $C_{\mathrm{top}}$ equals the Euler tensor of the displayed $I^{\mathrm{top}}_{\mathrm{scale}}$ entrywise, $490$ entries, and its $f''''$ coefficient in $E^{00}[f\,e_{00}]$ is $-\frac18=2\ell(v_{\mathrm{scale}})$).
\end{proposition}

\begin{remark}[Top contact versus full tensor; relation to XVI-L]
\label{rem:wpB-topfull}
The split between the cutoff-independent top component and the cutoff-dependent lower-degree components matches the residue left open by XVI-L, thm:L6: ``No explicit nonzero scaling residue (anomaly form) is claimed (Remark rem:Lressretch)'', where rem:Lressretch notes that ``a nonzero homogeneous distribution of degree $(-4)$ may still have vanishing spherical average against the radial derivative of $(\varpi_\mu)$''.
The value $r=-i/(8\pi^2)$ (Theorem~\ref{thm:M6-S2}) discharges only the top contact (the nonvanishing spherical average at $N=0$); the lower-degree, $\mu\,\mathrm d\varpi$-dependent contacts at $N\ge1$ are not discharged.
\end{remark}

\begin{corollary}[Time-only readings of the scale tensor]
\label{cor:wpB-timeonly}
Assume Gate~V.
In the time-only reading and in the time-only $\chiI\equiv1$ reading of Section~\ref{sec:to-timeonly}, for every $\mu>0$ and every admissible $\varpi$, the collapsed scale tensor $\Coll_t(C_{\mathrm{scale}}(\mu))$ is not variational.
(The proof combines the archived top-order coordinates with the exact kernels of Theorem~\ref{thm:to-timeonly} and Proposition~\ref{prop:to-chiplateau}; it is complete relative to the reported computations of Lemma~\ref{lem:to-comp}; audit~3 review.)
\end{corollary}
\begin{proof}
The top-order part $C_{\mathrm{top}}$ of $C_{\mathrm{scale}}$ is a nonzero vector of the $D=4$ block (all its slots have $|\gamma|=N$, hence $D=4-N+|\gamma|=4$), and its $N=0$ coordinate is $-2i\pi^2c^2\neq0$ (archived in \texttt{wpB/\_wpB\_state.json}; re-checked by \texttt{regression\_M.py}).
$H_t$ is homogeneous in $D$ (Lemma~\ref{lem:to-comp}: the blocks have disjoint rows), so the $D=4$ component of $H_t(C_{\mathrm{scale}})$ is $H_t(C_{\mathrm{top}})$, whatever the $\varpi$-dependent lower-order parts are.
In the time-only reading $\ker H_t$ is $0$ at $D=4$ (Theorem~\ref{thm:to-timeonly}), so $H_t(C_{\mathrm{top}})\neq0$.
In the time-only $\chiI\equiv1$ reading $\ker H_t$ at $D=4$ is spanned by vectors supported on $N=4$ (Proposition~\ref{prop:to-chiplateau}); $C_{\mathrm{top}}$ has a nonzero $N=0$ coordinate, so again $H_t(C_{\mathrm{top}})\neq0$.
By Lemma~\ref{lem:helm}, applied with the time-only adjoint of Lemma~\ref{lem:to-intertwine}, a collapsed tensor with nonzero adjoint defect is not an Euler tensor.
\end{proof}

\begin{corollary}[M-C$_{\mathrm{curv}}$ obstructed; broad M-C not achieved]
\label{cor:wpB-MC}
Assume Gate~V. For the specified two-edge graph and the XVI-L prescription:
\begin{enumerate}
\item \textbf{M-C$_{\mathrm{curv}}$} (Definition~\ref{def:MCcurv}) is \textbf{Proved obstructed} (graph-specific): for every $\mu$ and every admissible $\varpi$ the full collapsed scale variation does not lie in the span of the M.5 curvature targets, because its fixed $D=4$ component already lies outside.
\item \textbf{Broad M-C} is \textbf{not achieved} and remains \textbf{Unresolved}: criterion~(i) of Definition~\ref{def:MCcurv} holds (the top scale tensor has its own variational representative $I^{\mathrm{top}}_{\mathrm{scale}}$, Theorem~\ref{thm:wpB-dp}(i)), but criterion~(ii) fails because XVI-L leaves $\varpi$ unspecified (thm:L6). Their numerical values and the variationality of the resulting full tensor are not determined for the unspecified cutoff. \textbf{M-C$_{\mathrm{data}}$} is satisfied.
\end{enumerate}
This is not a claim about gravity generally. Adding non-gauge-invariant quadratic representatives to the ledger changes the target ledger of M-C$_{\mathrm{curv}}$ but may remain within broad Target~I; gauge-fixing/ghost or further graph contributions are another, independent enlargement.
\end{corollary}
\begin{proof}
(1) M-C$_{\mathrm{curv}}$ requires a matching in the curvature-target span together with the matched-scale data of Definition~\ref{def:MCcurv}.
The scale derivative equals the dilation defect at every grade ($B_\mu=0$; exact $\varpi$-independent top residues; Theorem~\ref{thm:wpB-B0}); the numerical values of the lower-degree components and the variationality of the resulting full tensor are not determined for the unspecified cutoff (XVI-L, thm:L6).
But for every $\mu$ and every admissible $\varpi$ the collapsed scale tensor is not in the span of the ledger targets ($E_{P_1}$, $E_{P_2}$, $E_{\mathrm{EH}}$, $E_{R^2}$, $E_{\mathrm{Ric}^2}$, with the reading's weight): in the formal, $\chi$-plateau and preparation-plateau readings and in both time-only readings its $D=4$ component is not variational (table above; Corollary~\ref{cor:wpB-timeonly}), whereas every target is self-adjoint; in the double plateau its $D=4$ component is nonzero and outside $\mathrm{span}(E_{R^2},E_{\mathrm{Ric}^2})$ (Theorem~\ref{thm:wpB-dp}, Proposition~\ref{prop:wpB-sep}), and targets of degree $D=0,2$ cannot cancel it because collapse is homogeneous in $D$.
The lower-order, $\varpi$-dependent part cannot repair this, for the same reason.
```

(2) Theorem~\ref{thm:wpB-dp}(i),(ii): the $D=4$ component is the Euler tensor of $I^{\mathrm{top}}_{\mathrm{scale}}$, which lies outside the curvature ledger but is a local quadratic functional, so criterion~(i) of broad M-C holds; criterion~(ii) requires an explicitly specified ϖ and a functional representation of the entire collapsed scale variation, which XVI-L, thm:L6, does not supply. M-C$_{\mathrm{data}}$ (matching + explicit prescription-dependent scaling formulas) is satisfied.
\end{proof}

\paragraph{Archives.}
\begin{sloppypar}\noindent
\texttt{wpB/higher_grade_scaling.py} (185 checks; output \texttt{higher_grade_scaling_out.txt}), \texttt{scale_derivative_coefficients.json}, \texttt{dilation_defect_coefficients.json}, \texttt{scale_defect_difference.json}, \texttt{full_scale_collapse_certificates.json}, \texttt{matched_scale_ledger.txt}, \texttt{_wpB_state.json}, \texttt{_wpB_sectionD.json}.
The script keeps c^2 symbolic; its status strings ‘‘CONDITIONAL on Input (S-bv)’’ predate Theorem~\ref{thm:wpA-Sbv} and are superseded by it.
\end{sloppypar}

\paragraph{M.6 status.}
\begin{itemize}
\item Dilation identity and $A_\mu=\mathcal A$ for $N=0$: \textbf{Constructed} (Proposition~\ref{prop:M6-dil}).
\item $A_\mu=\mathcal A_\mu$ ($B_\mu=0$) for all $N=0,\dots,4$ and every admissible ϖ: \textbf{Constructed} (Theorem~\ref{thm:wpB-BO}).
\item S^2 residue $r=-2i\pi^2 c^2=-i/(8\pi^2)\neq0$: \textbf{Constructed} (Theorem~\ref{thm:M6-S2}, with Theorem~\ref{thm:wpA-Sbv}).
\item Top-order residues, all grades: \textbf{Constructed}, exact and ϖ-independent (units $i\pi^2c^2=i/(16\pi^2)$); lower-order values: \textbf{Unresolved} (depend on $\nu_k[\varpi]$, $k=1,\dots,4$; ϖ not fixed by XVI-L; the radial comparison is a separately named specialization).
\item $N=0$ anomaly component C_{anom} (higher-grade coordinates set to zero for this probe): \textbf{Constructed} (Definition~\ref{def:M6-Canom}).
\item Collapse of the $N=0$ component: \textbf{Constructed} (Theorem~\ref{thm:M6-collapse}); it contributes no nonzero plateau matching (Corollary~\ref{cor:M6-compat}).
\item Collapse of the full scale tensor: \textbf{Constructed} in all readings. In the double plateau, the cutoff-independent $D=4$ component is variational and Lorentz covariant, but not linearized-gauge invariant; it lies outside $\mathrm{span}(E_{R^2},E_{\mathrm{Ric}^2})$ (Theorem~\ref{thm:wpB-dp}, Proposition~\ref{prop:wpB-sep}). The full scale tensor has cutoff-dependent lower-degree components and need not be variational; its Lorentz covariance is not established. Time-only readings non-variational (Corollary~\ref{cor:wpB-timeonly}).
\item Release: \textbf{M-C$_{\mathrm{curv}}$ Proved obstructed} (graph-specific); \textbf{broad M-C not achieved} (Unresolved; ϖ unspecified by XVI-L, thm:L6); \textbf{M-C$_{\mathrm{data}}$ Satisfied} (Corollary~\ref{cor:wpB-MC}); not a claim about gravity generally.
\end{itemize}
\begin{sloppypar}\noindent
Archives: \texttt{scaling_anomaly.py}, \texttt{scaling_anomaly_out.txt}, \texttt{scaling_anomaly_cert.json}, \texttt{scaling_anomaly_cert.txt}; \texttt{wpB/}.
\end{sloppypar}

\section{M.7 Covariance and Ward-compatibility constraints}
\label{sec:m7}

This section treats \emph{prescription} covariance of an extension $\RM=\RL+\Delta\RL$ on the flat background.
It is kept distinct from interacting-sector Ward/BV symmetry (M.8--M.9; Target~II).
Gate~S-bv is certified (Theorem~\ref{thm:wpA-Sbv}), so the (S-bv) dependence of this section is discharged; M-C$_{\mathrm{curv}}$ is Proved obstructed (graph-specific), broad M-C is not achieved (Corollary~\ref{cor:wpB-MC}), and M-D is not claimed.
The audited M.7 revision repaired three defects of the previous M.7 computation (Remark~\ref{rem:M7-audit}).

\paragraph{Flat-background Lorentz covariance conditions (LC0--LC3).}
\begin{enumerate}
\item[(LC0)] Lie algebra $\mathfrak{so}(1,3)\subset\mathfrak{gl}(4)$, $X=(X^\mu{}_\nu)$ acting on coordinates z^μ, with $X^{\mathsf T}\eta+\eta X=0$; generators $\{J_{12},J_{13},J_{23},K_{01},K_{02},K_{03}\}$ (K_{0i} symmetric, J_{ij} antisymmetric as matrices).
\item[(LC1)] \emph{Coordinate polynomials.} On functions of the difference variable, $(X\cdot f)(z)=-(Xz)^\mu\partial_\mu f(z)$.
By Theorem~\ref{thm:wpA-Sbv}, $\mathcal H_{4+N}\cong\mathrm{Poly}_N$ via numerators (Q is invariant), inducing matrices $\rho_N(X)$ on the selected basis B_N.
\item[(LC2)] \emph{Contact distributions (repaired).} $W_N=\mathrm{span}\{\partial_\gamma\delta:|\gamma|\le N\}$, where $\partial_\gamma=\partial_0^{\gamma_0}\cdots\partial_3^{\gamma_3}$ are lower-index coordinate derivatives in z (the same ∂^γ as in $\mathcal G(C)=\sum C\,\bar a\,\partial^\gamma\delta$ and in the collapse ∂_w^γ).
$\mathfrak{so}(1,3)$ acts by pull-back, $\langle X\cdot T,f\rangle:=-\langle T,X\cdot f\rangle=\langle T,(Xz)^\mu\partial_\mu f\rangle$; for $T=\sum_\nu c_\nu\partial_\nu\delta$ this gives $X\cdot\partial_\nu\delta=X^\mu{}_\nu\,\partial_\mu\delta$, i.e.\ $c\mapsto Xc$.
On coordinate polynomials $p=\sum_\nu c_\nu z^\nu$ the action (LC1) is $c\mapsto-X^{\mathsf T}c$.
These differ for boosts.
The intertwiner is the metric identification $z^\mu\leftrightarrow\partial^\mu\delta=\eta^{\mu\nu}\partial_\nu\delta$ (index raising by η, using $\eta X\eta=-X^{\mathsf T}$), extended multiplicatively: a coordinate polynomial $q(z)$ corresponds to $q(\partial^\cdot)\delta$, so the coefficient of $\partial_\gamma\delta$ equals $(-1)^{\gamma_1+\gamma_2+\gamma_3}$ times the coefficient of z^γ.
Writing $P=\mathrm{diag}((-1)^{\gamma_1+\gamma_2+\gamma_3})$, the action on W_N in the basis $\partial_\gamma\delta$ is $\rho^\delta_N(X)=P\,\rho^{\mathrm{poly}}_N(X)\,P$.

```
\item[(LC3)] A contact change $\Delta\RL(u)=\sum_{|\gamma|\le N}C_\gamma(u)\,\partial_\gamma\delta$ is infinitesimally
      covariant iff $C\circ\rho_N(X)=\rho^\delta_N(X)\circ C$ for all $X$.
Homogeneous solutions are the Lorentz-equivariant contact assignments.
The inhomogeneous correction of a concrete obstruction $\mathrm{Obs}_L(X,u)=X_*\RL(u)-\RL(X\cdot u)$ requires an explicit $\
      varpi$ and is \textbf{Unresolved} (a Euclidean-radial bump is not Lorentz-invariant; a boost obstruction is expected).
\end{enumerate}
A linear divergence identity of an Euler tensor is \textbf{not} a Ward identity.

\paragraph{Lorentz action on $B_N$ and on contact distributions.}
\begin{proposition}[so(1,3) action on $B_N$ and on $W_N$]
\label{prop:M7-rho}
For each $N=0,\ldots,4$ the maps $\rho_N(X)$ on $B_N$ and $\rho^\delta_N(X)$ on $\{\partial_\gamma\delta\}$ define Lie-algebra
       homomorphisms (in particular $[\rho(K_{01}),\rho(K_{02})]=\rho(J_{12})$ for both).
The $N=0$ action on $B_0$ is trivial.
\end{proposition}

\begin{proposition}[Boost verification on $\partial_\mu\delta$; exact]
\label{prop:M7-boost}
In the basis $(\partial_0\delta,\partial_1\delta,\partial_2\delta,\partial_3\delta)$ the pull-back action of every generator
      is the matrix $X$ itself; for $K_{01}$,
\[
\rho^\delta(K_{01})\big|_{|\gamma|=1}=\begin{pmatrix}0&1&0&0\\1&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix},
\qquad
\rho^{\mathrm{poly}}(K_{01})\big|_{|\gamma|=1}=-\begin{pmatrix}0&1&0&0\\1&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}.
\]
The second matrix is the coordinate matrix used by the previous revision for the SA intersection (it identified $\partial_\
      gamma\delta$ with $z^\gamma$).
For the three boosts the two differ by a sign; for the three rotations they coincide; in all cases $\rho^\delta=\eta\,,\rho^{\
      mathrm{poly}}\eta$ at first order and $\rho^\delta_N=P\rho^{\mathrm{poly}}_NP$ for all $N\le4$ (checked exactly for all
      six generators).
A symbolic test-function computation gives $\langle K_{01}\cdot\partial_1\delta,f\rangle=-\partial_0f(0)=\langle\partial_0\
      delta,f\rangle$, confirming $K_{01}\cdot\partial_1\delta=+\partial_0\delta$.
\end{proposition}

\begin{proof}
$z^\rho\partial_\nu\delta=-\delta^\rho_\nu\delta$ and $z^\rho\partial_\mu\partial_\nu\delta=-\delta^\rho_\nu\partial_\mu\delta
      -\delta^\rho_\mu\partial_\nu\delta$ give $-(Xz)^\mu\partial_\mu\partial_\nu\delta=X^\mu{}_\nu\partial_\mu\delta+(\mathrm
      {tr}X)\partial_\nu\delta$, and $\mathrm{tr}X=0$.
The higher-order statement follows because $q\mapsto q(\partial^\cdot)\delta$ is multiplicative and $\delta$ is invariant; it
      is also checked exactly by pairing with $z^\gamma/\gamma!$ for all $|\gamma|\le4$ (\texttt{covariance\_constraints.py}).
\end{proof}

Ultralocal Lorentz scalars $P_1=\int\eta^{\mu\alpha}\eta^{\nu\beta}h_{\mu\nu}h_{\alpha\beta}$ and $P_2=\int(\mathrm{tr}_\eta h
      )^2$, and the matched line $P_1-\tfrac{89}{140}P_2$, are annihilated by all six generators.

\paragraph{Compatibility matrix: solutions, constraint rank, cokernel.}
\begin{theorem}[Equivariant contact assignments]
\label{thm:M7-compat}
Write $\mathrm{Hom}_G(\mathcal H_{4+N},W_N)$ for $\mathfrak{so}(1,3)$-equivariant contact maps at grade $N$.
\begin{enumerate}
\item \emph{Abstract count.} $\mathrm{Poly}_N=\bigoplus_{k=0}^{\lfloor N/2\rfloor}H_{N-2k}$ with $\dim H_m=(m+1)^2$, $W_N\cong
      \mathrm{Poly}_{\le N}$ as representations (via the intertwiner of (LC2)), and
\[
\dim\mathrm{Hom}_G(\mathcal H_{4+N},W_N)=\frac{(\lfloor N/2\rfloor+1)(\lfloor N/2\rfloor+2)}{2}.
\]
Explicit basis: $C_{k,j}=P\circ\varphi_{k,j}$ with $\varphi_{k,j}:p\mapsto Q^j\square^k p$, $k=0,\ldots,\lfloor N/2\rfloor$, $
      j=0,\ldots,k$; in words, $Q^j\square^kp$ read as a polynomial in $\partial^\mu$ acting on $\delta$.
\item \emph{Coordinate verification.} Each $C_{k,j}$ satisfies $C\rho_N(X)=\rho^\delta_N(X)C$ exactly for all six generators
      and all $N\le4$.
The uncorrected maps $\varphi_{k,j}$ used previously fail exactly the three boost equations whenever $P$ acts nontrivially on
      their image, i.e.\ for $(N,k,j)\in\{(1,0,0),\allowbreak (2,0,0),\allowbreak (3,0,0),\allowbreak (3,1,0),\allowbreak
      (3,1,1),\allowbreak (4,0,0),\allowbreak (4,1,0),\allowbreak (4,1,1)\}$; the six maps whose image lies at $|\gamma|$ even
       with $P=+1$ ($N=0$; $(2,1,\cdot)$; $(4,2,\cdot)$) were unaffected.
For $N\le2$ the full constraint map was additionally solved by an exact verified kernel computation.
\item Graded compatibility matrix. Slots $=\dim\mathcal H_{4+N}\cdot\dim W_N$; the constraint map is $C\mapsto(C\rho_N(X)-\rho
      ^\delta_N(X)C)_{X}$ with codomain $\bigoplus_{6\text{ generators}}\mathrm{Hom}(\mathcal H_{4+N},W_N)$ of dimension $6\
      cdot$slots;\allowbreak\ rank $=$ slots $-\dim$ sol.\ $=$ codimension of the solution space; coker $=6\cdot$slots $-$
      rank.
\begin{center}
\begin{tabular}{crrrrrrl}
$N$ & $\dim H$ & $\dim W$ & slots & rank (=codim) & $\dim$ codomain & coker & $\dim$ sol./block\\
\hline
0 & 1 & 1 & 1 & 0 & 6 & 6 & 1 / K\\
1 & 4 & 5 & 20 & 19 & 120 & 101 & 1 / K\\
2 & 10 & 15 & 150 & 147 & 900 & 753 & 3 / K\\
3 & 20 & 35 & 700 & 697 & 4200 & 3503 & 3 / $\Creg$\\
4 & 35 & 70 & 2450 & 2444 & 14700 & 12256 & 6 / $\Creg$
\end{tabular}
\end{center}
The cokernel is dominated by the redundancy of the six generator equations and is not, by itself, an obstruction count.
```

```
Total equivariant freedom: $14=5$ (K-block, $N\le2$)$+9$ ($\Creg$, $N\ge3$) (unchanged by the repair, since $P$ is an isomorphism).
\item Separation: (a) $\Creg$ solutions ($N\ge3$) are strict $K$-regression-preserving ($\Delta\RL|_{\mathcal V_{\mathrm K}}=0$); (b) K-block solutions ($N\le2$) modify $\RL|_{\mathcal V_{\mathrm K}}$ and break XVI-K regression if used.
\item The inhomogeneous $\mathrm{Obs}_L(\RL)$ is \textbf{Unresolved}.
\end{enumerate}
\end{theorem}

\begin{proof}
Schur count from the harmonic decomposition; linear independence of $\{C_{k,j}\}$; equivariance against $\rho^\delta$ on all six generators; $N\le2$ exact kernel of the full constraint map (\texttt{m4\_common.block\_kernel}, verified on all rows).
For $N=3,4$ the rank follows from rank--nullity with the abstract kernel dimension and the explicit verified basis.
Certificates: \texttt{covariance\_matrix.txt}, \texttt{covariance\_certificates.txt}, \texttt{covariance\_summary.json}, \texttt{covariance\_exact\_certificates.json}.
\end{proof}

\paragraph{Intersection with the variational subspace.}
Let $\mathrm{Equiv}\subset\Ccoh$ be the $14$-dimensional span of the corrected $C_{k,j}$ and $V:=\mathrm{Equiv}\cap\mathrm{SA}$ (reading-dependent).
Write $\Ccoh=\mathcal C_{\mathrm K}\oplus\Creg$ by grade ($N\le2$ / $N\ge3$), with projections $\pi_K,\pi_{\mathrm{reg}}$.
Four distinct objects are reported: the projections $\pi_K(V)$, $\pi_{\mathrm{reg}}(V)$ (equivariant, but \emph{not} self-adjoint in general) and the intersections $V\cap\mathcal C_{\mathrm K}$, $V\cap\Creg$ (genuine SA assignments).
Always $\dim(V\cap\mathcal C_{\mathrm K})=\dim V-\dim\pi_{\mathrm{reg}}(V)$, $\dim(V\cap\Creg)=\dim V-\dim\pi_K(V)$, hence $\dim(V\cap\mathcal C_{\mathrm K})+\dim(V\cap\Creg)\le\dim V$.

\begin{proposition}[Equivariant $\cap$ SA, exact; Gate~V]
\label{prop:M7-inter}
$V$ is the exact rational kernel of the adjoint-defect map $x\mapsto E_{C(x)}^\dagger-E_{C(x)}$ on the $14$ equivariant coordinates:
\begin{center}
\begin{tabular}{lcccccc}
\hline
Reading & $\dim V$ & $\dim\pi_K(V)$ & $\dim V\cap\mathcal C_{\mathrm K}$ & $\dim\pi_{\mathrm{reg}}(V)$ & $\dim V\cap\Creg$ & $\dim\Coll(V)$\\
\hline
formal & $1$ & $0$ & $0$ & $1$ & $1$ & $1$\\
$\chi$-plateau & $1$ & $0$ & $0$ & $1$ & $1$ & $1$\\
double plateau & $12$ & $5$ & $3$ & $9$ & $7$ & $9$\\
\hline
\end{tabular}
\end{center}
\begin{enumerate}
\item Formal and $\chi$-plateau: $V$ is the line of $C_{2,0}$ at $N=4$ ($\square^2$, $\gamma=0$), contained in $\Creg$; $\Coll(V)$ is spanned by the Euler tensor of $P_1-\tfrac{89}{140}P_2$ (ratio $17920$ in the formal reading), i.e.\ the matched ultralocal line of Proposition~\ref{prop:match}.
\item Double plateau: $\dim V=12$, of which $3$ dimensions lie in $\ker\Coll$.
$\pi_K(V)$ is all of $\mathrm{Equiv}_K$ ($5$), but only $V\cap\mathcal C_{\mathrm K}$ ($3$) consists of SA assignments; the remaining projected directions have nonzero adjoint defect and are \emph{not} SA assignments.
Likewise $\pi_{\mathrm{reg}}(V)=\mathrm{Equiv}_{\mathrm{reg}}$ ($9$) while $V\cap\Creg$ has dimension $7$.
Check: $3+7=10\le12$.
\end{enumerate}
Every basis vector of $V$, $V\cap\mathcal C_{\mathrm K}$ and $V\cap\Creg$ is re-verified from scratch against all constraints: equivariance (six generators, all grades) and zero adjoint defect; \texttt{regression\_M.py} re-verifies them independently, including the Helmholtz route $\mathrm{EL}(\tfrac12\langle h,E_{\mathrm{Ch}}\rangle)=E_C$.
Mod-$p$ ranks at three primes agree and are supplementary only.
Prescription-covariant variational contacts exist but do not by themselves produce a Ward identity.
\end{proposition}

\paragraph{Reconciliation with previously stated numbers.}
The equivariant freedom $14=5+9$ and the formal/$\chi$-plateau value $\dim(\mathrm{Equiv}\cap\mathrm{SA})=1$ are confirmed (the formal SA vector sits at $|\gamma|=0$, where $P=1$).
The previous double-plateau values ``$5$ globally, $5$ on K, $3$ on $\Creg$'' were wrong on three counts: the coordinate action was wrong for boosts (Proposition~\ref{prop:M7-boost}); the grade-restricted numbers were $\dim(\mathrm{Equiv}_G\cap\pi_G(\mathrm{SA}))$ (intersection with a \emph{projection} of SA), not $V\cap\mathcal C_G$; and the modular reduction \texttt{int(Fraction(c) \% p)} truncated the $14639$ non-integer rational entries of the SA basis.
With the old vectors and a correct reduction the old routine gives $5/4/6$; it reproduces $5/3/5$ only with the truncating reduction (\texttt{m7\_legacy\_reconstruction.py}).

\paragraph{Linearized divergence identities (external data retained).}
\begin{proposition}[Linear divergence identities; not Ward identities]
\label{prop:M7-id}
Let $E^{\mu\nu}$ denote Gate~V Euler tensors; they carry \emph{upper} indices because they are paired with the lowered $h_{\mu\nu}$, and $(\operatorname{div}E)^\nu:=\partial_\mu E^{\mu\nu}$ involves no metric factor.
\begin{enumerate}
\item[(A)] Contracted Bianchi: $\eta^{\mu\rho}\partial_\rho G^{\mathrm{lin}}_{\mu\nu}=0$ identically in jets ($G$ with lower indices).
\item[(B$'$)] \emph{Curvature squares.} For $I_w=\int w\,(\alpha R_{\mathrm{lin}}^2+\beta R^{\mathrm{lin}}_{\mu\nu}R_{\mathrm{lin}}^{\mu\nu})$ with constant $\alpha,\beta$ and \emph{any} fixed weight $w$ (in particular $w=1$ in the double plateau, $w=\sigeps$ in the $\chi$-plateau reading, $w=\chiI\sigeps$ formally),
\[
```

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\partial_\mu E_{I_w}^{\mu\nu}=0\quad\text{identically.}
\]
Reason: $R_{\mathrm{lin}}$ and $R^{\mathrm{lin}}_{\mu\nu}$ are invariant \emph{pointwise} under $\delta h_{\mu\nu}=\partial_\
    mu\xi_\nu+\partial_\nu\xi_\mu$, so $0=\delta I_w=2\int E^{\mu\nu}\partial_\mu\xi_\nu=-2\int\xi_\nu\partial_\mu E^{\mu\nu
    }$; weight-derivative terms arise in the Euler tensor and cancel in its divergence.
(For the weighted Einstein--Hilbert action the analogous weight-derivative terms are $\sigma G^{\mu\nu}+q^{\mu\nu}\square\
    sigma-\nabla^\mu\nabla^\nu\sigma$ of XII-C, lem:weighted-curvature, eq:weighted-full-variation; no landmark constructs
    the weighted $R^2/\mathrm{Ric}^2$ Euler tensors, which are internal to XVI-M.)
This holds separately for $R^2$, $\mathrm{Ric}^2$ and for $R^2-\tfrac{188813}{88456}\mathrm{Ric}^2$ (exact assertions in all
    three readings).
\item[(EH)] For $I_w=\int w\,h^{\mu\nu}G^{\mathrm{lin}}_{\mu\nu}$: $\partial_\mu E_{I_w}^{\mu\nu}=(\partial_\mu w)\,G_{\mathrm
    {lin}}^{\mu\nu}[h]$ (indices raised with $\eta$), which vanishes identically for arbitrary $h$ iff $w$ is locally
    constant (double plateau).
Reason: $\delta(h\cdot G)=2\partial_\mu\xi_\nu G^{\mu\nu}$ and $\partial_\mu G^{\mu\nu}=0$.
\item[(B)] \emph{Ultralocal.} For the matched line $\int w(h_{\mu\nu}h^{\mu\nu}-\tfrac{89}{140}h^2)$: $\partial_\mu E^{\mu\nu
    }=2\partial_\mu[w(h^{\mu\nu}-\tfrac{89}{140}\eta^{\mu\nu}h)]\ne0$ in every reading; this nonvanishing is genuine,
    because the ultralocal term is not invariant under $\delta h=\partial\xi+\partial\xi$.
\end{enumerate}
The Euler tensors of $\int R_{\mathrm{lin}}^2$ and $\int\mathrm{Ric}_{\mathrm{lin}}^2$ are also built independently, from $g=\
    eta+\epsilon h$ by first-order expansion of the Christoffel symbols and Riemann tensor (no hand-written linearized
    formula), with \texttt{sympy}'s \texttt{euler\_equations} and the Gate~V weights; they coincide exactly with the targets
     used in M.4/M.5 (and $R_{\mathrm{lin}}$, $R^{\mathrm{lin}}_{\mu\nu}$ agree with the formulas of Proposition~\ref{prop:
    match} with sign $+1$).
The certified preimages of $E_{R2}$, $E_{RC2}$ (in $\Ccoh$) and of $E_{R2}-\tfrac{188813}{88456}E_{RC2}$ (in $\Creg$) collapse
     exactly to these independent targets and satisfy the identity.
These are linear (free-field gauge) identities; none of (A), (B), (B$'$), (EH) is an interacting Ward identity (no ghost/gauge
    -fixed interacting sector is present; Target~II unfinished).
\end{proposition}

\begin{remark}[Seeking $\RM=\RL+\Delta\RL$]
\label{rem:M7-RM}
Within the homogeneous constraint (LC3), any $\Delta\RL$ with contact map in $\mathrm{Hom}_G$ yields a prescription-covariant
    change.
Correcting a nonzero $\mathrm{Obs}_L(\RL)$ is Unresolved pending an explicit $\varpi$.
No claim is made that $\RL$ itself is Lorentz covariant (cf.\ XVI-L, non-properties of $\RL$).
\end{remark}

\begin{remark}[Audit log (M.7 audited revision; retained)]
\label{rem:M7-audit}
\begin{enumerate}
\item \emph{Curvature-square divergence (bug).} Old: \texttt{covariance\_constraints.py} computed $\sum_\mu\eta^{\mu\mu}\
    partial_\mu E^{\mu\nu}$, inserting a metric sign into the divergence of an upper-index Euler tensor, and recorded ``does
     not vanish'' through a check that always passed\linebreak[1] (\texttt{require(True,}\allowbreak\texttt{\ldots)}). The
    Euler tensor itself (\texttt{jet\_euler.euler}, factor $1/w$) and the off-diagonal multiplicities were correct. Fix: $\
    partial_\mu E^{\mu\nu}$ with no metric factor; hard assertions $=0$ for $R^2$, $\mathrm{Ric}^2$ and the combination in
    all readings; independent metric-based construction; targets and preimages checked. The matched coefficient
    $188813/88456$ is unchanged (the matching code never used the divergence routine).
\item \emph{Intersection dimensions (bug).} Old: grade-restricted SA \emph{projections} intersected with $\mathrm{Equiv}_G$,
    and a truncating modular reduction; reported $5/5/3$. Fix: exact kernel of the adjoint-defect map on $\mathrm{Equiv}$,
    four objects per reading (Proposition~\ref{prop:M7-inter}); legacy reconstruction archived.
\item \emph{Stage~2 fixed data (overstatement).} Old: criterion without the adjoint-defect map; transition bound $0\le\dim\ker
    \Coll_{\mathrm{fixed}}\le22$ unsupported. Fix: criteria (i)--(iii), time-only caveat, universal-versus-actual separation
    , domain-size bound $3321$ (Section~\ref{sec:m4-fixed}).
\item \emph{Contact representation (bug).} Old: $\partial_\gamma\delta$ identified with $z^\gamma$ ``by the same formula'',
    giving the wrong boost action; ``homogeneous cokernel $=$ codimension'' was a misnomer. Fix: explicit $\eta$ intertwiner
    , exact boost verification (Proposition~\ref{prop:M7-boost}), corrected equivariant vectors; rank/codimension and
    cokernel (with codomain) reported separately.
\item \emph{M.6 scope (overstatement).} $C_{\mathrm{anom}}$ renamed the $N=0$ anomaly component; the M-C sentence is replaced
    (Section~\ref{sec:m6}).
\item \emph{Status.} M.7 downgraded to ``Constructed after audit'' for exactly verified items; the former ``ALL CHECKS PASSED
    '' covering wrong expected identities is replaced by assertions of the corrected identities in \texttt{covariance\_
    constraints.py} and \texttt{regression\_M.py}.
\end{enumerate}
\end{remark}

\paragraph{M.7 status.}
\textbf{Constructed after audit} (Gate~V for the SA items; (S-bv) certified), exactly verified: the actions $\rho_N$, $\rho^\
    delta_N$ and the $\eta$ intertwiner; the boost verification; the abstract equivariant count $14=5+9$ and its coordinate
    verification; the exact $V=\mathrm{Equiv}\cap\mathrm{SA}$ with its four derived objects in the formal, $\chi$-plateau
    and double-plateau readings; the linear divergence identities (A), (B), (B$'$), (EH).
\textbf{Unresolved}: $\mathrm{Obs}_L(\RL)$ for a concrete $\varpi$; any interacting Ward/BV statement (M.8--M.9).


\section{M.8 Action, insertion, ghosts, and diagrammatic closure (stub)}
\label{sec:m8}
\textbf{Unresolved.} Target~II requires the action, gauge fixing, ghosts and antifields, the composite insertion, BRST/BV
    transformations, and a diagram inventory beyond the XVI-L two-edge graph.
No finite fully specified BV slice is available from predecessors; building a full BV complex here would be speculative.
\textbf{Recommendation:} M.8 should wait for XVI-N (or an explicitly finite BV slice).
Completion ledger stub: \texttt{completion\_ledger\_M89.json}.
\textbf{M-D and M-E are not claimed.} Prescription covariance (M.7) is not interacting-sector symmetry.
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1987
1988 \section{M.9 Conditional symmetry-restoration result (stub)}
1989 \label{sec:m9}
1990 \textbf{Unresolved.} Depends on M.8 (closed sector).
1991 Possible outcomes once M.8 is specified: Constructed / Conditionally constructed / Obstructed / Unresolved.
1992 M.4 already shows that not every collision term can be made into a bulk gravitational counterterm: in the formal reading, no
      derivative term can.
1993 M.7 supplies prescription-covariant variational contacts of low dimension; it does not restore interacting Ward identities.
1994
1995 %=====================================================================
1996 \section{Status}
1997 \label{sec:status}
1998
1999 \begin{remark}[Revision log (this cycle)]
2000 \label{rem:revlog}
2001 This cycle integrates three completed work packages; XVI-A--XVI-L and their artifacts are not modified.
2002 \begin{enumerate}
2003 \item \textbf{Work package A} (Section~\ref{sec:wpA-Sbv}): Gate~S-bv certified (Theorem~\ref{thm:wpA-Sbv}); $c=1/(4\pi^2)$,
      boundary side $Q-iO$, including across the nonzero null cone; convention ledger; position-space forms of $W_O$ and $E
      _O^{\mathrm A}$.
2004 Consequences applied: the M.2 presentation, the M.3/M.4 ranks and the M.7 constructions are no longer conditional on (S-bv) (
      only $c\neq0$ and the single side were used); $r=-i/(8\pi^2)$ (Theorem~\ref{thm:M6-S2}); the identification at $z=0$ is
      not established ($S$ itself is global, XVI-J).
2005 Every former ``Gate~S-bv conditional'' statement in the body was replaced by a reference to Theorem~\ref{thm:wpA-Sbv}; the
      Euclidean motivation (Remark~\ref{rem:Sbv-motivation}) is kept and is explicitly not the certification.
2006 \item \textbf{Work package B} (Section~\ref{sec:wpB}): $A_\mu=\mathcal A_\mu$ ($B_\mu=0$) at every grade $N\le4$; top-order
      residues $\varpi$-independent; lower orders through $\nu_k[\varpi]$; double-plateau scale tensor variational but not
      matched (superseded in audit~3: only the top $D=4$ component is variational; Theorem~\ref{thm:wpB-dp}); M-C \emph{Proved
       obstructed} in the graph-specific matching domain (superseded in audit~3 by: M-C$_{\mathrm{curv}}$ Proved obstructed,
      broad M-C not achieved; Corollary~\ref{cor:wpB-MC}).
2007 The former M.6 remark ``$A_\mu=\mathcal A$ unresolved for $N\ge1$'' is marked resolved.
2008 \item \textbf{Work package C} (Section~\ref{sec:to-timeonly}): universal time-only classification; $\chiI\equiv1$ time-only
      reading; route~1 $=$ route~2; the double-plateau curvature-square matching does not survive the time-only reading (
      Remark~\ref{rem:M5-timeonly}).
2009 Downgrade applied: the actual-data consequence is Proposition~\ref{prop:to-actual} with hypotheses (H1)--(H3), not a
      Constructed claim; the linear independence of the $\sigeps^{(k)}$ is a separately proved Lemma~\ref{lem:to-indep}.
2010 \item \textbf{Integration consistency checks.}
2011 (a) \emph{Normalization of $c$.} The fragments agree: wpA certifies $c=1/(4\pi^2)$ with side $Q-iO$; the motivational value of
       Remark~\ref{rem:Sbv-motivation} coincides; wpB works with symbolic $c^2$, and substituting gives the unit $i\pi^2c^2=i
      /(16\pi^2)$, with $N=0$ top coefficient $-2\,i\pi^2c^2=r$; wpC uses only $c\neq0$. No sign or $2\pi$ discrepancy was
      found; \texttt{conventions.json} records $c$ and the side.
2012 (b) \emph{Labels.} No clashes between the fragments and the manuscript; the wpC labels \texttt{sec:time-only}, \texttt{def:
      time-only}, \texttt{thm:time-only} were renamed into the \texttt{to-} prefix (\texttt{sec:to-timeonly}, \texttt{def:to-
      timeonly}, \texttt{thm:to-timeonly}); wpA/wpB use the prefixes \texttt{wpA-}, \texttt{wpB-}.
2013 (c) \emph{Fragment typos.} A malformed longtable column specification and a retarded/advanced slip in the wpA fragment were
      corrected; plain-text cross-references were turned into \verb|\ref|s.
2014 (d) \emph{Superseded statements.} The fragments' own ``conditional on (S-bv)'' or ``not discharged'' clauses (written before
      integration) are superseded by Theorem~\ref{thm:wpA-Sbv}; wpC keeps ``on the current coherent presentation''.
2015 (e) \emph{Time-only scale tensor.} wpB left open how the scale tensor behaves in the time-only readings; Corollary~\ref{cor:
      wpB-timeonly} (new, derived at integration from the archived top-order coordinate and the wpC kernels) closes this: it
      is non-variational there.
2016 (f) \emph{Historical outputs.} Earlier-cycle script outputs and certificates keep their historical status strings (e.g.\ ``S-
      bv CONDITIONAL'', ``M-C \dots unfinished''); they are reproduced verbatim and are superseded by this revision.
2017 \end{enumerate}
2018 \end{remark}
2019
2020 \begin{remark}[Revision log (audit 3)]
2021 \label{rem:revlog3}
2022 Integration and proof audit following expert review; no new work package, no expansion; XVI-A--XVI-L, their artifacts and the
      GitHub repository are not modified.
2023 \begin{enumerate}
2024 \item \emph{Gate~S-bv provenance.} XVI-J already prints the formula (eq:W, eq:DF: $h_O=-W_O$, $d=W_O-iE_O^{\mathrm A}$, $S=-d
      $; eq:rational: $d=-\frac1{4\pi^2}(Q_{\mathrm{geom}}-i0)^{-1}$ on $z\neq0$ including nonzero null points; prop:q uses it
      ); XI-E, eq:h, only sets $h=-W$ and the jump. These are quoted in Section~\ref{sec:wpA-identify}, and Theorem~\ref{thm:
      wpA-Sbv} is reframed as a written check of XVI-J, eq:rational (corroboration, not a frozen XVI-L premise; the direct
      Fourier proof and Lemma~\ref{lem:wpA-varreg} stand independently). New Lemma~\ref{lem:wpA-varreg} proves the boundary
      value of the variable regulator $Q-2i\tau z^0-\tau^2$ near nonzero null points. The damping parameter is renamed $\
      varepsilon\to\tau$ (and the Euclidean slice variable $\tau\to\tau_{\mathrm E}$) in the manuscript and the wpA fragment
      and script. The scope sentence on ``extension across $z=0$'' is replaced by: the predecessor $S$ is globally defined (
      XVI-J); what is not established here is its position-space identification at the origin.
2025 \item \emph{Top versus full scale tensor.} In the double plateau, the cutoff-independent $D=4$ component is variational and
      Lorentz covariant, but not linearized-gauge invariant. The full scale tensor has cutoff-dependent lower-degree
      components and need not be variational; its Lorentz covariance is not established (Theorem~\ref{thm:wpB-dp}). The
      relation to XVI-L, thm:L6 / rem:Lressretch, is recorded (Remark~\ref{rem:wpB-topfull}); $r=-i/(8\pi^2)$ discharges only
      the top contact.
2026 \item \emph{Separating functional.} Proposition~\ref{prop:wpB-sep}: $\ell=a+b-c+d+e$ annihilates $v_{R^2},v_{\mathrm{Ric}^2}$
      and gives $\ell(v_{\mathrm{scale}})=-\frac1{16}$; basis order and signs verified against \texttt{full\_scale\_collapse\_
      certificates.json}, \texttt{COV4} and \texttt{conventions.json}; pure-gauge interpretation verified with \texttt{sympy};
       the $f''''$ coefficient $-\frac18$ of $E^{00}[f\,e_{00}]$ is recomputed from the archived $C_{\mathrm{top}}$.
2027 \item \emph{M-C split.} Definition~\ref{def:MCcurv}: broad M-C is the matched-scale criterion (proved matching + explicit $\
      varpi$ and a functional representation of the entire collapsed scale variation); M-C$_{\mathrm{data}}$ (matching +
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      explicit prescription-dependent scaling formulas) is satisfied; M-C$_{\mathrm{curv}}$ Proved obstructed (graph-specific)
      ; broad M-C not achieved (Unresolved) because XVI-L leaves $\varpi$ unspecified (thm:L6). The implication that ghosts/
      diagrams are the only enlargement is removed (Corollary~\ref{cor:wpB-MC}; release ladder; abstract).
\item \emph{Time-only.} The step ``By Lemma~\ref{lem:to-indep} this reduces to linear independence of the occurring switch
      jets'' is removed; the sufficient condition is kept for the joint family $\{\sigeps^{(k)}\partial^\rho\chiI\}$, with the
       counterexample Remark~\ref{rem:to-joint}. ``Do not extend to any region where $\sigeps$ varies'' is replaced by ``do
      not admit the stated universal time-only completion; the actual-ramp conclusion holds under Proposition~\ref{prop:to-
      actual}''. Corollary~\ref{cor:wpB-timeonly} was reviewed; its proof is complete relative to the reported computations
      and it is kept as a corollary.
\item \emph{Proof corrections.} (a) Lemma~\ref{lem:wpB-dw}: the basis derivative is $\mu\partial_\mu f_\gamma=(E-|\gamma|)f_\
      gamma$, combined with $(|\gamma|-|\alpha|)$ from $M^{-1}$. (b) Lemma~\ref{lem:bv}(c): explicit $\widehat{uv}=\frac1{2\pi
      }\hat u*\hat v$ and Beta integral, with identification of the H\"ormander product by the local Fourier formula; the ``
      sequential continuity'' shortcut is removed. (c) Proposition~\ref{prop:M7-id}(B$'$): weight-derivative terms arise in
      the Euler tensor and cancel in its divergence (cf.\ XII-C, eq:weighted-full-variation). (d) Shell annotations in the run
       listing and README are \texttt{\#} comments. (e) \texttt{regression\_M.py} labels current-status versus historical-
      record checks and has an audit-3 section.
\end{enumerate}
\end{remark}

\begin{remark}[Revision log (final verification)]
\label{rem:revlog-final}
Final verification edits; no expansion (no M.8--M.9); XVI-A--XVI-L, their artifacts and the GitHub repository are not modified
      .
\begin{enumerate}
\item \emph{Time-only $\chi$-plateau kernel.} Proposition~\ref{prop:to-chiplateau}: $\ker H_t|_D$ is displayed by cases ($D
      =0,2,4$ versus $D=1,3$); odd spatial order is anti-self-adjoint ($E[h]=\sigma(t)K\partial_1 h$ gives $E^\dagger=-E$
      since $\partial_1\sigma=0$); the spacetime-formal non-self-adjointness remark refers to the extra $D=2,4$ directions;
      dimensions $(26,0,156,0,390)$ unchanged; regression asserts the odd-$D$ anti-self-adjointness exactly.
\item \emph{Broad M-C matched-scale criterion.} Definition~\ref{def:MCcurv}: broad M-C requires (i)~a proved functional
      matching and (ii)~an explicit cutoff $\varpi$ with a functional representation of the entire collapsed scale variation;
      M-C$_{\mathrm{data}}$ (matching + explicit prescription-dependent scaling formulas) is satisfied; broad M-C Unresolved
      because XVI-L leaves $\varpi$ unspecified (thm:L6). Replaced ``uncontrolled'' by: numerical values and variationality of
       the full tensor are not determined for the unspecified cutoff.
\item \emph{Predecessor provenance.} Pinned local/frozen files and archived corpus excerpts under \texttt{XVI-M-artifacts/\
      allowbreak provenance/}; XVI-J eq:rational marked as corroboration, not a frozen XVI-L premise; ``resting on a new input
      '' $\to$ ``an explicitly checked kernel identification''.
\item \emph{Wording.} Lemma~\ref{lem:M2comp}: independently spot-checks every relation on $(\partial^\alpha S)(\partial^\beta
      S)$ at three rational points. Proposition~\ref{prop:M7-id}(EH): vanishes identically for arbitrary $h$ iff $w$ is
      locally constant.
\item \emph{Full scratch rerun.} \texttt{XVIM\_FULL\_WP\allowbreak =1} regression recorded as \texttt{regression\_full\_\
      allowbreak out.txt} (artifacts and package); appendix listing updated.
\end{enumerate}
\end{remark}

\subsection{Constructed}
\begin{enumerate}
\item M.0: frozen dependency ledger; Targets I/II; normalization table (now with the certified $S$ row); Gates S-bv (certified
      ) and V; verification boundary; release ladder.
\item M.1 (unconditional): general-$h$ assembly; exact specialization to XVI-L ($1065/1065$); supports $255/255/235$.
\item Gate~S-bv: \textbf{certified} (Theorem~\ref{thm:wpA-Sbv}); $S=\frac1{4\pi^2}(Q-i0)^{-1}$ on $\Omega$, across the nonzero
       null cone; $W_0$ and $E_0^{\mathrm A}$ in position space.
\item M.2 (unconditional on $\Omega$): boundary-value calculus lemma; two-direction relation equivalence; $d_N=(1,4,10,20,35)
      $; $\dim\mathcal V_{\mathrm{adm}}=70$; $185$ certified relations; wave-operator bound $\le210$.
\item M.3: $\dim\Ccoh=3321$; $\mathcal G_{\mathrm{formal}}$ injective; background kernel $1750$; plateau kernel $21$.
\item M.4 (Gate~V): collapse injective (formal, $\chi$-plateau, time-only), kernel $21$ (prep.\ plateau) / $22$ (double
      plateau); SA dimensions $26/26/47/1347$ (time-only: $26$; time-only with $\chiI\equiv1$: $572$); variational images
      $26/26/26/1325$ (time-only: $26$ and $572$); Helmholtz lemma and certificates; $R^2$ and $\mathrm{Ric}^2$ matchings in
      the double plateau.
\item Stage~2: fixed-data criteria (i)--(iii), universal region comparisons, and the universal time-only classification (
      Theorem~\ref{thm:to-timeonly}, Proposition~\ref{prop:to-chiplateau}); explicit $\sigeps$ and the linear independence of
      its derivatives (Lemma~\ref{lem:to-indep}).
\item M.5: named theorem~\ref{thm:M5-plateau-R2} (double-plateau domain); ledgers~(1)--(2); completion-gate obstruction;
      ledgers~(3)--(4) Unresolved.
\item M.6: dilation identity; $A_\mu=\mathcal A_\mu$ for all $N\le4$ (Theorem~\ref{thm:wpB-BO}); $S^2$ residue $r=-i/(8\pi^2)
      $; top-order residues ($\varpi$-independent); $N=0$ component collapse; full scale-tensor collapse; in the double
      plateau, the cutoff-independent $D=4$ component is variational and Lorentz covariant, but not linearized-gauge invariant
      , and lies outside the curvature span (exact separating functional, Proposition~\ref{prop:wpB-sep}); the full scale
      tensor has cutoff-dependent lower-degree components and need not be variational (Theorem~\ref{thm:wpB-dp}).
\item M.7 (\textbf{Constructed after audit}; Gate~V for SA items; exactly verified items only): so(1,3) action on $B_N$ and
      pull-back action on $\partial_\gamma\delta$ with explicit $\eta$ intertwiner and exact boost verification; abstract
      equivariant count $1,1,3,3,6$ (K $5$ + $\Creg$ $9$) with coordinate verification; exact $V=\mathrm{Equiv}\cap\mathrm{SA
      }$ ($\dim V=1/1/12$) with $\pi_K(V)$, $V\cap\mathcal C_{\mathrm K}$, $\pi_{\mathrm{reg}}(V)$, $V\cap\Creg$ ($0,0,1,1$ /
      $0,0,1,1$ / $5,3,9,7$); linear divergence identities (A), (B), (B$'$), (EH). $\mathrm{Obs}_L(\RL)$ Unresolved.
\item M.8--M.9: light stubs; Target~II / BV Unresolved; M.8 recommended to wait for XVI-N.
\end{enumerate}
Release: \textbf{M-A} Constructed (presentation unconditional on $\Omega$). \textbf{M-B} Constructed in the formal reading (
      Gate~V convention). \textbf{M-C$_{\mathrm{curv}}$} Proved obstructed (graph-specific); broad \textbf{M-C} not achieved (
      Unresolved; $\varpi$ unspecified by XVI-L, thm:L6); \textbf{M-C$_{\mathrm{data}}$} Satisfied. \textbf{M-D/M-E} not
      claimed.

\subsection{Proved obstructed}
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\begin{enumerate}
\item \textbf{M-C$_{\mathrm{curv}}$ (graph-specific; Definition~\ref{def:MCcurv}).} For the specified two-edge graph, the XVI-L prescription and any admissible $\varpi$, the full collapsed scale variation does not lie in the span of the M.5 curvature targets, because its fixed $D=4$ component already lies outside (Theorem~\ref{thm:wpB-dp}, Proposition~\ref{prop:wpB-sep}, Corollaries~\ref{cor:wpB-timeonly},~\ref{cor:wpB-MC}). Not a claim about gravity generally, and not an obstruction of broad M-C.
\end{enumerate}

\subsection{Selected-image statements (Gate~V; formal-versus-fixed kept)}
\begin{enumerate}
\item Formal reading: the formal universal-jet collapse image of this selected graph contains no nonzero variational tensor of total derivative degree $D\ge1$ ($\mathrm{SA}_D=0$ for $D\ge1$).
This is a vanishing inside one selected image under one reading; it is not a global obstruction to derivative counterterms.
\item Double plateau: the specified quadratic Einstein--Hilbert Euler tensor is not in the selected double-plateau collapse image.
Membership outside that selected image, or after other diagrams, is not addressed.
\item Time-only readings: the variational image is the $26$ ultralocals ($\chiI$ formal) or $572$ spatial-only, non-Lorentz directions ($\chiI\equiv1$); only $P_1-\tfrac{89}{140}P_2$ among the probed targets is an image; the weighted EH, $R^2$ and $\mathrm{Ric}^2$ Euler tensors are not (Proposition~\ref{prop:to-match}).
\end{enumerate}
These do not claim absence of derivative counterterms in a completed gravitational sector, failure for every fixed preparation, or impossibility after adding other diagrams.

\subsection{Unresolved}
\begin{enumerate}
\item Broad M-C (Definition~\ref{def:MCcurv}: proved functional matching plus an explicitly specified cutoff $\varpi$ and a functional representation of the entire collapsed scale variation): not achieved, because XVI-L leaves $\varpi$ unspecified (thm:L6). The top scale tensor has its own variational representative $I^{\mathrm{top}}_{\mathrm{scale}}$, so the M-C$_{\mathrm{curv}}$ obstruction does not decide broad M-C; M-C$_{\mathrm{data}}$ is satisfied. Their numerical values and the variationality of the resulting full tensor are not determined for the unspecified cutoff (Corollary~\ref{cor:wpB-MC}).
\item Lorentz covariance and variationality of the full (not only top) double-plateau scale tensor for the actual, unspecified $\varpi$.
\item The position-space identification of the predecessor $S$ at the origin $z=0$ ($S$ itself is globally defined, XVI-J; not needed for the punctured-space results).
\item Actual fixed-data dimensions where the actual $\chiI$ is non-constant, and transition regions (only $0\le\dim\ker\Coll_{\mathrm{fixed}}\le3321$); whether the actual $\chiI$ satisfies the hypotheses of Proposition~\ref{prop:to-actual} on given regions. Sufficient test (recorded in Section~\ref{sec:to-timeonly}): on an open $U$, a full-rank exact evaluation matrix of the joint family of external-data coefficient functions $\sigeps^{(k)}(t)\partial^\rho\chiI(x)$ occurring in the graph map, in $\Coll_t$ and in the adjoint-defect map $H_t$; not verified for any concrete $\chiI$ (separate independence of the two factors does not suffice, Remark~\ref{rem:to-joint}).
\item Lower-order scale residues: their values depend on $\nu_k[\varpi]$, $k=1,\dots,4$; XVI-L does not fix $\varpi$; the radial comparison is a separately named specialization, not the XVI-L prescription.
\item Inhomogeneous $\mathrm{Obs}_L(\RL)$ for a concrete $\varpi$ (M.7); the closed sector (M.8); restoration (M.9).
\item EOM and permitted field redefinitions for ledgers~(3)--(4).
\item Powers of $\kappa$ and $\hbar$, the remaining factors of $i$, loop order, and the relation of $\chiI$ to the action weighting.
\item Physical (normalized) values of the matched coefficients.
\end{enumerate}

\subsection{Non-claims}
\begin{enumerate}
\item M-D and M-E are not claimed. The M-C$_{\mathrm{curv}}$ obstruction is graph-specific: it concerns the specified two-edge graph, the XVI-L prescription and the M.5 curvature targets; it is not a claim about gravity generally and not an obstruction of broad M-C. Enlarging the ledger (by non-gauge-invariant quadratic representatives, which may remain within broad Target~I, or by gauge-fixing/ghost or further graph contributions) gives a different matching domain, not covered here.
\item The certification of Gate~S-bv is a written check of XVI-J, eq:rational, on $\mathbb R^4\setminus\{0\}$ only: the predecessor $S$ is globally defined (XVI-J), and its position-space identification at the origin is not established here; no $i\varepsilon$ prescription of the predecessors is modified, and the Euclidean motivation is not used as a proof.
\item The $546$ extra variational directions of the time-only $\chiI\equiv1$ reading are spatial-only and non-Lorentz; they are not proposed as counterterms.
\item Proposition~\ref{prop:to-actual} is a conditional transfer, not a certification of actual fixed-data dimensions.
\item $B_\mu=0$ does not remove prescription dependence: lower-order residues still depend on $\varpi$.
\item The linear divergence identities of M.7 are free-field gauge identities, not interacting Ward identities; prescription covariance is not interacting-sector symmetry.
\item The double-plateau intersection numbers of the previous M.7 draft ($5/5/3$) and its statement that the curvature-square Euler tensor has nonzero divergence are withdrawn (Remark~\ref{rem:M7-audit}).
\item No $\varepsilon\to0$; no UV completion; no interacting BRST/Ward identity as proved; no all-orders renormalization.
\item The XI inverse $G_{\mathrm{gr}}^{\mathrm R}=-\Pc E_0^{\mathrm R}$ is not equated with the X-sheet $\Delta_R^{\mathrm{gr}}=-4\Pc E_0^{\mathrm R}$.
\item Preimage coefficients and collision dimensions are not physical Wilson coefficients.
\item Scale-dependent subtraction and scale-variation coefficients are not physical beta functions.
\item No release claim of XVI-L is altered.
\item Computational certificates support the stated identities; they do not replace the written proofs.
\end{enumerate}

\scopebox{Release.}{%
\textbf{XVI-M draft:}
\begin{itemize}
\item General-$h$ coefficients: Constructed; they specialize exactly to XVI-L.
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\item Gate~S-bv: certified (written check of XVI-J, eq:rational), $S=\frac1{4\pi^2}(Q-iO)^{-1}$ on $\mathbb R^4\setminus \{0\}$; graded presentation, contact maps, collapse and variational subspaces: Constructed, unconditional on (S-bv).
\item Formal, (universal) preparation-plateau and time-only readings: only ultralocal $\chiI\sigeps K(h,h)$ potentials are variational ($546$ further spatial-only directions if $\chiI\equiv1$ in the time-only reading).
\item Double plateau: linearized $R^2$ and $\mathrm{Ric}^2$ are matched (Theorem~\ref{thm:M5-plateau-R2}); the specified quadratic Einstein--Hilbert Euler tensor is not in the selected collapse image; the matching does not survive the time-only reading.
\item Fixed data: criteria, region comparisons and time-only classification constructed; actual-ramp transfer as a proposition with explicit hypotheses; non-constant $\chiI$ Unresolved.
\item M.6: $r=-i/(8\pi^2)$ (top contact only); $A_\mu=\mathcal A_\mu$ at all grades. In the double plateau, the cutoff-independent $D=4$ component is variational and Lorentz covariant, but not linearized-gauge invariant, and lies outside $\mathrm{span}(E_{R^2},E_{\mathrm{Ric}^2})$ ($\ell(v_{\mathrm{scale}})=-\frac1{16}$). The full scale tensor has cutoff-dependent lower-degree components and need not be variational.
\item M.7 (Constructed after audit): prescription Lorentz compatibility matrix (equivariant $1,1,3,3,6$) with corrected contact representation; exact $\mathrm{Equiv}\cap\mathrm{SA}$ ($1/1/12$); linear divergence identities; Obs$_L(\RL)$ Unresolved. M.8 waits for XVI-N.
\end{itemize}
M-A Constructed. M-B Constructed in the formal reading. M-C$_{\mathrm{curv}}$ Proved obstructed (graph-specific; not a claim about gravity generally). Broad M-C not achieved (Unresolved; $\varpi$ unspecified by XVI-L, thm:L6). M-C$_{\mathrm{data}}$ Satisfied. M-D/M-E not claimed.
}


\begin{thebibliography}{9}
\bibitem{BF} R.~Brunetti, K.~Fredenhagen, \emph{Microlocal analysis and interacting quantum field theories}, Commun.\ Math.\ Phys.\ 208 (2000) 623, arXiv:math-ph/9903028.
\bibitem{FR} K.~Fredenhagen, K.~Rejzner, \emph{Perturbative algebraic quantum field theory}, arXiv:1208.1428.
\bibitem{Hor} L.~H\"ormander, \emph{The Analysis of Linear Partial Differential Operators I}, 2nd ed., Springer, 1990 (Thms .~6.1.2, 8.1.6, 8.2.4, 8.2.10; theorem numbers as in that edition).
\bibitem{GS} I.~M.~Gel'fand, G.~E.~Shilov, \emph{Generalized Functions, Vol.~1}, Academic Press.
\bibitem{Olver} P.~J.~Olver, \emph{Applications of Lie Groups to Differential Equations}, 2nd ed., Springer (Thm.~5.92).
\bibitem{series} Series files XI-E, XII-C, XVI-I, XVI-J, XVI-K, XVI-L (quoted by label).
\end{thebibliography}

%=====================================================================
\appendix
\section{Computational artifacts}
\label{app:script}

Inventory of \texttt{XVI-M-artifacts/}:
\begin{sloppypar}
\begin{itemize}
\item Shared code and copies: \texttt{xvi\_common.py} (byte-identical to XVI-L); \texttt{m4\_common.py}; \texttt{jet\_euler.py}.
\item M.1: \texttt{graph\_general.py}, with \texttt{graph\_general\_out.txt} and \texttt{graph\_general\_table.txt}.
\item M.2: \texttt{kernel\_relations.py}, with \texttt{kernel\_relations\_out.txt}, \texttt{kernel\_basis\_table.txt} (reconstruction certificates and numerator rows), \texttt{kernel\_relation\_matrix.txt} (nullspace vectors) and \texttt{kernel\_basis.json}.
\item M.3: \texttt{contact\_space.py}, with \texttt{contact\_space\_out.txt} and \texttt{graph\_contact\_map.txt} ($u_m$, annihilators).
\item M.4: \texttt{contact\_collapse.py}, with \texttt{contact\_collapse\_out.txt}, \texttt{collapse\_kernel.json} and \texttt{collapse\_kernel\_certificates.txt}; \texttt{helmholtz\_check.py}, with \texttt{helmholtz\_check\_out.txt}, \texttt{helmholtz\_certificates.txt} and \texttt{helmholtz\_summary.json}; \texttt{matching\_probe.py}, with \texttt{matching\_probe\_out.txt} and \texttt{matching\_certificates.txt}.
\item Stage~2: \texttt{fixed\_data\_reading.py}, with \texttt{fixed\_data\_reading\_out.txt}, \texttt{fixed\_data\_reading.json}, \texttt{prep\_plateau\_sa.json}.
\item M.5: \texttt{counterterm\_functionals.py}, with \texttt{counterterm\_functionals\_out.txt}, \texttt{matching\_matrix.json}/\texttt{.txt}, \texttt{matching\_certificates\_M5.txt}, \texttt{ledger\_*.json}, \texttt{ledger\_tables.txt}, \texttt{m5\_ledger\_summary.json}.
\item M.6: \texttt{scaling\_anomaly.py}, with \texttt{scaling\_anomaly\_out.txt}, \texttt{scaling\_anomaly\_cert.json}, \texttt{scaling\_anomaly\_cert.txt}.
\item M.7 (audited): \texttt{covariance\_constraints.py} (imports \texttt{curvature\_independent.py}, the independent metric-linearization route), with \texttt{covariance\_constraints\_out.txt}, \texttt{covariance\_matrix.txt}, \texttt{covariance\_certificates.txt}, \texttt{covariance\_summary.json}, \texttt{covariance\_exact\_certificates.json} (exact rational bases of $V$, $\pi_K(V)$, $\pi_{\mathrm{reg}}(V)$, $V\cap\mathcal C_{\mathrm K}$, $V\cap\Creg$ and the corrected equivariant vectors); \texttt{m7\_legacy\_reconstruction.py} with \texttt{m7\_legacy\_reconstruction\_out.txt} and \texttt{.json} (reconstruction of the withdrawn numbers); completion stub \texttt{completion\_ledger\_M89.json}.
\item Work package A (\texttt{wpA/}, Gate~S-bv): \texttt{kernel\_identification.tex} (source fragment of Section~\ref{sec:wpA-Sbv}), \texttt{kernel\_identification\_checks.py} (symbolic/numerical cross-checks; writes \texttt{kernel\_identification\_out.txt}), \texttt{kernel\_conventions.json} (convention ledger, $c$, boundary side).
\item Work package B (\texttt{wpB/}, scaling): \texttt{scaling\_prescription\_identity.tex} (source fragment of Section~\ref{sec:wpB}), \texttt{higher\_grade\_scaling.py} with \texttt{higher\_grade\_scaling\_out.txt}, \texttt{scale\_derivative\_coefficients.json}, \texttt{dilation\_defect\_coefficients.json}, \texttt{scale\_defect\_difference.json}, \texttt{full\_scale\_collapse\_certificates.json}, \texttt{matched\_scale\_ledger.txt}, and the state files \texttt{\_wpB\_state.json}, \texttt{\_wpB\_sectionD.json}.
\item Work package C (\texttt{wpC/}, time-only): \texttt{time\_only\_section.tex} (source fragment of Section~\ref{sec:to-timeonly}), \texttt{time\_only\_helmholtz.py} with \texttt{time\_only\_helmholtz\_out.txt}, \texttt{time\_only\_collapse\_certificates.json}, \texttt{time\_only\_sa\_certificates.json}, \texttt{time\_only\_matching\_certificates.json}, \texttt{time\_only\_summary.json}, \texttt{time\_only\_specialization\_tests.txt}.
The fragments are archived as delivered, except the audit-3 renaming of the damping parameter $\varepsilon\to\tau$ (and of the Euclidean slice variable to $\tau_{\mathrm E}$) in \texttt{wpA/kernel\_identification.tex} and \texttt{wpA/kernel\_
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     identification\_checks.py}, whose new check section~7b is also audit-3; the manuscript sections are edited versions (
     Remarks~\ref{rem:revlog} and~\ref{rem:revlog3}).
\item Bookkeeping: \texttt{regression\_M.py}, \texttt{regression\_M\_out.txt} and \texttt{regression\_full\_\allowbreak out.
     txt} (full scratch rerun with \texttt{XVIM\_FULL\_WP\allowbreak =1}); \texttt{make\_manifest.py} and \texttt{provenance
     \_manifest.txt}; \texttt{provenance/} (predecessor pins and corpus excerpts); \texttt{conventions.json}; \texttt{README.
     txt}.
\end{itemize}
\end{sloppypar}
Run from the archive parent directory, \texttt{/workspace}:
\begin{lstlisting}
PY=/workspace/tmp/venv/bin/python ; cd /workspace ; A=XVI-M-artifacts
$PY $A/graph_general.py           > $A/graph_general_out.txt
$PY $A/kernel_relations.py        > $A/kernel_relations_out.txt
$PY $A/contact_space.py           > $A/contact_space_out.txt
$PY $A/contact_collapse.py        > $A/contact_collapse_out.txt
$PY $A/helmholtz_check.py         > $A/helmholtz_check_out.txt
$PY $A/matching_probe.py          > $A/matching_probe_out.txt
$PY $A/fixed_data_reading.py      > $A/fixed_data_reading_stdout.txt   # writes fixed_data_reading_out.txt
$PY $A/counterterm_functionals.py > $A/counterterm_functionals_out.txt
$PY $A/scaling_anomaly.py         > $A/scaling_anomaly_out.txt
$PY $A/covariance_constraints.py  > $A/covariance_constraints_out.txt  # imports curvature_independent.py
$PY $A/m7_legacy_reconstruction.py > $A/m7_legacy_reconstruction_out.txt
$PY $A/wpA/kernel_identification_checks.py   # writes wpA/kernel_identification_out.txt
$PY $A/wpB/higher_grade_scaling.py > $A/wpB/higher_grade_scaling_out.txt   # ~5.5 min; writes wpB/*.json
$PY $A/wpC/time_only_helmholtz.py   # ~2.5 min; writes wpC/time_only_helmholtz_out.txt, wpC/*.json
$PY $A/make_manifest.py
$PY $A/regression_M.py            > $A/regression_M_out.txt
# full scratch rerun of wpB/wpC (about 8 min); recorded output:
XVIM_FULL_WP=1 $PY $A/regression_M.py > $A/regression_full_out.txt
\end{lstlisting}
Exit status $0$ together with \texttt{ALL CHECKS PASSED} means every check passed.
XVI-L and XVI-K artifacts are not modified; \texttt{regression\_M.py} verifies their SHA-256.
\texttt{regression\_M.py} reruns the wpA script and compares its output byte for byte, checks the archived wpB/wpC results and
      recomputes the key ones independently; setting \texttt{XVIM\_FULL\_WP\allowbreak =1} also reruns the wpB and wpC
     scripts in a scratch copy and compares outputs, ignoring timing lines.
\emph{Historical output strings.} Outputs and certificates from earlier cycles (M.2--M.7, and the wp scripts written before
     integration) carry status strings such as ``CONDITIONAL on (S-bv)'', ``Gate S-bv not discharged'' or ``M-C \dots
     unfinished''. They are reproduced verbatim below as records of those runs and are superseded by this revision (Theorem~\
     ref{thm:wpA-Sbv}, Corollary~\ref{cor:wpB-MC}); the numbers they certify are unchanged. In \texttt{regression\_M.py},
     checks are labelled \texttt{[current status]} (assertions about the present manuscript and ledger) or \texttt{[
     historical record]} (checks that an archived output still carries its historical status string).

\section{Script outputs (verbatim copies of the current recorded outputs)}
\label{app:out}
\paragraph{M.1 (\texttt{graph\_general\_out.txt}).}
\begin{lstlisting}
# graph_general.py output   (python 3.13.5, sympy 1.14.0, exact QQ arithmetic via fractions.Fraction)
# M.1 GENERAL-h assembly of the XVI-L complete-source two-edge graph (independent h-jets)
CHECK frozen input present: XVI-L-artifacts/source_full_table.txt: True
# sha256(XVI-L-artifacts/source_full_table.txt) = e13ce0ae29c769ac3b154b338a7dba2eaa75c2f12fa3427c93bdbe1cfb08430f
CHECK frozen input present: XVI-K-artifacts/hessian_full_out.txt: True
# sha256(XVI-K-artifacts/hessian_full_out.txt) = ce840f99f6df6ba31f6d415d78ec050a5ba09bb0c3153192d94c829f585ce490
CHECK frozen input present: XVI-L-artifacts/two_edge_coeff_table.txt: True
# sha256(XVI-L-artifacts/two_edge_coeff_table.txt) = cf53538133e87da76fa126e884223bab2871ac3a177f740005f66df4d3e0b1c7
CHECK frozen input present: XVI-L-artifacts/xvi_common.py: True
# sha256(XVI-L-artifacts/xvi_common.py) = 5b7b0c483a31489548efc901e3388afae7a86efc376073ab25416b34367b6f3d
CHECK xvi_common.py is byte-identical to XVI-L-artifacts/xvi_common.py: True
CHECK P_A table has 5056 entries (XVI-L L.1): True
CHECK source placements 1/8/36 for C_2/M_2/J_3: True
CHECK P_C2 uses only sigma 2-jet symbols: True
CHECK P_M2 uses only sigma 1-jet symbols: True
CHECK P_J3 uses only sigma: True
CHECK L coefficient count 2988 (XVI-K): True
CHECK slot symmetry L_ij^{pq} = L_ji^{qp} (entrywise, symbolic): True
CHECK each L^{pq}_{ij} is linear in h-jets of order exactly 2-|p|-|q| (XVI-K prop:hess(1)): True
# distinct residual-metric jet symbols occurring in L: 122
CHECK 45 ordered vertex placements (symbolic L): True
# T_i^r = (-P B^i)^r ; nnz = 22/100
# === GENERAL-h ASSEMBLY (all sectors) ===
# sector C_2 assembled in 0.1s; ordered nonzero (placement,mn) contributions 450
# sector M_2 assembled in 0.5s; ordered nonzero (placement,mn) contributions 3600
# sector J_3 assembled in 0.5s; ordered nonzero (placement,mn) contributions 10644
# assembly wall time 1.1s
CHECK edge interchange: ordered contribution (A;a,b;p,q) equals (A;b,a;q,p) for EVERY ordered placement with distinct swap (
     exhaustive over all enumerated placements: 1000 unordered pairs): True
CHECK Lambda_adm has 255 labels (XVI-L L.3): True
CHECK admissible per sector 25/84/250 (XVI-L): True
CHECK every monomial is exactly (one sigma-jet symbol) x (one h-jet symbol): bilinear in prep jet and residual-metric jet:
     True
CHECK sector grading: prep-jet order = 2-s, h-jet order = 2-v, N = s+v (s=0,1,2 for C2,M2,J3): True
CHECK no unintended generators (only sector prep symbols and XVI-K h-jet symbols): True
```

```
CHECK all coefficients exact rationals: True
CHECK output tensor stored as symmetric (mu<=nu) components (P_A symmetric in (mu,nu), XVI-L L.1 (C3)): True
CHECK C_2 formal support lies in its admissible block: True
CHECK M_2 formal support lies in its admissible block: True
CHECK J_3 formal support lies in its admissible block: True
CHECK XVI-L two_edge_coeff_table.txt parsed: 1065 nonzero entries: True
CHECK general-h table specialized to h=(w^1)^2 eta has exactly 1065 nonzero (A,lambda,mn) entries: True
CHECK specialized nonzero key set equals XVI-L key set (sector, label, component): True
CHECK EXACT SPECIALIZATION: every grouped coefficient equals the XVI-L background coefficient (all 1065 entries,
      coefficientwise in prep jet and w^1): True
CHECK background support recomputed: C2 25, M2 84, J3 202, union 235 (XVI-L): True
# === SUPPORT DISTINCTIONS ===
# C_2: |Lambda_adm|=25 {0: 1, 1: 4, 2: 20} ; |Lambda_graph^formal|=25 {0: 1, 1: 4, 2: 20} ; |Lambda_graph^background|=25 {0:
      1, 1: 4, 2: 20}
# M_2: |Lambda_adm|=84 {1: 4, 2: 20, 3: 60} ; |Lambda_graph^formal|=84 {1: 4, 2: 20, 3: 60} ; |Lambda_graph^background|=84 {1:
       4, 2: 20, 3: 60}
# J_3: |Lambda_adm|=250 {2: 20, 3: 60, 4: 170} ; |Lambda_graph^formal|=250 {2: 20, 3: 60, 4: 170} ; |Lambda_graph^background
      |=202 {2: 20, 3: 32, 4: 150}
# complete: |Lambda_adm|=255 {0: 1, 1: 4, 2: 20, 3: 60, 4: 170} ; |Lambda_graph^formal|=255 {0: 1, 1: 4, 2: 20, 3: 60, 4: 170}
       ; |Lambda_graph^background|=235 {0: 1, 1: 4, 2: 20, 3: 60, 4: 150}
CHECK Lambda_graph^background subset Lambda_graph^formal subset Lambda_adm: True
# labels in Lambda_adm but NOT in Lambda_graph^formal (0):
# labels in Lambda_graph^formal but vanishing on the XVI-L background (20): {(), 0123} {3, 012} {33, 12} {33, 02} {33, 01} {2,
       013} {23, 11} {23, 01} {23, 00} {22, 13} {22, 03} {22, 01} {1, 023} {13, 02} {13, 00} {12, 03} {12, 00} {123, 0} {11,
      03} {11, 02}
# nonzero (lambda,mn) entries by sector: {'C_2': 250, 'M_2': 840, 'J_3': 2192}  bilinear terms: {'C_2': 4008, 'M_2': 8092, 'J
      _3': 6546}
# sector-summed a_lambda: nonzero (lambda,mn) entries 2322 ; bilinear terms 18646
# sample: a^{C_2,(0,0)}_{{0,0}} has 78 bilinear terms (sigma_2jet x d^2h)
#   first terms: 5*sig_11*h_00_11 ; 1*sig_11*h_00_22 ; 1*sig_11*h_00_33 ; -10*sig_11*h_01_01 ; -2*sig_11*h_02_02 ; -2*sig_11*h
      _03_03
# === INDEPENDENT C_2 ROUTE, GENERAL h (xvi_common.polar_s, all 25 labels x 10 mn) ===
CHECK GENERAL-h C_2 regression: all 25 labels x 10 components equal the independent polar_s route, coefficientwise in (sigma
      2-jet) x (h-jet) (250 comparisons, 23.2s): True
# wrote graph_general_table.txt : 3282 nonzero (A,lambda,mn) lines
# sha256(graph_general_table.txt) = 4cd29d3c9f691340b314aecbfe7d6aef3590b9cee21dcdb486fe9fbcdd679e49
# total runtime 27.1s
ALL CHECKS PASSED
\end{lstlisting}

\paragraph{M.2 (\texttt{kernel\_relations\_out.txt}).}
\begin{lstlisting}
# kernel_relations.py output   (python 3.13.5, sympy 1.14.0, exact QQ arithmetic)
# M.2 numerical-kernel relations: presentation C^{Lambda_adm} -> V_adm, degree by degree
# --- finite symbolic check behind the MOTIVATION for (S-bv) (Euclidean calculation; does not certify (S-bv)) ---
CHECK int_0^oo e^{-p tau} sin(p r) dp = r/(tau^2+r^2): True
CHECK XVI-K eq:W-fam integrand at z0=-i*tau gives 1/(4 pi^2 (tau^2+r^2)) (radial integration): True
CHECK d^alpha (1/Q) = p_alpha / Q^{1+|alpha|} with p_alpha homogeneous of degree |alpha| (all 70 |alpha|<=4): True
CHECK p_0 = 1: True
CHECK p_{e_0} = -2 z0  (coordinate derivative d_0 Q = 2 z0): True
CHECK p_{e_1} = +2 z1  (coordinate derivative d_1 Q = -2 z1): True
CHECK sign example: p_{e0} p_{e1} = -4 z0 z1: True
CHECK sign example: p_{e0+e1} = -8 z0 z1: True
CHECK sign example: (d0 S)(d1 S) = (1/2) S d0 d1 S  (numerators -4 z0 z1 = (1/2)(-8 z0 z1)): True
# labelled generators per N: [1, 4, 20, 60, 170]  total 255
CHECK label counts 1,4,20,60,170 (=Lambda_adm, XVI-L): True
CHECK qo_lambda = c^2 n_lambda (Q-i0)^{-2-N}, n_lambda=p_alpha p_beta homogeneous of degree N, for all 255 labels: True
CHECK N=0: greedy basis has rank-many elements (1): True
CHECK N=0: exact reconstruction n_lambda = sum_b R_{lambda b} n_b for all 1 labels: True
CHECK N=0: every exported relation vector annihilates the numerator map exactly (0 relations): True
CHECK N=0: dim ker pi_N = #labels - rank = 0: True
CHECK N=1: greedy basis has rank-many elements (4): True
CHECK N=1: exact reconstruction n_lambda = sum_b R_{lambda b} n_b for all 4 labels: True
CHECK N=1: every exported relation vector annihilates the numerator map exactly (0 relations): True
CHECK N=1: dim ker pi_N = #labels - rank = 0: True
CHECK N=2: greedy basis has rank-many elements (10): True
CHECK N=2: exact reconstruction n_lambda = sum_b R_{lambda b} n_b for all 20 labels: True
CHECK N=2: every exported relation vector annihilates the numerator map exactly (10 relations): True
CHECK N=2: dim ker pi_N = #labels - rank = 10: True
CHECK N=3: greedy basis has rank-many elements (20): True
CHECK N=3: exact reconstruction n_lambda = sum_b R_{lambda b} n_b for all 60 labels: True
CHECK N=3: every exported relation vector annihilates the numerator map exactly (40 relations): True
CHECK N=3: dim ker pi_N = #labels - rank = 40: True
CHECK N=4: greedy basis has rank-many elements (35): True
CHECK N=4: exact reconstruction n_lambda = sum_b R_{lambda b} n_b for all 170 labels: True
CHECK N=4: every exported relation vector annihilates the numerator map exactly (135 relations): True
CHECK N=4: dim ker pi_N = #labels - rank = 135: True
# === DEGREE-BY-DEGREE PRESENTATION (conditional on (S-bv)) ===
#  N | labelled generators | dim Poly_N | d_N = dim H_{4+N} | dim ker pi_N
#  0 |   1 |  1 |  1 |   0
#  1 |   4 |  4 |  4 |   0
```

```
#  2 |  20 | 10 | 10 |  10
#  3 |  60 | 20 | 20 |  40
#  4 | 170 | 35 | 35 | 135
CHECK d_N = dim Poly_N = C(N+3,3) = 1,4,10,20,35 : numerator map is ONTO homogeneous polynomials of degree N: True
# dim V_adm = 70 (vs 255 labels); dim ker pi = 185
CHECK dim V_adm = 70 (conditional on (S-bv)): True
CHECK every label with N<=2 is an XVI-K label (V_K block = C_2 block): True
# V_K = H_4 + H_5 + H_6 ; dim V_K = 15 = {0: 1, 1: 4, 2: 10} ; V_K cap H_7 = V_K cap H_8 = 0
CHECK dim V_K = 1+4+10 = 15 (25 XVI-K generators, 10 relations, all at N=2): True
CHECK selected basis of H_4+H_5+H_6 consists of XVI-K labels (basis extends a basis of V_K): True
# === UNCONDITIONAL STRUCTURAL RELATIONS (square S = 0 on R^4\{0}; no (S-bv)) ===
#  N=0: wave-operator relations generated 0, span dim 0 ; dim ker pi_N = 0 ; unconditional upper bound d_N <= 1
#  N=1: wave-operator relations generated 0, span dim 0 ; dim ker pi_N = 0 ; unconditional upper bound d_N <= 4
CHECK N=2: every wave-operator relation lies in ker pi_N (consistency with (S-bv)): True
#  N=2: wave-operator relations generated 1, span dim 1 ; dim ker pi_N = 10 ; unconditional upper bound d_N <= 19
CHECK N=3: every wave-operator relation lies in ker pi_N (consistency with (S-bv)): True
#  N=3: wave-operator relations generated 8, span dim 8 ; dim ker pi_N = 40 ; unconditional upper bound d_N <= 52
CHECK N=4: every wave-operator relation lies in ker pi_N (consistency with (S-bv)): True
#  N=4: wave-operator relations generated 36, span dim 36 ; dim ker pi_N = 135 ; unconditional upper bound d_N <= 134
# unconditional upper bound: dim V_adm <= 210 ; (S-bv) value: 70
CHECK wave-operator span is contained in the full kernel (dimension check): True
# relations NOT generated by the wave-operator family (need (S-bv)): {0: 0, 1: 0, 2: 9, 3: 32, 4: 99}
# sample N=2 relations (reconstruction form  e_lambda - sum_b R e_b = 0):
#   (-1)*q{(), 33} + (-1)*q{(), 22} + (-1)*q{(), 11} + (1)*q{(), 00}
#   (-1)*q{(), 33} + (-1/2)*q{(), 22} + (-1/2)*q{(), 11} + (1)*q{3, 3} + (1)*q{0, 0}
#   (-1/2)*q{(), 01} + (1)*q{1, 0}
#   (-1/2)*q{(), 02} + (1)*q{2, 0}
# dim D_{<=N} = {0: 1, 1: 5, 2: 15, 3: 35, 4: 70}
# dim C_coh = sum_N d_N dim D_{<=N} = 3321   (raw labelled slots 14321, XVI-L count 14321)
# dim C_reg (Delta R|_{V_K}=0) = d_3*35 + d_4*70 = 3150
CHECK raw slot count reproduces XVI-L 14321: True
CHECK dim C_coh = 3321, dim C_reg = 3150 (conditional on (S-bv)): True
# wrote kernel_basis_table.txt  sha256=267344b69a488ec7f99fa3a86180fe1cf25431c60a31beeec3d777da579d88e8
# wrote kernel_relation_matrix.txt  sha256=f25684da36b2f19709e344ab3cb599c093bf8d4e9b236ba3ef069ac2fc8dcbec
# wrote kernel_basis.json  sha256=fcbe2876abd2a6b410813f732356f829c970c22c3a333f7e4ec34909c0e0793b
# total runtime 2.9s
ALL CHECKS PASSED
\end{lstlisting}

\paragraph{M.3 (\texttt{contact\_space\_out.txt}).}
\begin{lstlisting}
# contact_space.py output   (python 3.13.5, exact QQ arithmetic via fractions.Fraction)
# M.3 coherent contact space, graph-contact map G, kernel/image (conditional on (S-bv) through M.2)
CHECK input present: graph_general_table.txt: True
# sha256(graph_general_table.txt) = 4cd29d3c9f691340b314aecbfe7d6aef3590b9cee21dcdb486fe9fbcdd679e49
CHECK input present: kernel_basis.json: True
# sha256(kernel_basis.json) = fcbe2876abd2a6b410813f732356f829c970c22c3a333f7e4ec34909c0e0793b
CHECK graph_general_table.txt: 3282 sector-resolved nonzero lines (M.1): True
CHECK d_N = 1,4,10,20,35 (M.2, conditional on (S-bv)): True
CHECK reconstruction available for all 255 labels: True
# dim D_{<=N} = {0: 1, 1: 5, 2: 15, 3: 35, 4: 70}  ; dim C_coh = 3321  ; dim C_reg = 3150  ; raw labelled slots = 14321
CHECK FORMAL N=0: exact rank 1 equals independent mod-p rank: True
CHECK FORMAL N=1: exact rank 4 equals independent mod-p rank: True
CHECK FORMAL N=2: exact rank 10 equals independent mod-p rank: True
CHECK FORMAL N=3: exact rank 20 equals independent mod-p rank: True
CHECK FORMAL N=4: exact rank 35 equals independent mod-p rank: True
# FORMAL                    dim U_N = [1, 4, 10, 20, 35] ; dim im G = 3321 ; dim ker G = 0 ; on C_reg: dim im = 3150, dim ker
      = 0  (0.1s)
CHECK TIME-ONLY N=0: exact rank 1 equals independent mod-p rank: True
CHECK TIME-ONLY N=1: exact rank 4 equals independent mod-p rank: True
CHECK TIME-ONLY N=2: exact rank 10 equals independent mod-p rank: True
CHECK TIME-ONLY N=3: exact rank 20 equals independent mod-p rank: True
CHECK TIME-ONLY N=4: exact rank 35 equals independent mod-p rank: True
# TIME-ONLY                 dim U_N = [1, 4, 10, 20, 35] ; dim im G = 3321 ; dim ker G = 0 ; on C_reg: dim im = 3150, dim ker
      = 0  (0.1s)
CHECK BACKGROUND N=0: exact rank 1 equals independent mod-p rank: True
CHECK BACKGROUND N=1: exact rank 4 equals independent mod-p rank: True
CHECK BACKGROUND N=2: exact rank 10 equals independent mod-p rank: True
CHECK BACKGROUND N=3: exact rank 20 equals independent mod-p rank: True
CHECK BACKGROUND N=4: exact rank 10 equals independent mod-p rank: True
# BACKGROUND                dim U_N = [1, 4, 10, 20, 10] ; dim im G = 1571 ; dim ker G = 1750 ; on C_reg: dim im = 1400, dim
     ker = 1750  (0.0s)
CHECK TIME+BG N=0: exact rank 1 equals independent mod-p rank: True
CHECK TIME+BG N=1: exact rank 4 equals independent mod-p rank: True
CHECK TIME+BG N=2: exact rank 10 equals independent mod-p rank: True
CHECK TIME+BG N=3: exact rank 16 equals independent mod-p rank: True
CHECK TIME+BG N=4: exact rank 10 equals independent mod-p rank: True
# TIME+BG                   dim U_N = [1, 4, 10, 16, 10] ; dim im G = 1431 ; dim ker G = 1890 ; on C_reg: dim im = 1260, dim
     ker = 1890  (0.0s)
CHECK PLATEAU(J_3 only, formal) N=0: exact rank 0 equals independent mod-p rank: True
CHECK PLATEAU(J_3 only, formal) N=1: exact rank 0 equals independent mod-p rank: True
```

```
CHECK PLATEAU(J_3 only, formal) N=2: exact rank 10 equals independent mod-p rank: True
CHECK PLATEAU(J_3 only, formal) N=3: exact rank 20 equals independent mod-p rank: True
CHECK PLATEAU(J_3 only, formal) N=4: exact rank 35 equals independent mod-p rank: True
# PLATEAU(J_3 only, formal)  dim U_N = [0, 0, 10, 20, 35] ; dim im G = 3300 ; dim ker G = 21 ; on C_reg: dim im = 3150, dim ker = 0  (0.0s)
CHECK C_2 only, formal N=0: exact rank 1 equals independent mod-p rank: True
CHECK C_2 only, formal N=1: exact rank 4 equals independent mod-p rank: True
CHECK C_2 only, formal N=2: exact rank 10 equals independent mod-p rank: True
CHECK C_2 only, formal N=3: exact rank 0 equals independent mod-p rank: True
CHECK C_2 only, formal N=4: exact rank 0 equals independent mod-p rank: True
# C_2 only, formal           dim U_N = [1, 4, 10, 0, 0] ; dim im G = 171 ; dim ker G = 3150 ; on C_reg: dim im = 0, dim ker = 3150  (0.0s)
CHECK M_2 only, formal N=0: exact rank 0 equals independent mod-p rank: True
CHECK M_2 only, formal N=1: exact rank 4 equals independent mod-p rank: True
CHECK M_2 only, formal N=2: exact rank 10 equals independent mod-p rank: True
CHECK M_2 only, formal N=3: exact rank 20 equals independent mod-p rank: True
CHECK M_2 only, formal N=4: exact rank 0 equals independent mod-p rank: True
# M_2 only, formal           dim U_N = [0, 4, 10, 20, 0] ; dim im G = 870 ; dim ker G = 2451 ; on C_reg: dim im = 700, dim ker = 2450  (0.0s)
CHECK specialized visible ranks are bounded by the formal ones (specialization cannot raise rank): True
CHECK N=0: S^2 direction is graph-visible (U_0 = H_4): True
CHECK coefficient monomials of different N are disjoint (grading prep order 2-s, h order 2-N+s): True
CHECK every reduced coefficient monomial has prep order + h order = 4 - N: True
CHECK N=0: annihilator of U_N has dimension d_N - dim U_N = 0: True
CHECK N=1: annihilator of U_N has dimension d_N - dim U_N = 0: True
CHECK N=2: annihilator of U_N has dimension d_N - dim U_N = 0: True
CHECK N=3: annihilator of U_N has dimension d_N - dim U_N = 0: True
CHECK N=4: annihilator of U_N has dimension d_N - dim U_N = 0: True
CHECK kernel certificate: for every annihilator vector x of U_N, sum_b x_b abar_b = 0 exactly (graph-invisible numerical direction): True
# wrote graph_contact_map.txt sha256=49b59c00dc788efcb48681f4bb711fa11350f251404977aba5d251df6018628e
# === COMPARISON: ambient prescription freedom vs graph-visible collision freedom (FORMAL) ===
#   raw labelled contact slots (XVI-L count)          : 14321
#   coherent assignments dim C_coh (M.2, (S-bv))       : 3321
#   graph-visible collision freedom dim im G (formal)  : 3321
#   graph-invisible coherent directions dim ker G      : 0
#   regression-preserving: dim C_reg 3150 -> dim im 3150, ker 0
# total runtime 0.3s
ALL CHECKS PASSED
\end{lstlisting}

\paragraph{M.4 collapse (\texttt{contact\_collapse\_out.txt}).}
\begin{lstlisting}
# contact_collapse.py output   (exact QQ arithmetic; conditional on (S-bv) through M.2)
# M.4 contact-collapse map: rank, kernel; injectivity of G does NOT imply injectivity after collapse
CHECK graph_general_table loaded (3282 lines; field indices a<=b): True
CHECK d_N = 1,4,10,20,35 (M.2): True
CHECK dim C_coh = 3321 contact variables: True
# sha256(graph_general_table.txt) = 4cd29d3c9f691340b314aecbfe7d6aef3590b9cee21dcdb486fe9fbcdd679e49
# sha256(kernel_basis.json) = fcbe2876abd2a6b410813f732356f829c970c22c3a333f7e4ec34909c0e0793b
# number of (D, parity) blocks: 40 ; max block size: 292
CHECK formal: rank-nullity: rank 3321 + ker 0 = 3321: True
# formal           dim im Collapse = 3321 ; dim ker Collapse = 0 ; nnz entries = 666507  (5.5s)
CHECK chi_plateau: rank-nullity: rank 3321 + ker 0 = 3321: True
# chi_plateau      dim im Collapse = 3321 ; dim ker Collapse = 0 ; nnz entries = 157549  (3.2s)
CHECK double_plateau: rank-nullity: rank 3299 + ker 22 = 3321: True
# double_plateau   dim im Collapse = 3299 ; dim ker Collapse = 22 ; nnz entries = 73320  (91.0s)
CHECK FORMAL: Collapse injective on C_coh (dim ker = 0): True
CHECK chi_plateau: Collapse injective: True
CHECK double_plateau: dim ker Collapse = 22: True
# NOTE: injectivity of G_formal (M.3) does not by itself imply injectivity of Collapse;
#       the formal reading happens to make Collapse injective as well (verified above).
# dim C_reg = 3150 (expected 3150)
CHECK dim C_reg = 3150: True
# formal           dim (ker Collapse cap C_reg) = 0 ; dim Collapse(C_reg) = 3150
# chi_plateau      dim (ker Collapse cap C_reg) = 0 ; dim Collapse(C_reg) = 3150
# double_plateau   dim (ker Collapse cap C_reg) = 0 ; dim Collapse(C_reg) = 3150
# wrote collapse_kernel.json collapse_kernel_certificates.txt
# === SUMMARY ===
# formal           dim im = 3321 ; dim ker = 0 ; ker_cap_C_reg = 0 ; perD (size,ker) = {0: [35, 0], 1: [160, 0], 2: [440, 0], 3: [944, 0], 4: [1742, 0]}
# chi_plateau      dim im = 3321 ; dim ker = 0 ; ker_cap_C_reg = 0 ; perD (size,ker) = {0: [35, 0], 1: [160, 0], 2: [440, 0], 3: [944, 0], 4: [1742, 0]}
# double_plateau   dim im = 3299 ; dim ker = 22 ; ker_cap_C_reg = 0 ; perD (size,ker) = {0: [35, 0], 1: [160, 0], 2: [440, 0], 3: [944, 4], 4: [1742, 18]}
# total runtime 105.4s
ALL CHECKS PASSED
\end{lstlisting}

\paragraph{M.4 Helmholtz (\texttt{helmholtz\_check\_out.txt}).}
\begin{lstlisting}
```

```
# helmholtz_check.py output   (exact QQ; conditional on (S-bv) through M.2)
# M.4 formal self-adjoint (variational) subspace under Gate V; Helmholtz certificates
CHECK dim C_coh = 3321: True
CHECK unit test: Euler derivatives of 30 random quadratic Lagrangians (independent sympy route) are self-adjoint under m4_common.adjoint, all readings, D=0..4: True
CHECK negative control: E^{00}=d_0 h_{01} is NOT self-adjoint in every reading: True
CHECK formal: every exported SA vector satisfies E_C = E_C^dag exactly: True
# formal           dim SA = 26 ; per D = {0: 26, 1: 0, 2: 0, 3: 0, 4: 0} ; defect-map nnz = 1482596 (37.2s)
CHECK chi_plateau: every exported SA vector satisfies E_C = E_C^dag exactly: True
# chi_plateau      dim SA = 26 ; per D = {0: 26, 1: 0, 2: 0, 3: 0, 4: 0} ; defect-map nnz = 645026 (16.6s)
CHECK double_plateau: every exported SA vector satisfies E_C = E_C^dag exactly: True
# double_plateau   dim SA = 1347 ; per D = {0: 26, 1: 0, 2: 271, 3: 12, 4: 1038} ; defect-map nnz = 60016 (28.2s)
CHECK formal: dim Collapse(SA) = dim SA - dim ker Collapse = 26 - 0 = 26 (mod-p rank lower bound meets the upper bound): True
# formal           dim (T_variational cap im Collapse) = 26 ; per D = {0: 26, 1: 0, 2: 0, 3: 0, 4: 0}
CHECK chi_plateau: dim Collapse(SA) = dim SA - dim ker Collapse = 26 - 0 = 26 (mod-p rank lower bound meets the upper bound): True
# chi_plateau      dim (T_variational cap im Collapse) = 26 ; per D = {0: 26, 1: 0, 2: 0, 3: 0, 4: 0}
CHECK double_plateau: dim Collapse(SA) = dim SA - dim ker Collapse = 1347 - 22 = 1325 (mod-p rank lower bound meets the upper bound): True
# double_plateau   dim (T_variational cap im Collapse) = 1325 ; per D = {0: 26, 1: 0, 2: 271, 3: 8, 4: 1020}
CHECK FORMAL: SA = 26-dimensional, supported only at D=0 (no variational collapse with any derivative): True
CHECK chi_plateau: same as formal: True
# formal           dim (SA cap C_reg) = 26
# chi_plateau      dim (SA cap C_reg) = 26
# double_plateau   dim (SA cap C_reg) = 1208
CHECK Helmholtz certificate A (formal, D=0): Euler derivative of I_C = (1/2) int <h,E_C[h]> equals E_C: True
CHECK Certificate A K-matrix satisfies CORRECT Gate-V identity w_mn K^{mn}_ab = w_ab K^{ab}_mn: True
CHECK Certificate A K-matrix FAILS the reversed identity w_ab K^{mn}_ab = w_mn K^{ab}_mn (as required): True
CHECK Certificate A numeric check: w00 K00_02 = -56/3 = w02 K02_00: True
CHECK Helmholtz certificate B (double plateau, D=4, fourth-order operator): Euler(I_C) = E_C: True
CHECK Helmholtz certificate B2 (double plateau, D=2, second-order operator): Euler(I_C) = E_C: True
CHECK negative control: for a non-self-adjoint collapse E_C, Euler(I_C) != E_C: True
# wrote helmholtz_certificates.txt
# --- certificate A (formal, D=0): explicit matched quadratic functional ---
#   C[N=4, b={(),0222}, gamma=()] = 1
#   E^{00} = -56*h_02_*s_*x_/3
#   E^{02} = -28*h_00_*s_*x_/3 - 28*h_22_*s_*x_/3
#   E^{22} = -56*h_02_*s_*x_/3
#   I_C = int d^4y L ,  L = -56*h_00_*h_02_*s_*x_/3 - 56*h_02_*h_22_*s_*x_/3
# --- certificate B (double plateau, D=4): 4 E-terms ; B2 (D=2): 4 E-terms (see certificates file)
# === SUMMARY ===
# formal           {'dim_SA': 26, 'perD': {0: 26, 1: 0, 2: 0, 3: 0, 4: 0}, 'dim_SA_cap_Creg': 26, 'dim_variational_image': 26, 'variational_perD': {0: 26, 1: 0, 2: 0, 3: 0, 4: 0}}
# chi_plateau      {'dim_SA': 26, 'perD': {0: 26, 1: 0, 2: 0, 3: 0, 4: 0}, 'dim_SA_cap_Creg': 26, 'dim_variational_image': 26, 'variational_perD': {0: 26, 1: 0, 2: 0, 3: 0, 4: 0}}
# double_plateau   {'dim_SA': 1347, 'perD': {0: 26, 1: 0, 2: 271, 3: 12, 4: 1038}, 'dim_SA_cap_Creg': 1208, 'dim_variational_image': 1325, 'variational_perD': {0: 26, 1: 0, 2: 271, 3: 8, 4: 1020}}
# total runtime 124.6s
ALL CHECKS PASSED
\end{lstlisting}

\paragraph{M.4/M.5 matching probes (\texttt{matching\_probe\_out.txt}).}
\begin{lstlisting}
# matching_probe.py output   (exact QQ; conditional on (S-bv) through M.2)
CHECK target construction: linearized Bianchi identity d^mu G_lin_{mu nu} = 0 in jets: True
CHECK formal/P1 target has Lorentz-reflection parity 0: True
CHECK formal/P1 target is self-adjoint under Gate V (Euler derivative): True
CHECK formal/P2 target has Lorentz-reflection parity 0: True
CHECK formal/P2 target is self-adjoint under Gate V (Euler derivative): True
CHECK formal/C_coh: explicit preimage of the intersection direction ['1', '-89/140'] reproduces it exactly: True
# formal          C_coh D=0 block(parity 0) size=10 targets=['P1', 'P2'] : dim(span cap im) = 1 ; members: {'P1': False, 'P2': False} ; direction(s) in target coords: [['1', '-89/140']] (block wall 0.0s; probe elapsed 0.0s)
CHECK formal/C_reg: explicit preimage of the intersection direction ['1', '-89/140'] reproduces it exactly: True
# formal          C_reg D=0 block(parity 0) size=10 targets=['P1', 'P2'] : dim(span cap im) = 1 ; members: {'P1': False, 'P2': False} ; direction(s) in target coords: [['1', '-89/140']] (block wall 0.0s; probe elapsed 0.0s)
CHECK chi_plateau/P1 target has Lorentz-reflection parity 0: True
CHECK chi_plateau/P1 target is self-adjoint under Gate V (Euler derivative): True
CHECK chi_plateau/P2 target has Lorentz-reflection parity 0: True
CHECK chi_plateau/P2 target is self-adjoint under Gate V (Euler derivative): True
CHECK chi_plateau/C_coh: explicit preimage of the intersection direction ['1', '-89/140'] reproduces it exactly: True
# chi_plateau     C_coh D=0 block(parity 0) size=10 targets=['P1', 'P2'] : dim(span cap im) = 1 ; members: {'P1': False, 'P2': False} ; direction(s) in target coords: [['1', '-89/140']] (block wall 0.0s; probe elapsed 0.0s)
CHECK chi_plateau/C_reg: explicit preimage of the intersection direction ['1', '-89/140'] reproduces it exactly: True
# chi_plateau     C_reg D=0 block(parity 0) size=10 targets=['P1', 'P2'] : dim(span cap im) = 1 ; members: {'P1': False, 'P2': False} ; direction(s) in target coords: [['1', '-89/140']] (block wall 0.0s; probe elapsed 0.0s)
CHECK double_plateau/P1 target has Lorentz-reflection parity 0: True
CHECK double_plateau/P1 target is self-adjoint under Gate V (Euler derivative): True
CHECK double_plateau/P2 target has Lorentz-reflection parity 0: True
CHECK double_plateau/P2 target is self-adjoint under Gate V (Euler derivative): True
CHECK double_plateau/C_coh: explicit preimage of the intersection direction ['1', '-89/140'] reproduces it exactly: True
```

```
# double_plateau  C_coh D=0 block(parity 0) size=10 targets=['P1', 'P2'] : dim(span cap im) = 1 ; members: {'P1': False, 'P2': False} ; direction(s) in target coords: [['1', '-89/140']] (block wall 0.0s; probe elapsed 0.0s)
CHECK double_plateau/C_reg: explicit preimage of the intersection direction ['1', '-89/140'] reproduces it exactly: True
# double_plateau  C_reg D=0 block(parity 0) size=10 targets=['P1', 'P2'] : dim(span cap im) = 1 ; members: {'P1': False, 'P2': False} ; direction(s) in target coords: [['1', '-89/140']] (block wall 0.0s; probe elapsed 0.0s)
CHECK double_plateau/EH target has Lorentz-reflection parity 0: True
CHECK double_plateau/EH target is self-adjoint under Gate V (Euler derivative): True
# double_plateau  C_coh D=2 block(parity 0) size=84 targets=['EH'] : dim(span cap im) = 0 ; members: {'EH': False} ; direction(s) in target coords: None (block wall 0.0s; probe elapsed 0.2s)
# double_plateau  C_reg D=2 block(parity 0) size=80 targets=['EH'] : dim(span cap im) = 0 ; members: {'EH': False} ; direction(s) in target coords: None (block wall 0.0s; probe elapsed 0.3s)
CHECK double_plateau/R2 target has Lorentz-reflection parity 0: True
CHECK double_plateau/R2 target is self-adjoint under Gate V (Euler derivative): True
CHECK double_plateau/RC2 target has Lorentz-reflection parity 0: True
CHECK double_plateau/RC2 target is self-adjoint under Gate V (Euler derivative): True
CHECK double_plateau/C_coh/R2: explicit preimage C reproduces the target Euler tensor exactly: True
CHECK double_plateau/C_coh/RC2: explicit preimage C reproduces the target Euler tensor exactly: True
CHECK double_plateau/C_coh: explicit preimage of the intersection direction ['1', '0'] reproduces it exactly: True
# double_plateau  C_coh D=4 block(parity 0) size=292 targets=['R2', 'RC2'] : dim(span cap im) = 2 ; members: {'R2': True, 'RC2': True} ; direction(s) in target coords: [['1', '0'], ['0', '1']] (block wall 38.6s; probe elapsed 40.0s)
CHECK double_plateau/C_reg: explicit preimage of the intersection direction ['1', '-188813/88456'] reproduces it exactly: True
# double_plateau  C_reg D=4 block(parity 0) size=265 targets=['R2', 'RC2'] : dim(span cap im) = 1 ; members: {'R2': False, 'RC2': False} ; direction(s) in target coords: [['1', '-188813/88456']] (block wall 5.9s; probe elapsed 45.9s)
# wrote matching_certificates.txt
# === SUMMARY (conditional on (S-bv)) ===
# formal          C_coh  P1+P2    D=0 : intersection dim 1 ; member: {'P1': False, 'P2': False} ; direction [['1', '-89/140']]
# formal          C_reg  P1+P2    D=0 : intersection dim 1 ; member: {'P1': False, 'P2': False} ; direction [['1', '-89/140']]
# chi_plateau     C_coh  P1+P2    D=0 : intersection dim 1 ; member: {'P1': False, 'P2': False} ; direction [['1', '-89/140']]
# chi_plateau     C_reg  P1+P2    D=0 : intersection dim 1 ; member: {'P1': False, 'P2': False} ; direction [['1', '-89/140']]
# double_plateau  C_coh  P1+P2    D=0 : intersection dim 1 ; member: {'P1': False, 'P2': False} ; direction [['1', '-89/140']]
# double_plateau  C_reg  P1+P2    D=0 : intersection dim 1 ; member: {'P1': False, 'P2': False} ; direction [['1', '-89/140']]
# double_plateau  C_coh  EH       D=2 : intersection dim 0 ; member: {'EH': False} ; direction None
# double_plateau  C_reg  EH       D=2 : intersection dim 0 ; member: {'EH': False} ; direction None
# double_plateau  C_coh  R2+RC2   D=4 : intersection dim 2 ; member: {'R2': True, 'RC2': True} ; direction [['1', '0'], ['0', '1']]
# double_plateau  C_reg  R2+RC2   D=4 : intersection dim 1 ; member: {'R2': False, 'RC2': False} ; direction [['1', '-188813/88456']]
CHECK double plateau: linearized R^2 and Ric^2 Euler tensors are collapse images (explicit preimages certified): True
CHECK double plateau: linearized Einstein tensor is NOT a collapse image: True
# total wall-clock runtime 46.4s (sum of sequential block walls; not parallel)
ALL CHECKS PASSED
\end{lstlisting}

\paragraph{Stage 2 fixed-data (\texttt{fixed\_data\_reading\_out.txt}).}
\begin{lstlisting}
# fixed_data_reading_out.txt -- Stage 2 summary (conditional on S-bv)
# Imported: sigma_eps = XII-C/XVI-L time-only ramp; chi_I = compact switch; Gate V unchanged
# S-bv: CONDITIONAL (not discharged)

STATUS: fixed-data criteria and universal region comparisons constructed; actual fixed-data dimensions unresolved
REGION 1 (full switched/prepared): status = sufficient criteria (i) G, (ii) Coll, (iii) adjoint-defect injectivity of ev_U
  ker G_formal subseteq ker G_fixed; (iii) needed for SA equality; time-only ramp: own Helmholtz analysis needed (Unresolved)
  comparison class formal: dim ker Coll = 0
REGION 2 (prep plateau, UNIVERSAL formal chi-jets): status = EXACT for the universal reading (actual chi_I: Unresolved)
  dim ker Coll = 21 ; SA_D0 = 26
  full SA dim = 47 ; variational image = 26 ; perD SA = {'0': 26, '1': 0, '2': 0, '3': 4, '4': 17}
REGION 3 (double plateau): status = EXACT (M.4)
  dim ker Coll = 22 ; SA = 1347 ; var image = 1325
REGION 4 (transitions): status = Unresolved (domain-size bound 0 <= dim ker Coll_fixed <= 3321)

DISTINCTIONS:
  support vanishing: C_2/M_2 die on {sigma=1}; chi-Leibniz dies on {chi=1}
  additional relations: possible for concrete ramp; none certified
  genuine SA: Gate V self-adjointness of collapsed operator

ULTRALOCAL: 26-dim family; 26 K-patterns in SA basis; Lorentz-scalar matched line: P1-(89/140)P2
ACCEPTANCE: criteria / exact-universal / exact / unresolved for regions 1..4
total runtime 120.3s
ALL CHECKS PASSED
\end{lstlisting}

\paragraph{M.5 ledger (\texttt{counterterm\_functionals\_out.txt}).}
\begin{lstlisting}
# counterterm_functionals.py output   (M.5 ledger; conditional on (S-bv))
CHECK matching_certificates.txt present: True
CHECK expected matching preimages from M.4 probes: True
CHECK 26-dim ultralocal SA basis: True
CHECK P1/P2 matching row present: True
CHECK R2 C_coh preimage recorded as non-regression-preserving (K-block): True
CHECK C_reg R2/RC2 combination is regression-preserving: True
CHECK completion gate obstruction recorded: True
```

```
CHECK EOM ledger empty because EOM undefined: True
CHECK field-redef ledger empty because undefined: True
# wrote matching_matrix.json/.txt matching_certificates_M5.txt ledger_*.json ledger_tables.txt
# wrote counterterm_functionals_out.txt m5_ledger_summary.json
# Named theorem: PlateauCurvatureSquareMatching
# total runtime 0.1s
ALL CHECKS PASSED
\end{lstlisting}

\paragraph{M.6 scaling anomaly (\texttt{scaling\_anomaly\_out.txt}).}
\begin{lstlisting}
# scaling_anomaly.py output
# Stage 4 / M.6: dilation identity, S^2 residue, anomaly assignment, collapse probe
# CONDITIONAL: residue evaluation uses Input (S-bv); Gate S-bv not discharged
CHECK dilation identity: mu d/dmu varpi_mu = E varpi_mu (rho=0): True
# identity (N=0): A_mu(u) = (E+4) R_{L,mu}(u) = - <u, E varpi_mu> delta
CHECK N=0 structural identity A_mu = (E+4) R_{L,mu} recorded (analytic; see M.6): True
CHECK contour residue at z_m tends to -1: True
CHECK angular u-integral Iu0 = -2 i pi: True
CHECK sphere integral of (Q-i0)^{-2} = -2 i pi^2: True
# CONDITIONAL ON (S-bv): int_{S^3} (Q-i0)^{-2} d sigma = -2 i pi^2
CHECK S^2 residue coefficient r = -2 i pi^2 c^2: True
CHECK S^2 residue nonzero for every c != 0 (S-bv): True
# CONDITIONAL ON (S-bv): A_mu(S^2) = r delta with r = -2 i pi^2 c^2 != 0
# Residue independent of mu and of radial profile of varpi (varpi(0)=1)
# Higher-N residues (N>=1): Unresolved in this dispatch (formula recorded in M.6)
CHECK N=0 coherent basis is the single S^2 label {(),()}: True
CHECK anomaly assignment sits at contact variable index 0 (N=0,gamma=()): True
# N=0 anomaly component C_anom: C[0] = r; higher-grade (N>=1) coordinates explicitly set to ZERO for this probe
# (higher-N residues Unresolved: C_anom is NOT the full graph scale variation)
# collapse[formal]: nnz=744 sa_defect_nnz=4602 variational_nonzero=False
# collapse[chi_plateau]: nnz=744 sa_defect_nnz=1938 variational_nonzero=False
# collapse[prep_plateau]: nnz=0 sa_defect_nnz=0 variational_nonzero=False
# collapse[double_plateau]: nnz=0 sa_defect_nnz=0 variational_nonzero=False
CHECK formal: S^2 anomaly collapse nonzero: True
CHECK formal: S^2 anomaly collapse NOT self-adjoint: True
CHECK formal: S^2 anomaly is NOT a nonzero variational tensor: True
CHECK chi-plateau: S^2 anomaly collapse nonzero: True
CHECK chi-plateau: S^2 anomaly NOT self-adjoint: True
CHECK prep-plateau: S^2 anomaly collapses to 0 (sigma-derivatives): True
CHECK double-plateau: S^2 anomaly collapses to 0 (matching domain): True
CHECK S^2 coefficient ahatbar supported only on |partial sigma|=2 (C_2 sector): True
CHECK double-plateau R2 and RC2 Euler tensors are linearly independent: True
CHECK double-plateau Collapse(C_anom)=0, so matched projection = 0: True
# Matched projection onto span(E_R2, E_RC2) in double plateau: (0,0)
# Unmatched remainder in double plateau: 0 (collapse itself vanishes)
# Unmatched remainder in formal reading: full Collapse(C_anom) retained (non-variational)
CHECK M-C not claimable: the full controlled matched scaling calculation is unfinished: True
# M-C remains incomplete because the full controlled matched scaling calculation is unfinished.
# The N=0 component alone contributes no nonzero plateau matching.
# wrote scaling_anomaly_cert.json
# wrote scaling_anomaly_cert.txt
ALL CHECKS PASSED
\end{lstlisting}

\paragraph{M.7 covariance (\texttt{covariance\_constraints\_out.txt}, \texttt{covariance\_matrix.txt}).}
\begin{lstlisting}
# covariance_constraints.py output  (AUDITED REVISION)
# Stage 5 / M.7: Lorentz action, equivariant contacts, exact SA intersections, linear divergence identities
# CONDITIONAL on Input (S-bv); Gate S-bv not discharged; M-C/M-D not claimed
# Prescription covariance != interacting Ward/BV; linear divergence identity != Ward identity
CHECK d_N=(1,4,10,20,35) under (S-bv): True
CHECK dim C_coh = 3321: True
CHECK W dims and C_coh: True
CHECK generator J12 is eta-skew: True
CHECK generator J13 is eta-skew: True
CHECK generator J23 is eta-skew: True
CHECK generator K01 is eta-skew: True
CHECK generator K02 is eta-skew: True
CHECK generator K03 is eta-skew: True
CHECK p_0=1: True
CHECK B_N basis of Poly_N, N=0: True
CHECK B_N basis of Poly_N, N=1: True
CHECK B_N basis of Poly_N, N=2: True
CHECK B_N basis of Poly_N, N=3: True
CHECK B_N basis of Poly_N, N=4: True
CHECK N=0: [rho(K01),rho(K02)]=rho(J12) on B_N: True
CHECK N=1: [rho(K01),rho(K02)]=rho(J12) on B_N: True
CHECK N=2: [rho(K01),rho(K02)]=rho(J12) on B_N: True
CHECK N=3: [rho(K01),rho(K02)]=rho(J12) on B_N: True
CHECK N=4: [rho(K01),rho(K02)]=rho(J12) on B_N: True
```

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CHECK N=0 trivial action: True
CHECK N=0 J12: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=0 J13: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=0 J23: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=0 K01: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=0 K02: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=0 K03: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=0: [rhoD(K01),rhoD(K02)]=rhoD(J12) on d_gamma delta: True
CHECK N=1 J12: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=1 J13: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=1 J23: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=1 K01: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=1 K02: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=1 K03: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=1: [rhoD(K01),rhoD(K02)]=rhoD(J12) on d_gamma delta: True
CHECK N=2 J12: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=2 J13: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=2 J23: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=2 K01: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=2 K02: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=2 K03: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=2: [rhoD(K01),rhoD(K02)]=rhoD(J12) on d_gamma delta: True
CHECK N=3 J12: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=3 J13: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=3 J23: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=3 K01: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=3 K02: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=3 K03: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=3: [rhoD(K01),rhoD(K02)]=rhoD(J12) on d_gamma delta: True
CHECK N=4 J12: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=4 J13: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=4 J23: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=4 K01: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=4 K02: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=4 K03: pull-back action on d_gamma delta = P rho^poly P (eta intertwiner): True
CHECK N=4: [rhoD(K01),rhoD(K02)]=rhoD(J12) on d_gamma delta: True
CHECK first-order basis order d_0..d_3: True
CHECK first order J12: X.(d_nu delta) = X^mu_nu d_mu delta (abstract action matrix = X): True
CHECK first order J12: old coordinate matrix = -X^T: True
CHECK first order J12 (rotation): abstract action = old coordinate matrix: True
CHECK first order J12: abstract = eta (old) eta: True
CHECK first order J13: X.(d_nu delta) = X^mu_nu d_mu delta (abstract action matrix = X): True
CHECK first order J13: old coordinate matrix = -X^T: True
CHECK first order J13 (rotation): abstract action = old coordinate matrix: True
CHECK first order J13: abstract = eta (old) eta: True
CHECK first order J23: X.(d_nu delta) = X^mu_nu d_mu delta (abstract action matrix = X): True
CHECK first order J23: old coordinate matrix = -X^T: True
CHECK first order J23 (rotation): abstract action = old coordinate matrix: True
CHECK first order J23: abstract = eta (old) eta: True
CHECK first order K01: X.(d_nu delta) = X^mu_nu d_mu delta (abstract action matrix = X): True
CHECK first order K01: old coordinate matrix = -X^T: True
CHECK first order K01 (boost): abstract action = -(old coordinate matrix): OLD MATRIX WRONG FOR BOOSTS: True
CHECK first order K01: abstract = eta (old) eta: True
CHECK first order K02: X.(d_nu delta) = X^mu_nu d_mu delta (abstract action matrix = X): True
CHECK first order K02: old coordinate matrix = -X^T: True
CHECK first order K02 (boost): abstract action = -(old coordinate matrix): OLD MATRIX WRONG FOR BOOSTS: True
CHECK first order K02: abstract = eta (old) eta: True
CHECK first order K03: X.(d_nu delta) = X^mu_nu d_mu delta (abstract action matrix = X): True
CHECK first order K03: old coordinate matrix = -X^T: True
CHECK first order K03 (boost): abstract action = -(old coordinate matrix): OLD MATRIX WRONG FOR BOOSTS: True
CHECK first order K03: abstract = eta (old) eta: True
# first-order boost K01 on (d0 delta,d1 delta,d2 delta,d3 delta): abstract = [['0', '1', '0', '0'], ['1', '0', '0', '0'], ['0', '0', '0', '0'], ['0', '0', '0', '0']]
#                                         old coordinate matrix = [['0', '-1', '0', '0'], ['-1', '0', '0', '0'], ['0', '0', '0', '0'], ['0', '0', '0', '0']]
CHECK symbolic test function: K01.(d_1 delta) = + d_0 delta (old matrix gave - d_0 delta): True
CHECK Poly_N = sum H_{N-2k} dims, N=0: True
CHECK Poly_N = sum H_{N-2k} dims, N=1: True
CHECK Poly_N = sum H_{N-2k} dims, N=2: True
CHECK Poly_N = sum H_{N-2k} dims, N=3: True
CHECK Poly_N = sum H_{N-2k} dims, N=4: True
CHECK N=0: number of phi_{k,j} = Schur count 1: True
CHECK N=0: corrected equivariant maps linearly independent: True
# N=0: corrected C = P o phi_{k,j} equivariant on all 6 generators (exact); maps [(0, 0)]
CHECK N=1: number of phi_{k,j} = Schur count 1: True
CHECK N=1: corrected equivariant maps linearly independent: True
# N=1: corrected C = P o phi_{k,j} equivariant on all 6 generators (exact); maps [(0, 0)]
CHECK N=2: number of phi_{k,j} = Schur count 3: True
CHECK N=2: corrected equivariant maps linearly independent: True
# N=2: corrected C = P o phi_{k,j} equivariant on all 6 generators (exact); maps [(0, 0), (1, 0), (1, 1)]
CHECK N=3: number of phi_{k,j} = Schur count 3: True
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CHECK N=3: corrected equivariant maps linearly independent: True
# N=3: corrected C = P o phi_{k,j} equivariant on all 6 generators (exact); maps [(0, 0), (1, 0), (1, 1)]
CHECK N=4: number of phi_{k,j} = Schur count 6: True
CHECK N=4: corrected equivariant maps linearly independent: True
# N=4: corrected C = P o phi_{k,j} equivariant on all 6 generators (exact); maps [(0, 0), (1, 0), (1, 1), (2, 0), (2, 1), (2, 2)]
CHECK N=0 old vector trivially equivariant: True
CHECK OLD (uncorrected) vectors: those changed by P fail exactly the three boost equations; unchanged ones (image only at even spatial order) pass: True
# old vectors failing boost equivariance: ['(1, 0, 0)', '(2, 0, 0)', '(3, 0, 0)', '(3, 1, 0)', '(3, 1, 1)', '(4, 0, 0)', '(4, 1, 0)', '(4, 1, 1)'] ; unchanged by P: ['(0, 0, 0)', '(2, 1, 0)', '(2, 1, 1)', '(4, 2, 0)', '(4, 2, 1)', '(4, 2, 2)']
CHECK equivariant dims 1,1,3,3,6: True
CHECK N=0: exact kernel of the equivariance constraint map (pull-back action) has dim 1, rank 0: True
CHECK N=1: exact kernel of the equivariance constraint map (pull-back action) has dim 1, rank 19: True
CHECK N=2: exact kernel of the equivariance constraint map (pull-back action) has dim 3, rank 147: True
# cokernel = dim codomain - rank, codomain = (+)_6 Hom(H_{4+N},W_N): {0: 6, 1: 101, 2: 753, 3: 3503, 4: 12256}
CHECK equivariant freedom 14 = 5 (K, N<=2) + 9 (C_reg, N>=3): True
CHECK P1, P2, P1-(89/140)P2 are so(1,3)-invariant: True
# --- exact V = Equiv cap SA (14 equivariant coordinates; adjoint defect map; exact QQ kernel) ---
CHECK formal: all 1 basis vectors of V re-verified from scratch: zero adjoint defect AND equivariance (6 generators, all N): True
CHECK formal: dim(V cap K)=dimV-dim pi_reg(V), dim(V cap C_reg)=dimV-dim pi_K(V): True
CHECK formal: dim(V cap K)+dim(V cap C_reg)=1 <= dim V=1: True
CHECK formal: V cap K <= pi_K(V), V cap C_reg <= pi_reg(V): True
CHECK formal: pi_K(V) cap SA = V cap K and pi_reg(V) cap SA = V cap C_reg (projections NOT relabelled as SA): True
# formal          dim V=1 | pi_K(V)=0  V cap K=0 | pi_reg(V)=1  V cap C_reg=1 | dim Coll(V)=1 (2.6s)
CHECK chi_plateau: all 1 basis vectors of V re-verified from scratch: zero adjoint defect AND equivariance (6 generators, all N): True
CHECK chi_plateau: dim(V cap K)=dimV-dim pi_reg(V), dim(V cap C_reg)=dimV-dim pi_K(V): True
CHECK chi_plateau: dim(V cap K)+dim(V cap C_reg)=1 <= dim V=1: True
CHECK chi_plateau: V cap K <= pi_K(V), V cap C_reg <= pi_reg(V): True
CHECK chi_plateau: pi_K(V) cap SA = V cap K and pi_reg(V) cap SA = V cap C_reg (projections NOT relabelled as SA): True
# chi_plateau     dim V=1 | pi_K(V)=0  V cap K=0 | pi_reg(V)=1  V cap C_reg=1 | dim Coll(V)=1 (0.7s)
CHECK double_plateau: all 12 basis vectors of V re-verified from scratch: zero adjoint defect AND equivariance (6 generators, all N): True
CHECK double_plateau: dim(V cap K)=dimV-dim pi_reg(V), dim(V cap C_reg)=dimV-dim pi_K(V): True
CHECK double_plateau: dim(V cap K)+dim(V cap C_reg)=10 <= dim V=12: True
CHECK double_plateau: V cap K <= pi_K(V), V cap C_reg <= pi_reg(V): True
CHECK double_plateau: pi_K(V) cap SA = V cap K and pi_reg(V) cap SA = V cap C_reg (projections NOT relabelled as SA): True
# double_plateau  dim V=12 | pi_K(V)=5  V cap K=3 | pi_reg(V)=9  V cap C_reg=7 | dim Coll(V)=9 (0.9s)
CHECK formal and chi-plateau: dim(Equiv cap SA) = 1: True
CHECK formal: the SA equivariant line lies in C_reg (N=4): True
CHECK formal: Coll(V) = span of E[P1-(89/140)P2] (ratio 17920); V = phi_{2,0} at N=4 (Box^2, gamma=0): True
CHECK formal: supplementary mod-p kernel dims [1, 1, 1] agree with the exact dim V (correct rational reduction): True
CHECK chi_plateau: supplementary mod-p kernel dims [1, 1, 1] agree with the exact dim V (correct rational reduction): True
CHECK double_plateau: supplementary mod-p kernel dims [12, 12, 12] agree with the exact dim V (correct rational reduction): True
# --- linear (gauge) divergence identities: d_mu E^{mu nu}, E with UPPER indices, no metric factor ---
CHECK Identity A: eta^{mu rho} d_rho G^lin_{mu nu} = 0 (G LOWER indices, metric factor correct here): True
CHECK double_plateau: d_mu E^{mu nu}[R2] == 0 identically (weight w=1): True
CHECK double_plateau: d_mu E^{mu nu}[RC2] == 0 identically (weight w=1): True
CHECK double_plateau: d_mu E^{mu nu}[R2-(188813/88456)RC2] == 0 identically (weight w=1): True
CHECK chi_plateau: d_mu E^{mu nu}[R2] == 0 identically (weight w=s_): True
CHECK chi_plateau: d_mu E^{mu nu}[RC2] == 0 identically (weight w=s_): True
CHECK chi_plateau: d_mu E^{mu nu}[R2-(188813/88456)RC2] == 0 identically (weight w=s_): True
CHECK formal: d_mu E^{mu nu}[R2] == 0 identically (weight w=s_*x_): True
CHECK formal: d_mu E^{mu nu}[RC2] == 0 identically (weight w=s_*x_): True
CHECK formal: d_mu E^{mu nu}[R2-(188813/88456)RC2] == 0 identically (weight w=s_*x_): True
CHECK diagnosis: the OLD formula sum_mu eta^{mu mu} d_mu E^{mu nu} is nonzero (this produced the false 'does not vanish'): True
CHECK independent metric linearization: R_lin(metric) = +1 x R_lin(manuscript formula): True
CHECK independent metric linearization: Ric_lin(metric) = +1 x Ric_lin(manuscript formula): True
CHECK independent route: d_mu E^{mu nu}[R2] == 0 (Functions, euler_equations): True
CHECK independent route: old eta^{mu mu}-weighted formula nonzero for R2: True
CHECK independent Euler tensor of int R2 equals the matching_probe/jet_euler target exactly: True
CHECK independent route: d_mu E^{mu nu}[RC2] == 0 (Functions, euler_equations): True
CHECK independent route: old eta^{mu mu}-weighted formula nonzero for RC2: True
CHECK independent Euler tensor of int RC2 equals the matching_probe/jet_euler target exactly: True
# independent curvature route done (13.8s); sign(R)=+1 sign(Ric)=+1 (squares are sign-insensitive)
CHECK preimage [double_plateau C_coh R2]: Coll(C) equals the INDEPENDENTLY built target exactly: True
CHECK preimage [double_plateau C_coh R2]: d_mu E_C^{mu nu} == 0: True
CHECK preimage [double_plateau C_coh R2]: support meets K-block (N<=2): True
CHECK preimage [double_plateau C_coh RC2]: Coll(C) equals the INDEPENDENTLY built target exactly: True
CHECK preimage [double_plateau C_coh RC2]: d_mu E_C^{mu nu} == 0: True
CHECK preimage [double_plateau C_coh RC2]: support meets K-block (N<=2): True
CHECK preimage [double_plateau C_reg (1)R2+(-188813/88456)RC2]: Coll(C) equals the INDEPENDENTLY built target exactly: True
CHECK preimage [double_plateau C_reg (1)R2+(-188813/88456)RC2]: d_mu E_C^{mu nu} == 0: True
CHECK preimage [double_plateau C_reg (1)R2+(-188813/88456)RC2]: support inside C_reg (N>=3): True
CHECK double_plateau: EH identity d_mu E^{mu nu} = (d_mu w) G_lin^{mu nu}[h]  (zero iff w constant): True
CHECK chi_plateau: EH identity d_mu E^{mu nu} = (d_mu w) G_lin^{mu nu}[h]  (zero iff w constant): True
CHECK formal: EH identity d_mu E^{mu nu} = (d_mu w) G_lin^{mu nu}[h]  (zero iff w constant): True
```

```
CHECK double_plateau: ultralocal P1-(89/140)P2: d_mu E^{mu nu} = 2 d_mu[w(h^{mu nu}-(89/140)eta^{mu nu} h)] != 0 (not gauge invariant): True
CHECK chi_plateau: ultralocal P1-(89/140)P2: d_mu E^{mu nu} = 2 d_mu[w(h^{mu nu}-(89/140)eta^{mu nu} h)] != 0 (not gauge invariant): True
CHECK formal: ultralocal P1-(89/140)P2: d_mu E^{mu nu} = 2 d_mu[w(h^{mu nu}-(89/140)eta^{mu nu} h)] != 0 (not gauge invariant): True
# wrote covariance_matrix.txt covariance_certificates.txt covariance_summary.json covariance_exact_certificates.json completion_ledger_M89.json
CHECK M-C/M-D not claimed: True
# total runtime 241.8s
ALL CHECKS PASSED
# ---------------- covariance_matrix.txt ----------------
# covariance_matrix.txt -- XVI-M M.7 compatibility matrix (CONDITIONAL on (S-bv)), AUDITED REVISION
# Contact maps C: H_{4+N} -> W_N = span{d_gamma delta : |gamma|<=N}, pull-back action (eta intertwiner)
# rank = rank of the constraint map C -> (C rho_V(X) - rho_delta(X) C)_X = codim of the solution space
# codomain = (+)_{6 generators} Hom(H_{4+N},W_N), dim 6*slots ; coker = dim codomain - rank
#
N  dimH  dimW  slots  rank(=codim)  dim_codomain  dim_coker  dim_sol  block
0     1     1      1             0             6          6        1  K
1     4     5     20            19           120        101        1  K
2    10    15    150           147           900        753        3  K
3    20    35    700           697          4200       3503        3  C_reg
4    35    70   2450          2444         14700      12256        6  C_reg
#
K_block_dim_sol = 5
C_reg_dim_sol = 9
total_equivariant = 14
#
# Exact V = Equiv cap SA and the four objects:
V formal dim_V=1 pi_K(V)=0 V_cap_K=0 pi_reg(V)=1 V_cap_Creg=1 dim_Coll(V)=1
V chi_plateau dim_V=1 pi_K(V)=0 V_cap_K=0 pi_reg(V)=1 V_cap_Creg=1 dim_Coll(V)=1
V double_plateau dim_V=12 pi_K(V)=5 V_cap_K=3 pi_reg(V)=9 V_cap_Creg=7 dim_Coll(V)=9
#
# Linear divergence identities (d_mu E^{mu nu}, E upper indices):
ID_A eta^{mu rho} d_rho G^lin_{mu nu} = 0 : HOLDS (not Ward)
ID_B ultralocal P1-(89/140)P2 : d_mu E^{mu nu} = 2 d_mu[w(h^{mu nu}-(89/140)eta^{mu nu}h)] != 0 (not gauge invariant)
ID_Bprime R2, RC2, R2-(188813/88456)RC2 : d_mu E^{mu nu} = 0 in formal, chi-plateau, double plateau (linear gauge identity; not Ward)
ID_EH : d_mu E^{mu nu} = (d_mu w) G_lin^{mu nu} ; zero in double plateau
#
# Gate S-bv CONDITIONAL. M-C not claimed. M-D not claimed.
\end{lstlisting}
\paragraph{M.7 audit legacy reconstruction (\texttt{m7\_legacy\_reconstruction\_out.txt}).}
\begin{lstlisting}
# m7_legacy_reconstruction.py output (supplementary; modular ranks)
CHECK double_plateau SA basis 1347 loaded: True
CHECK double_plateau SA basis has 14639 non-integer rational entries (truncation-sensitive): True
# double_plateau  vectors=old       reduction=buggy_int_truncation G=K     dim(Equiv_G cap pi_G(SA)) at 3 primes = [5, 5, 5]
# double_plateau  vectors=old       reduction=buggy_int_truncation G=C_reg dim(Equiv_G cap pi_G(SA)) at 3 primes = [3, 3, 3]
# double_plateau  vectors=old       reduction=buggy_int_truncation G=all   dim(Equiv_G cap pi_G(SA)) at 3 primes = [5, 5, 5]
# double_plateau  vectors=old       reduction=correct_reduction    G=K     dim(Equiv_G cap pi_G(SA)) at 3 primes = [5, 5, 5]
# double_plateau  vectors=old       reduction=correct_reduction    G=C_reg dim(Equiv_G cap pi_G(SA)) at 3 primes = [4, 4, 4]
# double_plateau  vectors=old       reduction=correct_reduction    G=all   dim(Equiv_G cap pi_G(SA)) at 3 primes = [6, 6, 6]
# double_plateau  vectors=corrected reduction=buggy_int_truncation G=K     dim(Equiv_G cap pi_G(SA)) at 3 primes = [5, 5, 5]
# double_plateau  vectors=corrected reduction=buggy_int_truncation G=C_reg dim(Equiv_G cap pi_G(SA)) at 3 primes = [3, 3, 3]
# double_plateau  vectors=corrected reduction=buggy_int_truncation G=all   dim(Equiv_G cap pi_G(SA)) at 3 primes = [5, 5, 5]
# double_plateau  vectors=corrected reduction=correct_reduction    G=K     dim(Equiv_G cap pi_G(SA)) at 3 primes = [5, 5, 5]
# double_plateau  vectors=corrected reduction=correct_reduction    G=C_reg dim(Equiv_G cap pi_G(SA)) at 3 primes = [9, 9, 9]
# double_plateau  vectors=corrected reduction=correct_reduction    G=all   dim(Equiv_G cap pi_G(SA)) at 3 primes = [12, 12, 12]
CHECK verbatim old computation reproduces the previously reported 5 / 3 / 5: True
CHECK old vectors with correct reduction: dim(Equiv_old cap SA) = 6 (old report 5 was a truncation artefact): True
CHECK corrected vectors with correct reduction: dim(Equiv cap SA) = 12 (agrees with exact): True
# wrote m7_legacy_reconstruction.json
ALL CHECKS PASSED
\end{lstlisting}
\paragraph{Work package A (\texttt{wpA/kernel\_identification\_out.txt}).}
\begin{lstlisting}
# kernel_identification_checks.py output
# xvi-M wpA: Gate S-bv identification of S with c (Q-i0)^{-1}
# python 3.13.5, sympy 1.14.0, mpmath 1.3.0, numpy 2.5.3
# signature eta=diag(1,-1,-1,-1); massless; z=x-y

# --- 1. Radial Fourier integral for W_0 (symbolic) ---
CHECK int_0^oo e^{-tau p} sin(p r) dp = r/(tau^2+r^2) for tau>0: True
CHECK analytic continuation spot (tau,t,r)=(0.2,0.5,1.0), err=1.88e-05: True
CHECK analytic continuation spot (tau,t,r)=(0.1,-1.2,0.7), err=5.25e-05: True
CHECK analytic continuation spot (tau,t,r)=(0.05,2.0,1.5), err=4.50e-04: True
CHECK damped radial W equals -1/(4 pi^2 (Q - 2 i tau t - tau^2)): True
CHECK Euclidean slice z0=-i tE: W=1/(4 pi^2 (tE^2+r^2)) (matches rem:Sbv-motivation): True
# => W_0 = -1/(4 pi^2) * 1/(Q - i0 sgn(z^0)) as the boundary value of the damped integral
```

```
2891
2892 # --- 2. Sign check for (s-i0)^{-1} = PV(1/s) + i pi delta(s) ---
2893 CHECK 1D Im~pi f(0) Gaussian#0, err=4.02e-04: True
2894 CHECK 1D Re~PV Gaussian#0, err=0.00e+00: True
2895 CHECK 1D Im~pi f(0) Gaussian#1, err=2.80e-04: True
2896 CHECK 1D Re~PV Gaussian#1, err=3.38e-04: True
2897 CHECK 1D Im~pi f(0) Gaussian#2, err=2.57e-04: True
2898 CHECK 1D Re~PV Gaussian#2, err=2.87e-04: True
2899 CHECK 1D Im~pi f(0) Gaussian#3, err=4.16e-05: True
2900 CHECK 1D Re~PV Gaussian#3, err=7.97e-05: True
2901 CHECK opposite side Im~-pi f(0): True
2902 # confirmed: (s-i0)^{-1}=PV(1/s)+i pi delta(s); (s+i0)^{-1}=PV-i pi delta
2903
2904 # --- 3. Algebraic combination of singular terms (symbolic coefficients) ---
2905 CHECK sgn(z0) dependence cancels: delta coeff same on both time sheets: True
2906 CHECK common delta coeff equals c * i pi = i/(4 pi): True
2907 CHECK PV coeff equals c = 1/(4 pi^2): True
2908 # S = c (PV(1/Q) + i pi delta(Q)) = c (Q - i0)^{-1},  c = 1/(4 pi^2)
2909
2910 # --- 4. Causal propagator delta_0 from W - W^T ---
2911 CHECK delta_0 = (1/(2 pi)) sgn(z0) delta(Q): True
2912 CHECK E_0^A = (1/(2 pi)) theta(-z0) delta(Q) on Omega: True
2913
2914 # --- 5. Off-cone pointwise normalization (spacelike & timelike) ---
2915 CHECK off-cone point#0 has Q!=0 (Q=-1.0): True
2916 CHECK off-cone S=-W=c/Q at point#0 z=(0.0, 1.0, 0.0, 0.0), Q=-1: True
2917 CHECK off-cone point#1 has Q!=0 (Q=-3.75): True
2918 CHECK off-cone S=-W=c/Q at point#1 z=(0.5, 2.0, 0.0, 0.0), Q=-3.75: True
2919 CHECK off-cone point#2 has Q!=0 (Q=-5.0): True
2920 CHECK off-cone S=-W=c/Q at point#2 z=(1.0, 2.0, 1.0, 1.0), Q=-5: True
2921 CHECK off-cone point#3 has Q!=0 (Q=3.75): True
2922 CHECK off-cone S=-W=c/Q at point#3 z=(2.0, 0.5, 0.0, 0.0), Q=3.75: True
2923 CHECK off-cone point#4 has Q!=0 (Q=7.0): True
2924 CHECK off-cone S=-W=c/Q at point#4 z=(3.0, 1.0, 1.0, 0.0), Q=7: True
2925 CHECK off-cone point#5 has Q!=0 (Q=3.5): True
2926 CHECK off-cone S=-W=c/Q at point#5 z=(-2.0, 0.0, 0.5, 0.5), Q=3.5: True
2927 CHECK off-cone point#6 has Q!=0 (Q=-2.4365534255462): True
2928 CHECK off-cone S=-W=c/Q at point#6 z=(0.30471707975443135, -1.0399841062404955, 0.7504511958064572, 0.9405647163912139), Q
     =-2.43655: True
2929 CHECK off-cone point#7 has Q!=0 (Q=1.994514293379158): True
2930 CHECK off-cone S=-W=c/Q at point#7 z=(-1.9510351886538364, -1.302179506862318, 0.12784040316728537, -0.3162425923435822), Q
     =1.99451: True
2931 CHECK off-cone point#8 has Q!=0 (Q=-2.105702756487829): True
2932 CHECK off-cone S=-W=c/Q at point#8 z=(-0.016801157504288795, -0.85304392757358, 0.8793979748628286, 0.7777919354289483), Q
     =-2.1057: True
2933 CHECK off-cone point#9 has Q!=0 (Q=-2.2232612075272082): True
2934 CHECK off-cone S=-W=c/Q at point#9 z=(0.06603069756121605, 1.1272412069680329, 0.4675093422520456, -0.8592924628832382), Q
     =-2.22326: True
2935 CHECK off-cone point#10 has Q!=0 (Q=-1.5576462298657727): True
2936 CHECK off-cone S=-W=c/Q at point#10 z=(0.36875078408249884, -0.9588826008289989, 0.8784503013072725, -0.049925910986252896), Q
     =-1.55765: True
2937 CHECK off-cone point#11 has Q!=0 (Q=-1.9479776365819612): True
2938 CHECK off-cone S=-W=c/Q at point#11 z=(-0.18486236354526056, -0.6809295444039414, 1.2225413386740303, -0.15452948206880215), Q
     =-1.94798: True
2939 CHECK spacelike unit: S=c/Q=-c: True
2940 CHECK timelike unit: S=c/Q=c: True
2941
2942 # --- 6. Numerical spatial Fourier for W vs closed form (damped) ---
2943 CHECK Fourier vs closed W at (z0,r)=(0.5,1.0), err=3.30e-05: True
2944 CHECK Fourier vs closed W at (z0,r)=(-0.5,1.0), err=3.30e-05: True
2945 CHECK Fourier vs closed W at (z0,r)=(1.5,0.8), err=3.30e-05: True
2946 CHECK Fourier vs closed W at (z0,r)=(-1.2,2.0), err=3.31e-05: True
2947 CHECK Fourier vs closed W at (z0,r)=(0.0,1.5), err=3.30e-05: True
2948 CHECK Fourier vs closed W at (z0,r)=(2.0,3.0), err=3.32e-05: True
2949 CHECK Fourier vs closed W at (z0,r)=(-2.5,1.1), err=3.32e-05: True
2950 CHECK Fourier vs closed W at (z0,r)=(0.3,0.3), err=3.30e-05: True
2951
2952 # --- 7. Null-cone chart s=Q: full one-variable distribution ---
2953 CHECK on {Q=0}\{0}, z0 != 0 (so sgn(z0) and theta(+/-) are smooth there): True
2954 # chart density future: Im=0.709646 vs pi*rho(0)=0.709704, err_im=5.84e-05, err_re=1.14e-12
2955 CHECK chart-future Im~pi rho(0), err=5.84e-05: True
2956 CHECK chart-future Re~PV, err=1.14e-12: True
2957 # chart density past: Im=0.709646 vs pi*rho(0)=0.709704, err_im=5.84e-05, err_re=1.14e-12
2958 CHECK chart-past Im~pi rho(0), err=5.84e-05: True
2959 CHECK chart-past Re~PV, err=1.14e-12: True
2960 CHECK dQ != 0 on the nonzero null cone (Q is a local coordinate): True
2961 # pullback of (s-i0)^{-1} therefore defines (Q-i0)^{-1} across the nonzero null cone
2962 # combined with ?3: S/c equals that single boundary value on both sheets
2963
2964 # --- 7b. Variable regulator s - tau^2 - i tau a(s), a of fixed sign (lem:wpA-varreg) ---
```

```
CHECK variable regulator a = 2 + sin s (z^0>0 sheet): |pairing - (PV + i pi sgn(a) phi(0))| = 1.20e-01, 1.24e-02, 1.24e-03 (tau=1e-2,1e-3,1e-4); decreasing: True
CHECK variable regulator a = -(1.5 + 0.5 cos s) (z^0<0 sheet): |pairing - (PV + i pi sgn(a) phi(0))| = 1.14e-01, 1.17e-02, 1.17e-03 (tau=1e-2,1e-3,1e-4); decreasing: True
CHECK regulator identity alpha^2 + r^2 = -(Q - 2 i tau z^0 - tau^2): variable coefficient a = 2 z^0: True
# => near each nonzero null point (z^0 of fixed sign) the damped kernel tends to (Q - i0 sgn z^0)^{-1}

# --- 8. Cross-checks: Box, homogeneity, transposition, M.6 residue ---
CHECK homogeneity degree -2 for 1/Q (off cone; BV inherits): True
CHECK Box(c/Q)=0 off the cone as rational identity (got 0): True
CHECK transposition: S^T=S (from W-W^T=-i delta_0 and E^R-E^A=delta_0): True
CHECK propagator norm: Box S = i delta on R^4 (hence 0 on Omega): True
CHECK M.6 residue with c=1/(4 pi^2): r = -i/(8 pi^2): True
# r = -I/(8*pi**2) = -i/(8 pi^2)
CHECK counterfactual Q+i0 would flip residue sign (excluded): True

# --- 9. Fourier-family vs position: radial closed-form consistency ---
CHECK off-cone -W_damped = c/(Q-2itau t-tau^2) at (z0,r)=(0.5,1.0), err=0.00e+00: True
CHECK off-cone -W_damped = c/(Q-2itau t-tau^2) at (z0,r)=(-0.5,1.0), err=0.00e+00: True
CHECK off-cone -W_damped = c/(Q-2itau t-tau^2) at (z0,r)=(1.5,0.8), err=3.47e-18: True
CHECK off-cone -W_damped = c/(Q-2itau t-tau^2) at (z0,r)=(-1.2,2.0), err=1.73e-18: True
CHECK off-cone -W_damped = c/(Q-2itau t-tau^2) at (z0,r)=(0.0,1.5), err=0.00e+00: True
CHECK off-cone -W_damped = c/(Q-2itau t-tau^2) at (z0,r)=(2.0,3.0), err=8.67e-19: True
CHECK off-cone -W_damped = c/(Q-2itau t-tau^2) at (z0,r)=(-2.5,1.1), err=8.67e-19: True
CHECK off-cone -W_damped = c/(Q-2itau t-tau^2) at (z0,r)=(0.3,0.3), err=7.04e-14: True
CHECK near-cone off-sheet agreement at t=1.05, rel=1.073e-02: True
CHECK near-cone off-sheet agreement at t=0.95, rel=9.229e-03: True
CHECK near-cone off-sheet agreement at t=-1.05, rel=3.024e-02: True
CHECK near-cone off-sheet agreement at t=-0.95, rel=2.974e-02: True
# Fourier/position consistency off cone confirmed via damped closed forms

# --- SUMMARY ---
{
  "outcome": "Certified",
  "identity": "S = (1/(4*pi^2)) * (Q - i0)^{-1}  on D'(R^4\\{0})",
  "c": "1/(4*pi^2)",
  "boundary_side": "Q-i0",
  "M6_residue_with_c": "-i/(8*pi^2)",
  "seeds": {
    "numpy_rng": [
      42,
      7,
      99,
      100
    ],
    "script_seed_doc": 20261006
  }
}
# ALL CHECKS PASSED
\end{lstlisting}

\paragraph{Work package B (\texttt{wpB/higher\_grade\_scaling\_out.txt}).}
\begin{lstlisting}
# higher_grade_scaling.py output  (XVI-M work package B; python 3.13.5, sympy 1.14.0)
# CONDITIONAL on Input (S-bv): S = c (Q-i0)^{-1}; c^2 symbolic. Gate V and single-switch collapse unchanged.

# === Section A: complete differentiation of the XVI-L subtraction formula (all N=0..4) ===
# M_{beta alpha}(lam) = binom(beta,alpha) lam^{|beta|-|alpha|} p_{beta-alpha},  p_delta = d^delta varpi(0)
CHECK A N=0 [symbolic]: M^{(rho,mu)} unit lower-triangular (Leibniz, varpi_mu(0)=1): True
CHECK A N=0 [symbolic]: M M^{-1} = 1 (exact): True
CHECK A N=0 [symbolic]: mu d(M^{-1})/dmu = -M^{-1}(mu dM/dmu)M^{-1} (exact): True
CHECK A N=0 [symbolic]: -M^{-1}(mu dM)M^{-1} = D M^{-1} - M^{-1} D, D=diag(|beta|) (dilation grading): True
CHECK A N=0 [symbolic]: mu d/dmu P_alpha (complete formula) = (E-|alpha|) P_alpha for all |alpha|<=N: True
CHECK A N=0 [symbolic]: w_alpha = (z^alpha/alpha!) T_{N-|alpha|}[1/varpi_mu] varpi_mu (closed form): True
CHECK A N=0 [symbolic]: d^beta w_alpha(0) = delta^beta_alpha (|alpha|,|beta|<=N): True
CHECK A N=0 [random_rational_0]: M^{(rho,mu)} unit lower-triangular (Leibniz, varpi_mu(0)=1): True
CHECK A N=0 [random_rational_0]: M M^{-1} = 1 (exact): True
CHECK A N=0 [random_rational_0]: mu d(M^{-1})/dmu = -M^{-1}(mu dM/dmu)M^{-1} (exact): True
CHECK A N=0 [random_rational_0]: -M^{-1}(mu dM)M^{-1} = D M^{-1} - M^{-1} D, D=diag(|beta|) (dilation grading): True
CHECK A N=0 [random_rational_0]: mu d/dmu P_alpha (complete formula) = (E-|alpha|) P_alpha for all |alpha|<=N: True
CHECK A N=0 [random_rational_0]: w_alpha = (z^alpha/alpha!) T_{N-|alpha|}[1/varpi_mu] varpi_mu (closed form): True
CHECK A N=0 [random_rational_0]: d^beta w_alpha(0) = delta^beta_alpha (|alpha|,|beta|<=N): True
CHECK A N=0: literal d^beta(z^alpha varpi_mu/alpha!)(0) equals M_of entries (explicit varpi, first 12x12 block): True
CHECK A N=0: explicit non-radial varpi=exp(q): mu d/dmu w_alpha^{(mu)} = (E-|alpha|) w_alpha^{(mu)} as functions: True
CHECK A N=1 [symbolic]: M^{(rho,mu)} unit lower-triangular (Leibniz, varpi_mu(0)=1): True
CHECK A N=1 [symbolic]: M M^{-1} = 1 (exact): True
CHECK A N=1 [symbolic]: mu d(M^{-1})/dmu = -M^{-1}(mu dM/dmu)M^{-1} (exact): True
CHECK A N=1 [symbolic]: -M^{-1}(mu dM)M^{-1} = D M^{-1} - M^{-1} D, D=diag(|beta|) (dilation grading): True
CHECK A N=1 [symbolic]: mu d/dmu P_alpha (complete formula) = (E-|alpha|) P_alpha for all |alpha|<=N: True
CHECK A N=1 [symbolic]: w_alpha = (z^alpha/alpha!) T_{N-|alpha|}[1/varpi_mu] varpi_mu (closed form): True
CHECK A N=1 [symbolic]: d^beta w_alpha(0) = delta^beta_alpha (|alpha|,|beta|<=N): True
```

```
CHECK A N=1 [random_rational_0]: M^{(rho,mu)} unit lower-triangular (Leibniz, varpi_mu(0)=1): True
CHECK A N=1 [random_rational_0]: M M^{-1} = 1 (exact): True
CHECK A N=1 [random_rational_0]: mu d(M^{-1})/dmu = -M^{-1}(mu dM/dmu)M^{-1} (exact): True
CHECK A N=1 [random_rational_0]: -M^{-1}(mu dM)M^{-1} = D M^{-1} - M^{-1} D, D=diag(|beta|) (dilation grading): True
CHECK A N=1 [random_rational_0]: mu d/dmu P_alpha (complete formula) = (E-|alpha|) P_alpha for all |alpha|<=N: True
CHECK A N=1 [random_rational_0]: w_alpha = (z^alpha/alpha!) T_{N-|alpha|}[1/varpi_mu] varpi_mu (closed form): True
CHECK A N=1 [random_rational_0]: d^beta w_alpha(0) = delta^beta_alpha (|alpha|,|beta|<=N): True
CHECK A N=1: literal d^beta(z^alpha varpi_mu/alpha!)(0) equals M_of entries (explicit varpi, first 12x12 block): True
CHECK A N=1: explicit non-radial varpi=exp(q): mu d/dmu w_alpha^{(mu)} = (E-|alpha|) w_alpha^{(mu)} as functions: True
CHECK A N=2 [symbolic]: M^{(rho,mu)} unit lower-triangular (Leibniz, varpi_mu(0)=1): True
CHECK A N=2 [symbolic]: M M^{-1} = 1 (exact): True
CHECK A N=2 [symbolic]: mu d(M^{-1})/dmu = -M^{-1}(mu dM/dmu)M^{-1} (exact): True
CHECK A N=2 [symbolic]: -M^{-1}(mu dM)M^{-1} = D M^{-1} - M^{-1} D, D=diag(|beta|) (dilation grading): True
CHECK A N=2 [symbolic]: mu d/dmu P_alpha (complete formula) = (E-|alpha|) P_alpha for all |alpha|<=N: True
CHECK A N=2 [symbolic]: w_alpha = (z^alpha/alpha!) T_{N-|alpha|}[1/varpi_mu] varpi_mu (closed form): True
CHECK A N=2 [symbolic]: d^beta w_alpha(0) = delta^beta_alpha (|alpha|,|beta|<=N): True
CHECK A N=2 [random_rational_0]: M^{(rho,mu)} unit lower-triangular (Leibniz, varpi_mu(0)=1): True
CHECK A N=2 [random_rational_0]: M M^{-1} = 1 (exact): True
CHECK A N=2 [random_rational_0]: mu d(M^{-1})/dmu = -M^{-1}(mu dM/dmu)M^{-1} (exact): True
CHECK A N=2 [random_rational_0]: -M^{-1}(mu dM)M^{-1} = D M^{-1} - M^{-1} D, D=diag(|beta|) (dilation grading): True
CHECK A N=2 [random_rational_0]: mu d/dmu P_alpha (complete formula) = (E-|alpha|) P_alpha for all |alpha|<=N: True
CHECK A N=2 [random_rational_0]: w_alpha = (z^alpha/alpha!) T_{N-|alpha|}[1/varpi_mu] varpi_mu (closed form): True
CHECK A N=2 [random_rational_0]: d^beta w_alpha(0) = delta^beta_alpha (|alpha|,|beta|<=N): True
CHECK A N=2: literal d^beta(z^alpha varpi_mu/alpha!)(0) equals M_of entries (explicit varpi, first 12x12 block): True
CHECK A N=2: explicit non-radial varpi=exp(q): mu d/dmu w_alpha^{(mu)} = (E-|alpha|) w_alpha^{(mu)} as functions: True
CHECK A N=3 [symbolic]: M^{(rho,mu)} unit lower-triangular (Leibniz, varpi_mu(0)=1): True
CHECK A N=3 [symbolic]: M M^{-1} = 1 (exact): True
CHECK A N=3 [symbolic]: mu d(M^{-1})/dmu = -M^{-1}(mu dM/dmu)M^{-1} (exact): True
CHECK A N=3 [symbolic]: -M^{-1}(mu dM)M^{-1} = D M^{-1} - M^{-1} D, D=diag(|beta|) (dilation grading): True
CHECK A N=3 [symbolic]: mu d/dmu P_alpha (complete formula) = (E-|alpha|) P_alpha for all |alpha|<=N: True
CHECK A N=3 [symbolic]: w_alpha = (z^alpha/alpha!) T_{N-|alpha|}[1/varpi_mu] varpi_mu (closed form): True
CHECK A N=3 [symbolic]: d^beta w_alpha(0) = delta^beta_alpha (|alpha|,|beta|<=N): True
CHECK A N=3 [random_rational_0]: M^{(rho,mu)} unit lower-triangular (Leibniz, varpi_mu(0)=1): True
CHECK A N=3 [random_rational_0]: M M^{-1} = 1 (exact): True
CHECK A N=3 [random_rational_0]: mu d(M^{-1})/dmu = -M^{-1}(mu dM/dmu)M^{-1} (exact): True
CHECK A N=3 [random_rational_0]: -M^{-1}(mu dM)M^{-1} = D M^{-1} - M^{-1} D, D=diag(|beta|) (dilation grading): True
CHECK A N=3 [random_rational_0]: mu d/dmu P_alpha (complete formula) = (E-|alpha|) P_alpha for all |alpha|<=N: True
CHECK A N=3 [random_rational_0]: w_alpha = (z^alpha/alpha!) T_{N-|alpha|}[1/varpi_mu] varpi_mu (closed form): True
CHECK A N=3 [random_rational_0]: d^beta w_alpha(0) = delta^beta_alpha (|alpha|,|beta|<=N): True
CHECK A N=3 [random_rational_1]: M^{(rho,mu)} unit lower-triangular (Leibniz, varpi_mu(0)=1): True
CHECK A N=3 [random_rational_1]: M M^{-1} = 1 (exact): True
CHECK A N=3 [random_rational_1]: mu d(M^{-1})/dmu = -M^{-1}(mu dM/dmu)M^{-1} (exact): True
CHECK A N=3 [random_rational_1]: -M^{-1}(mu dM)M^{-1} = D M^{-1} - M^{-1} D, D=diag(|beta|) (dilation grading): True
CHECK A N=3 [random_rational_1]: mu d/dmu P_alpha (complete formula) = (E-|alpha|) P_alpha for all |alpha|<=N: True
CHECK A N=3 [random_rational_1]: w_alpha = (z^alpha/alpha!) T_{N-|alpha|}[1/varpi_mu] varpi_mu (closed form): True
CHECK A N=3 [random_rational_1]: d^beta w_alpha(0) = delta^beta_alpha (|alpha|,|beta|<=N): True
CHECK A N=3: literal d^beta(z^alpha varpi_mu/alpha!)(0) equals M_of entries (explicit varpi, first 12x12 block): True
CHECK A N=3: explicit non-radial varpi=exp(q): mu d/dmu w_alpha^{(mu)} = (E-|alpha|) w_alpha^{(mu)} as functions: True
CHECK A N=4 [random_rational_0]: M^{(rho,mu)} unit lower-triangular (Leibniz, varpi_mu(0)=1): True
CHECK A N=4 [random_rational_0]: M M^{-1} = 1 (exact): True
CHECK A N=4 [random_rational_0]: mu d(M^{-1})/dmu = -M^{-1}(mu dM/dmu)M^{-1} (exact): True
CHECK A N=4 [random_rational_0]: -M^{-1}(mu dM)M^{-1} = D M^{-1} - M^{-1} D, D=diag(|beta|) (dilation grading): True
CHECK A N=4 [random_rational_0]: mu d/dmu P_alpha (complete formula) = (E-|alpha|) P_alpha for all |alpha|<=N: True
CHECK A N=4 [random_rational_0]: w_alpha = (z^alpha/alpha!) T_{N-|alpha|}[1/varpi_mu] varpi_mu (closed form): True
CHECK A N=4 [random_rational_0]: d^beta w_alpha(0) = delta^beta_alpha (|alpha|,|beta|<=N): True
CHECK A N=4 [random_rational_1]: M^{(rho,mu)} unit lower-triangular (Leibniz, varpi_mu(0)=1): True
CHECK A N=4 [random_rational_1]: M M^{-1} = 1 (exact): True
CHECK A N=4 [random_rational_1]: mu d(M^{-1})/dmu = -M^{-1}(mu dM/dmu)M^{-1} (exact): True
CHECK A N=4 [random_rational_1]: -M^{-1}(mu dM)M^{-1} = D M^{-1} - M^{-1} D, D=diag(|beta|) (dilation grading): True
CHECK A N=4 [random_rational_1]: mu d/dmu P_alpha (complete formula) = (E-|alpha|) P_alpha for all |alpha|<=N: True
CHECK A N=4 [random_rational_1]: w_alpha = (z^alpha/alpha!) T_{N-|alpha|}[1/varpi_mu] varpi_mu (closed form): True
CHECK A N=4 [random_rational_1]: d^beta w_alpha(0) = delta^beta_alpha (|alpha|,|beta|<=N): True
CHECK A N=4: literal d^beta(z^alpha varpi_mu/alpha!)(0) equals M_of entries (explicit varpi, first 12x12 block): True
CHECK A N=4: explicit non-radial varpi=exp(q): mu d/dmu w_alpha^{(mu)} = (E-|alpha|) w_alpha^{(mu)} as functions: True
# Section A consequence (exact, every N, every varpi):  mu d_mu w_alpha = (E-|alpha|) w_alpha ; hence  a_mu(u,alpha) = (-1)^{|
     alpha|+1} <u,(E-|alpha|) w_alpha>  and  calA coefficient (-1)^{|alpha|}<R u,(N-E) w_alpha> = (-1)^{|alpha|+1}<u,(E-|
     alpha|)w_alpha> (moments vanish).

# === Section B: finite exact end-to-end test (Euclidean toy homogeneous kernels; NOT the graph kernels) ===
CHECK B toy N=0 u=z^() r^{-4-N-|d|}: A_mu[phi] = mu d/dmu <u,W^mu phi> equals calA[phi]=<u,W^mu((N-E)phi)> exactly: True
CHECK B toy N=0 u=z^(): A_mu[phi] = sum_gamma a_gamma (-1)^|gamma| d^gamma phi(0) with a_gamma from the nu_k formula: True
CHECK B toy N=0: top-order (|gamma|=N) coefficients mu-independent: True
CHECK B toy N=0: a_gamma(mu) = (mu/mu0)^{N-|gamma|} a_gamma(mu0) for every gamma: True
# toy N=0 u=z^(): A_mu[phi] = 2*pi**2 ; coefficients {'()': '2*pi**2'}
CHECK B toy N=1 u=z^0 r^{-4-N-|d|}: A_mu[phi] = mu d/dmu <u,W^mu phi> equals calA[phi]=<u,W^mu((N-E)phi)> exactly: True
CHECK B toy N=1 u=z^0: A_mu[phi] = sum_gamma a_gamma (-1)^|gamma| d^gamma phi(0) with a_gamma from the nu_k formula: True
CHECK B toy N=1: top-order (|gamma|=N) coefficients mu-independent: True
CHECK B toy N=1: a_gamma(mu) = (mu/mu0)^{N-|gamma|} a_gamma(mu0) for every gamma: True
# toy N=1 u=z^0: A_mu[phi] = pi**2*(-23*lam - 3024)/6048 ; coefficients {'()': '-23*pi**2*lam/12096', '0': '-pi**2/2', '1':
     '0', '2': '0', '3': '0'}
CHECK B toy N=1 u=z^2 r^{-4-N-|d|}: A_mu[phi] = mu d/dmu <u,W^mu phi> equals calA[phi]=<u,W^mu((N-E)phi)> exactly: True
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CHECK B toy N=1 u=z^2: A_mu[phi] = sum_gamma a_gamma (-1)^|gamma| d^gamma phi(0) with a_gamma from the nu_k formula: True
CHECK B toy N=1: top-order (|gamma|=N) coefficients mu-independent: True
CHECK B toy N=1: a_gamma(mu) = (mu/mu0)^{N-|gamma|} a_gamma(mu0) for every gamma: True
# toy N=1 u=z^2: A_mu[phi] = pi**2*(140 - lam)/840 ; coefficients {'()': '-pi**2*lam/840', '0': '0', '1': '0', '2': '-pi
    **2/2', '3': '0'}
CHECK B toy N=2 u=z^01 r^{-4-N-|d|}: A_mu[phi] = mu d/dmu <u,W^mu phi> equals calA[phi]=<u,W^mu((N-E)phi)> exactly: True
CHECK B toy N=2 u=z^01: A_mu[phi] = sum_gamma a_gamma (-1)^|gamma| d^gamma phi(0) with a_gamma from the nu_k formula: True
CHECK B toy N=2: top-order (|gamma|=N) coefficients mu-independent: True
CHECK B toy N=2: a_gamma(mu) = (mu/mu0)^{N-|gamma|} a_gamma(mu0) for every gamma: True
# toy N=2 u=z^01: A_mu[phi] = pi**2*(lam*(5350 - 179969*lam) + 201600)/4838400 ; coefficients {'()': '-179969*pi**2*lam
    **2/4838400', '0': '-pi**2*lam/2016', '1': '59*pi**2*lam/96768', '2': '-pi**2*lam/13440', '3': '-pi**2*lam/32256', '00':
     '0', '01': 'pi**2/12', '02': '0', '03': '0', '11': '0', '12': '0', '13': '0', '22': '0', '23': '0', '33': '0'}
CHECK B toy N=2 u=z^00 r^{-4-N-|d|}: A_mu[phi] = mu d/dmu <u,W^mu phi> equals calA[phi]=<u,W^mu((N-E)phi)> exactly: True
CHECK B toy N=2 u=z^00: A_mu[phi] = sum_gamma a_gamma (-1)^|gamma| d^gamma phi(0) with a_gamma from the nu_k formula: True
CHECK B toy N=2: top-order (|gamma|=N) coefficients mu-independent: True
CHECK B toy N=2: a_gamma(mu) = (mu/mu0)^{N-|gamma|} a_gamma(mu0) for every gamma: True
# toy N=2 u=z^00: A_mu[phi] = pi**2*(lam*(17062838*lam + 575) - 3360000)/1612800 ; coefficients {'()': '8531419*pi**2*lam
    **2/2419200', '0': '23*pi**2*lam/32256', '1': '-pi**2*lam/2016', '2': 'pi**2*lam/6720', '3': '-pi**2*lam/16128', '00': '
    pi**2/8', '01': '0', '02': '0', '03': '0', '11': 'pi**2/24', '12': '0', '13': '0', '22': 'pi**2/24', '23': '0', '33': '
    pi**2/24'}

# === Section C: exact J(P,m) = int_{S^3} P (Q-i0)^{-m} d sigma (two independent routes) ===
CHECK C: int_{S^3}(Q-i0)^{-2} = -2 i pi^2 (recovers M.6 residue r = -2 i pi^2 c^2): True
CHECK C route B: int_{S^3}(Q-i0)^{-2} = -2 i pi^2: True
CHECK C numeric (Q - i eps)^{-2} check for (a,b)=(0,0): |extrap-exact|<1e-4: True
CHECK C numeric (Q - i eps)^{-4} check for (a,b)=(2,2): |extrap-exact|<1e-4: True
CHECK C numeric (Q - i eps)^{-5} check for (a,b)=(0,4): |extrap-exact|<1e-4: True
CHECK C numeric (Q - i eps)^{-6} check for (a,b)=(4,2): |extrap-exact|<1e-4: True

# === Section D: per-grade coefficients (N=0 audit, then N=1, 2, 3, 4) ===
CHECK d_N = (1,4,10,20,35): True
CHECK p_{e0} = -2 z0 (d^alpha S = c p_alpha (Q-i0)^{-1-|alpha|}): True
CHECK 185 relation vectors parsed from kernel_relation_matrix.txt: True
CHECK D N=0: reconstructed label coefficients sum_b R_{lambda b} a_b equal direct per-label evaluation (all 1 labels, all
    gamma): True
CHECK D N=0: every relation vector (0) annihilates the top-order, radial-specialization and general-varpi coefficient vectors:
     True
CHECK D N=0: B_mu = A_mu - calA_mu vanishes on top order |gamma|=N (route A vs route B, exact): True
# N=0: top-order residue slots nonzero = 1 of 1 ; B_mu = 0 (top exact; lower by Section A identity)
CHECK D N=1: reconstructed label coefficients sum_b R_{lambda b} a_b equal direct per-label evaluation (all 4 labels, all
    gamma): True
CHECK D N=1: every relation vector (0) annihilates the top-order, radial-specialization and general-varpi coefficient vectors:
     True
CHECK D N=1: B_mu = A_mu - calA_mu vanishes on top order |gamma|=N (route A vs route B, exact): True
# N=1: top-order residue slots nonzero = 4 of 16 ; B_mu = 0 (top exact; lower by Section A identity)
CHECK D N=2: reconstructed label coefficients sum_b R_{lambda b} a_b equal direct per-label evaluation (all 20 labels, all
    gamma): True
CHECK D N=2: every relation vector (10) annihilates the top-order, radial-specialization and general-varpi coefficient vectors
    : True
CHECK D N=2: B_mu = A_mu - calA_mu vanishes on top order |gamma|=N (route A vs route B, exact): True
# N=2: top-order residue slots nonzero = 22 of 100 ; B_mu = 0 (top exact; lower by Section A identity)
CHECK D N=3: reconstructed label coefficients sum_b R_{lambda b} a_b equal direct per-label evaluation (all 60 labels, all
    gamma): True
CHECK D N=3: every relation vector (40) annihilates the top-order, radial-specialization and general-varpi coefficient vectors
    : True
CHECK D N=3: B_mu = A_mu - calA_mu vanishes on top order |gamma|=N (route A vs route B, exact): True
# N=3: top-order residue slots nonzero = 68 of 400 ; B_mu = 0 (top exact; lower by Section A identity)
CHECK D N=4: reconstructed label coefficients sum_b R_{lambda b} a_b equal direct per-label evaluation (all 170 labels, all
    gamma): True
CHECK D N=4: every relation vector (135) annihilates the top-order, radial-specialization and general-varpi coefficient
    vectors: True
CHECK D N=4: B_mu = A_mu - calA_mu vanishes on top order |gamma|=N (route A vs route B, exact): True
# N=4: top-order residue slots nonzero = 194 of 1225 ; B_mu = 0 (top exact; lower by Section A identity)
CHECK D N=0 audit recovered: A_mu(S^2) = calA(S^2) = -2 i pi^2 c^2 delta: True
CHECK D test: Res(d_alpha u) = d_alpha Res(u) for every label of grade N<=3 and every alpha (340 identities): True
CHECK D test: top-order residue map is Lorentz-equivariant on all 6 so(1,3) generators (pull-back action), N=0..4: True
CHECK D test N=0: residue map = exact combination of audited equivariant maps phi_{k,k}: True
CHECK D test N=1: residue map = exact combination of audited equivariant maps phi_{k,k}: True
CHECK D test N=2: residue map = exact combination of audited equivariant maps phi_{k,k}: True
CHECK D test N=3: residue map = exact combination of audited equivariant maps phi_{k,k}: True
CHECK D test N=4: residue map = exact combination of audited equivariant maps phi_{k,k}: True
# residue map in equivariant coordinates (units i pi^2 c^2): {0: {'phi_00': '-2'}, 1: {'phi_00': '1/2'}, 2: {'phi_00':
    '-1/12', 'phi_11': '-1/48'}, 3: {'phi_00': '1/96', 'phi_11': '1/384'}, 4: {'phi_00': '-1/960', 'phi_11': '-1/3840', 'phi
    _22': '-1/61440'}}
# wrote scale_derivative_coefficients.json, dilation_defect_coefficients.json, scale_defect_difference.json
# Sections A-D done in 291.1s

# === Section E (B.4): C_scale(mu) collapse / adjoint defect / variationality / matching per reading ===
CHECK E: top-order (prescription-independent) part of C_scale sits in contact degree D=4: True
CHECK E: lower-order slots with k=N-|gamma| sit in D=4-k (k=1..4 -> D=3,2,1,0): True
CHECK E: radial specialization: odd-k lower-order coefficients vanish (parity): True
```

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CHECK E formal: collapse and adjoint defect of the top part are homogeneous of total derivative degree 4 (no cross-degree mixing): True
# formal         top D=4: collapse nnz=14112 (per N {'0': 744, '1': 4440, '2': 10278, '3': 10040, '4': 4604}), adjoint-defect nnz=16608, self-adjoint=False (1.1s)
CHECK E chi_plateau: collapse and adjoint defect of the top part are homogeneous of total derivative degree 4 (no cross-degree mixing): True
# chi_plateau    top D=4: collapse nnz=2522 (per N {'0': 744, '1': 1728, '2': 2386, '3': 1550, '4': 514}), adjoint-defect nnz=4090, self-adjoint=False (0.3s)
CHECK E prep_plateau: collapse and adjoint defect of the top part are homogeneous of total derivative degree 4 (no cross-degree mixing): True
# prep_plateau   top D=4: collapse nnz=4580 (per N {'0': 0, '1': 0, '2': 2274, '3': 4072, '4': 4604}), adjoint-defect nnz=4090, self-adjoint=False (0.5s)
CHECK E double_plateau: collapse and adjoint defect of the top part are homogeneous of total derivative degree 4 (no cross-degree mixing): True
# double_plateau top D=4: collapse nnz=490 (per N {'0': 0, '1': 0, '2': 474, '3': 514, '4': 514}), adjoint-defect nnz=0, self-adjoint=True (0.1s)
CHECK E double plateau: the collapsed prescription-independent scale tensor is NONZERO and self-adjoint (variational): True
CHECK E formal: collapsed top scale tensor is NOT self-adjoint (non-variational); full defect retained: True
CHECK E chi_plateau: collapsed top scale tensor is NOT self-adjoint (non-variational); full defect retained: True
CHECK E prep_plateau: collapsed top scale tensor is NOT self-adjoint (non-variational); full defect retained: True
CHECK E formal: N=0 component collapse has 744 nonzero keys (reproduces M.6 Theorem thm:M6-collapse(1)): True
CHECK E double plateau: N=0 and N=1 residue components collapse to 0 (N=0 consistent with M.6); N=2,3,4 carry the tensor: True
CHECK E double plateau: top scale tensor is NOT in span(E_R2, E_RC2) (exact rank test 2 -> 3): True
CHECK E double plateau: top scale tensor is NOT a multiple of E_{R2-(188813/88456)RC2}: True
CHECK E double plateau: top scale tensor is not a multiple of E_EH (degree mismatch D=4 vs 2): True
CHECK E: exact span solution verified: True
CHECK E double plateau: top scale tensor lies in the 5-dim span of Lorentz-covariant D=4 quadratic Euler tensors: True
CHECK E: exact span solution verified: True
CHECK E: exact span solution verified: True
CHECK E double plateau: d_mu E^{mu nu} != 0 for the top scale tensor (not linearized-gauge invariant): True
CHECK E double plateau: d_mu E^{mu nu} = 0 for E_R2, E_RC2 (control): True
# double plateau top tensor = Euler tensor of -71*L1/120 - 27*L2/80 - 11*L3/10 + 19*L4/30 - 13*L5/15  (covariant basis); R2 -> {'L1=h_{mn} Box^2 h^{mn}': '0', 'L2=h Box^2 h': '1', 'L3=(d^m h_{mn}) Box (d_r h^{rn})': '0', 'L4=h Box d^m d^n h_{mn}': '-2', 'L5=(d^m d^n h_{mn})^2': '1'} ; RC2 -> {'L1=h_{mn} Box^2 h^{mn}': '1/4', 'L2=h Box^2 h': '1/4', 'L3=(d^m h_{mn}) Box (d_r h^{rn})': '1/2', 'L4=h Box d^m d^n h_{mn}': '-1/2', 'L5=(d^m d^n h_{mn})^2': '1/2'}
CHECK E formal radial k=2: collapse homogeneous of degree D=2: True
CHECK E formal radial k=4: collapse homogeneous of degree D=0: True
CHECK E formal envelope k=1: ker Coll inside SA-parameter space (dimension check): True
# formal         envelope k=1 (D=3): params=120 rank Coll=120 dim SA=0 var image=0
CHECK E formal envelope k=2: ker Coll inside SA-parameter space (dimension check): True
# formal         envelope k=2 (D=2): params=84 rank Coll=84 dim SA=0 var image=0
CHECK E formal envelope k=3: ker Coll inside SA-parameter space (dimension check): True
# formal         envelope k=3 (D=1): params=56 rank Coll=56 dim SA=0 var image=0
CHECK E formal envelope k=4: ker Coll inside SA-parameter space (dimension check): True
# formal         envelope k=4 (D=0): params=35 rank Coll=35 dim SA=26 var image=26
CHECK E chi_plateau radial k=2: collapse homogeneous of degree D=2: True
CHECK E chi_plateau radial k=4: collapse homogeneous of degree D=0: True
CHECK E chi_plateau envelope k=1: ker Coll inside SA-parameter space (dimension check): True
# chi_plateau    envelope k=1 (D=3): params=120 rank Coll=120 dim SA=0 var image=0
CHECK E chi_plateau envelope k=2: ker Coll inside SA-parameter space (dimension check): True
# chi_plateau    envelope k=2 (D=2): params=84 rank Coll=84 dim SA=0 var image=0
CHECK E chi_plateau envelope k=3: ker Coll inside SA-parameter space (dimension check): True
# chi_plateau    envelope k=3 (D=1): params=56 rank Coll=56 dim SA=0 var image=0
CHECK E chi_plateau envelope k=4: ker Coll inside SA-parameter space (dimension check): True
# chi_plateau    envelope k=4 (D=0): params=35 rank Coll=35 dim SA=26 var image=26
CHECK E prep_plateau radial k=2: collapse homogeneous of degree D=2: True
CHECK E prep_plateau radial k=4: collapse homogeneous of degree D=0: True
CHECK E prep_plateau envelope k=1: ker Coll inside SA-parameter space (dimension check): True
# prep_plateau   envelope k=1 (D=3): params=120 rank Coll=120 dim SA=0 var image=0
CHECK E prep_plateau envelope k=2: ker Coll inside SA-parameter space (dimension check): True
# prep_plateau   envelope k=2 (D=2): params=84 rank Coll=84 dim SA=0 var image=0
CHECK E prep_plateau envelope k=3: ker Coll inside SA-parameter space (dimension check): True
# prep_plateau   envelope k=3 (D=1): params=56 rank Coll=56 dim SA=0 var image=0
CHECK E prep_plateau envelope k=4: ker Coll inside SA-parameter space (dimension check): True
# prep_plateau   envelope k=4 (D=0): params=35 rank Coll=35 dim SA=26 var image=26
CHECK E double_plateau radial k=2: collapse homogeneous of degree D=2: True
CHECK E double_plateau radial k=4: collapse homogeneous of degree D=0: True
CHECK E double_plateau envelope k=1: ker Coll inside SA-parameter space (dimension check): True
# double_plateau envelope k=1 (D=3): params=120 rank Coll=120 dim SA=0 var image=0
CHECK E double_plateau envelope k=2: ker Coll inside SA-parameter space (dimension check): True
# double_plateau envelope k=2 (D=2): params=84 rank Coll=84 dim SA=49 var image=49 targets cap image=0
CHECK E double_plateau envelope k=3: ker Coll inside SA-parameter space (dimension check): True
# double_plateau envelope k=3 (D=1): params=56 rank Coll=56 dim SA=0 var image=0
CHECK E double_plateau envelope k=4: ker Coll inside SA-parameter space (dimension check): True
# double_plateau envelope k=4 (D=0): params=35 rank Coll=35 dim SA=26 var image=26 targets cap image=1
# wrote full_scale_collapse_certificates.json (33.2s)
# wrote matched_scale_ledger.txt
# total runtime 324.3s ; 185 checks
ALL CHECKS PASSED
exit 0
\end{lstlisting}
```

```
\paragraph{Work package C (\texttt{wpC/time\_only\_helmholtz\_out.txt}).}
\begin{lstlisting}
# time_only_helmholtz.py output  (python 3.13.5, sympy 1.14.0; exact QQ; CONDITIONAL on (S-bv) via M.2 basis)
# XVI-M work package C: universal time-only Helmholtz classification (reading time_only)
CHECK dim C_coh = 3321, graph table 3282 lines: True
# input sha256: {'graph_general_table.txt': '4cd29d3c9f691340b314aecbfe7d6aef3590b9cee21dcdb486fe9fbcdd679e49', 'kernel_basis.
    json': 'fcbe2876abd2a6b410813f732356f829c970c22c3a333f7e4ec34909c0e0793b', 'm4_common.py': '932b716db827efc0cbb3d9eca91c
    633d9eecf932c50c1988008f98795337bf1e'}
CHECK dim C_reg = 3150 (N>=3), dim K block = 171 (N<=2): True
# === C.2 specialization tests ===
CHECK S_t(D_mu F) = D_mu^(t) S_t(F) for 400 random exact polynomials x 4 directions, and S_t(D^beta F) = D^(t)beta S_t(F) for
    random |beta|<=4: True
CHECK negative control: a D_1^(t) that differentiates sigma (sigma'(t) for d_1) violates the intertwining: True
CHECK sympy Function route: d_mu[F(sigma(t), chi(x), h(x))] = D_mu^(t) S_t(F) on 25 random polynomials (sigma a function of x0
     only): True
CHECK S_t(D_mu F) = D_mu^(t) S_t(F) on 960 actual collapse / adjoint-defect expressions (120 random contact variables x 4
    directions): True
# ######## reading time_only ########
# === C.3/C.4 blockwise: route 1 vs route 2, exact kernels ===
# blockwise pass 64.9s
CHECK [time_only] ROUTE 1 = ROUTE 2 for Coll_t: all 3321 columns equal exactly (nnz 446292): True
CHECK [time_only] ROUTE 1 = ROUTE 2 for H_t: all 3321 adjoint-defect columns equal exactly (nnz 573582): True
CHECK [time_only] grading: every time-only key in block (D, parity) has total derivative count D and parity of the block (
    blocks have disjoint rows): True
# supplementary mod-p ranks (all rows, p=1e9+7, 998244353) agree with exact ranks in every block: True
CHECK [time_only] ker Coll_t is contained in ker H_t (each ker Coll_t vector has zero time-only adjoint defect): True
CHECK [time_only] the 26 formal SA vectors (helmholtz_certificates.txt) lie in ker H_t (specialization: ker H_formal <= ker H_
    t): True
CHECK [time_only] D=0: dim Coll_t(ker H_t) = 26 (mod-p rank lower bound = dim ker H_t - dim ker Coll_t upper bound): True
CHECK [time_only] D=0: regression-preserving variational image dim Coll_t(ker H_t cap C_reg) = 26: True
CHECK [time_only] D=1: dim Coll_t(ker H_t) = 0 (mod-p rank lower bound = dim ker H_t - dim ker Coll_t upper bound): True
CHECK [time_only] D=1: regression-preserving variational image dim Coll_t(ker H_t cap C_reg) = 0: True
CHECK [time_only] D=2: dim Coll_t(ker H_t) = 0 (mod-p rank lower bound = dim ker H_t - dim ker Coll_t upper bound): True
CHECK [time_only] D=2: regression-preserving variational image dim Coll_t(ker H_t cap C_reg) = 0: True
CHECK [time_only] D=3: dim Coll_t(ker H_t) = 0 (mod-p rank lower bound = dim ker H_t - dim ker Coll_t upper bound): True
CHECK [time_only] D=3: regression-preserving variational image dim Coll_t(ker H_t cap C_reg) = 0: True
CHECK [time_only] D=4: dim Coll_t(ker H_t) = 0 (mod-p rank lower bound = dim ker H_t - dim ker Coll_t upper bound): True
CHECK [time_only] D=4: regression-preserving variational image dim Coll_t(ker H_t cap C_reg) = 0: True
# === CLASSIFICATION TABLE (time_only reading; conditional on (S-bv)) ===
# D | size | kerColl | imColl | kerH | Coll(kerH) | kerColl^Creg | kerColl^K | kerH^Creg | kerH^K | Coll(kerH^Creg)
# 0 | 35 | 0 | 35 | 26 | 26 | 0 | 0 | 26 | 0 | 26
# 1 | 160 | 0 | 160 | 0 | 0 | 0 | 0 | 0 | 0 | 0
# 2 | 440 | 0 | 440 | 0 | 0 | 0 | 0 | 0 | 0 | 0
# 3 | 944 | 0 | 944 | 0 | 0 | 0 | 0 | 0 | 0 | 0
# 4 | 1742 | 0 | 1742 | 0 | 0 | 0 | 0 | 0 | 0 | 0
# totals: {'size': 3321, 'size_Creg': 3150, 'size_K': 171, 'dim_ker_Coll_t': 0, 'dim_im_Coll_t': 3321, 'dim_ker_H_t': 26, 'dim
    _Coll_t_of_ker_H_t': 26, 'dim_ker_Coll_t_cap_Creg': 0, 'dim_ker_Coll_t_cap_K': 0, 'dim_ker_H_t_cap_Creg': 26, 'dim_ker_H
    _t_cap_K': 0, 'dim_regression_preserving_variational_image': 26}
# variational-image monomial structure per D: {"0": {"n_vectors": 26, "monomial_type_counts": {"sigma^(0) chi_0 h_0": 198}, "
    sigma_orders_present": [0], "has_chi_derivatives": false, "max_h_order": 0}}
# === C.5 functional probes (weight w = chi_I(x) sigma_eps(t); all preparation/switch derivatives retained) ===
CHECK [time_only] weighted P1 target: total derivative degree 0, parity 0: True
CHECK [time_only] weighted P1 target (time-only Euler tensor) is self-adjoint under the time-only Gate V adjoint: True
CHECK [time_only] weighted P2 target: total derivative degree 0, parity 0: True
CHECK [time_only] weighted P2 target (time-only Euler tensor) is self-adjoint under the time-only Gate V adjoint: True
CHECK [time_only] weighted EH target: total derivative degree 2, parity 0: True
CHECK [time_only] weighted EH target (time-only Euler tensor) is self-adjoint under the time-only Gate V adjoint: True
CHECK [time_only] weighted R2 target: total derivative degree 4, parity 0: True
CHECK [time_only] weighted R2 target (time-only Euler tensor) is self-adjoint under the time-only Gate V adjoint: True
CHECK [time_only] weighted RC2 target: total derivative degree 4, parity 0: True
CHECK [time_only] weighted RC2 target (time-only Euler tensor) is self-adjoint under the time-only Gate V adjoint: True
CHECK [time_only] divergence identity: d_mu E^{mu nu} = 0 for weighted R2 with w = chi sigma(t) (time-only jets): True
CHECK [time_only] divergence identity: d_mu E^{mu nu} = 0 for weighted RC2 with w = chi sigma(t) (time-only jets): True
CHECK [time_only] divergence identity: d_mu E_EH^{mu nu} = (1) (d_mu w) G_lin^{mu nu} for weighted EH (w = chi sigma(t)): True
CHECK [time_only] ultralocal weighted P1: divergence nonzero (not gauge invariant): True
CHECK [time_only] P1+P2/C_coh: explicit preimage of direction ['1', '-89/140'] reproduces the weighted target exactly: True
CHECK [time_only] P1+P2/C_coh: route-2 collapse of the preimage equals the target: True
CHECK [time_only] P1+P2/C_coh: preimage lies in ker H_t (variational): True
CHECK [time_only] P1+P2/C_coh: EL(1/2 <h, E_C[h]>) = E_C (independent sympy time-only jet route): True
CHECK [time_only] P1+P2/C_coh: EL(1/2 <h, E_C[h]>) = E_C via sympy euler_equations with sigma(x0), chi(x), h(x) Functions:
    True
# probe P1+P2   C_coh D=0 block=10 : dim(span cap im Coll_t)=1 ; members={'P1': False, 'P2': False} ; directions=[['1',
    '-89/140']] (0.1s)
CHECK [time_only] P1+P2/C_reg: explicit preimage of direction ['1', '-89/140'] reproduces the weighted target exactly: True
CHECK [time_only] P1+P2/C_reg: route-2 collapse of the preimage equals the target: True
CHECK [time_only] P1+P2/C_reg: preimage lies in ker H_t (variational): True
CHECK [time_only] P1+P2/C_reg: EL(1/2 <h, E_C[h]>) = E_C (independent sympy time-only jet route): True
CHECK [time_only] P1+P2/C_reg: EL(1/2 <h, E_C[h]>) = E_C via sympy euler_equations with sigma(x0), chi(x), h(x) Functions:
    True
```

```
# probe P1+P2   C_reg D=0 block=10 : dim(span cap im Coll_t)=1 ; members={'P1': False, 'P2': False} ; directions=[['1', '-89/140']] (0.0s)
# probe EH      C_coh D=2 block=84 : dim(span cap im Coll_t)=0 ; members={'EH': False} ; directions=[] (0.1s)
# probe EH      C_reg D=2 block=80 : dim(span cap im Coll_t)=0 ; members={'EH': False} ; directions=[] (0.0s)
# probe R2+RC2  C_coh D=4 block=292 : dim(span cap im Coll_t)=0 ; members={'R2': False, 'RC2': False} ; directions=[] (1.1s)
# probe R2+RC2  C_reg D=4 block=265 : dim(span cap im Coll_t)=0 ; members={'R2': False, 'RC2': False} ; directions=[] (0.6s)
# ######## reading time_only_chi_plateau ########
# === C.3/C.4 blockwise: route 1 vs route 2, exact kernels ===
# blockwise pass 36.4s
CHECK [time_only_chi_plateau] ROUTE 1 = ROUTE 2 for Coll_t: all 3321 columns equal exactly (nnz 91568): True
CHECK [time_only_chi_plateau] ROUTE 1 = ROUTE 2 for H_t: all 3321 adjoint-defect columns equal exactly (nnz 130110): True
CHECK [time_only_chi_plateau] grading: every time-only key in block (D, parity) has total derivative count D and parity of the block (blocks have disjoint rows): True
# supplementary mod-p ranks (all rows, p=1e9+7, 998244353) agree with exact ranks in every block: True
CHECK [time_only_chi_plateau] ker Coll_t is contained in ker H_t (each ker Coll_t vector has zero time-only adjoint defect): True
CHECK [time_only_chi_plateau] the 26 formal SA vectors (helmholtz_certificates.txt) lie in ker H_t (specialization: ker H_formal <= ker H_t): True
CHECK [time_only_chi_plateau] D=0: dim Coll_t(ker H_t) = 26 (mod-p rank lower bound = dim ker H_t - dim ker Coll_t upper bound): True
CHECK [time_only_chi_plateau] D=0: regression-preserving variational image dim Coll_t(ker H_t cap C_reg) = 26: True
CHECK [time_only_chi_plateau] D=1: dim Coll_t(ker H_t) = 0 (mod-p rank lower bound = dim ker H_t - dim ker Coll_t upper bound): True
CHECK [time_only_chi_plateau] D=1: regression-preserving variational image dim Coll_t(ker H_t cap C_reg) = 0: True
CHECK [time_only_chi_plateau] D=2: dim Coll_t(ker H_t) = 156 (mod-p rank lower bound = dim ker H_t - dim ker Coll_t upper bound): True
CHECK [time_only_chi_plateau] D=2: regression-preserving variational image dim Coll_t(ker H_t cap C_reg) = 156: True
CHECK [time_only_chi_plateau] D=3: dim Coll_t(ker H_t) = 0 (mod-p rank lower bound = dim ker H_t - dim ker Coll_t upper bound): True
CHECK [time_only_chi_plateau] D=3: regression-preserving variational image dim Coll_t(ker H_t cap C_reg) = 0: True
CHECK [time_only_chi_plateau] D=4: dim Coll_t(ker H_t) = 390 (mod-p rank lower bound = dim ker H_t - dim ker Coll_t upper bound): True
CHECK [time_only_chi_plateau] D=4: regression-preserving variational image dim Coll_t(ker H_t cap C_reg) = 390: True
# === CLASSIFICATION TABLE (time_only reading; conditional on (S-bv)) ===
# D | size | kerColl | imColl | kerH | Coll(kerH) | kerColl^Creg | kerColl^K | kerH^Creg | kerH^K | Coll(kerH^Creg)
# 0 | 35 | 0 | 35 | 26 | 26 | 0 | 0 | 26 | 0 | 26
# 1 | 160 | 0 | 160 | 0 | 0 | 0 | 0 | 0 | 0 | 0
# 2 | 440 | 0 | 440 | 156 | 156 | 0 | 0 | 156 | 0 | 156
# 3 | 944 | 0 | 944 | 0 | 0 | 0 | 0 | 0 | 0 | 0
# 4 | 1742 | 0 | 1742 | 390 | 390 | 0 | 0 | 390 | 0 | 390
# totals: {'size': 3321, 'size_Creg': 3150, 'size_K': 171, 'dim_ker_Coll_t': 0, 'dim_im_Coll_t': 3321, 'dim_ker_H_t': 572, 'dim_Coll_t_of_ker_H_t': 572, 'dim_ker_Coll_t_cap_Creg': 0, 'dim_ker_Coll_t_cap_K': 0, 'dim_ker_H_t_cap_Creg': 572, 'dim_ker_H_t_cap_K': 0, 'dim_regression_preserving_variational_image': 572}
# variational-image monomial structure per D: {"0": {"n_vectors": 26, "monomial_type_counts": {"sigma^(0) chi_0 h_0": 198}, "sigma_orders_present": [0], "has_chi_derivatives": false, "max_h_order": 0}, "2": {"n_vectors": 156, "monomial_type_counts": {"sigma^(0) chi_0 h_2": 1188}, "sigma_orders_present": [0], "has_chi_derivatives": false, "max_h_order": 2}, "4": {"n_vectors": 390, "monomial_type_counts": {"sigma^(0) chi_0 h_4": 2970}, "sigma_orders_present": [0], "has_chi_derivatives": false, "max_h_order": 4}}
# === C.5 functional probes (weight w = sigma_eps(t) [chi_I == 1]; all preparation/switch derivatives retained) ===
CHECK [time_only_chi_plateau] weighted P1 target: total derivative degree 0, parity 0: True
CHECK [time_only_chi_plateau] weighted P1 target (time-only Euler tensor) is self-adjoint under the time-only Gate V adjoint: True
CHECK [time_only_chi_plateau] weighted P2 target: total derivative degree 0, parity 0: True
CHECK [time_only_chi_plateau] weighted P2 target (time-only Euler tensor) is self-adjoint under the time-only Gate V adjoint: True
CHECK [time_only_chi_plateau] weighted EH target: total derivative degree 2, parity 0: True
CHECK [time_only_chi_plateau] weighted EH target (time-only Euler tensor) is self-adjoint under the time-only Gate V adjoint: True
CHECK [time_only_chi_plateau] weighted R2 target: total derivative degree 4, parity 0: True
CHECK [time_only_chi_plateau] weighted R2 target (time-only Euler tensor) is self-adjoint under the time-only Gate V adjoint: True
CHECK [time_only_chi_plateau] weighted RC2 target: total derivative degree 4, parity 0: True
CHECK [time_only_chi_plateau] weighted RC2 target (time-only Euler tensor) is self-adjoint under the time-only Gate V adjoint: True
CHECK [time_only_chi_plateau] divergence identity: d_mu E^{mu nu} = 0 for weighted R2 with w = chi sigma(t) (time-only jets): True
CHECK [time_only_chi_plateau] divergence identity: d_mu E^{mu nu} = 0 for weighted RC2 with w = chi sigma(t) (time-only jets): True
CHECK [time_only_chi_plateau] divergence identity: d_mu E_EH^{mu nu} = (1) (d_mu w) G_lin^{mu nu} for weighted EH (w = chi sigma(t)): True
CHECK [time_only_chi_plateau] ultralocal weighted P1: divergence nonzero (not gauge invariant): True
CHECK [time_only_chi_plateau] P1+P2/C_coh: explicit preimage of direction ['1', '-89/140'] reproduces the weighted target exactly: True
CHECK [time_only_chi_plateau] P1+P2/C_coh: route-2 collapse of the preimage equals the target: True
CHECK [time_only_chi_plateau] P1+P2/C_coh: preimage lies in ker H_t (variational): True
CHECK [time_only_chi_plateau] P1+P2/C_coh: EL(1/2 <h, E_C[h]>) = E_C (independent sympy time-only jet route): True
CHECK [time_only_chi_plateau] P1+P2/C_coh: EL(1/2 <h, E_C[h]>) = E_C via sympy euler_equations with sigma(x0), chi(x), h(x) Functions: True
# probe P1+P2   C_coh D=0 block=10 : dim(span cap im Coll_t)=1 ; members={'P1': False, 'P2': False} ; directions=[['1', '-89/140']] (0.1s)
```

```
CHECK [time_only_chi_plateau] P1+P2/C_reg: explicit preimage of direction ['1', '-89/140'] reproduces the weighted target exactly: True
CHECK [time_only_chi_plateau] P1+P2/C_reg: route-2 collapse of the preimage equals the target: True
CHECK [time_only_chi_plateau] P1+P2/C_reg: preimage lies in ker H_t (variational): True
CHECK [time_only_chi_plateau] P1+P2/C_reg: EL(1/2 <h, E_C[h]>) = E_C (independent sympy time-only jet route): True
CHECK [time_only_chi_plateau] P1+P2/C_reg: EL(1/2 <h, E_C[h]>) = E_C via sympy euler_equations with sigma(x0), chi(x), h(x) Functions: True
# probe P1+P2   C_reg D=0 block=10 : dim(span cap im Coll_t)=1 ; members={'P1': False, 'P2': False} ; directions=[['1', '-89/140']] (0.0s)
# probe EH      C_coh D=2 block=84 : dim(span cap im Coll_t)=0 ; members={'EH': False} ; directions=[] (0.1s)
# probe EH      C_reg D=2 block=80 : dim(span cap im Coll_t)=0 ; members={'EH': False} ; directions=[] (0.1s)
# probe R2+RC2  C_coh D=4 block=292 : dim(span cap im Coll_t)=0 ; members={'R2': False, 'RC2': False} ; directions=[] (1.5s)
# probe R2+RC2  C_reg D=4 block=265 : dim(span cap im Coll_t)=0 ; members={'R2': False, 'RC2': False} ; directions=[] (0.3s)
# === new time-only variational directions: reading time_only_chi_plateau, D>=1 ===
CHECK [time_only_chi_plateau] D=2: every new variational direction is sigma(t) x (purely SPATIAL derivative operator on h): no sigma derivative, no time derivative of h: True
CHECK [time_only_chi_plateau] D=2: EL(1/2 <h, E_C[h]>) = E_C for all 156 new directions (independent sympy time-only jet route) (2.2s): True
CHECK [time_only_chi_plateau] D=2: EL check via sympy euler_equations with sigma(x0), h(x) Functions on 3 sample directions: True
CHECK [time_only_chi_plateau] D=2: every new direction remains SA in the double-plateau reading (specialization sigma == 1): True
CHECK [time_only_chi_plateau] D=2: no new direction is SA in the (spacetime-formal sigma) chi_plateau reading: the directions are time-only-specific: True
# D=2 new directions: dim 156 ; h-derivative multi-indices used {'23': 198, '11': 198, '22': 198, '33': 198, '13': 198, '12': 198} ; support N-counts {4: 210} ; |gamma| counts {2: 210}
CHECK [time_only_chi_plateau] D=4: every new variational direction is sigma(t) x (purely SPATIAL derivative operator on h): no sigma derivative, no time derivative of h: True
CHECK [time_only_chi_plateau] D=4: EL(1/2 <h, E_C[h]>) = E_C for all 390 new directions (independent sympy time-only jet route) (6.3s): True
CHECK [time_only_chi_plateau] D=4: EL check via sympy euler_equations with sigma(x0), h(x) Functions on 3 sample directions: True
CHECK [time_only_chi_plateau] D=4: every new direction remains SA in the double-plateau reading (specialization sigma == 1): True
CHECK [time_only_chi_plateau] D=4: no new direction is SA in the (spacetime-formal sigma) chi_plateau reading: the directions are time-only-specific: True
# D=4 new directions: dim 390 ; h-derivative multi-indices used {'1123': 198, '2223': 198, '2333': 198, '1111': 198, '1122': 198, '1133': 198, '2222': 198, '2233': 198, '3333': 198, '1113': 198, '1223': 198, '1333': 198, '1112': 198, '1222': 198, '1233': 198} ; support N-counts {4: 525} ; |gamma| counts {4: 525}
CHECK [time_only_chi_plateau] the 26 ultralocal SA vectors are supported on N=4, gamma=0: True
CHECK [time_only_chi_plateau] D=2: span{ultralocal SA (26) x spatial gamma (6)} (156 independent vectors) lies in ker H_t and has dim = dim ker H_t = 156, hence equals it: True
CHECK [time_only_chi_plateau] D=4: span{ultralocal SA (26) x spatial gamma (15)} (390 independent vectors) lies in ker H_t and has dim = dim ker H_t = 390, hence equals it: True
CHECK supplementary (floating, 60 digits): Wronskian of rho, rho', ..., rho'''' (XII-C ramp profile) nonzero at s = 1/2, 1/3, 1/7: True
# Wronskian W(rho,...,rho^(4)) at s: {'1/2': '88060251340800.0', '1/3': '2.575905490314e+15', '1/7': '4.10243726191388e+17'}
# total runtime 154.1s
ALL CHECKS PASSED
\end{lstlisting}

\paragraph{Regression (\texttt{regression\_full\_out.txt}; \texttt{XVIM\_FULL\_WP=1}).}
\begin{lstlisting}
/workspace/XVI-M-artifacts/regression_M.py:828: SyntaxWarning: invalid escape sequence '\l'
  require("\label{rem:revlog-final}" in tex and "Revision log (final verification)" in tex,
/workspace/XVI-M-artifacts/regression_M.py:841: SyntaxWarning: invalid escape sequence '\m'
  require("M-C$_{\mathrm{data}}$" in tex and "explicitly specified cutoff" in tex,
/workspace/XVI-M-artifacts/regression_M.py:846: SyntaxWarning: invalid escape sequence '\m'
  require("D=0,2,4" in tex and "D=1,3" in tex and "mathcal K_{\mathrm{SA}}" in tex,
# regression_M.py output   (python 3.13.5, sympy 1.14.0)
CHECK provenance manifest lists the frozen predecessor files (18): True
CHECK frozen file unchanged: XVI-L.tex: True
CHECK frozen file unchanged: XVI-L-artifacts/README.txt: True
CHECK frozen file unchanged: XVI-L-artifacts/counterterm_map.py: True
CHECK frozen file unchanged: XVI-L-artifacts/counterterm_map_out.txt: True
CHECK frozen file unchanged: XVI-L-artifacts/kernel_inventory.py: True
CHECK frozen file unchanged: XVI-L-artifacts/kernel_inventory_out.txt: True
CHECK frozen file unchanged: XVI-L-artifacts/regression_L.py: True
CHECK frozen file unchanged: XVI-L-artifacts/regression_L_out.txt: True
CHECK frozen file unchanged: XVI-L-artifacts/source_full.py: True
CHECK frozen file unchanged: XVI-L-artifacts/source_full_out.txt: True
CHECK frozen file unchanged: XVI-L-artifacts/source_full_table.txt: True
CHECK frozen file unchanged: XVI-L-artifacts/two_edge_coeff_table.txt: True
CHECK frozen file unchanged: XVI-L-artifacts/two_edge_full.py: True
CHECK frozen file unchanged: XVI-L-artifacts/two_edge_full_out.txt: True
CHECK frozen file unchanged: XVI-L-artifacts/two_edge_support.txt: True
CHECK frozen file unchanged: XVI-L-artifacts/xvi_common.py: True
CHECK frozen file unchanged: XVI-K-artifacts/hessian_full_out.txt: True
CHECK frozen file unchanged: XVI-K-artifacts/xvi_common.py: True
CHECK graph_general_out.txt ends with ALL CHECKS PASSED: True
CHECK graph_general_out.txt: every CHECK line is True: True
```

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CHECK kernel_relations_out.txt ends with ALL CHECKS PASSED: True
CHECK kernel_relations_out.txt: every CHECK line is True: True
CHECK contact_space_out.txt ends with ALL CHECKS PASSED: True
CHECK contact_space_out.txt: every CHECK line is True: True
CHECK contact_collapse_out.txt ends with ALL CHECKS PASSED: True
CHECK contact_collapse_out.txt: every CHECK line is True: True
CHECK helmholtz_check_out.txt ends with ALL CHECKS PASSED: True
CHECK helmholtz_check_out.txt: every CHECK line is True: True
CHECK matching_probe_out.txt ends with ALL CHECKS PASSED: True
CHECK matching_probe_out.txt: every CHECK line is True: True
CHECK fixed_data_reading_out.txt ends with ALL CHECKS PASSED: True
CHECK fixed_data_reading_out.txt: every CHECK line is True: True
CHECK counterterm_functionals_out.txt ends with ALL CHECKS PASSED: True
CHECK counterterm_functionals_out.txt: every CHECK line is True: True
CHECK scaling_anomaly_out.txt ends with ALL CHECKS PASSED: True
CHECK scaling_anomaly_out.txt: every CHECK line is True: True
CHECK covariance_constraints_out.txt ends with ALL CHECKS PASSED: True
CHECK covariance_constraints_out.txt: every CHECK line is True: True
CHECK m7_legacy_reconstruction_out.txt ends with ALL CHECKS PASSED: True
CHECK m7_legacy_reconstruction_out.txt: every CHECK line is True: True
# relation spot-check seed=42 ; points=[['-6', '-1/2', '-1', '-1'], ['8', '9/4', '-8', '-7/2'], ['-2/5', '-9/5', '-1/2', '2']]
CHECK conventions.json records regression_M relation seed 42: True
CHECK conventions.json records contact_space seed 20261005: True
CHECK conventions.json records helmholtz_check seed 7: True
CHECK three off-cone rational points from seed 42: True
CHECK parsed 185 kernel relations + 45 wave-operator generators: True
CHECK every exported relation vanishes exactly on (d^alpha S)(d^beta S), S=1/Q, at 3 random rational points off the cone (
     independent route): True
CHECK dimensions consistent between M.2 and M.3 outputs: True
CHECK M.1 exact specialization check recorded: True
CHECK sign example (d0 S)(d1 S) = (1/2) S d0 d1 S with S = 1/Q: True
CHECK sign example numerators -4 z0 z1 and -8 z0 z1: True
CHECK M.4 matching: archived preimage collapses EXACTLY to the Gate V Euler tensor of int R_lin^2 (double plateau): True
CHECK M.4 matching: that collapsed tensor is self-adjoint: True
CHECK M.4 Helmholtz: formal SA vector #0 is the Gate V Euler derivative of (1/2) int <h, E_C[h]> (independent route): True
CHECK M.4 headline numbers recorded (formal Collapse injective; formal dim SA = 26): True
CHECK single-switch: every tested column matches d^gamma(chi ahatbar)=d^gamma(bar_a) (2622 columns): True
CHECK single-switch: implemented columns differ from d^gamma(ahatbar) (no-chi) on 2552 sample columns: True
CHECK single-switch: implemented columns differ from d^gamma(chi^2 ahatbar) (double-chi) on 2552 sample columns: True
CHECK single-switch: every tested column matches d(chi ahatbar), not d(chi^2 ahatbar) or d(ahatbar).: True
CHECK Certificate A variable C[N=4,{(),0222},()] located in contact variables: True
CHECK Cert A weighted symmetry: w_mn K^{mn}_ab = w_ab K^{ab}_mn holds: True
CHECK Cert A weighted symmetry: reversed w_ab K^{mn}_ab = w_mn K^{ab}_mn FAILS (as required): True
CHECK Cert A numeric: w00 K00_02 = -56/3 = w02 K02_00: True
CHECK matching_probe outs cover formal / chi-plateau / double-plateau weight readings: True
CHECK formal: dim ker Collapse = 0: True
CHECK formal: dim SA = 26: True
CHECK formal: dim variational image = 26: True
CHECK formal: variational D=3 dim = 0: True
CHECK chi_plateau: dim ker Collapse = 0: True
CHECK chi_plateau: dim SA = 26: True
CHECK chi_plateau: dim variational image = 26: True
CHECK chi_plateau: variational D=3 dim = 0: True
CHECK double_plateau: dim ker Collapse = 22: True
CHECK double_plateau: dim SA = 1347: True
CHECK double_plateau: dim variational image = 1325: True
CHECK double_plateau: variational D=3 dim = 8: True
CHECK Stage2 prep_plateau ker Coll = 21: True
CHECK Stage2 ultralocal dim 26: True
CHECK Stage2 transitions: Unresolved with domain-size bound 3321 (unsupported bound 22 removed): True
CHECK Stage2 transition upper bound = 3321 = dim C_coh: True
CHECK Stage2 criterion includes (iii) adjoint-defect map and time-only caveat: True
CHECK M.5 named theorem recorded: True
CHECK EOM/field-redef ledgers empty: True
CHECK [current status] manuscript has Stage2/M.5 labels: True
CHECK [current status] manuscript still references Input (S-bv) (now certified): True
CHECK [current status] S-bv not discharged by matching (certified analytically, thm:wpA-Sbv): True
CHECK Gate V: W((mu,mu))=1 and W((mu,nu))=2 for mu<nu: True
CHECK Gate V weights on Cert A ultralocal K: w=1 diag / w=2 off symmetry holds: True
CHECK Gate V weight w_00 = 1: True
CHECK Gate V weight w_01 = 2: True
CHECK Gate V weight w_02 = 2: True
CHECK Gate V weight w_03 = 2: True
CHECK Gate V weight w_11 = 1: True
CHECK Gate V weight w_12 = 2: True
CHECK Gate V weight w_13 = 2: True
CHECK Gate V weight w_22 = 1: True
CHECK Gate V weight w_23 = 2: True
CHECK Gate V weight w_33 = 1: True
CHECK conventions.json records eight third-order double-plateau variational directions: True
CHECK conventions.json: third-order dirs are not gravitational scalar counterterms: True
```

```
CHECK [current status] conventions.json: (S-bv) not discharged by matching: True
CHECK all three readings present in matching_probe_out: True
CHECK single switch factor: implemented collapse equals d^gamma(chi ahatbar): True
CHECK scaling_anomaly_out ends with ALL CHECKS PASSED: True
CHECK M.6 sphere integral certificate present: True
CHECK M.6 S^2 residue certificate present: True
CHECK M.6 double-plateau vanishing recorded: True
CHECK M.6 formal non-variational recorded: True
CHECK [historical record] M.6 keeps M-C unclaimed: True
CHECK [historical record] scaling_anomaly_cert.json: M_C_claimable is False: True
CHECK cert residue value: True
CHECK [historical record] archived M.6 cert records its (historical) (S-bv) input: True
CHECK independent contour residue simplifies to -1: True
CHECK independent contour: Iu0=-2 i pi: True
CHECK independent: sphere = pi * Iu0 = -2 i pi^2: True
CHECK [historical record] archived M.7 summary records its (historical) (S-bv) input: True
CHECK [historical record] M.7 keeps M-C/M-D unclaimed: True
CHECK [historical record] M.7 does not claim Target II: True
CHECK M.7 abstract equivariant dims = 1,1,3,3,6: True
CHECK M.7 rank = slots - dim sol (codim) and coker = dim codomain - rank, codomain specified: True
CHECK M.7 equivariant freedom 14 = 5 K + 9 C_reg: True
CHECK M.7 formal: (dim V, pi_K V, V cap K, pi_reg V, V cap C_reg) = (1, 0, 0, 1, 1): True
CHECK M.7 formal: dim(V cap K)+dim(V cap C_reg) <= dim V: True
CHECK M.7 formal: kernel/projection rank-nullity consistency: True
CHECK M.7 chi_plateau: (dim V, pi_K V, V cap K, pi_reg V, V cap C_reg) = (1, 0, 0, 1, 1): True
CHECK M.7 chi_plateau: dim(V cap K)+dim(V cap C_reg) <= dim V: True
CHECK M.7 chi_plateau: kernel/projection rank-nullity consistency: True
CHECK M.7 double_plateau: (dim V, pi_K V, V cap K, pi_reg V, V cap C_reg) = (12, 5, 3, 9, 7): True
CHECK M.7 double_plateau: dim(V cap K)+dim(V cap C_reg) <= dim V: True
CHECK M.7 double_plateau: kernel/projection rank-nullity consistency: True
CHECK M.7 formal: 1 exact rational V basis vectors archived: True
CHECK M.7 formal: every archived V vector re-verified: zero adjoint defect and Euler(1/2<h,E_C h>) = E_C (independent jet_
     euler route): True
CHECK M.7 formal: four objects recomputed from global rational vectors: True
CHECK M.7 chi_plateau: 1 exact rational V basis vectors archived: True
CHECK M.7 chi_plateau: every archived V vector re-verified: zero adjoint defect and Euler(1/2<h,E_C h>) = E_C (independent jet
     _euler route): True
CHECK M.7 chi_plateau: four objects recomputed from global rational vectors: True
CHECK M.7 double_plateau: 12 exact rational V basis vectors archived: True
CHECK M.7 double_plateau: every archived V vector re-verified: zero adjoint defect and Euler(1/2<h,E_C h>) = E_C (independent
     jet_euler route): True
CHECK M.7 double_plateau: four objects recomputed from global rational vectors: True
CHECK archived equivariant vector #1 is supported on N=1: True
CHECK M.7 independent test-function check: corrected N=1 contact map is equivariant under K01,K02,K03,J12 on d_mu delta: True
CHECK M.7 independent test-function check: the old (no-eta) identification FAILS boost equivariance: True
CHECK M.7 linear divergence identity: d_mu E^{mu nu}[R2] == 0 (double plateau): True
CHECK M.7 diagnosis: old eta^{mu mu}-weighted formula nonzero for R2: True
CHECK M.7 linear divergence identity: d_mu E^{mu nu}[RC2] == 0 (double plateau): True
CHECK M.7 diagnosis: old eta^{mu mu}-weighted formula nonzero for RC2: True
CHECK M.7 linear divergence identity: d_mu E^{mu nu}[R2-(188813/88456)RC2] == 0 (double plateau): True
CHECK M.7 diagnosis: old eta^{mu mu}-weighted formula nonzero for R2-(188813/88456)RC2: True
CHECK M.7: certified C_reg preimage collapses to a divergence-free tensor: True
CHECK M.7 summary: divergence identity recorded as holding for R2/RC2/combination in all readings: True
CHECK M.7: matched coefficient 188813/88456 unchanged: True
CHECK covariance_matrix: 14 total; rank/codim and coker columns distinguished: True
CHECK M.7 legacy: old 'dp_all=5' reproduced only with the truncating reduction: True
CHECK M.8/M.9 completion ledger stub present: True
CHECK [historical record] completion ledger recommends waiting on M.8: True
CHECK [historical record] M.7 status downgraded to 'Constructed after audit' (exactly verified items only): True
CHECK [historical record] M.6: C_anom is the N=0 component (archived scope): True
CHECK [current status] conventions.json: broad M-C Not achieved (Unresolved); M-C_data Satisfied; M-C_curv Proved obstructed:
     True
CHECK [historical record] conventions.json keeps the audit-2 M-C status string (superseded): True
CHECK [current status] manuscript carries the (retained) M.7 audit log remark: True
CHECK [current status] manuscript no longer claims a nonzero curvature-square divergence: True
# M.7 covariance regression checks OK (corrected identities)
# === work packages A/B/C ===
CHECK wpA archive file present: kernel_identification.tex: True
CHECK wpA archive file present: kernel_identification_checks.py: True
CHECK wpA archive file present: kernel_identification_out.txt: True
CHECK wpA archive file present: kernel_conventions.json: True
CHECK wpB archive file present: scaling_prescription_identity.tex: True
CHECK wpB archive file present: higher_grade_scaling.py: True
CHECK wpB archive file present: higher_grade_scaling_out.txt: True
CHECK wpB archive file present: _wpB_state.json: True
CHECK wpB archive file present: _wpB_sectionD.json: True
CHECK wpB archive file present: scale_derivative_coefficients.json: True
CHECK wpB archive file present: dilation_defect_coefficients.json: True
CHECK wpB archive file present: scale_defect_difference.json: True
CHECK wpB archive file present: full_scale_collapse_certificates.json: True
CHECK wpB archive file present: matched_scale_ledger.txt: True
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CHECK wpC archive file present: time_only_section.tex: True
CHECK wpC archive file present: time_only_helmholtz.py: True
CHECK wpC archive file present: time_only_helmholtz_out.txt: True
CHECK wpC archive file present: time_only_summary.json: True
CHECK wpC archive file present: time_only_collapse_certificates.json: True
CHECK wpC archive file present: time_only_sa_certificates.json: True
CHECK wpC archive file present: time_only_matching_certificates.json: True
CHECK wpC archive file present: time_only_specialization_tests.txt: True
CHECK wpA script reruns with exit 0: True
CHECK wpA rerun output is byte-identical to the archived kernel_identification_out.txt: True
CHECK wpA output ends with ALL CHECKS PASSED: True
CHECK wpA: every CHECK line is True: True
CHECK wpA ledger: Certified, c=1/(4 pi^2), side Q-i0, across nonzero null cone: True
CHECK [current status] conventions.json gate_Sbv_status: Certified with c=1/(4 pi^2), side Q-i0 (agrees with wpA ledger): True
CHECK [current status] conventions.json normalization table: S row certified with c and boundary side: True
CHECK r = -2 i pi^2 c^2 = -i/(8 pi^2) with certified c: True
CHECK scale unit i pi^2 c^2 = i/(16 pi^2) with certified c: True
CHECK damped W_0 closed form 1/(4 pi^2 (a^2+r^2)) (independent sympy integral): True
CHECK a^2 + r^2 = -((z0 - i delta)^2 - r^2): W_0 = -(1/4pi^2)(Q - i0 sgn z0)^{-1}: True
CHECK sgn cancellation for sgn z0 = +1: delta(Q) coefficient = c i pi (side Q-i0): True
CHECK sgn cancellation for sgn z0 = -1: delta(Q) coefficient = c i pi (side Q-i0): True
CHECK wpB output ends with ALL CHECKS PASSED (followed by 'exit 0'): True
CHECK wpB: every CHECK line is True: True
CHECK wpB indep. rho=0 (symbolic): M M^{-1} = 1: True
CHECK wpB indep. rho=0 (symbolic): mu d(M^{-1})/dmu = D M^{-1} - M^{-1} D: True
CHECK wpB indep. rho=1 (symbolic): M M^{-1} = 1: True
CHECK wpB indep. rho=1 (symbolic): mu d(M^{-1})/dmu = D M^{-1} - M^{-1} D: True
CHECK wpB indep. rho=2 (symbolic): M M^{-1} = 1: True
CHECK wpB indep. rho=2 (symbolic): mu d(M^{-1})/dmu = D M^{-1} - M^{-1} D: True
CHECK wpB indep. rho=3 (random rational): M M^{-1} = 1: True
CHECK wpB indep. rho=3 (random rational): mu d(M^{-1})/dmu = D M^{-1} - M^{-1} D: True
CHECK wpB double plateau: rank(R2,RC2)=2 -> rank(R2,RC2,scale)=3 (scale tensor not in span(E_R2,E_Ric2)): True
CHECK conventions.json records the double-plateau scale 5-vector: True
CHECK wpB: double-plateau top scale tensor is self-adjoint/variational (certificate): True
CHECK wpB: top scale tensor not variational in formal / chi-plateau / prep-plateau readings: True
CHECK wpB: C_top has 289 entries, all in contact degree D=4: True
CHECK wpB: C_top N=0 coordinate = -2 (units i pi^2 c^2), i.e. r: True
CHECK wpB: residue map in audited equivariant coordinates as stated in the manuscript: True
CHECK wpC output ends with ALL CHECKS PASSED: True
CHECK wpC: every CHECK line is True: True
CHECK wpC: ker Coll_t = 0 for all D: True
CHECK wpC: ker H_t by D = (26,0,0,0,0): True
CHECK wpC: regression-preserving variational image = 26 ultralocals: True
CHECK wpC: chi==1 time-only ker H_t by D = (26,0,156,0,390), total 572: True
CHECK wpC: route 1 = route 2 (both readings): True
CHECK wpC time-only: only P1-(89/140)P2 is an image among P1,P2 (C_coh and C_reg): True
CHECK wpC time-only: weighted EH and R2/RC2 are not images (double-plateau matching does not survive time-only): True
CHECK wpC time-only chi==1: only P1-(89/140)P2 is an image among P1,P2 (C_coh and C_reg): True
CHECK wpC time-only chi==1: weighted EH and R2/RC2 are not images (double-plateau matching does not survive time-only): True
CHECK wpC: chi==1 time-only D=4 kernel (dim 390) supported on N=4 only (used in cor:wpB-timeonly): True
CHECK wpC independent: Wronskian W(rho,...,rho^(4))(1/2) = 88060251340800.0 (archived 8.80602513408e13); nonzero: True
CHECK XVIM_FULL_WP: wpB rerun reproduces higher_grade_scaling_out.txt modulo timing: True
CHECK XVIM_FULL_WP: wpC rerun reproduces time_only_helmholtz_out.txt modulo timing: True
# === audit 3 ===
CHECK [current status] conventions.json: separating functional basis order and ell=(1,1,-1,1,1): True
CHECK audit3: ell(v_scale) = -71/120-27/80+11/10+19/30-13/15 = -1/16 != 0 (archived conventions vector): True
CHECK audit3: ell(v_R2) = 0 and ell(v_Ric2) = 0: True
CHECK audit3: ell annihilates archived conventions vector R2: True
CHECK audit3: ell annihilates archived conventions vector RC2: True
CHECK audit3 sympy: pure gauge h_00=f(t): EL_f(L_1..L_5) = 2*(1,1,-1,1,1) f'''' (so L_j == (1,1,-1,1,1) f f'''' mod total
      derivatives): True
CHECK audit3 sympy: R_lin^2 and Ric_lin^2 vanish on pure gauge h_00=f(t): True
CHECK audit3 sympy: scale functional on pure gauge: EL_f = -(1/8) f'''' = 2*(-1/16) f'''' (units i/(16 pi^2)): True
CHECK audit3: basis check: (0,1,0,-2,1) in the L_1..L_5 basis gives the Gate V Euler tensor of R_lin^2: True
CHECK audit3: recollapsed C_top (double plateau, 490 entries) equals EXACTLY the Euler tensor of the displayed I_top_scale (
      units i pi^2 c^2): True
CHECK audit3: f'''' coefficient of E^00[f e_00] of the recollapsed C_top = -1/8 = 2*ell(v_scale): True
CHECK audit3: lem:bv(c) Beta-integral convolution (1/2pi) hat u_m * hat u_n = hat u_{m+n}, 1<=m,n<=3: True
CHECK [current status] audit3: wpA damping parameter renamed epsilon -> tau (script and fragment): True
CHECK audit3: wpA output contains the variable-regulator check (lem:wpA-varreg): True
CHECK [current status] manuscript defines \label{thm:wpA-Sbv}: True
CHECK [current status] manuscript defines \label{lem:wpA-varreg}: True
CHECK [current status] manuscript defines \label{thm:wpB-B0}: True
CHECK [current status] manuscript defines \label{thm:wpB-dp}: True
CHECK [current status] manuscript defines \label{prop:wpB-sep}: True
CHECK [current status] manuscript defines \label{rem:wpB-topfull}: True
CHECK [current status] manuscript defines \label{cor:wpB-MC}: True
CHECK [current status] manuscript defines \label{def:MCcurv}: True
CHECK [current status] manuscript defines \label{cor:wpB-timeonly}: True
CHECK [current status] manuscript defines \label{thm:to-timeonly}: True
CHECK [current status] manuscript defines \label{prop:to-actual}: True
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CHECK [current status] manuscript defines \label{lem:to-indep}: True
CHECK [current status] manuscript defines \label{rem:to-joint}: True
CHECK [current status] manuscript defines \label{rem:revlog}: True
CHECK [current status] manuscript defines \label{rem:revlog3}: True
CHECK [current status] manuscript defines \label{rem:revlog-final}: True
CHECK [current status] manuscript carries 'Revision log (audit 3)' and the retained cycle log: True
CHECK [current status] manuscript states the top-versus-full scale-tensor sentence (abstract, status box, wpB box, theorem): True
CHECK [current status] manuscript no longer calls the full double-plateau scale tensor variational/Lorentz covariant: True
CHECK [current status] manuscript: M-C_curv Proved obstructed; broad M-C not achieved; graph-specific wording: True
CHECK [current status] manuscript no longer marks broad M-C Proved obstructed: True
CHECK [current status] manuscript: time-only switch-jet reduction removed; joint-family condition kept: True
CHECK [current status] manuscript: corrected S scope sentence: True
CHECK [current status] manuscript: weighted curvature-square divergence wording corrected: True
CHECK [current status] manuscript: bv(c) shortcut removed; damping parameter is tau: True
CHECK [current status] run listing: annotations are # comments: True
CHECK [current status] manuscript no longer states that S-bv is conditional: True
CHECK [current status] manuscript keeps 'Euclidean motivation is not the certification': True
CHECK [current status] manuscript states the double-plateau matching does not survive the time-only reading: True
# work package A/B/C regression checks OK
# === final verification ===
CHECK [current status] provenance pin file exists: XVI-J_local.tex -> /workspace/XVI-M-artifacts/provenance/XVI-J_local.tex: True
CHECK [current status] provenance SHA-256 match for XVI-J_local.tex: True
CHECK [current status] provenance pin file exists: XVI-J_github_fetch.tex -> /workspace/XVI-M-artifacts/provenance/XVI-J_github_fetch.tex: True
CHECK [current status] provenance SHA-256 match for XVI-J_github_fetch.tex: True
CHECK [current status] provenance pin file exists: XVI-J_corpus_eqW_DF_rational_propq.txt -> /workspace/XVI-M-artifacts/provenance/XVI-J_corpus_eqW_DF_rational_propq.txt: True
CHECK [current status] provenance SHA-256 match for XVI-J_corpus_eqW_DF_rational_propq.txt: True
CHECK [current status] provenance pin file exists: XI-E_corpus_eq_h.txt -> /workspace/XVI-M-artifacts/provenance/XI-E_corpus_eq_h.txt: True
CHECK [current status] provenance SHA-256 match for XI-E_corpus_eq_h.txt: True
CHECK [current status] provenance pin file exists: XII-C_corpus_weighted_curvature.txt -> /workspace/XVI-M-artifacts/provenance/XII-C_corpus_weighted_curvature.txt: True
CHECK [current status] provenance SHA-256 match for XII-C_corpus_weighted_curvature.txt: True
CHECK [current status] provenance pin file exists: XVI-L_corpus_thm_L6.txt -> /workspace/XVI-M-artifacts/provenance/XVI-L_corpus_thm_L6.txt: True
CHECK [current status] provenance SHA-256 match for XVI-L_corpus_thm_L6.txt: True
CHECK [current status] provenance pin file exists: XVI-K_corpus_prop_hess.txt -> /workspace/XVI-M-artifacts/provenance/XVI-K_corpus_prop_hess.txt: True
CHECK [current status] provenance SHA-256 match for XVI-K_corpus_prop_hess.txt: True
CHECK [current status] provenance pin file exists: ../XVI-L.tex -> /workspace/XVI-L.tex: True
CHECK [current status] provenance SHA-256 match for ../XVI-L.tex: True
CHECK [current status] provenance pin file exists: ../XVI-K.tex -> /workspace/XVI-K.tex: True
CHECK [current status] provenance SHA-256 match for ../XVI-K.tex: True
CHECK [current status] predecessor_pins note: XVI-J eq:rational is corroboration, not frozen XVI-L: True
CHECK [current status] XVI-J eq:rational marked corroboration in pins: True
CHECK [current status] archived corpus excerpt present: XVI-J_corpus_eqW_DF_rational_propq.txt: True
CHECK [current status] archived corpus excerpt present: XI-E_corpus_eq_h.txt: True
CHECK [current status] archived corpus excerpt present: XII-C_corpus_weighted_curvature.txt: True
CHECK [current status] archived corpus excerpt present: XVI-L_corpus_thm_L6.txt: True
CHECK [current status] archived corpus excerpt present: XVI-K_corpus_prop_hess.txt: True
CHECK [current status] manuscript carries Revision log (final verification): True
CHECK [current status] manuscript: explicitly checked kernel identification: True
CHECK [current status] manuscript body (before final revlog) has no 'resting on a new input': True
CHECK [current status] manuscript: M.2 spot-check wording: True
CHECK [current status] manuscript: prop:M7-id (EH) locally constant: True
CHECK [current status] manuscript body (before final revlog) has no 'uncontrolled': True
CHECK [current status] manuscript: broad M-C matched-scale criterion and M-C_data: True
CHECK [current status] manuscript cites XVI-L thm:L6 and the undetermined-cutoff sentence: True
CHECK [current status] manuscript: prop:to-chiplateau cases display with K_SA: True
CHECK [current status] manuscript: odd-order anti-self-adjoint reason: True
CHECK [current status] wpC summary: odd-D=1 anti-self-adjoint exact, ker H_t=0: True
CHECK [current status] wpC summary: odd-D=3 anti-self-adjoint exact, ker H_t=0: True
CHECK [current status] appendix lists XVIM_FULL_WP full rerun writing regression_full_out.txt: True
CHECK [current status] manuscript has predecessor provenance table: True
# final verification checks OK
ALL CHECKS PASSED
\end{lstlisting}

\end{document}
```

3.6 RELEASE_MANIFEST

```
XVI-N r12 RELEASE MANIFEST (PRE-PDF; embedded in PDF; NO PDF hash / final page count)
date: 2026-10-07
revision: r12
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```
massless_theorem: conditional_on_inherited_identities (Route A)
route_B: not_claimed_discharged
arbitrary_prescribed_m_gt_0: unresolved
XVI-M.tex sha256 (unchanged): 41aa109935f40ddee3109de62c46329c189b5408007970cc90ed5611dc4739ea
output_root: build/outputs
interpreter: sys.executable
note: final page count and PDF sha256 appear ONLY in RELEASE_MANIFEST_EXTERNAL.txt
regression_math: 44 pass / 0 fail
regression_pack: 37 pass / 0 fail
small-mass: 0 <= m < 1.3715e-4 status=certified

69d60556798927bfd5c0c6dc9fcfcdd1754fff2609a48d2710cc77b58add1538  manuscript/XVI-N.tex
0846f9fec6f023ddd66b45ad6f26abec4a2716a9394cdb2c010fb07a146b8bbf  manuscript/body/XVI-N-body.tex
7a3601164609aa4d1e0466086258c6a95c722f523a5a34d7ae52c72ab8cc710a  manuscript/notes/small_mass_estimate.tex
41aa109935f40ddee3109de62c46329c189b5408007970cc90ed5611dc4739ea  provenance/XVI-M.tex
```